

Safeguarding Our Oceans

Strengthening marine protection
in New Zealand

Kate Mulcahy, Raewyn Peart
and Abbie Bull

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in New Zealand**

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chapter one



Introduction

Context

Historically, New Zealand has achieved some significant management successes in marine protection, and has been recognised on the international stage as a trailblazer in the field. Notably, the Marine Reserves Act 1971 and the introduction of the fisheries quota management system in 1986, were ahead of their time. However, in recent years, progress has stalled and numerous marine species are threatened. Today, New Zealand is falling behind international best practice in oceans management.

In 1999, the Parliamentary Commissioner for the Environment noted that it has become “*something of a cliché*” to draw attention to the benefits provided by New Zealand’s marine environment and the need to manage it better.¹ Unfortunately, 13 years later, little progress has been achieved.

When it was promulgated in 1971, the Marine Reserves Act was an innovative piece of legislation, but it is now out-of-date and not reflective of international best practice. The Act provides for the preservation of marine areas for the purpose of scientific study, with extractive activities effectively excluded. The inflexibility of the legislation has meant that a range of other tools, which do not have the protection of the marine environment as their core purpose, have been used to achieve less restrictive protection measures in New Zealand’s marine area.

In 2000, the Department of Conservation released the ‘New Zealand Biodiversity Strategy’ (Biodiversity Strategy) which emphasised the need for the establishment of a representative network of marine protected areas. It committed the government to achieving a target of protecting ten per cent of New Zealand’s marine environment by 2010 “*in view of establishing a network of representative protected marine areas*”. The strategy also highlighted the need to use a range of tools in addition to the Marine Reserves Act (such as those under the Fisheries Act 1996) to achieve this network. In doing so, greater integration between these different tools would be required.²

In the Biodiversity Strategy, the government also committed to reviewing the Marine Reserves Act 1971, “*to better provide for the protection of marine biodiversity, including extending its jurisdiction to protect marine biodiversity within and beyond the 12 mile limit.*”³ The Department of Conservation soon embarked on the review, holding numerous hui and public meetings between November 2000 and February 2001. It received 256 submissions on a discussion document.⁴ The Department identified a number of deficiencies with the Marine Reserves Act which rendered it inconsistent with current government policy. Key criticisms included:⁵

- The Act is intended to preserve areas in their natural state for the purpose of scientific study only and does not recognise the importance of other benefits derived from protecting marine life
- The Act is limited to the territorial sea and it is not possible to establish marine reserves in the exclusive economic zone
- The Act provides no guidelines regarding Treaty of Waitangi obligations
- The Act is devoid of appropriate linkages to recent marine management legislation
- Timeframes and processes are vague or unspecified, causing delays in processing applications

As a result of the review, replacement legislation was drafted. A Marine Reserves Bill was introduced into Parliament on 7 June 2002 and passed its first reading in the House of Representatives on 15 October that year. The Bill was referred to the Local Government and Environment Select Committee where it received 165 submissions. The Select Committee conducted a hearings process from February 2003.⁶ Progress on the Bill then stalled. This was due to the controversy over Māori ownership of the foreshore and seabed, which erupted in 2003, and which brought all significant marine policy initiatives to a halt. Subsequent to this, the Labour-led minority government was unable to garner sufficient votes for the Bill's passage, and it languished in Parliament.

In 2005, the Department of Conservation and the Ministry of Fisheries jointly published the 'Marine Protected Areas Policy and Implementation Plan' (MPA Policy).⁷ This was intended to give effect to the obligations set out in the Biodiversity Strategy and to employ a strategic approach to the establishment of marine protected areas. The objective of the MPA Policy is to "*Protect marine biodiversity by establishing a network of MPAs that is comprehensive and representative of New Zealand's marine habitats and ecosystems.*"⁸ The network is intended to protect representative examples of all marine habitats and ecosystems, as well as outstanding, rare, distinctive or internationally or nationally important marine habitats and ecosystems.

In 2008, the MPA Policy was further elaborated upon by the 'Marine Protected Areas: Classification, Protection Standard and Implementation Guidelines' (MPA Guidelines).⁹ The guidelines provided for the establishment of 14 community-based marine protection planning fora, which were to be tasked with identifying representative networks of marine protected areas in each coastal biogeographic region. The networks could potentially incorporate marine reserves as well as

areas protected under other legislation such as the Fisheries Act, the Resource Management Act 1991, and the Marine Mammals Protection Act 1978.

Only two fora were established before the further implementation of the MPA Policy was put on hold. The West Coast Marine Protection Forum was set up in 2005, to identify prospective marine protected areas along the west coast of the South Island. In 2008, the Sub-Antarctic Regional Marine Protection Planning Forum was established, to focus on protection in the territorial seas surrounding the Sub-Antarctic Islands. Both fora reported their recommendations to government in 2010, but at the time of writing these had yet to be implemented. Whilst the two fora met with some success, their experiences have highlighted the inadequacies of the current management framework, and the need for a wider overhaul of the system.

As a result, significant gaps in New Zealand's system of marine protection remain. In August 2012 a broad scale analysis of existing marine protection in New Zealand's territorial sea, undertaken by the Department of Conservation, found that "... *there are many large gaps in the current network of MPAs.*" It found that "[i]n some bioregions, very few, if any habitats are" in marine protected areas which meet the protection standard required.¹⁰

The National-led minority government has stated its intention to review the Marine Reserves Bill later in 2012. This provides an excellent opportunity to overhaul the current fragmented system of marine protection and to replace it with a comprehensive piece of legislation that reflects international best practice, provides for flexible management tools, ensures consistency of management, and creates a coherent, transparent establishment process.

Figure 1.1 Summary of the historical context to marine protected areas legislation in New Zealand

1971	Marine Reserves Act promulgated
1993	New Zealand ratifies the Convention on Biological Diversity
2000	New Zealand Biodiversity Strategy sets a target of protecting 10 per cent of New Zealand's marine environment by 2010 and indicates the need to review the Marine Reserves Act
2002	Marine Reserves Bill introduced to Parliament
2003	Progress on Bill stalls
2005	Marine Protected Areas Policy and Implementation Plan finalised
2008	Marine Protected Areas Classification, Protection Standard and Implementation Guidelines finalised Further roll-out of MPA Policy around the country put on hold after only 2 fora established out of 14 coastal biogeographic regions
2012	Government indicates an intention to revise the Marine Reserves Act

Purpose of report

This Environmental Defence Society policy report is designed to inform constructive debate on new marine protected areas legislation for New Zealand. It reviews the country's current marine protection framework, developments in other countries, and international best practice. It identifies weaknesses in New Zealand's current approaches, and develops recommendations for the design of new marine protection legislation.

Structure and methodology

The report is structured in three parts. The first part provides a description of marine protected areas; a review of their value and what they can achieve; and an overview of international experience and best practice. The second part contains an assessment of New Zealand's performance; a discussion of matters to be considered in the design of new legislation; and the identification of some legislative options. The third part of the report sets out eight case studies which cover experiences of establishing marine protected areas in New Zealand and other countries.

The international case studies include English marine conservation zones, marine protection under the California Marine Life Protection Act 1999, Australian Commonwealth marine reserves and New South Wales marine parks. The New Zealand case studies include approaches to marine protection by the Guardians of Fiordland, the West Coast Marine Protection Forum, the Sub-Antarctic Regional Marine Protection Planning Forum and Ngātiwai.

The material for the case studies was obtained from literature reviews and, in some cases, interviews with people involved in the marine protected area processes. The case study material has been synthesised and lessons learnt identified. The key findings from the case studies have been incorporated into the main body of the report and have informed the development of recommendations for new legislation in New Zealand.

Acknowledgements

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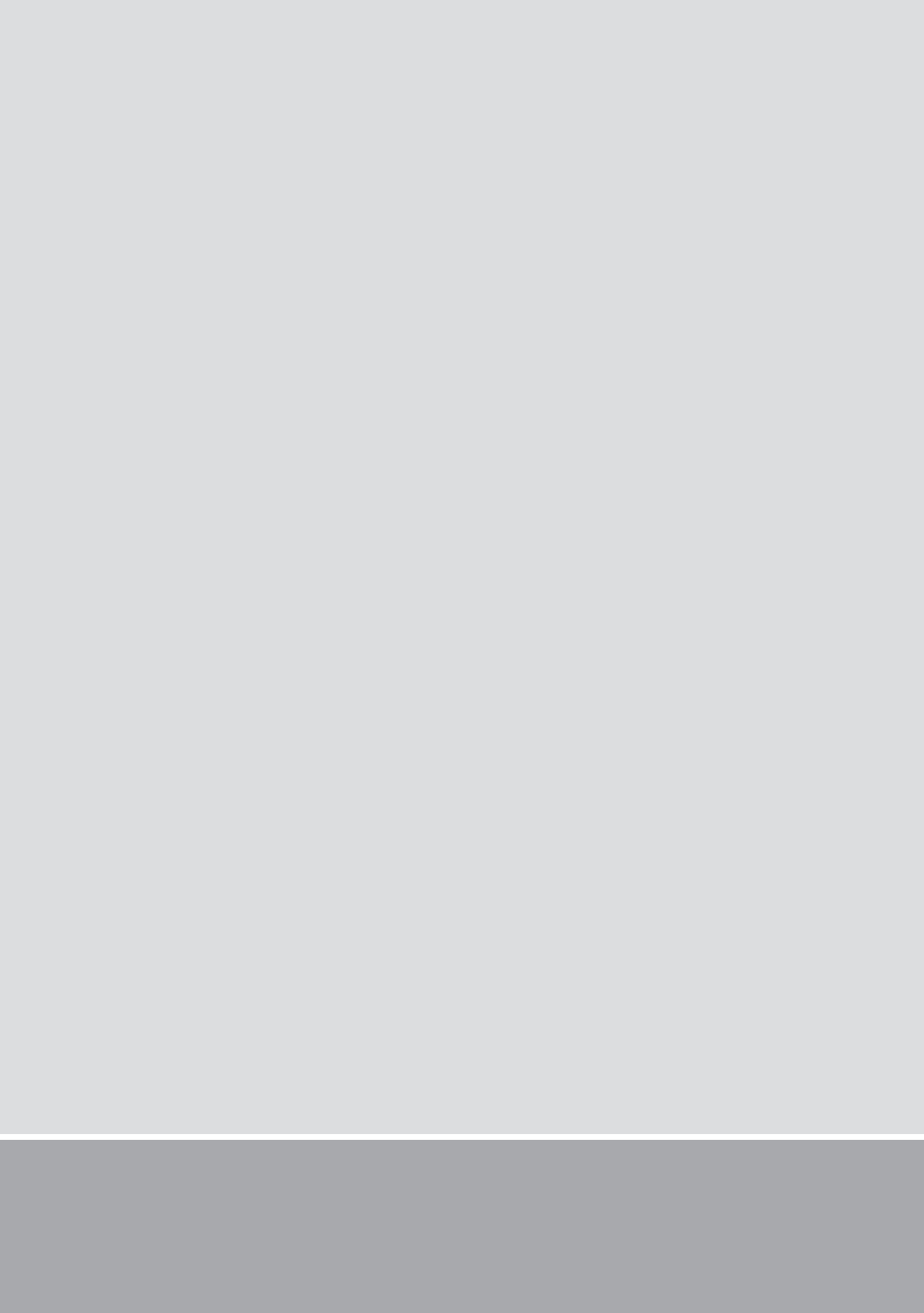
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PART ONE

INTERNATIONAL REVIEW



chapter two



International evolution of spatial
marine protection

Ideas about how marine environments should be managed and protected changed significantly over the course of the twentieth and early twenty-first centuries. This evolution of thinking internationally, helps to inform New Zealand's own development in marine protection law and policy, and provides the context for considering future changes.

The genesis of marine protection

Although the late nineteenth and early twentieth century saw growing awareness of human impacts on the environment, and the need to conserve areas in a natural state, concerns about the terrestrial environment dominated. The world's first terrestrial national park was established in 1872 at Yellowstone in the United States. This was followed in 1887 by New Zealand's first national park at Tongariro, the result of a generous gift of land to the Crown by Te Heuheu Tukino IV. At this time, the protection of marine ecosystems was not considered a priority.

Early spatial restrictions on activity in marine areas were generally directed towards protecting particular species such as fur seals, or the management of specific fish stocks, rather than the protection of the environment itself. Such techniques have been employed in many different places around the world for centuries. For example, in Oceania (Polynesia, Melanesia and Micronesia) rulers could close an area to replenish fish stocks or because they felt that the area had become over-exploited.¹ In 1957, Beverton and Holt provided the first formal numerical analysis and justification for the use of closed areas in fisheries management. Their work was inspired by observations of increasing fish stocks in the North Sea, after World War Two, when the fishing grounds were inaccessible due to the presence of mine fields.²

Nevertheless, despite localised attempts at stock management, a view persisted well into the twentieth century that the ocean was such a bountiful resource that nothing that humans did could significantly affect it. This view was famously expressed by Thomas Huxley in his 1883 address to the International Fisheries Exhibition in London when he said “... *probably all the great sea fisheries are inexhaustible; that is to say, that nothing we do seriously affects the number of fish. And any attempt to regulate these fisheries, seems consequently, from the nature of the case, to be useless ...*”³ The lack of concern for the health of the marine environment was exacerbated by the fact that most of it was out of sight and poorly understood.⁴

During the 1950s and 1960s, there was a growing realisation that better management of the impact of human activities on the marine environment was required, as it became apparent that human activity could indeed cause very real damage. Improvements in science and technology allowed for more effective study of the ocean and consequent improvements in understanding. However, such developments also facilitated increased fishing effort, as new technologies enabled fishing vessels to harvest fish more effectively, and to access previously un-fished areas. High profile collapses of some stocks occurred during this period, such as North Sea herring in the 1960s.

In 1958, the United Nations Convention on the Law of the Sea extended state jurisdictions beyond the customary three nautical mile territorial sea. It provided a legal framework that was designed to mitigate against the 'tragedy of the commons' that had characterised human exploitation of ocean resources. It sought to do this by enabling states to assert their jurisdiction over a greater proportion of the world's oceans. The boundary of the territorial sea was expanded out to 12 nautical miles and new, much larger, exclusive economic zones extended out to 200 nautical miles from land.

At the same time, there was a growing consensus amongst the international scientific community that marine protected areas should be established, not simply to manage fish stocks, but to play a role similar to that of national parks on land. The areas would preserve ecosystems in their natural state as insurance against the destructiveness of human activity.

Thus the First World Conference on National Parks in 1962 recognised that marine areas, as well as terrestrial ones, should be protected as national parks.⁵ Further work was subsequently undertaken on the international stage to develop measures to protect the world's oceans. In 1972, the Governing Council of the United Nations Environment Programme established the Regional Seas Programme. This adopted action plans, in each region around the world, for the protection of marine living resources from pollution and over-exploitation.⁶

The Convention on Wetlands of International Importance Especially as Waterfowl Habitat (the Ramsar Convention) and the Convention for the Protection of the World Cultural and Natural Heritage (the World Heritage Convention), concluded in 1971 and 1972 respectively, were amongst the instruments developed at an international level during this period to recognise the importance of, and need to protect, marine ecosystems.⁷

Establishing marine protected areas

From the 1970s onwards, nations in all parts of the world developed a greater interest in the establishment of marine protected areas. De Silva *et al* calculated that in 1970, there were 118 marine protected areas in 27 nations. By 1985, 430 marine protected areas had been proclaimed by 69 nations, with another 298 proposals under consideration.⁸ Efforts tended to focus on the protection of isolated individual areas, which were selected because of their particularly rich biodiversity, or the presence of rare species. Areas identified for protection were often small, because governments were reluctant to constrain the activities of extractive users of the marine area.⁹

Although a small percentage of the protected areas prohibited all extractive activities (often described as ‘marine reserves’) many did not. The term ‘marine protected area’ gained currency as a label for any type of spatial protection within the marine environment. It was used in many different jurisdictions to refer to a variety of management arrangements. As such, there is no universally accepted definition of the term ‘marine protected area.’ Nations and international organisations have adopted a range of interpretations, including the following:

- The Australian Marine Science Association defines ‘marine protected areas’ as *“areas of the ocean or coastal seas, securely reserved and effectively protected from at least some threats.”*¹⁰
- The United States Presidential Executive Order 13158 describes ‘marine protected areas’ as *“any area of the marine environment that has been reserved by federal, state, territorial, tribal or local laws or regulations to provide lasting protection for all or part of the natural and cultural resources therein.”*¹¹
- The Oslo/Paris convention (for the Protection of the Marine Environment of the North-East Atlantic) (OSPAR), covering the north-east Atlantic, states that the term ‘marine protected area’ *“means an area within the maritime area for which protective, conservation, restorative or precautionary measures, consistent with international law, have been instituted for the purpose of protecting and conserving species, habitats, ecosystems or ecological processes of the marine environment.”*¹²
- The California Marine Managed Areas Improvement Act 2000 provides that a ‘marine managed area’ under the Act *“is a named, discrete geographic*

marine or estuarine area seaward of the mean high tide line or the mouth of a coastal river, including any area of intertidal or subtidal terrain, together with its overlying water and associated flora and fauna that has been designated by law or administrative action to protect or conserve marine life and habitat.”¹³

- The International Union for the Conservation of Nature (IUCN), which has been instrumental in developing international thinking about marine protected area design, defines a ‘marine protected area’ as *“any area of the intertidal or sub-tidal terrain, together with its overlying water and associated flora, fauna, historical and cultural features, which has been reserved by law or other effective means to protect part or all of the enclosed environment.”¹⁴*

Whilst these definitions differ, what underpins them all is the requirement that the protection of the marine environment, or components of it, is the central objective. Similarly, the precise management measures employed to achieve protection are not relevant. Marine protected areas are not required to exclude activities, if such exclusion is not required to achieve the management objective(s).

Whilst protection of the environment must be the primary objective, marine protected areas may be designed with secondary purposes in mind. There is widespread support for the notion that the design of marine protected area systems should provide for the realisation of socio-economic objectives, as well as environmental ones.¹⁵ This idea recognises the importance of providing for human relationships with the marine area, which in turn increases support for management measures and thus compliance, and creates a resource of value to the community. Thus, opportunities for recreation, tourism, scientific study or even sustainable use might be appropriate secondary objectives.

Move to ecologically representative networks

Since the first small protected areas were established many decades ago, approaches to marine protection have evolved significantly. Today, best practice is informed by our understanding of the marine environment as a hierarchical, inter-dependent structure. Interference with some elements of the ecosystem may have far-reaching consequences, even altering the very nature of the ecosystem itself.¹⁶ The focus of marine protection measures has thus shifted from the protection of

individual sites identified on the basis of their location (a *spatial* focus), to the consideration of the marine ecosystem as a whole (a *systems* focus).

Nations, including New Zealand, have committed to the establishment of an ‘ecologically representative’ global marine protection system via a range of international instruments. Specifically, in 2004, the seventh meeting of the Conference of the Parties to the Convention on Biological Diversity set out targets for the establishment of “*comprehensive, effectively managed, and ecologically representative national and regional systems of protected areas*.”¹⁷ Then, the Eighth Conference of the Parties in 2006 adopted the following target: “*At least 10 per cent of each of the world’s marine and coastal ecological regions effectively conserved*”.¹⁸

The ‘ten per cent target’ was refined in 2010 at the tenth Conference of the Parties, which emphasised the importance of an integrated marine management approach to ensure that by 2020, “*at least 10 per cent of coastal and marine areas, especially areas of particular importance for biodiversity and ecosystem services, are conserved through effectively and equitably managed, ecologically representative and well connected systems of protected areas and other effective area-based conservation measures, and integrated into the wider seascapes*”.¹⁹

Although it provides a measurable goal to which nations can aspire, the adoption of the ten per cent target was somewhat arbitrary in scientific terms. The impact of protecting ten per cent of the marine area will differ in various jurisdictions, dependent on the characteristics of the marine environment and the pressures on it.²⁰ In addition, there is some flexibility in the measures that can be employed to meet the target: coastal and marine areas must be conserved in effective protected areas and “*other effective area-based conservation measures*.”

Many national governments have reflected the target in domestic policy. Initiatives for the establishment of new marine protected areas have been undertaken in many nations including the United States, Australia, the United Kingdom and New Zealand. These have sought to develop connected networks of marine protected areas, which together aim to protect a full range of ecosystem features (species and habitats), occurring within each nation’s waters.

Building linkages with broader marine management

It is now well-recognised that the protection of a small percentage of marine space, whilst ‘business as usual’ continues elsewhere, will not in itself safeguard

the health of marine ecosystems.²¹ This is for several reasons. Without effective management of the broader marine environment, marine protected areas will continue to be subject to the effects of significant harmful activities which occur outside their boundaries, such as pollution and run-off from land.

In addition, spatial protection measures are likely to have greater benefits for sedentary species than for those which move around or have dispersed lifecycles, meaning their effect is predominantly local.²² Furthermore, whilst well-designed networks may protect areas with high biodiversity values, the act of establishing linear boundaries will always be arbitrary. The marine environment is dynamic and interconnected, with linkages at different scales.²³ The distribution of marine species is also constantly shifting, a fact that is heightened by climate change.²⁴

Thus, there is now recognition that the establishment of marine protected areas alone should not be seen as a complete solution to the challenge of effective marine management.²⁵ Rather, they should be thought of as a vital part of a management system which is applied to the entire marine ecosystem.²⁶ Ecosystems-based management seeks to replace the fragmented, sector-based approach to marine management with an integrated system. This recognises that, just as ecosystems are interconnected, management of marine activities must be integrated.²⁷

Marine spatial planning processes, which identify marine protected areas within the context of a broader planning system, are one way of working towards ecosystems-based management. They enable the consideration of all activities and their impacts across larger areas of marine space.²⁸ If proposals to establish marine protected areas fail to consider this broader picture, they can have unintended impacts on other parts of the marine system, such as when fishing activity is displaced and increases pressure elsewhere.²⁹

Conclusion

Thinking about marine protection has changed significantly over the past century or so, from the early belief that the resources of the oceans were limitless, to the current acceptance that we need to manage these resources much more carefully. The creation of marine protected areas is now a widely accepted and applied approach, to achieve better protection of marine biodiversity, and to maintain ecosystem health. However, there is growing realisation that they cannot succeed in isolation. Increasingly, it is recognised that they should be embedded within a broader ecosystems-based approach to oceans management.

Figure 2.1: Summary of the international evolution of spatial marine protection

- Marine protection was considered unimportant until the mid-twentieth century. Early marine protection efforts focused on specific species or fish stocks.
- During the early 1960s, there was international recognition that marine areas should be spatially protected in a manner similar to terrestrial national parks.
- During the 1970s and 80s numerous marine protected areas were established in many countries around the world. These were largely individual isolated areas.
- During the 1990s, there was a shift in focus from the establishment of individual marine protected areas, to the establishment of networks.
- During the 2000s, there was growing recognition that marine protected areas needed to be incorporated into broader ecosystems-based marine management regimes, and that marine spatial planning was a useful tool to help achieve this.

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chapter three



Potential benefits of marine
protected areas

A review of the international literature indicates that marine protected areas can have a range of ecological, social, cultural and economic benefits. Some of these are described in the sections below. The extent to which these can be achieved in practice is largely dependent on the location, design and on-going management of the area.

It is difficult to predict what the precise effects of protecting an area might be, for a number of reasons. Globally, relatively few marine protected areas have been in existence for long enough to allow a full understanding of their socio-ecological benefits. Even fewer have been the subject of thorough data gathering which allows a proper before and after comparison to be undertaken.¹

Furthermore, data derived from existing marine protected areas can only be of limited assistance. The ecological communities to be protected at any given site are subject to unique environmental conditions and histories, which set the context in which responses to conservation measures can emerge.² The complexity of the many linkages and feedbacks that make up a socio-ecological system is such, that we can never predict with certainty, what the exact response will be to any intervention in the system. As a result, the effects of marine protection can sometimes be disappointing or even surprising.

For example, scientists originally believed that highly mobile species such as snapper would not benefit from the establishment of marine reserves in New Zealand, because they were too wide-ranging to be protected in one small area. Nevertheless, it was discovered that snapper display complex small-scale movement patterns during some stages of their life cycle, effectively adopting a 'home range'. This enabled them to remain protected from fishing in relatively small reserves.³

The current uncertainty and contestation around the impacts of marine protected areas highlights the urgent need for better monitoring systems. This is essential, so that current experience can help inform the better design of marine protection tools in the future.

Protection of biodiversity

Experience from New Zealand and overseas indicates that individual marine protected areas can provide significant benefits for the biodiversity contained within them. There is strong scientific evidence to indicate that where extractive activities are prohibited, the abundance of previously exploited marine species

can increase dramatically, and that this can occur within relatively short periods of time.⁴

For example, densities of yellowtail snapper within sanctuary zones established in Florida were found to have increased 15-fold only four years after the protection measures were implemented. After nine years of protection, scallops in New England protected areas became 25 times more abundant.⁵ At Cape Rodney-Okakari Point marine reserve near Leigh, which was established in 1975, snapper large enough to be legally caught were roughly fourteen times more abundant inside the reserve than outside it by 2002.⁶

Increased productivity

An increase in the abundance of previously fished species can have positive flow-on effects throughout the food web. For example, the increase in snapper and rock lobster within the Cape Rodney-Okakari Point marine reserve was associated with a marked decrease in the abundance of kina or sea urchins (on which snapper and rock lobster prey), and a regeneration of kelp beds (on which the sea urchins prey). This, in turn, is likely to increase overall biodiversity and productivity within the marine reserve as kelp forests support a much richer faunal community than the ‘urchin barrens’ which occur when sea urchins are present in large numbers.⁷

Increased resilience and maintenance of ecosystem services

We are only just beginning to understand the complexity of the interactions that link the oceans to the Earth’s processes and the ways in which we are reliant on their effective function.⁸ For example, the oceans regulate our climate and provide the oxygen we breathe, and fish contribute to nutrient recycling as well as providing protein for human consumption. Coastal habitats provide shoreline protection, water filtration, and biodiversity habitat. All these functions have a value to society, whether individually or as a part of the larger whole.

Societies depend on these ‘ecosystem services’⁹ for their existence, and value their ability to pass them on to future generations, although their importance is rarely explicitly recognised.¹⁰ Despite an enduring public discourse which

erroneously pits the health of the economy against the health of the marine environment, this new understanding is contributing to calls for a greater focus on ensuring the *resilience* of marine systems. This refers to their ability to withstand poor management, cumulative damage and catastrophic natural events and still retain their basic function, including the delivery of the ecosystem services upon which we rely.¹¹ Healthy ecosystems are able to absorb and adapt to stresses (up to a point), because they are complex systems, which by their very nature provide some buffering against change.

However, our current approach to natural resource management fails to take into account the manner in which ecosystems respond to change. In many instances our actions reduce the ability of these systems to respond normally. Management measures rarely take into account the impact that activities will have on the entire ecosystem, and the failure to do this, means that the system can become increasingly vulnerable to shocks and disturbances. For example, individual fish stocks are managed with only limited reference to the role of the species in the wider ecosystem.¹² More broadly, the cumulative impacts of the range of extractive activities that occur in the marine environment, are rarely considered at all.

The protection of biodiversity is one important way in which the resilience of ecosystems can be bolstered.¹³ As the Royal Society of New Zealand notes, the relationship between biodiversity and resilience is complex, but “[e]cosystem functions, such as nutrient regulation, are provided by the traits of organisms within that ecosystem. Greater genetic diversity provides a greater reservoir of traits that can replace traits lost if particularly important species are lost. More diversity also provides more opportunity for functions to operate across a broader range of conditions. In this way, biodiversity provides the insurance value that future environmental changes will not reduce services.”¹⁴

Well-designed marine protected areas can help to protect biological variation across habitats and species, and the maintenance of genetic variation within species, through protecting different populations across their geographic range.¹⁵ They can protect parts of the marine environment from stressors which are spatial in nature (i.e. the activity can be prohibited in protected areas) and have localised impacts.¹⁶

However, the establishment of marine protected areas cannot address those stressors which are large scale (such as climate change), outside the marine environment (such as sediment run-off), or which are not excluded by spatial boundaries (such as marine pollution). A broader, ecosystem-based management

plan, into which marine protected areas are embedded, is required to address those issues.¹⁷

Benchmarking of environmental health

The complexity of marine ecological dynamics makes it hard for scientists to assess the impact of anthropogenic pressures on the marine environment. It can be difficult to separate natural and anthropogenic disturbance and the direct effects of one activity over another.¹⁸ Modelling can assist, but cannot provide all the answers. The establishment of areas in which damaging activities are excluded, can play an important role in providing 'benchmarks' or 'controls', against which human impacts and management of the rest of the marine area can be evaluated. This enables scientists to assess the effects of pressure from human activities in the context of prevailing environmental conditions.¹⁹

Fisheries spill-over

It is possible that properly designed marine protected areas can benefit commercial and recreational fishers by helping to repopulate fish stocks outside the protected area. Much has been made of this potential, with proponents of marine protection seeking to use it to bolster their cause. Conversely, fishers often deny that marine protected areas are necessary or useful in the management of fish stocks, and resent the potential decrease in size of their fishing grounds. Clearly, marine protected areas will only directly benefit a fishery if harvestable-size stock moves from the closed area to open areas where they can be caught, or if eggs and larvae cross the boundary and establish outside the closed area.

The likelihood of 'spill-over' from a marine protected area into the fishable area depends on the mobility and habits of the species concerned, conditions inside and outside the marine protected area, and the area's design (for example, longer edges enhance the likelihood of spill-over).

The extent of spill-over is also dependent upon factors such as prevailing currents and spawning patterns. For example, studies of the Georges Bank protected area on the north-west coast of the Atlantic ocean, found that spill-over was enabled because fish larvae were transported out of the protected area by a well-established ocean current, an effect which was relatively easy to track.²⁰

Some spill-over effects may be localised around the area adjacent to the protected area.²¹ This has been documented in many areas, for example at Sumilon Island marine sanctuary in the Philippines,²² the Su Pallosu²³ and Columbretes Islands²⁴ marine reserves in the western Mediterranean, the Soufriere marine management area in St Lucia and the Merritt Island national wildlife refuge in Florida.²⁵ The effect is most easily illustrated by the presence of fishers around the edges of the protected area. For example, satellite tracking of vessels off the Georges Bank shows that commercial trawlers now concentrate much of their activity on the borders of closed areas.²⁶ Similarly, in New Zealand, cray pots often line the reserve boundaries around the marine reserve at Leigh.²⁷

Where recruitment is below the maximum possible outside the marine protected area, spill-over may be most apparent. Thus the most significant effects have been observed in locations where protected areas have been implemented in highly overfished ecosystems, particularly where the stock is heading towards extinction.²⁸

Arguments about the effectiveness or otherwise of marine protected areas in improving fish stocks should be kept in perspective, as marine protected areas are valuable for many other reasons, and improving fish stocks is not usually the prime reason for their establishment.

Protection of geological processes

Marine protected areas can be used to protect geological features and processes identified as particularly unique or typical. This may be closely interlinked with the protection of the biodiversity they support. For example, seamounts are valued both as rare geological features and for the unique ecological communities which are dependent on them.

In England, legislation providing for the creation of ‘marine conservation zones’ provides for their designation for the purpose of conserving features of geological or geomorphological interest, amongst other things.²⁹

Protection of cultural values

The protection of marine space can be of considerable cultural value. For example, the establishment of the Whanganui A Hei – Cathedral Cove marine reserve was supported by Ngāti Hei, because the area included tapu sites associated with tribal massacre.

Ngāti Hei believed that the establishment of the reserve enhanced its mana. Likewise, the Sugar Loaf Islands marine protected area includes tapu sites for at-sea burials.

Increased recreational and tourism opportunities

Marine protected areas can contribute significantly to tourism and recreational opportunities in the marine space and coastal area. The extent of these benefits varies, depending on the location of the protected area. Clearly, accessible coastal locations may provide the most significant benefits in this regard. The increased abundance and range of species, as well as the larger size of individuals, which results from many marine protected areas can be attractive to scuba divers and snorkelers. In New Zealand, as overseas, the opportunity to dive in a marine reserve is a draw-card in itself, as protected areas are generally associated with good diving. This creates additional market opportunities for businesses associated with recreational diving and undersea viewing.

Studies of the impacts of marine reserves in New Zealand have shown a range of tourism and recreation benefits. For example, the establishment of the Tonga Island marine reserve adjacent to the Abel Tasman National Park increased the recreational value of the area. This in turn attracted greater numbers of visitors to the area. In response, new accommodation enterprises have established, and there are growing numbers of water taxi operators. There has also been a significant growth in the number of kayak companies, from the two operating prior to the establishment of the marine reserve, to at least 13 currently.³⁰

A recent study estimated the value of marine recreation in the Hauraki Gulf at \$550 million. Nature-based tourism in the Gulf was estimated to be worth \$1,672 million, whilst recreational fishing was worth \$104 million.³¹ Similarly, the establishment of the Great Barrier Reef Marine Park in Australia has made a significant contribution to recreational opportunities and tourism revenue. The Park generates an estimated “[AU]\$2.8 billion a year in tourism and the local growth it supports in lodging, restaurants, transport, retail fishing gear, boat hire, charter fishing and guide businesses, scuba businesses and so forth.”³²

At the Tonga Island marine reserve, it was reported that a majority of the local community was opposed to the reserve prior to its establishment, fearing impacts on commercial and recreational fishing and doubting that any significant benefits could arise. After its establishment, many changed their minds, having seen significant benefits in terms of recreation opportunities and new tourism

businesses, and having discovered that the feared negative impacts were not nearly as bad as expected.³³

Similar benefits have been experienced from the Cape Rodney - Okakari Point marine reserve. An economic impact analysis undertaken in 2008 identified an estimated 375,000 annual visits to the reserve and the contribution of \$18.6 million a year into the local economy. This was compared to the cost to the Department of Conservation of managing the reserve of around \$70,000 per year.³⁴

Conclusion

Marine protected areas can potentially provide a range of benefits. Some of these benefits, such as biodiversity protection, are well-established by scientific research. Others, such as fisheries spill-over, are still subject to dispute and may vary according to local circumstances.

Fishers have often argued that the application of fisheries management tools, such as the quota management system, is sufficient to protect the marine environment and that additional tools, such as marine protected areas, are not required. However, it needs to be recognised that fisheries managers have been slow to apply ecosystems-based approaches to their management task. Fisheries management tools may be the most appropriate means by which to manage fish stocks in the short term, but marine protected areas are a vital tool to protect marine biodiversity and ecosystems, thereby sustaining fisheries in the longer term.

Figure 3.1: Summary of potential benefits of marine protected areas

- Protection of biodiversity
- Increased productivity of ecosystems
- Increased resilience of the marine environment to damage from human activity, thereby safeguarding ecosystem services
- Provision of 'benchmarks' against which the impacts of human activities in other parts of the marine environment can be compared
- Improved fisheries through spillover of eggs, larvae and adult fish
- Protection of geological processes
- Protection of cultural values
- Increased opportunities for tourism and recreational activities and maintenance of New Zealand's 'green' international image

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chapter four



International approaches
to marine protection

In recent years, many nations around the world have implemented legislation aimed at providing for effective spatial protection in the marine area. In this report we have considered initiatives undertaken by England (marine conservation zones under the Marine and Coastal Access Act 2009), California (marine protected areas under the Marine Life Protection Act 1999), Australia (Commonwealth marine reserves under the Environment Protection and Biodiversity Conservation Act 1999) and New South Wales (marine parks under the Marine Parks Act 1997). These experiences are described in detail in Part Three of this report and are summarised here.

English marine conservation zones

The Marine and Coastal Access Act, passed in 2009, aims to completely overhaul management of the United Kingdom's marine area. One of the key measures contained in the Act is provision for a new type of marine protected area which is intended to help achieve an ecologically coherent network of marine protected areas in United Kingdom's waters.

The legislation provides for different mechanisms to be applied in the various parts of the United Kingdom. In England, the legislation provides for the creation of 'marine conservation zones' designated for the purpose of conserving marine flora or fauna, marine habitats or types of marine habitat, and features of geological or geomorphological interest.¹

The objective is that the marine conservation zones, together with existing conservation sites, will form a network which satisfies the following conditions:

- that the network contributes to the conservation or improvement of the marine environment in the United Kingdom marine area;
- that the features which are protected by the sites comprised in the network represent the range of features present in the United Kingdom marine area; and
- that the designation of sites comprised in the network reflects the fact that the conservation of a feature may require the designation of more than one site.²

The legislation did not specify a process for establishment of marine conservation zones. A process for their establishment was developed by the government's statutory nature conservation advisory bodies, which are the Joint Nature

Conservation Committee and Natural England. The process provides for the identification of new marine conservation zone sites by stakeholder-led groups.

Four regional projects were created in which local stakeholders worked together to identify proposed marine conservation zone locations. This was in accordance with government guidance which set out the form that the project groups should take and the process that they should follow.

Each regional project was required to have a 'project board' with ultimate responsibility for the effective delivery of the project, and a 'regional project team' with responsibility for the day-to-day running of the projects. These full-time staff members came from a variety of backgrounds, in project management, marine protected area planning, marine ecology, Geographic Information Systems (GIS), stakeholder liaison, economics, communications and social science, in order to encompass the full range of skills required.

Each project team was required to contract an independent facilitator who was tasked with designing and running a stakeholder engagement process. This included undertaking a stakeholder analysis to ensure that all key stakeholders were represented, achieving consensus on membership of the stakeholder groups, developing the decision-making process, resolving disputes and facilitating constructive discussion.

The regional stakeholder groups were responsible for the substantive work in identifying proposed marine conservation zone sites. The groups comprised representatives from all significant sectors, and were assisted by government officials who provided technical expertise and information.

The regional groups were required to work consistently with detailed 'ecological network guidance' provided by the government's statutory nature conservation authorities. This set out what the proposed networks needed to achieve including protecting a representative range of habitats and species. Examples of each habitat and species were to be protected in at least one area where all extractive activity was prohibited. Management measures for the rest of the proposed marine conservation zones were to be designed to ensure that the only prohibited activities are those which could damage the particular features at which protection was aimed.

A relatively short time frame of 18 months was allowed for completion of the projects, including forming the groups, gathering data and developing recommendations. Three iterations of the proposed network were submitted for comment to the government's statutory science advisory bodies.

An independent ‘science advisory panel’ and the statutory nature conservation advisory bodies then assessed the recommendations against the requirements of the ecological network guidance and their own informed views about how such a project should be undertaken. The nature conservation advisory bodies reported to government in July 2012, and it is expected that public consultation on the proposals will take place in December 2012.

The careful design of the stakeholder groups, and the detailed design advice they were given, helped them to function effectively and to deliver proposals on time. The science advisory panel found that *“the majority of targets that were set to achieve an ecologically coherent network of MPAs have been met.”*³

However, the panel also reported that there were some shortcomings in the proposals, particularly in relation to the data used to support decision-making, the proposed conservation objectives, the number and size of proposed reference areas, and the weight given to socio-economic concerns in the decision-making process. This reflected a disjunct between the orientation of the panel, which expected decision-making to be science based, and the stakeholder-led process which was driven by a range of interests and was therefore more subjective.

Following the panel’s report, the government recommended that further work be undertaken to investigate the scientific basis underpinning the proposals. It also announced that the recommendations would be implemented in stages, rather than all at once, as previously planned. It appears that the least contentious areas will be designated first. As a result, whether or not the marine conservation zone project, as a whole, is ultimately a success will depend on whether the government exhibits continuing commitment to give effect to the full package of recommendations.

California Marine Life Protection Act

The Marine Life Protection Act was passed by the California State legislature in 1999. The Act requires the development of a statewide system of marine protected areas within state waters along California’s coastline.

Between 1999 and 2004 there were two abortive efforts at implementing the legislation. In the first attempt, the Department of Fish and Game and a ‘master plan team’ comprised of marine scientists and managers, developed a set of initial proposals for a statewide network of marine protected areas. These covered the entire 1,770 kilometres of California’s coast. They were developed without

significant stakeholder input and failed to achieve public support. The second attempt was aborted when it became clear that the resources allocated to the project would not be enough to achieve its goals.

Finally, the California Marine Life Protection Act Initiative was launched in August 2004 pursuant to a memorandum of understanding between the California Natural Resources Agency, the California Department of Fish and Game, and the Resources Legacy Fund Foundation. These agencies formed a public-private partnership known as the Marine Life Protection Act Initiative. The Initiative brought stakeholders, scientists, and policy-makers together in a public process designed to culminate in the adoption of marine protected areas by the California Fish and Game Commission and the passing of implementing regulations.

The Initiative adopted a region-by-region approach to implementation, enabling learning and adaptation from each successive planning process. The legislation provided for the inclusion of areas with various levels of protection. These include 'marine life reserves', which are areas where all extractive activities are prohibited, as well as areas where specified forms of commercial and/or recreational fishing are allowed. The legislation established the following guidelines:

- (1) Each marine protected area shall have identified goals and objectives. Individual marine protected areas may serve varied primary purposes while collectively achieving the overall goals and guidelines.
- (2) Marine life reserves in each bioregion shall encompass a representative variety of marine habitat types and communities, across a range of depths and environmental conditions.
- (3) Similar types of marine habitats and communities shall be replicated, to the extent possible, in more than one marine life reserve in each biogeographical region.
- (4) Marine life reserves shall be designed, to the extent practicable, to ensure that activities that upset the natural ecological functions of the area are avoided.
- (5) The marine protected area network and individual marine protected areas shall be of adequate size, number, type of protection, and location to ensure that each marine protected area meets its objectives and that the network as a whole meets the goals and guidelines.

The preferred siting alternatives were required to be designed in accordance with these guidelines, and also had to be consistent with the broader goals of the marine protected area programme.

Because the Marine Life Protection Act did not define the various different types of marine protected area, in 2000, the State of California adopted the Marine Managed Areas Improvement Act to achieve this. Prior to this legislation, there were 18 classifications and sub-classifications of marine areas within the state. The new legislation provided that there would be only six classifications: marine reserve, marine park, marine conservation area, marine cultural preservation area, marine recreational management area and water quality protection area.

Regional stakeholder groups were responsible for the identification of proposed marine protected area sites, assisted by full-time project teams and guided by a 'master plan science advisory team' which was appointed in accordance with the requirements of the Marine Life Protection Act. The master plan itself, which was prepared by the California Department of Fish and Game, guided the adoption and implementation of the marine protected areas programme. The master plan contains recommendations on the siting of new marine protected areas, modifications of existing marine protected areas and management, enforcement, monitoring and funding of California's marine protected areas.

A 'Blue Ribbon Task Force' was established to guide the development of alternative proposals presented to it by the regional stakeholders group. The task force also recommended a range of alternative proposals, and a preferred alternative proposal for each region, to the California Fish and Game Commission. In this way, the task force provided a 'buffer' between the stakeholders and the political decision-makers. The task force was to be composed of members "*known for their integrity, intellect and experience in public policy and concern for common good*"⁴ who were selected by the California Natural Resources Agency.

At the time of writing, marine protected area networks had been adopted for three of the five study regions (north central coast, central coast and south coast). The marine protected area network proposed for the north coast is pending adoption by the Commission. On the basis of the north coast preferred alternative, the statewide coastal network includes 123 marine protected areas, covering approximately 2,196 square kilometres (16 per cent) of state waters (excluding San Francisco Bay). They include 48 no-take marine reserves, which represent approximately nine per cent of state waters (excluding San Francisco Bay). Options for the final region, San Francisco Bay, are currently under development.

After the implementation of the Marine Life Protection Act, all regions experienced a substantial increase in the areal coverage of both marine protected areas and state marine reserves which are fully protected. Therefore, at a substantive level, the Initiative has undoubtedly achieved the requirement of the Marine Life Protection Act to implement an ‘improved marine life reserve component’.

Australian Commonwealth marine reserves

Under the Environment Protection and Biodiversity Conservation Act 1999, the Governor General is authorised to create ‘Commonwealth reserves’ within the Australian exclusive economic zone. The Act also empowers the Governor General to create ‘conservation zones’ for the purpose of protecting biodiversity in an area whilst it is being assessed for inclusion in a Commonwealth reserve.

The management measures implemented in Commonwealth reserves, and consequently the level of protection from human activity, vary depending on the particular feature identified for protection. As a result, some areas allow multiple uses and some are no-take zones. An individual Commonwealth reserve may contain a number of different areas permitting different levels of activity.

The Environment Protection and Biodiversity Conservation Act (and the Environment Protection and Biodiversity Conservation Regulations 2000) require that each reserve, and zone within it, should be allocated to a particular management category. The Commonwealth government has essentially adopted the IUCN set of seven management categories (although the numbering system is different). The categories are not arranged in a hierarchy of levels of protection, but describe the purpose of protection for a particular area, and the types of activity that will be permitted in there.

Although the creation and management of marine protected areas within Australian waters is the responsibility of various government bodies (i.e. states, territories and the Commonwealth government), the Commonwealth government has put in place measures to coordinate them into a coherent marine protected area network. The Commonwealth marine protected areas program, initiated in 1991, is intended to develop a national representative system of marine protected areas across Commonwealth, state and territory jurisdictions.

The primary goal of the programme is *“to establish and manage a comprehensive, adequate and representative system of marine protected areas to*

contribute to the long-term ecological viability of marine and estuarine systems, to maintain ecological processes and systems, and to protect Australia's biological diversity at all levels.”⁵

The following secondary goals are designed to be compatible with the primary goal:

- To promote the development of marine protected areas within the framework of integrated ecosystem management
- To provide a formal management framework for a broad spectrum of human activities, including recreation, tourism, shipping and the use or extraction of resources, the impacts of which are compatible with the primary goal
- To provide scientific reference sites
- To provide for the special needs of rare, threatened or depleted species and threatened ecological communities
- To provide for the conservation of special groups of organisms, e.g. species with complex habitat requirements or mobile or migratory species, or species vulnerable to disturbance which may depend on reservation for their conservation
- To protect areas of high conservation value including those containing high species diversity, natural refuges for flora and fauna and centres of endemism
- To provide for the recreational, aesthetic and cultural needs of indigenous and non-indigenous people

Since 2005, when regional marine planning was brought under the Environment Protection and Biodiversity Conservation Act, the national representative system of marine protected areas programme has been incorporated into the marine bioregional planning process. Plans for the Commonwealth marine area are now created as bioregional marine plans

Marine bioregional planning is being implemented in six large bioregions – the south-east, south-west, north-west, north, temperate east and coral sea marine regions. In broad terms, the bioregional plans set out the Australian government's objectives for conservation and environmental management measures such as marine protected areas, listed marine species protection and sustainable fisheries. The purpose of the marine plans is to guide the Environment Minister, sectoral

managers and industry about the key issues and priorities in the region, rather than to act as a regulatory tool.⁶ Through the planning process, the Commonwealth government identifies candidate areas for marine protected areas which can then be created as Commonwealth reserves.

In 2007, the Department of the Environment and Water Resources released goals and principles for the establishment of the national representative system of marine protected areas in Commonwealth waters.⁷ These set out four goals and principles which related to location, selection, size, shape and zoning.

The planning process involves the creation of a bioregional profile, and the development of a draft marine reserve plan, which goes out for public consultation before its finalisation. The bioregional profiles describe the ecological and biophysical features and the conservation values of each region, together with the human activities that occur there. In some regions, they also identify 'areas for further assessment'. The purpose of these is to narrow down the areas that warrant closer examination in the reserve identification process, because they are representative of the conservation values of the region.

In May 2011 draft plans were released for five regions, (the plan for the south-east region having been completed in 2004). Formal 90-day public consultations on the draft plans were undertaken in each region between May 2011 and February 2012. Two hundred and forty five public and stakeholder meetings were held across Australia, and 566,377 submissions received. After considering the public submissions, together with detailed socio-economic assessments, the government released final proposals in June 2012.

A further 60-day public consultation process was then undertaken. After the final marine bioregional plans have been approved, the proposed network of new marine protected areas will be proclaimed through a separate statutory process. The proclamation of the proposed national marine reserves network will increase the total number of Commonwealth marine reserves to 60, representing examples of all of Australia's different marine ecosystems and habitats.⁸ The largest of the proposals is a marine reserve covering 989,842 square kilometres of the Coral Sea. All mining and petroleum developments are to be prohibited in this area, whilst a sanctuary zone in which commercial fishing is prohibited, will cover approximately half of the reserve area.

New South Wales marine parks

The Australian Commonwealth government's marine protected areas program prompted the New South Wales government take action on marine protected areas, and it subsequently released a regional-scale policy, setting out goals for marine biodiversity conservation. This ultimately led to the passage of the Marine Parks Act 1997.

This legislation, which provides for the establishment of marine parks, has as its objectives:

1. To conserve marine biological diversity and marine habitats by declaring and providing for the management of a comprehensive system of marine parks
2. To maintain ecological processes in marine parks
3. Where consistent with the preceding objects:
 - a. to provide for ecologically sustainable use of fish (including commercial and recreational fishing) and marine vegetation in marine parks, and
 - b. to provide opportunities for public appreciation, understanding and enjoyment of marine parks.

The Marine Parks Act provides that the Governor General may declare an area, which may include any part of the marine area, land adjacent to the marine area, and islands within the marine area, to be a marine park.⁹ If land above mean high water springs is to be included in the park, the Governor General must seek the consent of the owner or the Minister administering the Crown Lands Act 1989.

The Marine Parks Act provides for the establishment of a 'Marine Parks Authority', a statutory body tasked with:

- Investigating, assessing and recommending where marine parks should be established or where variations to marine parks should be established
- Making recommendations on zoning measures within marine parks
- Preparing an operational plan in respect of each marine park
- Managing activities in marine parks so as to protect the marine environment
- Providing for and regulating the 'ecologically sustainable use' (including commercial and recreational fishing) of marine parks

- Public education
- Carrying out and encouraging research on how marine parks affect the marine environment and nearby communities

The Authority reports jointly to the Minister for Primary Industries and the Minister for the Environment. It is chaired by the Director General of the Department of Premier and Cabinet. The other two members are the Director General of the Department of Industry and Investment, and the Chief Executive of the Office of Environment and Heritage in the Department of Premier and Cabinet.

The Marine Parks Act provides for the establishment of a 'Marine Parks Advisory Council', consisting of the Director General of the Department of Environment and Climate Change, the Director General of the Department of Industry and Investment, one member representing the Commonwealth government, two members representing the interests of marine conservation, one member with expertise in marine science, and one member each to represent the interests of Aboriginal people, the tourism industry, commercial fishers, recreational fishers and scuba divers.

The Advisory Council, on the request of the relevant Ministers or the Authority, advises on a range of matters from a state-wide perspective including on proposals for marine parks, the conservation of marine biological diversity within parks, and their use.

The Marine Parks Act provides that the regulations will provide for the establishment of a zoning plan which will govern the use and management of the marine park. The Marine Parks Authority is required to draft a zoning plan for the marine park within 12 months of its establishment. The Authority submits the draft to the relevant ministers, together with any comments from the Advisory Council. The Ministers then have three months to either undertake public consultation or return the draft back to the Authority for amendment. The primary focus of public consultation in relation to marine parks is on the zoning plans.¹⁰

The Marine Parks (Zoning Plans) Regulation 2009 (which replaces the Marine Parks Zoning Regulations 1999) sets out the measures governing the zoning system. Under the regulations, there are a number of zones which may be included in a marine park. Sanctuary zones provide the highest level of protection, excluding all potentially damaging activities. Habitat protection zones, general use zones and special purpose zones provide for less strict management arrangements which protect particular features of the area whilst allowing some appropriate activities.

In 2001, the New South Wales Government prepared a document setting out how it intended to implement the Marine Parks Act and create a representative system of marine protected areas. The document articulated the government's goal which was "*to establish a comprehensive, adequate and representative system of marine protected areas that includes a full range of marine biodiversity at ecosystem, habitat, and species levels.*"¹¹

Six marine parks have been established under the legislation, ranging from 220 square kilometres to 987 square kilometres in size and covering a total area of 3473.5 square kilometres. Of this area, 24 per cent is protected in no-take zones.

In May 2011, the new coalition government abolished the changes to the zoning, arguing that decision-making about marine park zoning had been 'politicised' by the former government. "*The New South Wales Government is committed to common-sense marine parks policy that is based on science, not politics,*" and "*We will disallow these most recent changes, overturning Labor's politically motivated plans ...*"¹²

The government supported a bill proposed by the Shooters and Fishers Party which culminated in the Marine Parks Amendment (Moratorium) Act 2011. This imposes a moratorium on the declaration of additional marine parks, or the alteration or creation of sanctuary zones within existing marine parks, for a period of five years. It also requires Ministers to arrange for an independent scientific audit of the New South Wales marine parks.

The scientific audit was a commitment of the Liberal and National Parties prior to the election, and aimed to ensure that the government "*strike the right balance between sustainable fishing and conservation of marine biodiversity in our Marine Parks policy.*"¹³ Associate Professor Robert Beeton from the University of Queensland was appointed chair of the audit team. It was tasked with reviewing the effectiveness of marine parks in protecting different habitat types, providing recommendations for further action and, if appropriate, alternative management approaches.

The audit report was released in February 2012. Overall, the audit team supported the system of marine parks in New South Wales and also recommended that mechanisms be found to address gaps in coverage of two bioregions.¹⁴

However, the audit team recommended a fundamental shakeup of the institutional structure applying to the New South Wales marine area. It formed the view that the entire area, including the marine parks, should be managed by a single body under a new consolidated piece of legislation. The audit team's rationale for this recommendation was that "*the best design for a system that can*

achieve the sustainable management of the New South Wales coast is one that is managed as a single and continuous system” and that this is “a better way of co-ordinating many competing interests and fostering a stewardship ethic whilst managing for a plurality of legitimate concerns.”

Conclusion

As shown in Figure 4.1, each of the jurisdictions studied has adopted legislation to protect marine ecosystems. For each, the protection of the marine environment is an end in itself.

Legislative framework

The jurisdictions surveyed have used different combinations of law and policy to implement marine protected area networks. California and England have *legislation* requiring the establishment of a representative network of marine protected areas. The English legislation also provided for a new tool (marine conservation zones) to achieve this, whilst in California, further legislation reconciled existing marine protection tools and provided for a system of ‘marine managed areas’. Conversely, both the New South Wales and the Australian Commonwealth governments have a *policy* for the establishment of a representative network of marine protected areas, but use *existing legislation* to implement the policy.

The legislative framework established in California and England undoubtedly provided strong impetus for the achievement of their respective marine protection goals. However, equally, the weaker policy-level status of the Australian and New South Wales initiatives does not appear to have been an obstacle to achievement, as strong commitment from the respective governments have ensured that policy has been implemented.

Management measures

New South Wales, the Australian Commonwealth and California have a range of marine protection ‘categories’ which have corresponding appropriate management measures attached. The Californian system reconciled all the existing management categories into a simple six-category framework. Similarly, the Australian Commonwealth and New South Wales governments use a category system derived from IUCN guidance to design marine protected areas. In all of

these jurisdictions, this system has worked well to provide certainty to stakeholders, and to provide for a range of interests consistent with marine protection.

Conversely, the English Marine and Coastal Access Act provides for only one category of 'marine conservation zone', with the expectation that appropriate management measures will be identified after the areas for protection are confirmed, with reference to the specific requirements of each area. In contrast to the formulaic approach adopted by the other jurisdictions, this approach has contributed to uncertainty for stakeholders, confusion about what marine conservation zones actually are, and concern that when finally implemented, marine conservation zones will be no more than 'paper parks'. Furthermore, the inclusion of a requirement for no-take 'reference areas' in the guidance, but not in the legislation, exacerbated opposition to the establishment of marine conservation zones. Some stakeholders felt that they were being asked to agree to something for which there was no evidence of clear direction from government.

Design and implementation

Two approaches to the design and implementation of marine protected area networks were evidenced in the jurisdictions surveyed. A conventional 'top-down' approach was undertaken by the Australian Commonwealth and New South Wales administrations, which enabled government to maintain a strong grip on the decision-making process. In contrast, the Californian and English legislation provided for much more significant stakeholder involvement in the design of the marine protected area network.

The Californian initiative has met with significant success, but was the subject of several abortive attempts and required a very large commitment of resources. Nevertheless, it has only been completed in three of the five regions for which it was planned. England's marine conservation zone initiative was undertaken simultaneously in each of the four regions covering English waters and recommendations were made by all groups within the specified timeframe. However, following a review of the recommendations, the government is proposing to implement new marine conservation zones in stages.

Whilst the Australian Commonwealth and New South Wales governments were able to undertake marine protected area design using decision-making rooted in government policy and scientific advice, the Californian and English stakeholder-driven processes were required to reconcile evidence-based decision-making with stakeholder control. This was particularly challenging when coupled with

the inherent complexity of the marine environment and the absence of complete data. As a result, the recommendations reflected a compromise between the achievement of marine protection objectives and stakeholder interests.

Furthermore, it was apparent that the collaborative process did not sit well with England's conventional legislative process, to which the stakeholders' recommendations were subject, after being finalised. The key problem was that there was no certainty that the recommendations would survive public consultation and Ministerial discretion. The benefits of collaborative processes may be lost when it is not certain that the recommendations will be taken forward. Furthermore, stakeholders relied upon the ability to challenge the decision at a later stage rather than fully committing to reaching a collaborative agreement.

Each of the initiatives studied has met with challenges, but also many successes. The California Marine Life Protection Act has enabled the establishment of a network of marine protected areas, the design of which has been used as a blueprint for other jurisdictions. The New South Wales system is older, and the 2012 review recommends some alterations to the model. However the review found that the protection measures achieved by the system were appropriate and should be maintained, and that in fact, management measures should be broadened. The Australian Commonwealth initiative is ambitious but at the time of writing has resulted in final recommendations for the establishment of a significant number of new marine protected areas, representing Australia's key marine habitats. Similarly, the English initiative has resulted in proposals for the establishment of many new marine protected areas, demonstrating what can be achieved under a strong legislative and policy framework.

Figure 4.1: Summary of international approaches to marine protection

Legislative framework	England Marine Conservation Zones Marine and Coastal Access Act 2009	California Marine Life Protection Act California Marine Life Protection Act 1999; Marine Managed Areas Improvement Act 2000	Australia Commonwealth Marine Reserves Environmental Protection and Biodiversity Conservation Act 1999	New South Wales Marine Parks Marine Parks Act 1997
Purpose of marine protected areas under the legislation	The conservation of marine flora or fauna; marine habitats or types of marine habitat; features of geological or geomorphological interest	The conservation of biological diversity and the health of marine ecosystems; recovery of wildlife populations; improvements to recreational and educational opportunities consistent with biodiversity conservation; protection of representative and unique habitats for their intrinsic value	The protection and maintenance of biological diversity and of natural and associated cultural resources	The conservation of marine biological diversity and marine habitats and to maintain ecological processes in marine parks Where consistent with the preceding objects, to provide for ecologically sustainable use of fish (including commercial and recreational fishing) and marine vegetation and marine parks, and to provide opportunities for public appreciation, understanding and enjoyment of marine parks

England Marine Conservation Zones	California Marine Life Protection Act	Australia Commonwealth Marine Reserves	New South Wales Marine Parks
Organisations	Statutory science advisory bodies; regional stakeholder groups supported by full-time project team	Department of Fish and Game; Natural Resources Agency; Resources Legacy Fund Foundation; Blue Ribbon Task Force; regional stakeholder groups	Marine Parks Authority; Marine Parks Advisory Council
Scope	Mean high water springs to 200 nautical miles	State waters from mean high water springs to three nautical miles	State waters from mean high water springs to three nautical miles
Process adopted	Regional stakeholder groups identify marine protected area networks	Regional stakeholder groups identify marine protected area networks	Marine Parks Authority identifies proposed marine park sites
Design guidance	Ecological design guidance drafted by science advisory bodies set minimum requirements for network design	Blue Ribbon Task Force produced planning guidance assisted by science advisory team	Design undertaken by the Marine Parks Authority in accordance with 2001 design guidance
Management measures and zoning	Management measures to be designed on a case-by-case basis	State marine reserves, state marine parks and state marine conservation areas allow different levels of activity	Sanctuary zones, habitat protection zones, general use zones and special purpose zones

England Marine Conservation Zones	California Marine Life Protection Act	Australia Commonwealth Marine Reserves	New South Wales Marine Parks
Role of science Statutory science advisory bodies provided assistance, including economic, social and environmental science. Independent science advisory panel reviewed proposals	Independent science advisory team appointed	National Oceans Office led the design process	Marine Parks Authority led the design process
Indigenous rights Not relevant	No specific legal mechanisms for protecting indigenous rights but tribal interests represented in stakeholder groups	The development of the marine bioregional plan for the north region incorporated the preparation of three 'sea country' plans which enabled traditional owners to provide input into the planning process	Special purpose zones for customary use. Agreements with indigenous groups to provide for cultural resource use, and special permits provided from time-to-time for community events
Outcomes Each of the stakeholder groups met their targets in most key areas, although no-take zone proposals were inadequate. Proposals to be improved by statutory science advisory bodies	The statewide coastal network includes 123 marine protected areas. They cover approximately 2,196 square kilometres (16 per cent) of state waters (excluding San Francisco Bay). There are 48 state marine reserves which represent approximately nine per cent of state waters (excluding San Francisco Bay)	Government has been successful in completing draft plans for all Australia's bioregions. The five draft bioregional plans released over seven years later show a wide divergence in the proportion of each region which is proposed to be fully protected. This ranges from 3 per cent in the north to 51 per cent in the Coral Sea.	3,473.5 square kilometres is protected in marine parks, of which 24.5% is full no-take reserve. Recent review proposes changes to the system to fill gaps in biodiversity protection and develop broader spatial planning system

Endnotes

1. Section 117(1), Marine and Coastal Access Act 2009
2. Section 123 (3), Marine and Coastal Access Act 2009
3. Science Advisory Panel, 2011a, 2
4. California Department of Fish and Game, 2008
5. <http://www.environment.gov.au/coasts/mpa/nrsmpa/index.html> (accessed 31 July 2012)
6. See <http://www.environment.gov.au/coasts/mbp/index.html>
7. Australian Department of Environment and Water Resources, 2007
8. Australian Department of Sustainability, Environment, Water, Population and Communities, 2012
9. Section 6, Marine Parks Act 1997
10. New South Wales Marine Parks Authority, 2007, 30
11. New South Wales Marine Parks Authority, 2001
12. 'NSW Government takes the politics out of marine parks', 2011, Media release, 25 May, at <http://www.mpa.nsw.gov.au/pdf/mr-25-05-11.pdf> (accessed 31 July 2012)
13. 'Environmental expert to lead audit of NSW marine parks', 2011, Media release, 7 June, at http://www.dpi.nsw.gov.au/__data/assets/pdf_file/0005/394025/Environmental-expert-to-lead-audit-of-NSW-marine-parks.pdf (accessed 31 July 2012)
14. Beeton R *et al*, 2012, ix

chapter five



Elements of international
best practice

The design and operation of marine protected areas by numerous countries around the world has generated a wealth of experience from which New Zealand can draw. This chapter synthesises this experience to identify the elements of current international best practice, particularly focusing on the design of marine protected area networks, the identification of management tools, use of stakeholder processes and ongoing monitoring and enforcement.

Marine protected area design

When designing new marine protected areas there are a range of matters which need to be considered including the overall goals to be achieved, the scale of the areas, and the design principles to be applied.

Goals

There is general international agreement that marine protected areas should be designed so that they are capable of protecting a full range of the habitats, species and ecosystem processes present within regional or national jurisdictions. For example, the countries which are party to the OSPAR Convention for the north-east Atlantic have set out what the establishment of marine protected areas should achieve:¹

- a) *Protect, conserve and restore species, habitats and ecological processes which are adversely affected as a result of human activities*
- b) *Prevent degradation of and damage to species, habitats and ecological processes, following the precautionary principle*
- c) *Protect and conserve areas that best represent the range of species, habitats and ecological processes in the OSPAR area.*

Scale

Two main approaches have been adopted to achieve the goals sought for marine protection. The first has been to create large marine protected areas, and to incorporate a range of management measures within their boundaries, depending on the sensitivity of particular areas. The second has been to create a network of smaller protected areas which together aim to protect all the features of the ecosystem.

Very large marine protected areas are becoming increasingly common – the larger the area, the more likely that an adequately broad range of ecosystem features and processes will be protected. For example, at the time of writing, the Australian government has published proposals to create the world's largest marine protected area in the Coral Sea, extending over approximately 989,842 square kilometres. The proposal includes about half the total area in no-take reserves, with the rest being incorporated into multi-use, single-use and wilderness conservation areas where some recreational and commercial fishing may take place.²

Work is also being undertaken to develop marine protected areas in the Cook Islands (one million square kilometres) and in international waters in the Sargasso Sea (five million square kilometres). The United States-based Pew Environment Group is currently campaigning for the establishment of a 6,200 square kilometre 'ocean sanctuary' around the Kermadec Islands within New Zealand's exclusive economic zone.³

Significant benefits can be derived from developing large marine protected areas. As noted by the United States National Research Council, “[l]arger reserves protect more habitats and populations, providing a buffer against losses from environmental fluctuations or other natural factors that may increase mortality and reduce population growth.”⁴ Furthermore, studies have indicated that the establishment and management cost per unit area, at least up to a certain point, can decrease substantially as the size of the marine protected area increases.⁵

Large marine protected areas can also serve to expand protective measures into the open ocean, and in doing so protect not only what we know, but also all the species and ecosystems in the area that are not yet fully understood by science.⁶ In addition, they can result in the protection of remote, largely undisturbed areas, such as the lesser used outer parts of exclusive economic zones. This is the case with New Zealand's benthic protection areas which exclude trawling and dredging from 17 areas, totalling 1.2 million square kilometres of seabed, with seven of the areas abutting the outer limits of the country's exclusive economic zone.⁷ Areas like these are increasingly of scientific and management interest as technological developments enable greater access to these areas, both for scientific study and for exploitation.

Nevertheless there are practical and political barriers to the establishment of very large marine protected areas in many locations, as the needs of users must also be provided for. Policing and enforcement of the rules may also be difficult. The establishment of networks of smaller areas, chosen for their ability to protect different parts of the ecosystem, may be a more practical alternative. Thus the concept of an 'ecologically coherent network' has become the benchmark against

which to assess the effectiveness of marine protection. The IUCN states that *“An MPA network is an organised collection of individual sites, designed to link individual areas and to comprehensively represent the region's spectrum of marine life characteristics.”*⁸

Network design principles

The concept of ‘ecological coherence’ is still evolving, but a number of bodies have produced guidance for marine protected area projects on how to achieve it. The IUCN has developed criteria for the design of marine protected area networks, emphasising eight principal ecological design criteria:⁹

- **Representativeness** – Networks should represent the range of marine and coastal biological diversity (from genes to ecosystems) and the associated physical environment within the given area. Whereas previously the focus tended to be on habitats and species that are rare or unusual, best practice suggests that design should aim to protect both the range of species and habitats present in the region as well as species and habitats that are rare, threatened or declining.
- **Replication** – All habitats in each region should be replicated within the network and be distributed spatially throughout it. For example, the ecological network guidance for the English marine conservation zones requires the identification of two or more examples of each protected feature.
- **Viability** – Networks should incorporate self-sustaining geographically dispersed component sites of sufficient extent to ensure population persistence through natural cycles of variation. These sites should be independent (as far as possible) of activities in surrounding areas.
- **Precautionary design** – Network designers should base their decisions on the best available information, rather than delaying measures to await more and better information. Where information is limited, designers should adopt a precautionary approach.
- **Permanence** – Network design should provide long-term protection to effectively conserve diversity and replenish resources.
- **Maximum connectivity** – Network design should seek to maximise and enhance the linkages among individual marine protected areas,

groups of marine protected areas within a given region, or networks in the same and/or different regions.

- **Resilience** – Networks should be designed to maintain the natural states of ecosystems and to absorb shocks, particularly in the face of large scale and long term changes (such as climate change). A network will be resilient through the inclusion of replicates of representative habitats within the network, connectivity between sites within the network, ensuring all sites are of a viable size, and effective protection of features within the network.
- **Size and shape** – Individual marine protected area units within the network must be of sufficient size to minimise adverse impacts from activities outside the protective area (avoiding what is called the ‘edge effect’).

The aim of achieving ‘representativeness’ is likely to be an aspirational goal, as our understanding of the complexity of the marine environment likely precludes the establishment of marine protected areas which protect examples of all its features. Furthermore, some species are more likely to be effectively protected by spatial measures than others.

The question of how to ensure the long-term resilience of marine protected areas needs to be carefully considered. Allison *et al* have shown that marine protected area design should include buffers which provide insurance against catastrophic events, such as natural disasters and oil spills, which have the potential to prevent marine protected areas from achieving their goals. They argue that although catastrophic events may appear unlikely, in fact over the long term they are common, and thus should be considered in design. Mitigation can be achieved by reducing the risk (for example, through reducing the likelihood of oil spills by carefully managing oil extraction) or by expecting losses, for example by compensating for them through increasing the scale of marine protection by a certain magnitude, to reflect the risk that some will be lost.¹⁰

In recent times, driven by targets set out at international fora, there has been a focus on the need to establish marine protected areas over specified percentages of marine space. Whilst a commitment to achieve these goals provides useful impetus for improved marine protection, it is not the ‘be all and end all’. Rather, the focus needs to be on the establishment of marine protection measures which reflect the values of the marine area in question.

Management measures

Whilst ‘traditional’ notions might envisage that marine protected areas must be spaces in which all extractive activities are prohibited, current thinking suggests that more flexible measures may be appropriate. This can allow for the protection of larger areas whilst minimising the impact that this has on users. The identification of different areas or ‘zones’ within marine protected areas, to which different management rules apply, can also fit well within broader marine spatial planning approaches applied to the larger marine system. This approach can also enable marine protected areas to achieve ‘secondary’ purposes which reflect cultural and social values, so that in addition to the main purpose of protecting the environment, the areas can be managed for the achievement of other goals.

Zoning

Zoning has become a well-established tool for providing effective protection of marine ecosystems whilst also allowing for sustainable use. Such an approach permits different levels or types of activity to occur in different areas, in accordance with what the ecosystem can tolerate. Thus a ‘recreational use zone’ might prohibit commercial fishing but allow all non-extractive activities and some recreational fishing. Importantly however, taken together, the zones should provide for effective protection of the ecosystem as a whole.

In fact zoning has been used for many years. Despite its tiny size, the first statutory marine reserve in the United Kingdom, Lundy Island, incorporated a zoning system when it was established in 1986. Nowadays, zoning might be incorporated into a network of marine protected areas, where different parts of the network are assigned to different categories. It might also be applied to enable the establishment of large marine protection areas, in a situation where this would be unachievable, if marine protection meant that all activities had to be excluded from the area. The Great Barrier Reef Marine Park provides an excellent example of how zoning can be used to provide protection for a large marine area whilst also enabling sustainable use of the marine environment. Similarly, New South Wales operates six marine parks, where each park is divided into separate zones providing for different levels of use.

Classification systems

The design of management measures for each marine protected area can be done on a case-by-case basis. For example, the English legislation provides for

the development of detailed guidance about how marine protected areas should be designed and what management measures should achieve, but the actual management measures are to be established at a later date, once the identification of areas for protection has taken place. This approach has been criticised by stakeholders for failing to provide certainty about what the establishment of a marine protected area might mean for the activities that take place there.

Alternatively, a simple and transparent means by which to assign different levels of protection, is to establish pre-determined categories to be applied marine protected areas, which reflect the management measures to be implemented in them.

Thus the IUCN classifies six sub-categories of marine protected areas according to the purposes for which they are managed, keeping in mind that “*only those areas where the main objective is conserving nature can be considered protected areas; this can include many areas with other goals as well, at the same level, but in the case of conflict, nature conservation will be the priority.*”¹¹ These categories have been applied to the Australian Commonwealth marine protected area programme:

- *Category I – Protected area managed mainly for science or wilderness protection (Strict Nature Reserve/Wilderness Area);*
- *Category II – Protected area managed mainly for ecosystem protection and recreation (National Park);*
- *Category III – Protected area managed mainly for conservation of specific natural features (Natural Monument);*
- *Category IV – Protected area managed mainly for conservation through management intervention (Habitat/Species Management Area);*
- *Category V – Protected area managed mainly for landscape/seascape conservation and recreation (Protected Landscape/Seascape);*
- *Category VI – Protected area managed mainly for the sustainable use of natural ecosystems (Managed Resource Protected Area).*¹²

Under this system the appropriate management measures in each zone category will vary according to the purpose of the zone. For example, in areas managed mainly for wilderness protection, all extractive activities and perhaps public access will be prohibited. In areas managed for seascape conservation and recreation, non-extractive activities may be permitted. A particular marine protected area may have zones in different categories within its boundaries. For example, the

Great Barrier Reef Marine Park has been described as category VI in the IUCN system for the overall area, but it contains zones in categories Ia, Ib, II, III and IV.¹³

Importance of no-take zones

Even with the use of flexible zoning systems, no-take zones need to be at the core of marine protection. Such areas can provide unique benefits to scientists, marine managers and society as a whole. They are important for providing a benchmark for assessing the state of the environment and against which the health of the rest of the marine area and the success of management regimes can be measured. They can assist scientists to better understand how ecosystems function. They contribute significantly to the recovery and protection of marine ecosystems. In addition, they have an important educational and inspirational function, demonstrating what our marine space could be like with better management or lower levels of anthropogenic interference.

Care is needed to ensure that no-take zones are adequate in size and design, and are adequately enforced. A requirement for the identification of no-take zones has been included in the design of a number of marine protected area networks overseas. For example, the ecological network guidance which informed the English marine conservation zone identification project demanded the inclusion of two 'reference areas' where extractive activities are prohibited for each identified species and habitat within the exclusive economic zone.¹⁴

Integration with marine spatial planning

To achieve the best possible value from the establishment of marine protected areas, they need to be designed in the context of an ecosystem-based approach applied to the management of the broader marine area. A means of achieving this is through undertaking marine spatial planning, a process which provides for all uses of the marine area, including marine protection, considering the needs of each activity and their impact on the marine environment in an integrated manner.

In this way, managers can ensure provision for a range of activities, both present and future. This approach can also ensure that the potential for activities outside marine protected areas to damage areas within them is minimised. Marine spatial planning, whilst a relatively novel concept, has been, and is currently being undertaken in a range of different locations including England, Massachusetts,

British Columbia, the Netherlands, Sweden, Belgium, Norway and prospectively the Hauraki Gulf in New Zealand. An independent review of marine parks in New South Wales has recently recommended that the marine parks system be replaced by a holistic system for managing all activities in the marine estate.¹⁵

Marine spatial planning does not necessarily mean parcelling up all marine space into zones for different activities. Rather, such planning can be at a strategic level. For example, the Australian Commonwealth government's marine bioregional planning process provides for the making of plans for large parts of the marine area. These set out the Australian government's objectives for conservation and environmental management measures such as marine protected areas, listed marine species protection and sustainable fisheries.

Some of the most significant impacts on the marine area are terrestrial in origin. For example, in New Zealand, many coastal marine areas suffer from damage caused by sediment runoff and pollution originating from land. The loss of coastal habitat can cause additional problems. Managing such issues continues to pose a challenge for marine managers and effective tools are required to integrate catchment and marine management.

There have been some efforts overseas to deal with the issue, notably in the Monterey Bay national marine sanctuary, where the non-legislative management plan addresses the issue of sediment runoff from farm land and attempts to include farmers in dialogue about how the impacts should be managed.¹⁶ Closer to home, the Hauraki Gulf Marine Park Act 2000 recognises the linkage between catchments and the marine park through bringing together management agencies with jurisdiction over both areas in the Hauraki Gulf Forum and requiring them all to apply the same management objectives to the area. The proposed marine spatial plan being developed for the Gulf is intended to address impacts on the marine area derived from catchments, as well as those originating from marine uses. Nevertheless, it is clear that in most jurisdictions, significant work is required to address the impacts of terrestrial activity on the marine environment.

Process for designing new marine protected areas

Many nations, including New Zealand, have expressed an intention to establish a representative network of marine protected areas within a specified time period. Thus, they have adopted initiatives aimed at meeting these goals in a timely manner, rather than relying solely on the establishment of marine protected areas

on an *ad hoc* basis. There are some important lessons that can be learnt from these experiences.

Leadership and decision-making

Experiences overseas indicate that success is dependent on strong commitment being demonstrated by the relevant government agencies.¹⁷ This commitment is particularly important given the diverse range of interests potentially affected by the implementation of spatial regulation in the marine environment. Initiatives may meet with significant opposition from certain stakeholders, so the government agencies must show strong leadership in managing this.

Another key success factor is the development of clear policy and effective legislative instruments which set out how the network design and establishment process should be undertaken. Government agencies involved in the initiative must be in agreement about how such instruments should be interpreted, not least to ensure that a clear message is received by stakeholders.

Resourcing

The provision of adequate resources is vital. In California, the first two attempts at the establishment of a marine protected area network collapsed, at least in part because of a failure to resource the initiative adequately. Success was achieved once it was recognised that much greater levels of resources were required.¹⁸ Each of the initiatives surveyed in this paper reported the need for a huge amount of work in data collection and processing, public engagement, project management and science advice.

Stakeholder involvement

Historically, marine protected areas in economically developed jurisdictions with mature legal, bureaucratic and political systems, have tended to employ a government-led approach to the establishment of marine protected areas.¹⁹ However, in the last decades, the notion that stakeholders should be involved in their design and management has been growing in strength.²⁰

Stakeholder involvement may be achieved in a number of ways. Central government may undertake the identification and implementation of marine protected areas alone, seeking periodic stakeholder involvement. This 'consultative' approach has the advantage of increasing the likelihood that decisions will be made on the best available evidence, because those involved are

more likely to be experts with a good understanding of the scientific and technical aspects of the issue, and able to see the bigger picture.

Some jurisdictions have adopted collaborative initiatives which aim to incorporate stakeholder-led decisions into conventional government processes. These processes may be characterised as both 'top-down' and 'bottom-up', in the sense that they are led by central government, but provide for stakeholders to take key decisions.

There are a number of reasons for encouraging greater stakeholder involvement in, and ownership of, marine protected area decision-making. First, the introduction of spatial regulation in our oceans is relatively recent, and it is absent from most areas of marine space. The notion of the ocean as a commons, to which we all have rights, still underpins the public consciousness. Fishing, in particular, is a 'way of life' as well as being undertaken for employment and recreation purposes.

As a result, in many places including New Zealand, discourse about the marine area and how it should be used is characterised by fraught, polarised argument. Conservationists care passionately about protecting marine species and habitats from human impacts, whilst users of the oceans defend their right to undertake their activity equally passionately. Theoretically, collaborative processes can help stakeholders to move past the 'conservationists versus users' dichotomy by uniting diverse interests towards a common goal.

Furthermore, stakeholders can contribute valuable data, knowledge and experience which is critical to making effective management decisions in an inaccessible, poorly understood environment. The involvement of stakeholders can also ensure that socio-economic concerns are better addressed.²¹ Equally, where marine protected areas include management measures that involve restricting the activities of ocean users, implementation and enforcement of the rules is theoretically easier when stakeholders have been engaged in their design and as a consequence understand their meaning and the reasons for them. Politicians may be more likely to approve marine protected area proposals if they are confident that stakeholders' views have been heard and accommodated where possible. This helps to reduce the political risk of taking action.

Collaborative marine protection initiatives have met with some success, producing proposals for networks of marine protected areas in New Zealand, England and California, amongst others. However, ambitious collaborative processes can be expensive, as demonstrated by the Californian experience which only succeeded once significant financial support from a private foundation was secured.²²

Where lay participants take responsibility for decision-making, science-based priorities may be compromised. Stakeholders with no scientific expertise, and who are also concerned for their own interests, may be unable to make robust decisions which are supported by the available science.²³ Such problems have led some to argue that, whilst these types of processes are most effective in relation to small-scale, local marine protected area projects, they are not appropriate where there are significant economic activities to be considered, or where complex multi-activity management is necessary.²⁴

Furthermore, whilst collaborative processes seek to tap into benefits not normally available in traditional top-down governance, there is a limit to the amount of responsibility that governments are prepared (and able) to cede to the collaborative process. There is no guarantee that a collaborative group, given free rein, will produce decisions which are palatable to political sensibilities. The use of collaborative decision-making goes against the grain of traditional governance in Anglophone countries such as New Zealand, where government decision-making processes and consultation requirements are well established in existing legislation. If recommendations achieved through collaborative decision-making are rejected or altered significantly by the government, the resources invested in the process are wasted and stakeholder commitment to the initiative will be undermined.

Information

The absence of complete scientific data about the marine area is one of the key challenges facing marine managers wishing to develop new marine protected areas. In most areas, understanding of the ecosystem and the ecological processes and values which should be protected, is partial at best. Accordingly, it is difficult for managers to make informed decisions about effective management.

Critics may argue that the absence of complete data sets is a reason not to make, or to postpone, management decisions. However, in many areas enough is known for it to be apparent that better protection for the ecosystem is required. If a decision is made to proceed with the development of marine protected areas, all parties involved in the process need to accept that, whilst the available knowledge is not perfect, a decision will be made on the basis of the available data. In this situation, a feedback mechanism should be deployed, to allow for periodic review and amendment of measures – a form of adaptive management.

Monitoring and enforcement

Efforts focused on the establishment of new marine protected areas, in order to reach percentage-based goals, risk the establishment of ‘paper parks’ – areas which are technically marine protected areas but where activity continues as normal. This can occur either because the regulatory measures associated with the protected area are weak, or because the rules are not adhered to by users.

The design of marine protected areas needs to reflect the capacity of the administering authority to enforce the rules, and ensure that the rules are simple enough so that users can understand them. Equally, a robust monitoring and review programme is required, so that management measures can be evaluated on an ongoing basis, and adapted where appropriate. However, review programmes need to be carefully designed, to ensure that management measures are robust enough to endure, and that they cannot be arbitrarily weakened in the future by those who oppose the measures.

Figure 5.1: Summary of key elements of international best practice

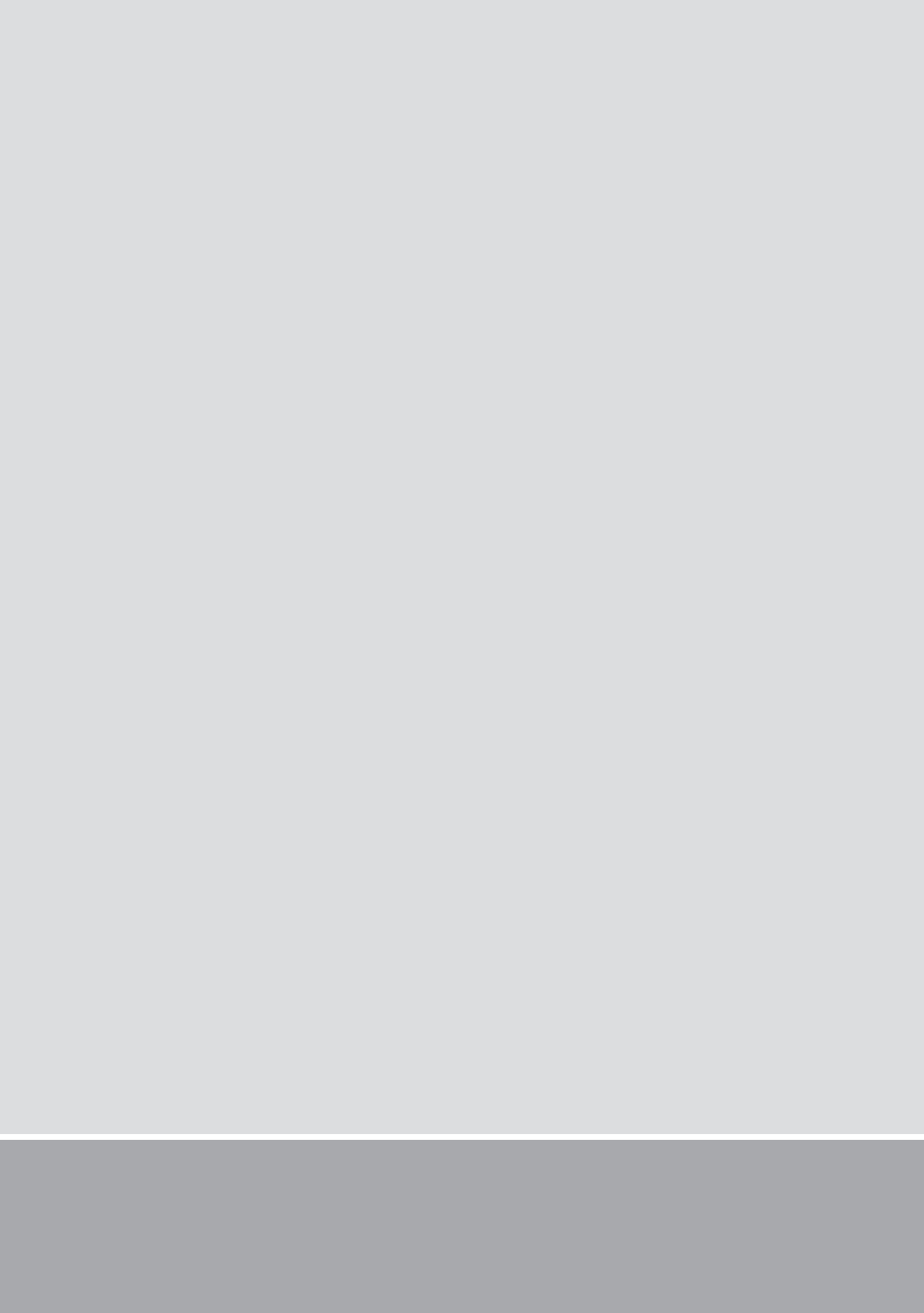
- Marine protected area networks should aim to protect the full range of habitats, species and processes present within the marine ecosystem
- Marine protected areas typically take the form of either large areas which incorporate finer-scaled management measures, or networks of smaller but more highly-protected areas
- Ecological design principles normally include the concepts of representativeness, replication, viability, precaution, permanence, connectivity and resilience
- Zoning is a well-established tool to enable both protection and sustainable use within marine protected areas
- A classification system can set out a range of protection levels which can be applied to marine protected areas, provided that a minimum area is protected in no-take marine reserves
- Marine protected areas may be most effective when incorporated into wider marine spatial planning approaches
- Strong leadership and commitment from government is essential for success
- Stakeholders should be involved in marine protected area design from the outset
- There are pros and cons to the use of collaborative initiatives to identify marine protected area networks, but if collaborative decision-making is to be used, the process must be carefully designed and well-resourced
- Marine protected areas need to be designed to facilitate compliance, effective enforcement and ongoing monitoring and review

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PART TWO

NEW ZEALAND'S PERFORMANCE





Why New Zealand needs
marine protection

New Zealand's management challenge is significantly larger than that of many coastal nations, as it is responsible for one of the largest marine jurisdictions in the world. This expansive area is a rich resource for the country, and is extremely important in both economic and cultural terms. As a result, the pressures on the environment are significant and good management is extremely important.

Large and biodiverse marine area

At more than 4.2 million square kilometres, the territorial sea and exclusive economic zone are over fourteen times the size of New Zealand's land area, thanks to the large number of offshore islands and islets that are included in the country's territory.¹ The extended continental shelf adds a further 1.7 million square kilometres, although New Zealand's right to manage this area is much more limited.^{2,3}

New Zealand's marine environment is host to a highly unusual array of unique and diverse ecosystems. This is partly due to its latitudinal spread. The area spans a latitude of 40 degrees, from subtropical Raoul Island (29°15'S), to Sub-Antarctic Campbell Island (52°30'S). This results in a wide range of oceanographic conditions, ranging from the tropical waters to the north, to the Sub-Antarctic waters of the south. It also has a unique and tumultuous geological history, in which it developed in isolation from other land masses for many millennia.

The islands of New Zealand are the visible surface of a large submerged continent, known as *Zealandia*. Zealandia began to split away from Australia and Antarctica 83 million to 100 million years ago, in the late Cretaceous period. Large fluctuations in sea level over the following millennia meant that Zealandia became mostly submerged, and it was only relatively recent volcanic activity and tectonic movement which brought about New Zealand's land mass as we know it today. This volcanic history has contributed to the undulating nature of the coastline with its many inlets, harbours, spits, fiords, sounds and estuaries. The complex topography creates a wide variety of coastal marine habitats and also makes it very difficult to measure the length of the coastline accurately – estimates have ranged between 15,000 and 19,000 kilometres.⁴

Offshore tectonic and volcanic activity has created a wide variety of seabed habitats, including seamounts, hydrothermal vents, mud plains and deep canyons. For example, the meeting point of the Pacific and Australian crustal plates is associated with the 10,047 metre deep Kermadec Trench, the volcanic Kermadec Ridge (which is part of the Pacific Ring of Fire) and the Puysegur Trench south-west of New Zealand

(which is up to 6,200 metres deep). South-east of New Zealand the Kaikoura Canyon, created by the movement of the tectonic plates towards each other, is an area of deep water (plunging down to 1,000 metres) unusually close to the coastline.

New Zealand's offshore waters are very deep. The limited availability of nutrients means that in a global context, offshore fishing grounds are relatively unproductive, accounting for less than one per cent of the global catch.⁵ However, the biodiversity of the region is unusually high. The Census of Marine Life's 2010 report on marine biodiversity in New Zealand, which studied marine species found within the 200 nautical mile boundary of New Zealand's exclusive economic zone, identified 17,135 living species, including 4,135 which were held in collections but remained undescribed.⁶

The list includes 148 species of cartilaginous fish, three species of pinnipeds and 43 species of cetaceans (comprising nearly half of the world's cetacean species). Of these cetaceans, nine are baleen whales, 17 are members of the dolphin family, and 12 are species of beaked whale about which little is known. Of New Zealand's 286 bird species, 122 are marine species, or spend some part of their lives in the maritime area. More sea birds breed in New Zealand than anywhere else in the world.

Figure 6.1: New Zealand's marine jurisdiction

Source: Ministry for the Environment



Many marine species yet to be discovered

The large size, geological variety and relative remoteness of New Zealand's marine jurisdiction mean that much of it remains unexplored, especially the 50 per cent of the exclusive economic zone that is deeper than 2,000 metres. It is thus estimated that there are around 17,000 undiscovered marine species in New Zealand's waters.⁷ This would put the country's total biodiversity at equal to, or in excess of, the region covered by the European Register of Marine Species, which is around 5.5 times larger and includes multiple state jurisdictions.⁸

Significant discoveries of previously unknown species are not uncommon in New Zealand. For example, in 2012 seven examples of a 'supergiant' shrimp or amphipod, were discovered deep in the Kermadec Trench. These were around 30 centimetres in length, three times larger than any previously known amphipod.⁹ 'Bioblitzs', where scientists and members of the public work together to undertake a biological survey of an area over a short period of time, can also yield new species. For example, a Bioblitz undertaken in the vicinity of the Wellington Taputeranga Marine Reserve in 2007, resulted in the discovery of new tube anemone, nudibranch and bryozoan species, amongst others.¹⁰

High social and cultural values

The oceans are of particular cultural importance to Māori. The marine area provides kaimoana (seafood) for sustenance, for ceremonial occasions and for trade. For example, tāngata whenu often refer to the Hauraki Gulf as 'pataka kai' or food-basket being "*a place where spiritual and physical sustenance is gained*".¹¹ Many cultural practices are associated with the exercise of kaitiakitanga over the marine area. New opportunities are being provided through the creation of taiāpure and mātaitai which enable traditional fisheries to be managed by tāngata whenua. For example, at the Maketu taiāpure in the Bay of Plenty, kaitiaki have undertaken stock assessment surveys, funded the removal of an invasive starfish species and initiated a breeding and planting programme to restore whitebait.¹²

New Zealanders are more generally ocean-loving people, and many cultural and recreational activities are based around the water. It is estimated that 19.5 per cent of the New Zealand population participates in recreational fishing,¹³ taking an estimated annual catch of 25,000 tonnes.¹⁴ Sport New Zealand's most recent 'Active New Zealand' national monitoring survey indicates that New Zealanders

participate in a range of ocean based sports – including swimming (34.4 per cent), canoeing and kayaking (6.4 per cent), surfing (4.4 per cent), diving (3.7 per cent) and sailing (2.4 per cent).¹⁵ Coastal marine areas, in particular, are popular for their intrinsic natural values and New Zealand is world-renowned for its beautiful coastlines.

High commercial values

Three of the four largest export earners in New Zealand are reliant, at least in part, on marine resources.

Tourism

In the year ending March 2011, tourism was New Zealand's second largest export earner. International tourist expenditure accounted for \$9.7 billion or 16.8 per cent of New Zealand's total export earnings. Total tourism expenditure (including both domestic and international tourism) was \$23 billion dollars, or 8.6 per cent of New Zealand's gross domestic product.¹⁶

Tourism New Zealand's marketing strategy is heavily reliant on the country's clean environment and international perceptions of New Zealand as an Eden-like escape.¹⁷ Coastal and marine activities form a key part of many tourist experiences.¹⁸ In 2007, 111,420 international tourists went dolphin watching and 98,110 went whale watching. Other popular activities include boating (698,380), kayaking (153,166), fishing (120,483) and swimming (104,592).¹⁹

Oil and gas extraction

New Zealand's third largest export earner is oil and gas extraction, which has been undertaken in New Zealand waters since the late 1950s. Activity has been focused off the coast of Taranaki.²⁰ In 2009, nearly 20 million barrels of oil and over 140 billion cubic feet of gas were generated in New Zealand, of which most came from offshore areas. It is believed that the continent of Zealandia is likely to have potential for many more hydrocarbon resources. The National Institute for Water and Atmospheric Research (NIWA) estimates that New Zealand's recoverable petroleum reserves could be around 10 to 20 billion barrels.²¹ The country's oil and gas reserves are highly significant, both in terms of potential export revenues,

and medium term energy security. In recent years, the government has prioritised the growth of the oil and gas sector,²² and this has stimulated exploration activity.

Seafood industry

The seafood industry, including aquaculture, is worth around \$4 billion to the New Zealand economy each year (2009 figures),²³ and is New Zealand's fourth largest export earner. At present, some 600,000 tonnes of seafood is harvested from New Zealand waters each year, and one hundred and thirty of the species found in New Zealand waters are fished commercially. Around 63 per cent of this harvest is comprised of mid and deep water fish, 12 per cent is pelagic fish (that inhabit the water column), 10 per cent is inshore species and 15 per cent is from aquaculture. Aquaculture is currently worth \$320 million per year, and is likely to expand significantly in the future.²⁴

Seabed mining

Seabed mining is a key potential growth area. Activity is currently limited to the extraction of sand from coastal marine areas for use in building construction and beach nourishment projects. However, in recent times there has been growing interest in the possibility of extracting other materials from the seabed. It is estimated that once extracted, they could be worth billions of dollars.

Ironsand found off the west coast of the North Island is one such resource. It is used for steel manufacture domestically and is also exported as a raw material. Ironsand has been mined onshore from the Taranaki coastline for many years, but increasing global prices, make mining it offshore a more economically viable prospect.

Deep water phosphorite nodules may be a valuable source of phosphates, used in fertiliser. They are found at depths of around 400 metres, in sandy silt located within the top 70 centimetres of the seabed. It is estimated that there are around 100 million tonnes of phosphates in the Chatham Rise area, located south-east of New Zealand. There are indications that commercial extraction is realistic and could commence in the next few years.

Gas hydrates, described by NIWA as “*an ice like form of water, with methane gas molecules caught in the ice crystal*”²⁵ are found east of the North Island at the Hikurangi Margin. They could become a significant source of energy in the future, but until recently, it was thought that they would be too difficult to extract.

Seabed massive sulphide deposits typically contain copper, lead, zinc and some silver and gold. They are formed on seamounts, where superheated fluids spurt through thermal chimneys. When the fluid hits the water, it hardens into such deposits. They were first discovered on the Kermadec Arc by NIWA and GNS Science researchers in the 1990s. Since then, several companies have investigated their extraction, and there is the potential for commercial exploitation in the future.²⁶

Significant pressures

The government estimates that about 30 per cent of New Zealand's marine environment is 'experiencing some degree of disturbance from human activities'. The largest impact is thought to be from commercial fishing and trawling.²⁷

Furthermore, land-based activities in catchments discharging into the coastal marine environment are also a key concern.²⁸ Sedimentation caused by changes in land use, sewage discharge, increased nitrogen and phosphorus loading, and heavy metal pollution are believed to be key causes of the degradation of New Zealand's inshore marine environment. All these impacts take place in a context where climate change has the potential to significantly alter New Zealand's marine environment, for example, through species shifting their range and ocean acidification.²⁹

Current management has resulted in some successes but there is evidence that the overall health of New Zealand's marine environments may be in decline. The Royal Society of New Zealand has noted (in respect of New Zealand's biodiversity generally) that "*the past fifty years have seen a substantial and largely irreversible loss of biodiversity. New Zealand's unique endemic biodiversity has similarly seen serious decline – an unknown but large loss of common wealth and natural heritage.*"³⁰ The Ministry for the Environment reported in 2007 that "*of the almost 16,000 known marine species in New Zealand, 444 are listed as threatened*".³¹

Trends in the abundance of keystone marine species reflect the likely overall decline. Keystone species are those which have a disproportionately large effect on their environment relative to their abundance, such as marine mammals, seabirds and sharks.³² Of New Zealand's three endemic marine mammal species and subspecies, the New Zealand sea lion and Maui's dolphin are classified as 'nationally critical' and the Hector's dolphin as 'nationally endangered' by the Department of Conservation.³³ Sixty-two per cent of New Zealand's ocean-going

seabirds are listed as threatened.³⁴ Although two species have shown some signs of recovery in recent times, another seven species have become increasingly threatened.³⁵ Many species of sharks, skates and dogfish are at risk, with the great white shark and basking shark being identified as chronically threatened.³⁶

Declining marine health has been documented on a regional scale: for example, the Hauraki Gulf State of the Environment Report 2011 notes that indicators of rare and threatened fish species, fish diversity, fish size and productivity 'displayed negative trends' in the Firth of Thames and most central Gulf areas.³⁷

New Zealand's fisheries are carefully managed under the quota management system, and thus it is sometimes argued that this means that New Zealand has no need for marine protected areas. However, the system is principally addressed at the management of target stocks rather than the protection of the marine environment as a whole. Furthermore, management of all existing activities in the marine area is hampered by an absence of complete data about the marine environment and the impact that human activity has on it. For example, in 2010, the Ministry for the Environment noted that there was sufficient information to assess fish stocks for 119 of the 633 stocks in the quota management system.³⁸ The long term impacts of activities such as sea-bed trawling, are not well understood.³⁹

The development of new activities such as minerals mining in areas of the marine space which are unique, fragile and little explored, brings with it increased risks that human activity will have unintended consequences. As a result, more than ever, an improved framework for marine protection is required.

Figure 6.2: Summary of reasons why New Zealand needs marine protection

- New Zealand has one of the largest marine jurisdictions in the world
- New Zealand's marine area has an unusually high biodiversity, and it is believed that a large proportion of the resident marine life is yet to be discovered
- New Zealand's marine area is very important to New Zealanders, providing for a range of economic, cultural and recreational activities
- The marine environment is experiencing significant pressure from human activity. Key activities include fishing, shipping, aquaculture, oil and gas extraction, sand mining and tourism
- There is growing interest in new extractive activities including ironsand mining off the west coast of the North Island, rock phosphate mining on the Chatham Rise, gas hydrate extraction and seabed massive sulphide mining on seamounts in the Kermadec Arc
- The impacts of anthropogenic pressures on the marine environment are poorly understood, whilst marine biodiversity is declining

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chapter seven



New Zealand's experience with
collaborative processes

Generally individual marine protected areas have been established in New Zealand on an *ad hoc* basis. In some areas, however, projects have been undertaken to implement marine protection measures on a broader scale.

Two areas, off the Sub-Antarctic Islands and the west coast of the South Island, have been the subject of marine protected area identification processes initiated by the Department of Conservation and Ministry of Fisheries under the MPA Policy. These two processes, together with the community-led initiative established by the Fiordland Marine Guardians to improve management measures in their local marine area, are set out in detail as case studies in Part Three of this report and are summarised in this chapter.

Fiordland Marine Guardians

In 1995 the ‘Guardians of Fiordland’s Fisheries’ was established by representatives of commercial fishing, recreational fishing, charter boat operators and iwi in the Fiordland area. They were motivated by concerns amongst the local community about growing tourism in the region and the sustainability of local fisheries. Their vision was “... *that the quality of Fiordland’s marine environment and fisheries, including the wider fishery experience, be maintained or improved for future generations to use and enjoy.*”¹

Resident fishers observed that a 1994 working group approach in Southland had successfully developed a fisheries plan for the Paterson Inlet (Stewart Island), assisted by the Ministry of Fisheries. Representatives from the fishing community approached the Ministry to assist with a similar initiative in Fiordland. The Guardians were thus initially supported by the Ministry of Fisheries, but this support was subsequently replaced by funding from the Ministry for the Environment through the Sustainable Management Fund.

Initially, the group focussed on developing agreements on fish stock management and measures to protect the fisheries, rather than on conservation of the marine environment more broadly. Representatives of conservation interests were not included in the group, as members felt that the inclusion of such ‘extreme’ views, would have prevented the cooperation and cohesion that the Guardians achieved and would have slowed down the process.

Over time this approach was modified and protection of the marine environment became the key focus of the group. Scientists and environmentalists were brought into the group and in May 1995, the organisation changed its name

to 'Guardians of Fiordland's Fisheries and Marine Environment Incorporated', reflecting its broader scope.

Using an independent facilitator, the group met every two months to develop the Fiordland Marine Conservation Strategy, completed in 2003. Gaps in available data were filled and supplemented by information provided by members about their own fishing activities. Further anecdotal evidence was gathered through communication with local communities. This kind of knowledge was exceptionally valuable and enabled the Guardians to develop an intimate knowledge of their marine space.

The strategy identified around 45 issues that affect fisheries and the marine environment. These issues were grouped into four general components:

- Provisions to address fisheries management
- Commitments to protect values of special significance
- Identification of potential risks and measures to respond to them
- Expression of kaitiakitanga, recognising Māori ties to the area

The Strategy set out a range of measures that were to be used to achieve the Guardians' aims. Members were able to identify areas of high biodiversity and those with fragile species, termed 'china shops'. Proposed protection measures for these areas included anchoring prohibitions and diving regulations.

Restrictions on commercial and recreational fishing were established, including catch limits and closed areas. There were also areas designated as no-take, although in the early years, marine reserves were not identified as the appropriate tool to achieve this.

The Strategy was given effect in the Fiordland (Te Moana o Atawhenua) Marine Management Act 2005. This formalised the Guardians' strategy into law. The legislation included the creation of eight new marine reserves within the fiords to protect the representative areas identified in the Strategy. As a result, the total marine area within Fiordland protected as marine reserves now totals 103 square kilometres. However, this comprises only 1.01 per cent of the Fiordland marine biogeographic region (which totals 10,246 square kilometres) and there are no marine reserves in the coastal marine area outside the fiords.

Through the Act, the Fiordland Marine Guardians are assigned an enduring role as government-appointed advisors responsible for monitoring and management of the area. They can continue to monitor the health of their environment and ensure that measures taken are having the desired effect. They can also apply

adaptive management and continue to make adjustments that reflect changing demands and environmental conditions. The Fiordland Marine Guardians do not see their work as a marine protected areas initiative. They take into consideration the ecological health of the whole area and apply their own approach.

In 2007, and then again in 2010, the Guardians conducted a user survey to investigate what activities people are conducting in Fiordland and to understand their perceptions and values associated with the area. The Guardians, in conjunction with support agencies, have a number of monitoring projects underway. The 'Fiordland marine area monitoring programme' involves research commissioned in several areas over both biological and social issues.

Broad stakeholder involvement during the early stages has been important to the ongoing success of this model. If not from the beginning, the Guardians model now involves wide representation from across the district including commercial and recreational fishers, divers, charter boat and tourism operators, environmentalists, scientists, community representatives and tangata whenua (Ngāi Tahu). This community-led approach, with strong local involvement, ensured stakeholder 'buy in' and agreement on major issues. Enforcement requires resources, and the more 'self-enforcement' that occurs, the more efficient the system will be.

West Coast Marine Protection Forum

The West Coast Marine Protection Forum was established in 2005 as a non-statutory body by the West Coast Conservator of the Department of Conservation, essentially as a test case for the implementation of the new MPA Policy. The area was chosen for a number of reasons – it is less densely populated than many other regions and there were no existing marine reserves or any other form of marine protection. In addition, staff from the Department of Conservation's Conservancy office, based in Hokitika, were enthusiastic about getting a process underway and were optimistic that it could work.

The Forum first convened in April 2005, prior to the release of the MPA Policy, which was published in December that year. The Forum was issued with terms of reference to assist with procedural matters. When the MPA Guidelines were published in 2008, new terms of reference were released, but the Forum decided to retain its original terms of reference for continuity and consistency.²

Stakeholder representatives were identified by Conservancy staff, who approached individuals in the community. The individuals invited to join were those who represented key interests, who had influence and connections in the community, and preferably who represented more than one interest. The result was that the group did not necessarily represent all users or interest groups in the area. The composition of the Forum would have been slightly different if the 2008 terms of reference had been issued from the outset, as these require the inclusion of other stakeholders (specifically science and mining industry representatives) that were not in fact included.³

Based on a consultative process, the Forum was expected to evaluate options for achieving a representative network of marine protected areas. These options would include mechanisms for statutory protection available under fisheries regulations, the Marine Reserves Act 1971 or other legislation if appropriate.⁴

The process was undertaken in three phases:

- Phase One: Information gathering and facilitation of stakeholder engagement
- Phase Two: Evaluation of the advice and viewpoints provided in order to develop options
- Phase Three: Reporting options back to the Department of Conservation, the Ministry of Fisheries and other agencies as required⁵

The Forum first met in April 2005 and then around another 29 times before producing its 'recommendation report' in July 2010.⁶ Meetings would typically last one to two days and would often include a tour or field trip. The Forum was able to use an independent facilitator to assist in progressing discussion, decision-making and achieving consensus.

The Forum was allocated assistance from a number of organisations and individuals to inform or support the process and aid decision-making. As the Department of Conservation and the Ministry of Fisheries (now Ministry for Primary Industries) are jointly responsible for developing and implementing the MPA Policy, both assigned multiple staff members (six to seven each) in a support role. Their staff provided science and policy advice where requested and were regarded as *ex officio* members of the Forum. The Department of Conservation largely funded the process.

The Department of Conservation and Ministry of Fisheries had very different approaches towards marine protection. The Ministry was more reluctant to support the creation of marine reserves, preferring instead Fisheries Act mechanisms.

Disagreement between the two agencies on the interpretation of the MPA policy, and on its implementation in practice, trickled down to the stakeholders causing confusion and frustration.

The Forum was ultimately unable to reach consensus. Although the Forum's 'consultation document' (August 2009) gave the impression that consensus had been reached, providing for one set of proposals with no options or alternatives, some of the commercial fishing representatives on the Forum did not agree with these proposals. Not all commercial fishers had equal interests, however, with the small owner-operator commercial fishers having different views to the large corporate entities. The lack of consensus did not 'officially' become apparent until after the submission phase was completed, and the Forum had to make final recommendations to Ministers.

The Forum made recommendations on four primary locations for marine protected areas. Having been unable to reach consensus, it presented alternative protection options which gave greater weight to either maximising the benefits for biodiversity protection or minimising impacts on users. Secondly, the Forum presented five additional educational showcase sites, which were intended to increase public understanding and appreciation of the marine area.

On receiving the recommendation report, the Ministers requested that their officials draft an advice paper, before making a decision.⁷ This paper was completed on 2 June 2011 and the Ministers' decision was released through a media statement on 26 August 2011.⁸

The officials' advice was that the smallest options presented by the Forum, in each of the primary sites, provided the best balance between biodiversity protection and minimising adverse impacts on existing users of the West Coast marine environment. They also advised that the educational sites were additional to the requirements of the MPA Policy but that the Ministers could consider them on their merits. The Ministers accepted most of the recommendations set out in the advice paper and opted to establish one educational site.

A total of 175.28 square kilometers of marine reserve was endorsed by the Ministers. Once established, Kahurangi and Ōkārito will be the two largest marine reserves on New Zealand's mainland coastline. Less restrictive marine protected areas, covering another 95.57 square kilometers, will adjoin two of the marine reserves using Fisheries Act tools to protect them from bottom trawling, dredging and Danish seining.⁹ The overall effect was to take the area fully protected within the west coast South Island bioregion (which covers 13,112 square kilometres) from zero to 1.3 per cent.

Following provision of the recommendations, the stakeholder group was disbanded and is not intended to have a role in the ongoing management of the marine protected areas.

The lack of direct implementing power for the MPA Policy proved a problem. The stakeholder forum was established only to make recommendations to Ministers. Once the respective Ministers made a decision on the recommendations, the full statutory process to implement the protection tools under both the Marine Reserves Act and Fisheries Act was initiated, including consultation.

This significant duplication of effort further delayed progress, and provides no certainty that the recommendations will be implemented. The West Coast Forum took five years to develop recommendations, and although the Department of Conservation expected to establish the marine reserves within 12 months, they have not yet been established. Despite the thorough process undertaken, the Ministers did not opt to establish the marine reserves by deeming legislation as had been done for Fiordland and which is in process for the Sub-Antarctic Islands.

Sub-Antarctic Marine Protection Planning Forum

The Sub-Antarctic Marine Protection Planning Forum was established as a non-statutory body in February 2008. The Forum's purpose was to provide options and recommendations to government for marine protection in the Sub-Antarctic biogeographic region. It was expected to meet the requirements of the MPA Policy and Guidelines and comply with the terms of reference provided by the supporting agencies.

The specific matters the Forum was to consider and present in its report to the Ministers were:

- Classification and inventory information
- Consultation with existing users and interests in the area
- Sites and potential tools for area-based protection of biodiversity
- Any consensus on proposed areas to be set aside as protected areas
- Consultation conducted on protection options and recommendations to the Ministers

The Chairperson and members of the Forum were appointed by the Minister of Conservation and Minister of Fisheries and were provided with terms of

reference by which they were to conduct themselves. The Forum consisted of 14 members representing commercial fishing, environment/conservation, Māori and scientific interests. Members of the Forum were expected to have strong links with the region, be able to negotiate, work well with others and be able to engage with their respective interest sectors and put forward these views during decision-making. An independent chairperson was appointed to ensure the delivery of recommendations to the Ministers. His main responsibility was to convene and facilitate meetings, but also to ensure that the requirements of the MPA Policy and the terms of reference were met.

The Forum convened in February 2008 and met seven times for full-day meetings during that year. By the end of the year it had fallen behind schedule, due to the time taken to understand and process the available information, and to begin network design work.

At this point, the Forum members decided to elect a smaller working group including four representatives of the Forum and the independent Chairperson to prepare the consultation document.

Representatives from the Department of Conservation and the Ministry of Fisheries were regarded as *ex officio* members of the Forum, but were also there to provide information and support where required. Officials provided national policy guidance to the Forum, along with planning and technical advice, and their support was consistent with the MPA Policy.

There was an expectation that members of the Forum would actively engage with their stakeholder groups, but also with those who may have an interest in, or could be affected by, marine protected areas. There was no residential community to consult within the Sub-Antarctic islands. Therefore a formal consultation process to reach the general public was undertaken, which allowed an opportunity for written submissions.

The members of the Forum were unable to reach consensus and resolved to present six options, two for each island. In each island scenario, the first option provided a combination of marine protected area tools for partial protection of the territorial sea, and the second option provided full protection of the territorial sea through the establishment of a marine reserve. The recommendation report advised the Ministers that they could select any combination of the six options. Following the making of recommendations to the government, the Forum disbanded and will have no future role in management of the new marine protected areas.

In January 2011, the government announced that three large marine reserves would be established around the 6,885 square kilometres of territorial seas of the Sub-Antarctic islands. Marine reserves were proposed to cover the entire territorial sea around Antipodes Island, 58 per cent of the territorial sea around the Bounty Islands, and 39 per cent of the territorial sea around Campbell Island. This means that a total area of 4,351 square kilometres will be protected in marine reserves. Prohibitions on Danish seining, bottom trawling and dredging under the Fisheries Act cover the remaining territorial sea areas.¹⁰

The marine reserves are to be implemented through special legislation. The government reported that the Marine Reserves Act 1971 is only designed to process single marine reserve applications and the lack of statutory time frames within the Act leads to protracted processes (the average time for implementing a marine reserve is 12 years). Special legislation would enable the reserves to be implemented as a package, which would be more efficient and less costly. It also recommended that the Fisheries Act regulations should follow the usual statutory process, as this would be consistent with existing restrictions in place, and provide greater flexibility and responsiveness for future adjustment.

The Sub-Antarctic Islands Marine Reserves Bill was introduced into Parliament on 7 July 2011. The Bill establishes the Moutere Mahue–Antipodes Island Marine Reserve; the Moutere Hauriri–Bounty Islands Marine Reserve, and the Moutere Ihupuku–Campbell Island Marine Reserve. In addition, the Bill requires the Department of Conservation, in consultation with the Ministry for Primary Industries, to commission an independent review of the Moutere Ihupuku–Campbell Island Marine Reserve at least five years after the Bill's commencement. This is to consider the effects on conservation and fisheries of extending the boundaries of the marine reserve. At the time of writing the Bill had not yet been passed.

Conclusions

As shown in Figure 7.1, each of the initiatives studied has resulted in proposals for the establishment of new marine protected areas.

Legislative framework

Initially the Guardians of Fiordland was an organisation with no statutory basis. However, its work was given statutory recognition through the passage of special

legislation creating the Guardians of Fiordland as a statutory body. Marine reserves in Fiordland were established under the same special legislation. Fisheries management measures were established separately under the Fisheries Act.

The other marine protection fora were established in accordance with the MPA Policy, and thus had no statutory basis, but could make recommendations for the establishment of marine reserves or spatial protection measures using Fisheries Act tools. Although marine reserves recommended by the West Coast Marine Protection Forum are to be established under the Marine Reserves Act, special legislation is being used to create marine reserves in the Sub-Antarctic Islands, in recognition of the fact that the Marine Reserves Act does not fit easily with the establishment of reserve networks under the MPA Policy.

Management measures

In developing proposals for management measures, both the Guardians of Fiordland and the marine protection fora were restricted to the use of fisheries management measures under the Fisheries Act and no-take marine reserves under the Marine Reserves Act. The use of fisheries tools for marine protection encouraged division between stakeholders, about whether marine reserves were required at all, as fisheries management tools were argued to be sufficient to achieve marine protection.

Design and implementation

The Guardians of Fiordland's initiative was a 'pure' collaborative process, started from the bottom up by interested stakeholders. Its outcomes reflected stakeholders concerns. Conversely, the initiatives implemented under the MPA Policy represent a combination of top-down and bottom-up approaches, as collaborative processes that were initiated and managed by government.

The collaborative approach employed in Fiordland has been successful, both in achieving the implementation of new marine protected areas, and developing a strong unified management body which retains a continued sense of ownership. However, the focus on marine protection was by no means assured. In the early days the Guardians were concerned with fisheries management rather than marine protection. Local initiatives benefit from strong stakeholder ownership, but are subject to local sensibilities, which may not be consistent with best practice marine protection.

The MPA Policy has also only been implemented in two coastal regions although the Policy envisages its application throughout New Zealand waters. The fora which were established struggled to some extent, partially because they were dealing with a new concept, but also because stakeholder agreement was difficult to achieve. In contrast to the Fiordland Marine Guardians, government involvement meant that there was not the same sense of ‘voluntariness’ about the process. It was anticipated that agreement would be easily achieved in the Sub-Antarctic Islands, where the isolation and absence of a resident population was believed to reduce the complexity of stakeholder needs. In fact, extractive users (in particular commercial fishers) were concerned about the potential precedent effect of agreeing to large marine protected areas in this region, as a similar approach may then be applied to other areas where there were established fisheries.

This experience illustrates that collaborative processes initiated by government operate in a significantly different context to collaborative processes developed from the ‘bottom up’. The possibility that agreement will not be reached must be taken into account in the planning stages, with sufficient resources invested in the initiative to limit the risk of failure. Measures such as the use of independent facilitation were reported to be very useful, but stakeholders in both the initiatives undertaken under the MPA Policy indicated that more guidance could have been provided to the groups, on both the process and what they were expected to achieve.

Ongoing management

In contrast to the Fiordland Marine Guardians, the Sub-Antarctic and West Coast fora will not have an ongoing role in management of the relevant marine area. The ongoing benefits that could have been achieved through the implementation of a collaborative process were lost when the fora were disbanded having made their recommendations.

New Zealand’s MPA Policy has resulted in proposals for new marine protected areas, but it has not generated the strong stakeholder ownership characteristic of the Guardians of Fiordland’s initiative. Stakeholders found it difficult to agree, and it is arguable whether the MPA Policy has resulted in the development of much of the ‘social capital’ that is one of the key potential benefits of collaborative processes. The stark choice between Fisheries Act tools and no-take marine reserves compounded that discord.

Figure 7.1: Summary of New Zealand's experience with collaborative processes for identifying marine protected areas

Key characteristics	Guardians of Fiordland	West Coast Marine Protection Forum	Sub-Antarctic Marine Protection Planning Forum
Date established	1995	2005	2008
Purpose	To maintain or improve the quality of Fiordland's marine environment and fisheries, including the wider fishery experience, for future generations to use and enjoy	To give effect to the MPA Policy	To give effect to the MPA Policy
Representation	Initially dominated by fishers; scientists and conservationists later permitted membership	A range of stakeholders included. Mining interests left out	A range of stakeholders included
Independent facilitation	Yes	Yes	Yes
Science input	At later stages	Yes but limited social science/economics input	Yes but limited social science/economics input
Public consultation	Public consultation on draft strategy 2003	Public consultation following forum recommendations	Public consultation during forum process
Consensus reached	Yes	No	No
Date of recommendations	2003	2010	2010
Level of protection	Brought full protection up to 1.01% of Fiordland bioregion	Will bring full protection up to 1.3% of West Coast South Island bioregion	Will bring full protection up to 42% of Sub-Antarctic bioregion
Implementation mechanism	Special legislation and Fisheries Act processes	Marine Reserves Act and Fisheries Act processes	Special legislation and Fisheries Act processes

Key characteristics	Guardians of Fiordland	West Coast Marine Protection Forum	Sub-Antarctic Marine Protection Planning Forum
Ongoing role for Forum	Yes	No	No
Strengths	• Ongoing involvement • Strong community and stakeholder buy-in • Achieved and maintains consensus on decisions • Speedy implementation through special legislation (although ad hoc) • Independent facilitator	• Broad range of stakeholders represented • Local fishers able and willing to provide valuable information about commercial trawl • Independent facilitator • Good commitment and continuity provided by members • Good technical support and policy advice provided by agency representatives at local level • Stakeholders selected had good standing in the community	• A good balance between fishing (extractive users) and environmental interests • Broad representation from non-governmental organisations • Independent chairperson skilled in environmental law and legal proceedings • Good technical support including GIS • Strong policy guidance • Adequate resourcing • Use of special legislation and introduction of tools as a package • Good commitment and continuity by stakeholder members

Key characteristics	Guardians of Fiordland	West Coast Marine Protection Forum	Sub-Antarctic Marine Protection Planning Forum
Weaknesses	• Marine protected areas small and none on the outer coast • Protection of marine mammals over-looked • Slow and reluctant to include environmental groups • Slow to include science representative	• Representation weighted towards extractive users • Marine science and mining industry not represented • Lack of priority given to marine science and data • Background information prepared after process underway causing delay • Lack of clarity provided by agencies at higher level when interpreting policy • No independent science advice • Constraints on information provided by Ministry of Fisheries • Took longer than expected • Consensus not achieved • Duplicated effort in implementation • Lack of integration of customary approaches • No ongoing involvement once recommendations made	• No representation from some interests including tourism and mining industries • Took longer than expected to complete • Forum was polarized between fishing interests in favour of mixed protection tools and environmental advocates in favour of no take reserves • No consensus achieved • No ongoing involvement once recommendations made • Duplicated effort in implementation

Endnotes

- 1 Ministry for the Environment, 2005
- 2 Bruce Hamilton, August 2011, *pers comm*
- 3 Campbell Robertson and Don Neale, August 2011, *pers comm*
- 4 <http://www.westmarine.org.nz/objectives.aspx> (accessed 31 July 2012)
- 5 <http://www.westmarine.org.nz/objectives.aspx> (accessed 31 July 2012)
- 6 West Coast Marine Protection Forum, 2010, 11
- 7 <http://www.doc.govt.nz/conservation/marine-and-coastal/marine-protected-areas/other-protected-areas/proposed-west-coast-marine-protected-areas/>
- 8 'Five marine reserves announced for West Coast', 2011, Media release, 26 August 2011 <http://www.beehive.govt.nz/release/five-marine-reserves-announced-west-coast> (accessed 31 July 2012)
- 9 'Five marine reserves announced for West Coast', 2011, Media release, 26 August 2011 <http://www.beehive.govt.nz/release/five-marine-reserves-announced-west-coast> (accessed 31 July 2012)
- 10 'Subantarctic Islands to become marine reserves', 2011, Media release, 29 January, <http://www.beehive.govt.nz/release/subantarctic-islands-become-marine-reserves> (accessed 31 July 2012)



New Zealand's performance compared
with international best practice

Historical context

New Zealand was one of the first nations to consider the development of legislation that would provide for the spatial protection of the marine environment. In 1965, Professor Valentine Chapman, Chairman of the University of Auckland's Leigh Laboratory Committee, wrote to the government's Marine Department to suggest that a no-take marine reserve in front of the Laboratory should be established. He argued that minimising the impact of human activities would greatly improve the effectiveness of scientific study there.¹

Chapman received an unhelpful response from the Marine Department, which indicated that there were no legal mechanisms available to enable such a measure. Accordingly, the scientific community embarked on a campaign to persuade the government to pass marine reserve legislation.² Six years later, in 1971, the Marine Department released a draft bill and the Marine Reserves Act was subsequently passed.

Following the passage of the Act, the University of Auckland made an application for the establishment of a marine reserve in front of the Leigh Laboratory. The first application was rejected and another submitted in 1973. The application was finally accepted in 1975, making the Cape Rodney-Okakari Point marine reserve the first to be established in New Zealand. It was also one of the first no-take marine reserves to be established under specially designed legislation anywhere in the world.

The Act reflects the fact that it was brought about by intensive lobbying from the scientific community, rather than as a result of government having a specific desire to implement measures for marine protection. At that time, the Marine Department was not enthusiastic about taking on a new responsibility to create marine reserves.³ Thus, in addition to enabling the Secretary of Marine to propose reserves, the legislation provided that a limited number of entities were also empowered to do so. These are currently:

- A university
- A body appointed to administer land which fronts the coast under the Reserves Act 1977
- An organisation engaged in, or having as one of its objects, the scientific study of marine life or natural history
- Māori hapū or iwi with tāngata whenua status over the area

The proponent is responsible for designing the proposal and undertaking notification and public consultation. The Marine Department would consider the application and results of the public consultation, and decide whether or not to recommend acceptance of the proposal to its Minister. Upon the establishment of the Department of Conservation, through the Conservation Act 1987, responsibility for administering the Act rested with that Department. The Director General, rather than the Secretary of Marine, can now initiate marine reserve proposals.

A proposed replacement for the Marine Reserves Act was introduced into Parliament on 7 June 2002 and passed its first reading in the House of Representatives on 15 October that year. The Bill was then referred to the Local Government and Environment Select Committee.⁴ The Bill aimed to resolve what were seen at the time as the key problems with the Marine Reserves Act 1971, and which rendered it inconsistent with current government policy, in particular the Biodiversity Strategy.

The Bill provided that marine reserves could be established between mean high water springs and the full extent of the 200-mile exclusive economic zone.⁵ The fact that the Marine Reserves Act applies only to the territorial sea was identified as a key barrier to successful implementation of the Biodiversity Strategy.

After progress of the Bill through Parliament stalled, the Department of Conservation and Ministry of Fisheries jointly published the MPA Policy in 2006. This was intended to give effect to the obligation set out in the Biodiversity Strategy, which required a plan to be developed and implemented in order to establish a network of areas to protect New Zealand's marine biodiversity.⁶ It was followed by the MPA Guidelines in 2008.

The MPA Policy has no legislative status. In the absence of legislative reform, it was an attempt to provide some policy coherence to the myriad of spatial protection tools which could be deployed in the marine area. It also sought to provide a more collaborative process for the identification of candidate areas, thereby reducing the level of conflict around marine reserve proposals.

Current marine protection tools

A range of spatial marine management tools can be used in New Zealand's waters as summarised in Figure 8.1. Current and proposed legislative and policy tools, which have been primarily directed at spatial marine protection in New Zealand, are the:

- Marine Reserves Act 1971, which only applies to the territorial sea
- Marine Reserves Bill, which is still before Parliament, so has yet to be implemented
- MPA Policy, which has only been implemented within the territorial sea in two bioregions, and for which further implementation is on hold
- Fisheries Act 1996, which is the mechanism largely through which marine protection has been achieved in the exclusive economic zone, in the absence of marine protected area legislation applying there. It has also been applied in the territorial sea because of the inflexible approach of the Marine Reserves Act, which effectively excludes all extractive activities from within marine reserves

The efficacy of these tools, when compared with international best practice, is discussed in the following sections.

Figure 8.1: Current spatial marine management measures

Source: Ministry for the Environment⁷

Management tool	Instrument	Restrictions	Area covered 2008 (km ²)	Application
Marine reserves	Marine Reserves Act 1971	All extractive activities are prohibited	12,792	Territorial sea
Marine mammal sanctuaries	Marine Mammals Protection Act 1978	A range of restrictions necessary to protect marine mammals, depending on the marine mammal sanctuary	6,180	Territorial sea
Marine parks	Hauraki Gulf Marine Park Act 2000 and 2001 amendment, Fisheries Act 1996, Sugar Loaf Islands Marine Protected Area Act 1991	A range of restrictions depending on the marine park	20,536	Territorial sea

Management tool	Instrument	Restrictions	Area covered 2008 (km ²)	Application
Submarine cables and pipeline protection zones	Submarine Cables and Pipeline Protection Order 1992	No fishing or anchoring except for ships being used for research by or for the Ministry for Primary Industries, as long as the research is undertaken without directly or indirectly attaching a vessel to the seabed	1,732	Territorial sea and exclusive economic zone
Mātaimai reserves	Fisheries Act 1996 Declaration of Mātaitai Reserve and appointment of Tāngata Kaitiaki/tiaki Notice	In general, commercial fishing is prohibited, and recreational/customary fishing is regulated by the tāngata kaitiaki/tiaki	204	Territorial sea
Taiapure	Fisheries Act 1996 Fisheries Order	Special management region for areas which have special customary significance to iwi or hapū as a source of food	388	Territorial sea
Section 186 temporary closures	Fisheries Act 1996 Fisheries (Temporary Closure) Notice	A range of restrictions depending on the particular area which will prohibit the removal of one or more species during a particular time period	769	Territorial sea
Benthic protection areas	Fisheries Act 1996 Fisheries Act (Benthic Protection Areas Regulations 2007)	Prohibition on the use of dredge and restrictions on the use of trawl nets within 100 metres of the seabed	1,250,000	Territorial sea and exclusive economic zone
Seamount Closures	Fisheries Act 1996 Fisheries Regulations	Prohibits seabed trawling	100,997	Exclusive economic zone

Purpose of marine protection

An effective system of marine protection will use tools which have the protection of the marine environment as their main objective. This ensures that management measures have a clear purpose, are effective and coherent, and are not compromised by other interests. Such a system can provide for 'secondary objectives' relating to desirable public uses of the marine environment (such as scientific study, recreational and cultural uses) providing that the main objective is not compromised.

The legislation in New Zealand which has been applied to marine protection has a range of different objectives. Consistent with its genesis, the purpose of the Marine Reserves Act is narrow, providing "*for the setting up and management of areas of the sea and foreshore as marine reserves for the purpose of preserving them in their natural state as the habitat of marine life for scientific study.*"⁸ The protection of marine biodiversity is not mentioned in the legislation.

The legislation does state that "*the public shall have freedom of access and entry to the reserves, so that they may enjoy in full measure the opportunity to study, observe, and record marine life in its natural habitat.*"⁹ However, the Act does not recognise other secondary objectives such as cultural and recreational uses.

The purpose of the Marine Reserves Bill is to "*conserve indigenous marine biodiversity in New Zealand's foreshore, internal waters, territorial sea and exclusive economic zone for current and future generations, by preserving and protecting within marine reserves:*

- a) representative examples of the full range of marine communities and ecosystems that are common or widespread; and*
- b) outstanding, rare, distinctive, or internationally or nationally important marine communities and ecosystems; and*
- c) natural features that are part of the biological and physical processes of the marine communities and ecosystems referred to in paragraphs a and b, in particular those nature features that are outstanding, rare, unique beautiful or important.*"¹⁰

Therefore the Bill contains as its primary purpose the protection of marine biodiversity, which is consistent with international best practice. The Bill does not specifically recognise secondary objectives for marine protected areas although it does refer to protecting historic material in a marine reserve and the importance of protecting undisturbed areas for science and educational purposes. In addition

it states that “*the use and enjoyment of marine reserves should be allowed, if consistent with the purpose of this Act, and appropriate provision should be made to facilitate that use and protect the quality of the experience.*”¹¹

The objective of the MPA Policy is to “*protect marine biodiversity by establishing a network of marine protected areas that is representative of New Zealand’s marine habitats and ecosystems*”. This objective is reflective of international best practice, New Zealand’s international obligations and the goal set out in the Biodiversity Strategy. It directly refers to networks and representativeness. However, this objective has yet to be incorporated into any legislation.

The purpose of the Fisheries Act is to provide for the utilisation of fisheries resources while ensuring sustainability. ‘Utilisation’ is defined in the Act as “*conserving, using, enhancing, and developing fisheries resources to enable people to provide for their social, economic, and cultural wellbeing.*” ‘Ensuring sustainability’ means:

- a) maintaining the potential of fisheries resources to meet the reasonably foreseeable needs of future generations; and
- b) avoiding, remedying, or mitigating any adverse effects of fishing on the aquatic environment.

The Act also includes a set of environmental principles which decision-makers are required to take into account. These include:

- a) associated or dependent species should be maintained above a level that ensures their long-term viability;
- b) biological diversity of the aquatic environment should be maintained;
- c) habitat of particular significance for fisheries management should be protected.

The main objective of the legislation is therefore utilisation of fisheries resources rather than marine protection. However, the Act does provide for the conservation and enhancement of groups or species of marine plants, animals and seabirds (which are included in the definition of ‘fisheries resources’) along with the management of the impacts of fishing activity on the marine environment. In addition, an environmental principle refers to the maintenance of biological diversity, although this is not a mandatory requirement.

New Zealand therefore currently lacks a clear legislative purpose for marine protected areas which reflects international best practice.

Marine protected area design

Best practice requires that marine protected areas are designed to protect ecosystems, either by protection of large areas or, more commonly, via a network of marine protected areas which protects representative examples of ecosystem features.

The Marine Reserves Act provides only for the protection of individual areas of “*such distinctive quality*”, or which are “*so typical, or beautiful, or unique, that their continued preservation is in the national interest*”.¹² The provision includes reference to ‘typical’ areas, providing for the protection of sites other than those which are particularly unique, but it does not explicitly recognise the contemporary approach to marine protection, which is to achieve *representativeness*. In addition, the absence of provision for marine reserves to be established in the exclusive economic zone means that the Act is not able to provide for the protection of many of New Zealand’s marine species and habitats.

The Marine Reserves Act does not recognise the need to consider any relationship between reserve sites – rather they are established individually. In practice this has largely been on an *ad hoc* basis. As a result, most marine reserves were not designed with the formation of an ecologically coherent system of marine protection in mind.

The Marine Reserves Bill recognises the need to protect representative examples of the full range of marine communities and ecosystems in marine protected areas. However, there is no direct reference to establishing ‘networks’ of marine protected areas and there is no provision in the legislation for a process to implement such a network-based approach.

The MPA Policy aims to address such failings, to achieve the integrated development of a coherent network of marine protected areas, which recognises and protects entire ecosystems. It sets out six network design principles which include that “*the marine protected area network will protect examples of the full range of natural marine habitats and ecosystems*” and that “*marine protected areas should be designated based on a consistent approach to classification of habitats and ecosystems.*”¹³

The MPA Guidelines provide more detailed information on how representativeness should be achieved. This has been informed by international best practice and experiences overseas. It identifies a number of ‘site identification and protected area design guidelines’ which include such considerations as the desirability of protecting whole habitats and ecosystems, the need for

marine protected areas to be of sufficient size to provide for the maintenance of populations of plants and animals, maximising connectivity, and representing latitudinal and longitudinal variation. The design guidelines which are to be used to identify and select potential protected areas are set out in Appendix 1.

Thus, the approach taken in the MPA Policy is far ahead of that in the Marine Reserves Act and subsequent Bill. However, the policy does not have any statutory status. It has only been implemented in two bioregions, and further implementation is currently on hold. It also lacks clarity, which resulted in for recommending options to Ministers which provided very different levels of protection.

The Fisheries Act contains no reference to the design of marine protected areas. They come under the definition of ‘sustainability measure’ which is defined in the Act as meaning “*any measure set or varied under Part 3 [sustainability measures] for the purpose of ensuring sustainability*”.

Management measures

A well accepted approach to managing marine protected areas is to establish a range of categories in which management measures differ according to the ‘secondary purpose’ of the area. Different zones can be established where specific types of activity are permitted, providing always that an adequate amount of space is allocated the highest level of protection.

The Marine Reserves Act does not provide for a flexible range of management measures. Although the Act does not specifically prohibit all fishing within marine reserves, any fishing that occurs must be authorised by particular provisions in the establishment notice, and must be consistent with the requirement that “*the marine life of the reserves shall as far as possible be protected and preserved.*”¹⁴ This is a high standard to meet, so in practice, marine reserves are generally areas where all fishing is prohibited.¹⁵ The Act does not specifically state that other extractive activities are prohibited, but it is unlikely that they would be deemed consistent with the preservation objective.

The Marine Reserves Bill provides that fishing is to be prohibited in marine reserves but other extractive activities such as mining may be allowed. Specifically petroleum extraction in marine reserves in the exclusive economic zone, and mining in the territorial sea, would be permitted and managed under the Crown

Minerals Act. Mining in marine reserves within the exclusive economic zone would be permitted by concession.

The MPA Policy provides a more flexible approach. The ‘protection standard,’ set out in the MPA Guidelines, provides assistance with the assessment of areas for inclusion in the network. Not all extractive activities need to be excluded from a marine protected area, but the management regime as a minimum must provide for the maintenance and recovery at the site of:¹⁶

- a) Physical features and biogenic structures that support biodiversity
- b) Ecological systems, natural species composition (including all life-history stages), and trophic linkages; and
- c) Potential for the biodiversity to adopt and recover in response to perturbation.

The protection standard describes three levels of marine protection and identifies the management tools that could serve to meet the required protection level. All three of these types are regarded as contributing to the Biodiversity Strategy target, but *only Type 1 and 2* are considered to be marine protected areas under the MPA Policy. They are:

- (1) Marine reserves which are ‘Type 1 MPAs’ (established under the Marine Reserves Act 1971);
- (2) Other marine protected areas which are ‘Type 2 MPAs’ (Fisheries Act prohibitions plus other tools in combination); and
- (3) Other marine protection tools.

The MPA Policy requires that, first, existing marine areas subject to restrictive management measures are to be assessed to establish whether they meet the protection standard and are therefore suitable for inclusion in the network. A gap analysis is then to be undertaken to identify gaps in coverage and thus where new marine protected areas should be designated.

The new marine protected areas are to be created via a combination of the available marine management tools (as set out in Figure 8.1), in order to achieve appropriate levels of protection within the network.

Therefore, the focus of the policy is on the *effect* of the management measures in a particular case, rather than on their purpose. Thus the definition of a marine protected area under the MPA Policy is “*an area of the marine environment especially dedicated to, or achieving, through adequate protection, the maintenance*”

and/or recovery of biological diversity at the habitat and ecosystem level in a healthy functioning state”.

This creates the situation where the unintended consequences of a particular management measure may be deemed adequate to call an area a marine protected area. This has the potential to cause problems where changes to the level of activity in the area are planned or proposed – for example, new fishing methods, mining in areas protected under the Fisheries Act, or cable management activities in cable protection zones.

In addition, the MPA Policy cannot achieve a coherent system of marine protection categories such as that established by the IUCN. Thus it is noted in the MPA Policy that the protection of marine historic or cultural heritage, or protection for non-extractive use (e.g. diving) or values, tourism or recreational opportunities are not addressed in the Policy, and that “[s]uch issues will be considered following the development of the [NZ] Oceans Policy.”¹⁷

Process

Leadership and decision-making

An effective marine protection strategy cannot be achieved without the commitment of all government agencies that have jurisdiction in the marine area.

The Marine Reserves Act is administered by the Department of Conservation. Marine reserve proposals are managed by the Department of Conservation and may also be initiated by the Director General. Although the Minister of Conservation is empowered to decide whether to grant an application, concurrence is required from the Ministers of Transport and for Primary Industries.

In practice, this system has proved challenging. The Department of Conservation and the Ministry for Primary Industries (and its predecessors) have different cultures and corporate goals. The time taken to establish marine reserves has in contentious cases been lengthy¹⁸ and in some cases, such as with the Aotea (Great Barrier Island) marine reserve proposal, a great deal of effort and resource has been expended only for concurrence to be withheld at the final stage of the process. The Director General of Conservation has publicly announced that he will not be initiating any more marine reserve proposals.

The Marine Reserves Bill proposes to remove the concurrence requirement. The Minister of Conservation becomes the final decision-maker and is required

to consult with the full range of ministries which might be affected by a proposed marine reserve (primary industries, transport, energy, foreign affairs and defence). In cases where the Director General of Conservation is the proponent of the marine reserve, the Minister must first receive and consider an independent report on the application.¹⁹ Anecdotally, the removal of the concurrence requirement proved contentious, and was one of the reasons why the Bill's progress in Parliament was stalled.

The MPA Policy is a joint project undertaken by the Department of Conservation and the Ministry for Primary Industries (with involvement from other interested authorities such as the Ministry of Transport and regional councils, as necessary). This approach aims to involve all the relevant stakeholders in the process and recognises the fact that the management of fisheries activity is a key element of successful marine protection. However, this means that two entities with different objectives are tasked with implementing one policy. This has resulted in a lack of clear direction being given to the marine protection planning fora, which in turn, made it more difficult for them to reach consensus.

Stakeholder involvement

As the case studies show, overseas jurisdictions have dealt with the issue of stakeholder engagement in different ways. Some undertake collaborative processes similar to those recently undertaken in New Zealand, and some choose more traditional government-led methods. Regardless of the method chosen, experience suggests that excellent lines of communication with stakeholders must be established very early on, and that all stakeholders should be consulted and their views considered.

The Marine Reserves Act does not set out a process for the identification or design of prospective marine reserves. However the implication is that the entity proposing the marine reserve will undertake the design process. The party initiating a marine reserve proposal must prepare the application, then put the proposal out for a public consultation, and *"call upon all persons wishing to object to the making of the Order to send their objections in writing, specifying the grounds thereof, to the Director-General within 2 months from the date of first publication of the notice"*.²⁰

Submissions are considered by the Minister of Conservation, who must decide whether to uphold any objections received, taking into account that the proposed reserve should not.²¹

- Interfere unduly with an estate or interest in land adjoining the proposed reserve
- Interfere unduly with any existing right of navigation
- Interfere unduly with commercial fishing
- Interfere unduly with or adversely affect any existing usage of the area for recreational purposes
- Otherwise be contrary to the public interest

If an objection is upheld, the area cannot be declared a marine reserve. Thus, the establishment process favours conservation interests, whilst the relationship of other stakeholders with the process is framed negatively. They are invited to object to the proposal, and if they can show undue effects as a result of the proposed marine reserve, the Minister must decline to grant the application. In essence this is an adversarial system which pits conservationists against other stakeholders.

Under the Marine Reserves Bill, proposals for the establishment of marine reserves may be made by the Director General of Conservation and any other person. The Director General has 20 days to consider whether a proposal should proceed as an application. The application is subject to a 40-day public consultation period, following which the Director General prepares a report and submits it to the Minister of Conservation for consideration. Marine reserves may not be created if they have an undue adverse effect on any other marine activity, including customary, recreational or commercial fishing.

The MPA Policy recognises the shift in thinking about iwi and stakeholder involvement that has occurred since the Marine Reserves Act was passed, and establishes a collaborative approach for the design of new marine protected area networks. Regional stakeholder groups are responsible for the majority of the design process, and are encouraged to reach consensus on a proposal which minimises negative impacts on each of their interests.

A key aim of this approach is to overcome the adversarial approach which has defined marine protection since the Marine Reserves Act was passed, so that all stakeholders are positively engaged in the process. However, the Marine Reserves Act, which has a divisive approach, is still the key implementation tool available.

The identification of inshore marine protected areas for inclusion in the network was to be undertaken by 14 regional stakeholder fora. Planning for offshore areas (the exclusive economic zone) was intended to commence in 2013 at a national level with an expert panel.

The MPA Guidelines describe how the marine protection planning fora should operate and give them five specific tasks:

1. Consider the classification and inventory information
2. Consult with existing users and interests in the area
3. Identify sites and potential tools for area-based protection of biodiversity
4. Seek to establish consensus on proposed areas to be set aside as protected areas
5. Consult on protection options and make written recommendations to Ministers

The fora are instructed to aim for consensus on their final recommendations to the Ministers. If this is not possible, a range of options can be presented. The MPA Policy and Guidelines contain some direction regarding the planning principles to which the groups should adhere and some guidance relating to the structure of the groups.

Thus far, the MPA Policy has been implemented in only two bioregions – the West Coast and the Sub-Antarctic Islands. The West Coast Marine Protection Forum was established as a pilot project prior to the publication of all the guidance, so it was implemented slightly differently. Further implementation of the MPA Policy has been put on hold.

If a collaborative approach is to be undertaken, significant resources and planning must be invested in the process. There were some weaknesses in the implementation of the MPA Policy which led to its achievements being less than might have been expected.

Formal stakeholder analyses were not undertaken prior to the establishment of either process, which would have ensured that the views of all stakeholders were included. The West Coast Marine Protection Forum, in particular, did not represent all users or interest groups in the area. A successful collaborative planning process should take into account all interests, and the presence of a range of perspectives helps to diffuse adversarial relationships.

In addition, the stakeholder groups were unable to reach consensus. In each case the Ministers were provided with a range of options from which they could choose, with the proposals for higher protection representing the views of the scientists and environmentalists and the proposals for lesser protection representing the views of the commercial users. Ideally the final proposal should have represented a synthesis of these differing views, which is what collaborative processes are intended to

achieve. Faced with a lack of consensus, the Ministers opted to adopt the option providing for the weaker protection measures in most cases.

Provision for tāngata whenua interests

The Marine Reserves Act is silent on the Treaty of Waitangi. It also has minimal requirements for the engagement of tāngata whenua with marine reserves. “*Māori iwi or hapū who have tāngata whenua status over an area*” are identified as a party which can initiate a marine reserve proposal. However, when it comes to involvement in the decision-making process for a proposal initiated by another party, there is no requirement for tāngata whenua to be consulted and no direct requirement for their interests to be considered. Section 5(6) of the legislation provides that the Minister ‘shall uphold’ an objection if satisfied that declaring a marine reserve would ‘interfere unduly’ with a range of interests including commercial fishing and recreational purposes. However, there is no mention of customary or cultural interests.

The Conservation Act under which the Department of Conservation was established and operates, and which provides for the Department’s administration of the Marine Reserves Act, does contain a requirement in section 4 that it be “*interpreted and administered so as to give effect to the principles of the Treaty of Waitangi*”. Thus, the Treaty obligations are incorporated indirectly into the Department’s administration of the Marine Reserves Act. In practice, tāngata whenua interests are considered when making decisions on whether to declare a marine reserve, and were the prime basis for declining the Aotea (Great Barrier Island) marine reserve proposal by the Minister of Fisheries.²²

There is no provision for tāngata whenua involvement in the ongoing management of marine reserves, which is solely the prerogative of the Director General of Conservation.²³ There is also no specific provision which would enable customary fishing to be undertaken within a marine reserve. This could only be achieved through the Minister of Conservation exercising his or her general discretion under section 3(3) of the Act to authorise fishing within the reserve through a *Gazette* notice. However, the Minister can only do so after having regard to the purpose of marine reserves which is ‘preserving’ the areas ‘for the scientific study of marine life’.

The Fisheries Act provides for the obligations set out in the Treaty of Waitangi Fisheries Claims Settlement 1992 to develop policies to recognise use and

management practices of Māori in the exercise of non-commercial fishing rights. Customary fishing provisions and spatial tools have been established under the legislation and associated regulations to help achieve this.

- *Mātaitai reserves* can be created in traditional fishing grounds for the purpose of providing for customary fishing practices. A local committee of tāngata whenua regulates the non-commercial harvest of seafood. Commercial fishing is prohibited unless the management body expressly requests that it be permitted
- *Taiāpure* are local fisheries in coastal waters that are of special significance to an iwi or hapū as a source of food or for spiritual and cultural reasons. A committee nominated by the local Māori community makes recommendations to the Minister for Primary Industries on regulations to control all types of fishing in the area
- *Temporary closure* of fishing areas or restriction on fishing methods to recognise and make provision for the use and management practices of tāngata whenua, otherwise known as rāhui

The Marine Reserves Bill provides much greater recognition of Māori interests. It contains a Treaty of Waitangi clause.²⁴ It recognises the importance of Māori traditional and contemporary knowledge (Te Ira Tangaroa). Iwi or hapū must be consulted during the preparation of a marine reserve proposal ‘if practicable’ and must be personally served with the public notification of the application. When recommending the making of an Order in Council to establish the reserve, the Minister of Conservation must be satisfied that it ‘will have no undue adverse effect’ on:²⁵

- (i) The relationship of iwi or hapū who are tāngata whenua or who have customary access, and their culture and traditions, with the marine area concerned
- (ii) The ability of iwi or hapū who are tāngata whenua, or who have customray access, to undertake customary food gathering to the extent authorised by any enactment

The Bill provides for a management body to be appointed to manage a marine reserve and this can consist “*of persons that the Minister thinks are fit for that purpose*”, so could include tāngata whenua members. There are similar provisions for the appointment of an advisory body.²⁶

The prime Māori criticism of the Bill was that it does not recognise the possibility that traditional kaitiakitanga could be provided for in marine protected areas. The Bill takes a harder line than the current legislation on excluding customary fishing, prohibiting the taking of marine life from a reserve unless for management, biosecurity purposes, scientific research or research contributing to Te Ira Tangaroa.²⁷ In addition, it specifically states that a concession may not be granted for the customary take of marine life within a reserve.²⁸ Similarly, the Bill does not provide for the integration of customary marine management tools such as mātaihai and taiapure with marine reserves.

The MPA Policy does recognise the potential of mātaihai and taiapure to contribute to marine protection. It notes that neither of these tools has biodiversity protection as its main objective, as their primary purpose relates to the support of customary fishing practices. But that if *“tāngata whenua so wish, it is possible that these tools could be applied in such a way that they can contribute to the MPA network”*.²⁹

Planning Principle 3 in the MPA Policy refers to provision for ‘the special relationship between the Crown and Māori’ including ‘kaitiakitanga, customary use and mātauranga Māori.’ The explanation to the principle refers to the need to ensure ‘effective participation of tāngata whenua in relevant processes’ and to consider the impacts of marine protected areas on ‘customary use and management practices’.³⁰

The MPA Guidelines state that tāngata whenua should be represented on the marine protection planning fora. Fora are also instructed to *“engage fully with tāngata whenua as key regional stakeholders”*.³¹ In practice, tāngata whenua representatives were included on the West Coast and Sub-Antarctic fora. However, a disjunct between the consideration of marine protected area tools and customary fisheries tools remained, as described in the West Coast Marine Protection Forum case study set out in chapter 17.

Current legislation, in the form of the Marine Reserves Act, does not adequately recognise Māori cultural interests in the marine area. The Marine Reserves Bill rectifies this to some extent, but takes a harder line in totally excluding customary fishing from marine reserves. The MPA Policy and Guidelines provide recognition that Māori customary fishing tools may be included as part of a marine protected area network. However, this was not achieved in practice in the two planning processes undertaken under the policy.

Integration with wider ecosystems-based management

As described in chapter 2, there is a growing recognition that effective marine protection and management involves more than just regulating some small areas, whilst pursuing business as usual in the rest of the marine space. Where possible, marine protected areas should be nested within a broader ecosystem-based management system, which addresses the entire marine area within a country's jurisdiction.

None of the current legislative tools or policies available in New Zealand make provision for this to occur. Management of marine activities is primarily split between fisheries management (by the Minister for Primary Industries under the Fisheries Act), marine protection (by the Minister of Conservation under the Marine Reserves Act) and coastal management (by regional councils under the Resource Management Act). There is no overarching management framework to bring these different management regimes together.

Equally, there is generally poor integration of the management of terrestrial activities and recognition of their impacts on the marine environment.³² Fragmented management has been highlighted as a contributor to the failure to “halt or reverse the decline in the Hauraki Gulf’s natural resources.”³³ This has led to the initiation of a process to prepare an integrated marine spatial plan for the Hauraki Gulf, which is intended to consider all the major stressors on the Gulf’s natural environment, as well as make provision for a range of cultural, commercial and recreational uses. The plan will be non-statutory, and in the absence of a legislative reform in this area, will need to be implemented through existing tools.

Marine protection outcomes

Taking into account all spatial protection measures, New Zealand has exceeded its target of 10 per cent protection of the marine area, with a total of 1,393,598 square kilometres having some form of protection, or around 33 per cent of New Zealand’s total marine area. A global inventory of marine protected areas prepared in advance of the Tenth Conference of the Parties to the Convention on Biological Diversity found that, of 190 countries or territories with marine jurisdictions, New Zealand is one of only 12 that have 10 per cent or more of their marine environment protected within marine protected areas,³⁴ and one of just

eight that has established extensive marine protected areas across their exclusive economic zone.

However, this record reflects the slow progress of marine protection efforts elsewhere. The global inventory found that the total area of marine protected areas world-wide covered just 1.31 per cent of the world's ocean surface. Furthermore, too great a focus on the overall spatial extent of marine protected area coverage risks obscuring the more important issue, which is the quality of the network design and the protection offered.

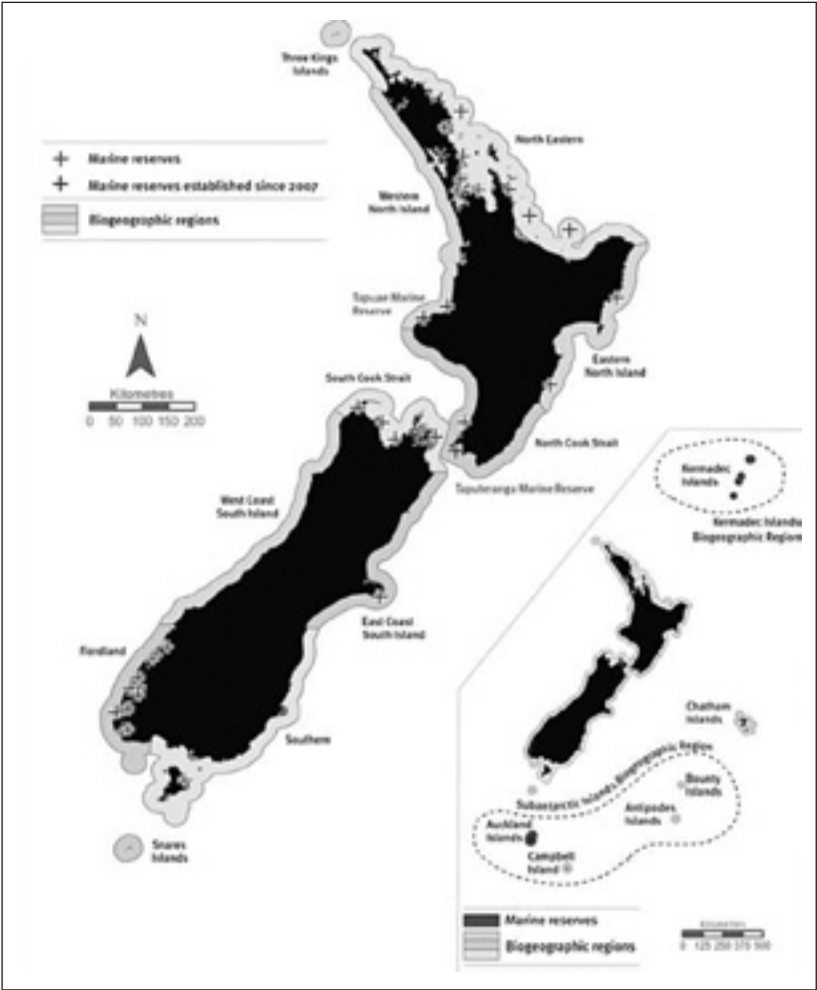
A more nuanced analysis of existing protection measures indicates that New Zealand is far from having a 'comprehensive and ecologically representative' national system of marine protected areas. A representative range of habitats is not protected. Marine protection around the mainland coast is particularly sparse, and there are no fully protected areas within the exclusive economic zone. The level of protection provided by the vast majority of New Zealand's marine protected areas is low, directed only at certain types of fishing activity. As interest in new extractive activities such as seabed mining grows, this issue will be of increasing concern.

New Zealand's territorial sea has been categorised into 14 biogeographic regions as shown in Figure 8.2. Analysis of the levels of protection in each of these bioregions helps to provide an indication of the representativeness and coverage of current marine protection measures in this area. In 2011, the Department of Conservation and Ministry for Primary Industries completed a gap analysis of marine protection within the territorial sea. The report was publicly released in August 2012 and included a summary of the percentage of mainland coastal biogeographic regions which were protected by marine reserves and Type 2 marine protected areas (see Figure 8.3).³⁵

Currently, 6.9 per cent of the territorial sea is protected by marine reserves. However much of this protection is located around the offshore islands in the Kermadec and Sub-Antarctic bioregions and the protection around the New Zealand mainland is much smaller.³⁶

Figure 8.2: New Zealand's marine biogeographic regions

Source: Ministry for the Environment



Around the mainland, the greatest level of protection is currently in the Fiordland bioregion where just over one per cent of the area is protected. Levels of full protection elsewhere are exceedingly small or non-existent. An additional 1.14 per cent of the territorial sea is protected by Type 2 marine protected areas (which

mainly includes areas where bottom trawling, Danish seining and dredging are prohibited). The gap analysis concluded that:

“With the exception of the Kermadec Islands Bioregion, there are many large gaps in the current MPA network for a range of habitats. In some bioregions, very few, if any, habitats are protected in Type 1 marine reserves or Type 2 MPAs. In other bioregions, the number of habitats represented in potential MPAs ranges up to around half of the total number of habitats in the bioregion. In most bioregions, however, only a very limited number of habitats have more than a few percent of their area within MPAs.”³⁷

There are currently no marine reserves in the exclusive economic zone as there is no legislation in place to provide for them (the Marine Reserves Act only applying to the territorial sea). However spatial protection measures have been put in place under the Fisheries Act. There are currently 17 benthic protection areas and 19 seamount closures which extend over 1.2 million square kilometres of seabed and protect 31 per cent of the exclusive economic zone from bottom trawling, dredging and netting within 100 metres of the seabed.³⁸ They include 52 per cent of New Zealand’s seamounts and 88 per cent of active hydrothermal vents.

The benthic protection areas are recognised in the World Database of Protected Areas³⁹ as falling under category VI (managed resource protected area) in the IUCN system. However, the level of biodiversity protection they provide remains an issue of scientific debate,⁴⁰ particularly as it has been argued that the areas protected have low biodiversity values.⁴¹ In addition, because the protective measures have been created under fisheries legislation, they do not restrict other activities which may damage the seabed such as minerals mining. Prospecting and exploration licences for seabed mining have been granted in the Chatham Rise benthic protection area amongst others.⁴² Furthermore, they are focused on the management of only some fishing activities. The Sub-Antarctic benthic protection areas assessed as part of the 2011 gap analysis of the territorial sea were found not to meet the protection standard required under the MPA Policy, because they permit Danish seining and amateur dredging to occur.⁴³ Such an assessment of the benthic protection areas in the exclusive economic zone has not yet been published.

Figure 8.3 Percentage of coastal biogeographic regions in marine reserves

Source: Department of Conservation and Ministry of Fisheries, 2011c

%total area of bioregion	Type 1 MPA marine reserve	Type 2 MPA cable or pipeline zone	Type 2 MPA marine park	Type 2 MPA Fiordland Marine Management Area	Type 2 MPA no trawl, Danish seine or dredge	Total MPA Type 2	Total MPA
Chatham Islands							
East Coast South Island	0.02						0.02
Eastern North Island	0.2						0.02
Fiordland	1.0			3.7		3.7	4.7
Kermadec Islands	100						100
North Cook Strait	0.2	1.7			0.01	1.8	2.0
North Eastern	0.2	2.3	0.06			2.4	2.6
Snares Islands							
South Cook Strait	0.3	1.1				1.1	1.4
Southern South Island	0.05				0.4	0.4	0.5
Sub-Antarctic Islands	42						42
Three Kings							
West Coast South Island							
Western North Island	0.2	2.2	0.03		0.03	2.3	2.5
% of area in NZ territorial waters	6.9	0.9	0.01	0.2	0.05	1.1	8.1

Conclusion

The Marine Reserves Act reflects its historical context and age, and does not embody current international best practice. It has a narrow purpose and scope and does not adequately provide for *tāngata whenua* and stakeholder engagement.

The Marine Reserves Bill addresses some of the key limitations of the Act but does not provide for a comprehensive system that would represent international best practice. Importantly, whilst recognising the need to establish a representative network of marine protected areas, the Bill does not propose a means for this to be achieved. Rather, it sets up a system for establishing individual marine protected areas in a similar way to the system under the Marine Reserves Act.

The involvement of stakeholders is provided for in the Bill in a similar way to the Marine Reserves Act – whilst anyone can propose the creation of a marine reserve, the involvement of other stakeholders is limited to a public consultation process. In the same way as the Marine Reserves Act, stakeholder involvement is framed negatively – if stakeholders can show that the proposal would have an undue adverse effect on their activity the proposal shall not be granted. This system encourages the adversarial, ‘conservationists versus users’ approach rather than moving towards more collaborative management.

Lastly, the Bill does not provide for flexible management measures or a range of categories of marine protected area. In practice, this means that tools under other legislation would need to continue to be used for marine protection in areas where more flexible management measures are desired.

Although the Fisheries Act has been utilised to achieve spatial protection measures, the legislation does not have marine protection as its purpose, and nor does it contain guidance on the appropriate design or management of marine protected areas. Applying only to fishing activity, protections achieved under this legislation can be undermined by other activities, such as mining.

The MPA Policy does largely reflect international best practice, but it has not been given statutory effect in legislation. It also has some significant shortcomings including the lack of a comprehensive and coherent system of marine protection categories, only limited guidance on design issues, lack of ongoing role for fora, and no mechanism to integrate marine protection with a wider ecosystems-based approach to marine management. One of the greatest difficulties is the need to implement the recommendations of fora through the existing inadequate legal mechanisms.

The lack of an adequate legal framework for marine protection is reflected in poor outcomes on the ground. Despite the existence of an extensive network of marine protected areas ‘on paper’, the reality is that existing protection measures are concentrated in some areas to the exclusion of others, and levels of protection are generally low. Only one mainland coastal biogeographic region currently has more than one per cent of its area in full protection (Fiordland) and several regions have no protection. There are no areas with high protection in the exclusive economic zone, and existing protection measures address only some of the activities which pose a threat to the marine environment.

Figure 8.4: Summary of New Zealand’s performance compared with international best practice

- The Marine Reserves Act has a narrow purpose and scope, pits conservationists against users, and fails to adequately address tāngata whenua interests
- The Marine Reserves Bill aims to address some of the limitations of the Act, but does not propose significant changes to the current system
- The Fisheries Act has been used as a *de facto* spatial marine protection tool, in the absence of other suitable legislation, but lacks a marine protection purpose and appropriate legislative provisions
- The MPA Policy reflects international best practice to a large extent, but also has some significant shortcomings, and does not have an adequate legislative foundation
- None of the current legal and policy tools effectively link marine protected areas with wider marine management
- The lack of properly designed marine protected areas tools is reflected in poor outcomes on the ground.

Endnotes

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- 6 Department of Conservation and Ministry of Fisheries, 2005
- 7 <http://www.mfe.govt.nz/environmental-reporting/oceans/protected-areas/management-tools.html> (accessed 27 August 2012)
- 8 Marine Reserves Act 1971
- 9 Section 3(2)(d), Marine Reserves Act 1971
- 10 Section 7, Marine Reserves Bill 2002
- 11 Section 9, Marine Reserves Bill 2002

- 12 Section 3, Marine Reserves Act 1971
- 13 Department of Conservation and Ministry of Fisheries, 2005, 16-17
- 14 Section 3(2)(b), Marine Reserves Act 1971
- 15 In some cases, some types of fishing have been permitted in marine reserves. For example, when the Poor Knights marine reserve was established in 1981, recreational fishing using drift lines, trolling or spear guns was permitted in the majority of the reserve. In 1998 the provisions permitting this activity expired and were not renewed, so the reserve is now fully protected from fishing activity.
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- 23 Section 11, Marine Reserves Act 1971
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- 40 McGinnis M, 2012
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chapter nine



Legislative design considerations

This chapter addresses the key factors that should be considered in the design of new marine protected areas legislation for New Zealand. These include the scope of the legislation, what its purpose should be, whether a classification system should be used, how the establishment process should be undertaken and linkages with broader marine management. In the concluding section, we have located the various approaches taken in New Zealand and the comparator countries, in respect of each factor, along a continuum of adopting a narrower or broader approach.

Scope

The scope of the legislation could be either narrow with a focus on no-take marine reserves or broader in accommodating different levels of protection.

Narrow scope

This is the approach taken in the Marine Reserves Act and also proposed in the Marine Reserves Bill, which retains the narrow focus on the establishment of no-take marine reserves.

In this arrangement, the MPA Policy would continue to be the instrument empowering the establishment of a representative network of marine protected areas, of which marine reserves under the new legislation would form one element. Tools under the various other legislative instruments would continue to be used in the establishment of marine protected area networks.

The strength of this approach is that it reinforces the important role that no-take marine reserves play in a representative network, and ensures that protection of the marine environment remains the core focus of protection measures.

However, the difficulty with this approach is that it does not address the absence of properly targeted legislative instruments that allow for the establishment of a full range of marine protection measures as part of an integrated network. Marine protected areas where some extractive activities are permitted would need to continue to be created under other legislation such as the Fisheries Act, which are not specifically designed for that purpose and have different priorities. Further, Māori perspectives on an integrated approach to marine management consistent with kaitiakitanga are not provided for.

In addition, this approach does not reflect the developments in international best practice that have occurred in recent years. A system which is inflexible and unable to provide for a range of interests is likely to alienate stakeholders and encourage the adversarial politics that have hampered marine reserve establishment in the past. Successful marine protected areas legislation provides for flexible measures, that can take account of the needs of stakeholders, whilst recognising that a healthy marine environment is beneficial to everyone.

Broader scope

This is the approach taken by all the jurisdictions included in the international case studies (England, California, Australian Commonwealth and New South Wales). In this case, the legislation would provide for the full range of flexible measures required to implement a representative network of marine protected areas. This could recognise and provide for a wide range of values (recreational, cultural, heritage and sustainable use) underpinned by a central goal of protecting the marine environment.

This approach would require the replacement of the Marine Reserves Act 1971 with new legislation and the replacement of, or significant amendments to, the MPA Policy. The 'Protection Standard' set out in the MPA Policy would need to be overhauled – all marine protected areas would have as their primary purpose the protection of marine biodiversity. As a result, areas where marine biodiversity protection is incidental to their primary purpose, such as cable protection zones, would not necessarily form part of the network.

The strength of this approach is that it enables the establishment of a coherent system of marine protected areas that can be managed as a whole for a common primary purpose. It also enables a tailor-made system of marine protection categories to be developed, which can provide for a range of activities, where consistent with the secondary objective of each category (see below).

Potential problems with applying this approach in the New Zealand situation include the risk that the focus on no-take marine reserves may be lost. If marine protection mechanisms are available which provide for lower levels of protection, decision-makers may opt for these because it is easier to achieve support for them. This could be addressed by requiring a minimum level of no-take protection, for example, by requiring that a comprehensive range of species and habitats must be protected in at least one no-take zone.

If a broader approach was adopted, transitional provisions could be included in new legislation to bring existing marine protected areas, such as benthic protection areas and seamount closures, within the new marine protected area network under the new legislative regime. Each existing spatial protection area could be allocated a category based on the values protected and management response.

Purpose

The purpose of marine protected areas within the legislation could narrowly focus on biodiversity protection or could more broadly recognise the wide range of potential benefits that can be achieved through the establishment of marine protected areas. Individual marine protected areas can be designed to generate recreational, cultural, scientific, and aspirational benefits for society. Marine protected area networks can safeguard ecosystem processes and contribute to the resilience of the marine environment.

Narrow purpose

The Marine Reserves Act has a narrow focus aimed solely at the protection of the marine environment for the purpose of scientific study. The Marine Reserves Bill proposes alterations to this narrow focus to reflect international thinking that suggests that indigenous marine biodiversity should be preserved as an end in itself, but the purpose remains narrow.

The purpose of the MPA Policy is directed towards the formation of a network of marine protected areas that is comprehensive and representative of New Zealand's marine habitats and ecosystems, but the Policy does not provide for a wider range of objectives to be incorporated into the network.

The benefit of a narrow purpose is that the simpler the legislation is, the easier it is for the public to understand what it seeks to achieve. Having one clear purpose emphasises the importance of marine protection and ensures that it is not obscured by multiple and potentially competing purposes.

The limitations of this approach are evident in New Zealand's current system of marine protection. A focus that does not recognise the value that marine protected areas can provide to society is more likely to alienate people and encourage a 'conservationists versus users' culture. This approach fails to recognise that

marine protected areas can provide a range of benefits, including the promotion of cultural and recreational values.

Broader purpose

In this case, the protection of biodiversity could be the main purpose, but a range of secondary purposes could also be adopted. These could include scientific study, habitat protection, protection of special features, cultural use, sustainable use, ocean health, recreation, tourism, resilience, and risk management.

The strength of this approach is that it provides the framework for a flexible system which recognises the needs of society, and the values that marine protected areas can provide, whilst underpinning all management measures with the need to protect the marine environment.

Potential problems with this approach include the difficulty of taking into account multiple purposes and the possibility that multiple purposes could obscure the primary objective of protection of the marine biodiversity.

Regardless of the option chosen, the stated purpose of the legislation should reflect New Zealand's intention to establish a representative network of marine protected areas.

Management measures

A system of classification can be applied to marine protected areas, which sets out a standard framework of marine protected area types reflecting a range of secondary purposes and appropriate management measures.

Management measures may be strictly defined for each category – with some or all activities expressly prohibited. Alternatively, one type of marine protected area can be provided for in which management measures are flexible, decided on a case-by-case basis dependent on the needs of the area in question.

No classification system

The Marine Reserves Act 1971 provides for only one type of marine protected area to be established – where all activities that are not consistent with the preservation of the marine environment are prohibited. In practice this means that all extractive activities are excluded.

The Marine Reserves Bill similarly provides for only one type of marine protected area – a marine reserve – but explicitly prohibits fishing whilst providing for the potential for mining activities to occur at some sites.

The English marine conservation zones system does not incorporate a classification system but does provide for a range of management measures to be implemented. Management measures are to be established on a case-by-case basis, dependent on the measures necessary to protect the particular features that have been identified for protection.

If only one type of marine protected area is to be established, which provides for a strict set of management measures, it should be easy for most users to understand the rules. However, the establishment of a representative network would likely require the additional use of a range of management measures, as set out in the MPA Policy. If management measures are defined on a case-by-case basis, they can be targeted towards the particular needs of the area concerned. However, this approach is likely to be confusing for users and makes it more difficult to ensure that management measures are meaningful and effective.

Classification system

The IUCN has developed a set of seven categories which reflect the different management measures that can be applied to marine protected areas to achieve different goals (providing that the primary goal is marine biodiversity protection). The Australian Commonwealth and states have adopted this system into domestic legislation, so that every marine protected area established is assigned one of these categories.

The MPA Policy classifies marine management measures into three groups (Type 1, Type 2 and 'other marine protection tools'). These categories bear no relationship to any internationally recognised classification system.

The Reserves Act 1977 provides for a system of classification for terrestrial reserves. Section 16 of the Act provides that reserves must be classified into one of seven principal categories: Recreation, Historic, Scenic, Nature, Scientific, Government Purpose and Local Purpose. The category into which a reserve is placed reflects the present values of the reserve, the future potential values and the possible future uses of the area. This classification system aims to ensure that the control, management, development, use and preservation of reserves is appropriate.

These categories are not directly applicable to the marine environment, but demonstrate a useful approach. An analogous classification system more closely based on the IUCN categories could be developed.

The main benefit of a classification system is that it provides a coherent, purpose-designed system which enables each area to be assigned a specific management objective which recognises and provides for all desirable uses and values. This results in certainty for stakeholders, together with flexibility, which makes the creation of marine protected areas easier. On the other hand, it could possibly make the development of tailor-made objectives for a particular area more difficult.

Linkages with wider ecosystem-based marine management

Initiatives overseas indicate that ecosystem-based marine management of entire marine jurisdictions, using tools such as marine spatial planning, is becoming a well-established objective. This raises the issue of the extent to which marine protected area design initiatives should be folded into wider management measures.

Identification undertaken independently

This approach is essentially business as usual, in which marine protected areas are established in parts of the marine area whilst activities in the rest of the marine environment are managed separately. The Marine Reserves Act, the Marine Reserves Bill and the MPA Policy operate on the basis of this approach.

The Californian system has not moved beyond the establishment of a network of marine protected areas, partly because the marine protected area identification process was so resource intensive that there is no political appetite to embark on further work. Similarly, the New South Wales system does not include wider management of the marine area, although a recent review commissioned by the government recommends that the system should be overhauled to enable a broader planning process to occur.

The benefit of this approach is that it avoids complicating the process, focusing effort on the core goal of creating marine protected areas. In addition, there is currently no statutory provision for integrated management in New Zealand's marine environment, or for tools which might provide a means to achieve it

such as marine spatial planning.¹ However, ecosystems-based management is becoming widely accepted around the world as an effective way to manage competing pressures on the marine environment. It seems likely to be adopted in New Zealand in the future. Therefore marine protection legislation which is not flexible enough to provide for a linkage with future integrated management processes may become quickly outdated.

Identification undertaken as part of marine spatial planning

There are a number of options for undertaking a marine spatial planning process which incorporates marine protected areas. First, marine protected areas could be identified as part of the planning process itself. This is the approach adopted by the Australian Commonwealth government in its bioregional planning process, although the bioregional plans only partially deal with other ocean uses.

The benefit of this approach is that it enables all activities and pressures to be considered at the same time, providing more certainty for ocean users, communities and the environment. On the other hand, it may result in the dilution of marine protected area proposals, as the focus may move from biodiversity protection to provision for other oceans uses and activities.

The second approach is to incorporate marine protected areas that have already been identified into marine spatial planning, so that broader oceans management can address impacts on marine protected areas from outside their boundaries. This is the approach being undertaken in England, where the enabling legislation provides for both the establishment of marine protected area networks and marine spatial planning which are to be undertaken in separate processes.

The benefit of this second approach is that it helps to ensure that the process for the establishment of marine protected areas remains focused on marine protection, and does not result in excessive emphasis being placed on user interests. However, this may prove confusing for stakeholders, who may not appreciate the difference between the processes. It would also utilise extra resources through the undertaking of two different projects.

Given that marine spatial planning is very likely to become more widely adopted, as pressures on the marine area increase, it would appear sensible for new marine protected areas legislation to recognise and provide for the establishment of marine protected areas within a wider marine spatial planning programme. For example, the legislation could make provision for the subsequent adoption of marine protected areas identified as part of a marine spatial planning

process, avoiding the need go through a separate public process. It could also provide for subsequent planning processes to recognise and incorporate existing marine protected areas into wider marine spatial planning.

Stakeholder involvement in marine protected area planning

The Marine Reserves Act provides for both certain interested parties and the Director General of Conservation to propose individual new marine reserves. A key benefit of this approach is that it provides for a mechanism by which interested parties can establish marine protected areas which they support, and over which local communities feel a sense of ownership. However, reliance on community proposals for new marine protected areas is clearly inadequate if the establishment of a representative network of marine protected areas is the goal: reserves proposed by the public are established on an *ad hoc* rather than strategic basis and are very likely to be established only in coastal areas. Accordingly, in addition to this arrangement, a government-led initiative to ‘fill in the gaps’ is required.

If an initiative to complete the establishment of a representative network of marine protected areas is undertaken, a key consideration is the extent to which stakeholders should be involved. Whilst conventional environmental management expects that management decisions will be undertaken by government entities, in recent times the benefits of involving stakeholders more fully have been recognised. In the context of the marine environment, stakeholders hold useful information that can inform management decisions, and the challenges of enforcement in the marine area make stakeholder support extremely valuable. Stakeholder involvement can be achieved in a number of different ways. At opposite ends of the spectrum, government may consult with stakeholders bilaterally, or alternatively establish a collaborative decision-making process in which stakeholders make all the key recommendations which are then adopted by government.

Government-led approach

This approach would follow the conventional model in which government leads the process for the identification, design and implementation of marine protected

areas, guided by government policy and advice from experts. Bilateral consultation with tangata whenua and key stakeholders could be undertaken in order that their needs are taken into account. The government would work up a draft proposal upon which public submissions would be sought.

This is the approach adopted for the Australian Commonwealth government marine bioregional planning programme, for the design and establishment of the Great Barrier Reef marine park, and in the establishment of the New South Wales marine parks. The Marine Reserves Act provides for a similar system, although a small number of stakeholders are permitted to put forward marine reserve proposals themselves.

The principal benefit of a government-led system is that it is much easier for the government to control the process and provide strong leadership, and to ensure that final recommendations are consistent with government policy. In addition, it is easier to ensure that the final product is more firmly rooted in the scientific information available, as experts who understand the detail of the issues are involved directly in decision-making.

In Australia this has been successful, as evidenced by the 2004 rezoning of the Great Barrier Reef Marine Park (where the level of protection in no-take zones was increased from 5 per cent to 33 per cent) and the progress made in the marine bioregional planning process. The Australian government's willingness to allocate 'structural adjustment packages' to affected stakeholders has assisted in the implementation of this approach.

Nevertheless, this approach can fail to address, and even encourage opposition to, marine protected area establishment. Stakeholders may feel that their interests and concerns are not being addressed by government decision-makers. This problem was experienced in New South Wales, where the public objected to the 'politicisation' of the marine park establishment process, ultimately resulting in a moratorium on the establishment of marine parks and an independent audit of their effectiveness.

Another risk is that without an established and transparent process for stakeholder engagement, government will be influenced during the political process by the most powerful stakeholders to the detriment of others. This issue has been observed in the Australian bioregional planning process where protection measures in some areas have been diluted to appease particular interests.

Collaborative approach

Collaborative decision-making delegates the task of identifying and designing new marine protected areas to stakeholder groups that are specially established for the purpose. The stakeholder groups can function in accordance with government guidance and aim to meet detailed ecological requirements. This is the approach taken in California and England and in the New Zealand MPA Policy.

Such an approach can be beneficial in a number of ways. It can provide a transparent and effective forum for stakeholder involvement and allow for consideration of all interests. The knowledge and experience of users of the marine area can make a useful contribution to design decisions. The approach may help to build support for marine protected area proposals, because the process helps stakeholders to understand the issues and the proposals, and to feel they have a stake in them.

The approach also provides a venue for diverse stakeholder groups which do not normally work together to build relationships, potentially breaking with the adversarial approach that usually characterises marine management issues. Collaborative processes can result in the establishment of a group which can continue to play a positive role in ongoing management of the area. A good example is the ongoing role adopted by the Guardians of Fiordland.

In recent times collaborative approaches have become more popular in New Zealand, with the establishment of successful initiatives such as the Land and Water Forum and the Upper Waitaki Shared Vision Forum. A similar approach is proposed for the Hauraki Gulf Marine spatial planning initiative and may also be introduced to plan-making processes under the Resource Management Act in the near future.

The collaborative approach carries some risks. Consensus may not be reached, as happened with the West Coast Marine Protection Forum and the Sub-Antarctic Marine Protection Planning Forum. Theoretically the approach would provide for scientific guidance to set out parameters within which lay stakeholders make their decisions. This would help to ensure that the resulting recommendations are scientifically robust, and meet minimum protection requirements, as well as reflecting socio-economic concerns. However, the marine environment is complex, with gaps in the available information. As a result, decision-making about marine protection is not simple. Thus, providing for stakeholders to do it means that there is significant potential for scientific integrity to be sacrificed in favour of socio-economic concerns.

Collaborative approaches are time consuming and resource intensive. Stakeholders need time to establish working relationships and understand the issues. Government must commit significant resources to ensure that the necessary information on the marine environment is available, and that the groups are well supported with guidance and expert advice. New Zealand has only limited resources for such projects, so the achievability of a collaborative approach for all New Zealand's marine areas should be carefully considered.

If a collaborative process is undertaken, careful consideration needs to be given to the process for implementation of the group's recommendations. Often, a conventional statutory process involving public consultation is undertaken. These processes may not combine easily with collaborative decision-making. For example, if public consultation is held *following* a collaborative process, this can destroy carefully achieved consensus, pitching different interests against each other again. In addition, the knowledge that a further consultation will be undertaken may have the effect of discouraging complete cooperation with the collaborative process, as participants are aware that they will have an opportunity to state their views again later.

Experiences of collaborative processes both in New Zealand and overseas also indicate that problems can arise where there is a disjunct between the expectations of participants, and the government's intentions, in terms of whether the group's recommendations will be taken forward by the government. Where Ministers must retain ultimate discretion, this should be made clear to participants, and the limits of their decision-making authority clearly communicated to them at an early stage.

The issue of compensation

The establishment of marine protected areas in Australia is often accompanied by financial support for those adversely affected, particularly fishers. Such compensation has not been paid in New Zealand on the creation of marine reserves. The issue of compensation was raised by the West Coast Marine Protection Forum, on the basis that if compensation was available, it would make reaching consensus between fishing and conservation interests more achievable. However, government made it very clear that it was not willing to offer any compensation to fishers.

On the positive side, the availability of compensation could help to reduce opposition to marine protected areas by those who may be adversely affected.

However, on the negative side, if such compensation is expected or required, it could make the establishment of a comprehensive and representative network of marine protected areas unaffordable and therefore unachievable.

Who decides on new marine protected areas?

The final decision about whether to establish a marine protected area could be undertaken jointly by a number of parties, to reflect the range of interests relevant to the issue, or could be undertaken by one decision-maker. To ensure objectivity, a single decision-maker could be selected on the basis of the lack of vested interest and thus an ability to consider the competing interests objectively.

Joint decision-making

This approach envisages that the consent of two or more Ministers will be required to establish a marine protected area. Under the Marine Reserves Act, the Department of Conservation has responsibility for granting marine reserve applications, but both the Minister for Primary Industries and the Minister of Transport must concur with the proposal. This has considerably slowed down the process of marine reserve creation and has led to a situation where a large amount of work has gone into developing a proposal, only for a Minister to veto it. The MPA Policy also relies on a joint approach. The Department of Conservation and the Ministry for Primary Industries were jointly tasked with implementing the policy, but they have different priorities, and implementation has now been put on hold.

The approach does have benefits, however. It helps to ensure that both conservation and other interests (such as fisheries and transport) are properly considered and taken into account when a decision is made to create a marine protected area.

Sole decision-maker

The alternative is to provide for one Minister or other entity to make the final decision. The Marine Reserves Bill provides for the Minister of Conservation to have sole decision-making power, following consultation with any relevant ministers. This is possibly one of the reasons why the passage of the Bill has

not been achieved, as it was thought by some parties to give too much power to conservation interests.

The main benefit of using a sole decision-maker is that it helps ensure a streamlined process and the efficient use of resources.

A potential problem is that it may fail to give sufficient weight to all of the relevant interests. This problem can be mitigated by ensuring that an appropriate Minister or entity has decision-making power, one that is not aligned to a particular sector. Concerns about vesting too much power in conservation interests could be alleviated by adopting an independent review process, for example along the lines of the board of inquiry process used in the preparation of national policy statements under the Resource Management Act. Key features of this process are the review of the draft plan by an independent body and a public hearing where submitters are heard and can present evidence in support of their submissions. The independent body should be made up of expert members, with a judicial chair. Further appeals could be made on points of law only.

In this process, the Minister of Conservation could make a final decision without requiring concurrence from the Minister for Primary Industries. The independent board of inquiry process provides an alternative means through which any concerns of the Ministry for Primary Industries can be fully taken into account externally to the Department of Conservation.

Providing for sole decision-making power should considerably speed up the process and is consistent with the principle that the power to make a decision should be aligned with the person who has accountability for the outcome of that decision.

Conclusion

Figure 9.1 locates the various approaches taken in New Zealand and in comparator countries along a continuum from a narrower to a broader approach. From this analysis, it is evident that the Marine Reserves Act and Bill take a narrow approach to marine protection, and as such are out of step with the international comparator countries. The MPA Policy takes a broader approach, but is still out of step with comparator countries in terms of failing to adopt a classification system. It also does not provide linkages with broader marine spatial planning which is something that is being undertaken in the United Kingdom and partially in Australia.

Figure 9.1: Summary of legislative design considerations and existing legislative systems

	Narrow Broad		
Scope	Marine Reserves Act Marine Reserves Bill	MPA Policy	New South Wales Marine Parks Act Australian Commonwealth MPA Program English Marine and Coastal Access Act California Marine Life Protection Act
Purpose	Marine Reserves Act Marine Reserves Bill		MPA Policy New South Wales Marine Parks Act Australian Commonwealth MPA Program UK Marine and Coastal Access Act California Marine Life Protection Act
Classification System	Marine Reserves Act	Marine Reserves Bill UK Marine and Coastal Access Act MPA Policy	New South Wales Marine Parks Act Australian Commonwealth MPA Program California Marine Life Protection Act
Linkages with Marine Spatial planning	Marine Reserves Act Marine Reserves Bill MPA Policy California Marine Life Protection Act New South Wales Marine Parks Act	Australian Commonwealth MPA Program	UK Marine and Coastal Access Act
Stakeholder involvement	Marine Reserves Act Marine Reserves Bill	Australian Commonwealth MPA Program New South Wales Marine Parks Act	MPA Policy UK Marine and Coastal Access Act California Marine Life Protection Act

Endnotes

- 1 Although there has been some movement toward a more ecosystem-based approach to *fisheries* management through the Ministry for Primary Industries' *Strategy for Managing the Environmental Effects of Fishing*



This chapter sets out three options for new legislation: to promulgate the Marine Reserves Bill, draft new dedicated marine protected areas legislation or draft new framework oceans legislation which incorporates marine protected areas within a broader marine spatial planning framework. The pros and cons of each of these options are explored in the following sections.

Option 1: Promulgate the Marine Reserves Bill 2002

Chapter 8 contained a detailed description and critique of the Marine Reserves Bill. This established that the Bill has some good features. It extends the ability to establish marine reserves into the exclusive economic zone. Its primary purpose, with a focus on the protection of marine biodiversity *as an end in itself*, reflects current thinking about marine protection.

However, there are some potential problems with the Bill as it is currently drafted. These primarily reflect the age of the Bill (it is now almost 10 years old) and the rapid development of international thinking in the field of marine protection over the past decade. These include the failure of the Bill to:

- Give effect to the government's intention to establish a representative network of marine protected areas, as set out in the Biodiversity Strategy and the MPA Policy. Given the central importance of this target, this is a significant omission
- Provide for a range of marine protected areas which provide different levels of protection. As described earlier in this report, all the other jurisdictions studied have provided for this. Failure to provide for a range of protection levels will necessitate continued reliance on other ill-targeted tools such as those under the Fisheries Act
- Adequately provide for the exercise of kaitiakitanga within marine reserves. There is no explicit provision for tūngata whenua input into the management of marine reserves. In addition, the legislation excludes customary fishing activities from marine reserves in all cases. This could result in customary rights being effectively extinguished, and not able to be reasserted, if the reserve status was revoked at a later date
- Include a classification system for different marine protected areas. Instead, some activities will be regulated strictly (fishing), whilst others will potentially

be permitted on an *ad hoc* basis (mining). This approach does not provide for certainty about the level of protection which marine reserves will enjoy, and also necessitates the continued use of poorly targeted Fisheries Act tools to establish protected areas which allow some fishing activity

- Provide for cultural or recreational objectives. The discussion paper accompanying the Bill suggests that these will be addressed during wider oceans reform, but it seems that the review of the Marine Reserves Act would be a more opportune time to do so
- Provide for a potential link to wider marine management tools such as marine spatial planning

The proposed establishment process continues the adversarial approach employed under the Marine Reserves Act, in which public consultation invites stakeholders to provide reasons why a proposed marine reserve should not be established. This system is a poor fit with the collaborative decision-making process set out in the MPA Policy.

The Bill provides for a sole Ministerial decision-maker, which could help streamline the process for marine reserve creation. However, it is questionable whether the legislation would work in practice, as other ministries may well object to the loss of their power of veto in favour of the Department of Conservation particularly where the process does not provide for strong independent scrutiny.

Option 2:

Draft new marine protected areas legislation which provides for an integrated approach to marine protected areas

Description

This approach would replace the Marine Reserves Bill with new legislation which provides for the establishment of a comprehensive and representative marine protected areas network. Legislation under this approach could have as its central purpose the achievement of a representative system of marine protected areas.

The legislation would provide for a full range of marine protected area categories, in order that the marine protected area network is managed under

one cohesive system. This is the approach adopted in all of the international case studies considered.

All existing marine protected areas would be incorporated into the system under transitional provisions. New marine protected areas would be established using a classification system analogous to that employed under the Reserves Act 1977 and similar to the IUCN's marine protected area classification which would set out management objectives and management measures.

The legislation could provide for an integrated process for the establishment of a representative network of marine protected areas and set out a time frame for its completion. This could take the form of a collaborative process, where appropriate. If a collaborative approach were adopted, the legislation should make provision for the development of detailed guidance on process and network design, which could be adopted as regulations. This would enable the current MPA Policy to be further developed in light of the experience of its implementation and developments in international thinking.

In addition, it would be appropriate for the legislation to provide a process for the establishment of individual marine protected areas which may be established primarily to 'fill in the gaps' following the completion of the network design process.

The Act could be administered by the Department of Conservation (which would have responsibility for ongoing management of marine protected areas), and the final decision on the establishment of new marine protected areas could be made by the Minister of Conservation on the basis of recommendations from the independent board of inquiry.

The legislation could provide that where the recommendations for new marine protected areas are the result of a collaborative process, the board of inquiry would be responsible for considering the group's final recommendations. The board of inquiry's terms of reference could be limited to those matters which have not been agreed by consensus, to provide an incentive for stakeholders to collaborate effectively.

Assessment

This approach would give legislative recognition to the marine protection goals set out in the Biodiversity Strategy and would provide all the tools required to develop a representative network of marine protected areas. It would enable a clear distinction to be made between measures which are aimed at marine protection (those available under the new legislation) and measures that are aimed at other

goals but may achieve marine protection indirectly, such as Fisheries Act tools which would not be used for marine protection within the network.

The approach would ensure that one agency is responsible for managing the network and that all management measures are underpinned by the objective of marine protection. The approach would also enable the establishment of a flexible system which provides for the realisation of a range of benefits, including recreational and cultural ones.

This option would overcome the main procedural problems with the current approach under the MPA Policy, where the legislative tools available have their own establishment processes, which do not interface well with the MPA Policy.

It would also allow for the development of more detailed policy and processes building on experience with the MPA Policy, which would enable any new collaborative processes to improve upon previous experiences. New processes could be tailored to the resources available and developed over time.

A board of inquiry process would enable proposals to undergo independent scrutiny outside of the Department of Conservation, and provide all stakeholders with the opportunity to make submissions and be heard in person. In circumstances where the proposals are the result of a collaborative process, limiting the Board's power to matters which are not agreed by consensus would help to ensure that the collaborative decision-making process is meaningful and that stakeholders will remain committed to achieving agreement.

Option 3:

Draft new framework oceans legislation which provides for marine spatial planning incorporating marine protected areas (oceans reform)

Description

This option comprises broader reform of New Zealand's marine management system. Provision for new marine protected areas would be contained within a new Oceans Act which would have as its purpose the integrated management of the territorial sea and exclusive economic zone. This approach would recognise that marine protected areas should be nested within a wider system designed to manage the marine area in a strategic and integrated manner.

This is the approach taken in the United Kingdom and Australia where broad marine management legislation governs decision-making about where activities in the marine area should take place, and incorporates both marine spatial planning and marine protected areas.

In New Zealand, new legislation could comprise a common set of management principles; detailed policies (expressed in a marine policy statement); spatial plans, including those taking in the interface between the terrestrial catchment and the coastal marine area (which could be incorporated as part of the marine policy statement); and marine protected areas which apply to the country's entire oceans domain including the territorial sea and exclusive economic zone. Decision-makers under other legislation, such as the Resource Management Act, Fisheries Act, Maritime Transport Act, Reserves Act and the Exclusive Economic Zone and Continental Shelf (Environmental Effects) Act, could be required to make decisions which are consistent with this legislation.

Such an approach enables an integrated management approach whilst many activities remain regulated under sectoral legislation. It would need to be well-designed to avoid creating uncertainty and a more cumbersome regulatory system.

The legislation could provide for an 'Oceans Agency' similar to the United Kingdom's marine management organisation. It would be responsible for marine management including overseeing marine spatial planning and marine protected area design.

Under this option the Ministry for the Environment could be responsible for the development of marine policy, whilst the Department of Conservation, Ministry for Primary Industries and other relevant ministries are responsible for advocating for their own particular areas of concern.

The legislation and/or regulations could set out processes for the design of the marine protected area network and for the establishment of individual marine protected areas. This could be undertaken separately from the marine spatial planning initiative, with identified marine protected areas then folded into the broader marine plans. Alternatively marine protected areas could be identified as part of the marine spatial planning process.

Assessment

Oceans reform, and the establishment of a marine spatial planning system, would clearly be a wider project than the establishment of marine protected area

legislation alone. But it would provide the best means of achieving an effective marine management system reflecting ecosystem-based approaches.

An over-arching legislative framework providing for marine spatial planning could help achieve a more integrated planning framework, without requiring additional environmental consents. Such an approach has already been applied to the Hauraki Gulf Marine Park and associated catchments, where a common set of matters of national significance and management objectives applies to fisheries, resource management and conservation management (although not to management under the Maritime Transport Act).

This approach could contribute significantly to the effectiveness of marine protection measures, as it would enable managers to plan for and control the impacts of activities which occur outside the marine protected area. In addition, it may provide a means of addressing the displacement of users from the protected area, as the wider spatial planning process should enable all interests to be provided for. The approach would enable managers to plan for future activities and environmental effects and to identify the best ways to manage them to ensure ocean health. Providing for integration with coastal planning should help to manage the threat of land-based activities on the marine area.

The establishment of an Oceans Agency with responsibility for marine management measures would have a range of benefits: it would replace the current fragmented, sector-based system with one where there is a clear separation between environmental regulation and resource extraction, and would provide an independent decision-making system that can cut across the current adversarial, interest-based approach.

The adoption of a new Oceans Act would recognise the fact that, on a global scale, marine spatial planning and integrated marine management are becoming well established as the best means by which to manage competing activities in the ocean and to ensure that ocean health is maintained. Marine spatial planning initiatives overseas have often been precipitated by particular pressures on the local marine environment. As activities in New Zealand waters diversify, and activity increases, there will be greater pressure to undertake reform. However, experience overseas indicates that although it may seem easier to wait until reform is urgently needed, in actual fact it would be easier to act earlier, so that a planning and management framework is in place that can address problems before they arise.

Conclusion

Our analysis indicates that the Marine Reserves Bill is no longer adequate and should not be promulgated in its current form. New dedicated marine protected areas legislation could readily provide for many of the elements of a world-leading approach to marine protection. We therefore recommend that such legislation be promulgated without delay.

However, such dedicated legislation would not enable marine protected areas to be integrated within a broader marine management system, and this would need to be achieved through broader oceans reform which we recommend should be pursued in the longer term.

Figure 10.1: Summary of legislative options

	Promulgate Marine Reserves Bill	New legislation providing for integrated approach	New framework oceans legislation
Enables the establishment of marine protected areas in entire marine jurisdiction	Yes	Yes	Yes
Provides for establishment of representative network of marine protected areas	No	Yes	Yes
Provides for the establishment of marine protected areas which protect biodiversity as an end in itself	Yes but Fisheries Act tools and others would still be used as well	Yes	Yes
Recognises all relevant values including recreational and cultural values	No	Yes	Yes
Provide for range of marine protected areas with different levels of protection	No	Yes	Yes
Manages the impacts on marine reserves of activities occurring outside the protected area including on land	No	No	Yes
Provides for collaborative decision making	No	Potentially	Potentially

chapter eleven



Recommendations

As a result of the investigation and analysis undertaken in the preceding chapters and the case studies set out in Part Three of this report, we have developed a series of recommendations for government on how we consider legislative reform in this area should proceed and what it should address. These are set out below along with the rationale behind each recommendation. They address both the short-term urgent need to replace the Marine Reserves Act as well as the longer term need to provide a legislative framework through which marine protected areas can be integrated into broader oceans management through an oceans reform process.

Legislative framework

Recommendation 1

New legislation replacing the Marine Reserves Act should be developed without delay.

The current legislation is outdated, cumbersome and has become unworkable. It does not reflect current thinking about marine protection and as such, should be replaced.

Recommendation 2

New legislation should have as its primary purpose the protection of marine biodiversity. The legislation should also provide for a range of secondary purposes, including cultural and recreational benefits.

New legislation should recognise the value of protecting biodiversity as an end in itself, as well as providing for the other benefits that marine protected areas can generate.

Recommendation 3

New legislation should enable marine protected areas to be created within the territorial sea and the exclusive economic zone.

A well-known limitation of the existing legislation is the absence of a mechanism to establish marine reserves in the exclusive economic zone. This should be addressed so that the new mechanisms can be applied in the same way in both the territorial sea and the exclusive economic zone.

Recommendation 4

The legislation should provide for a comprehensive system of marine protection measures, under which all elements of New Zealand's marine protected areas network would be incorporated.

New Zealand's existing marine protected area network includes areas managed under a range of instruments and by a number of different authorities. New legislation should consolidate them all into one system, managed for the same overall purpose, by the same entity.

Recommendation 5

Transitional provisions should be developed to incorporate areas protected under existing legislation into the new regime.

New legislation would need to provide for a mechanism by which existing marine protected areas, for example, benthic protection areas and marine reserves, can be moved across to the new regime.

Recommendation 6

New legislation should provide for a set of categories of marine protection, ranging from a reserve/no-take zone to areas where some sustainable extractive activities are permitted, based on the IUCN system. There should be a minimum requirement that no-take zones protect at least one example of all identified habitat types.

This approach would allow for a range of levels of protection whilst providing a comprehensive framework within which marine protection can be undertaken. It would also recognise the range of benefits that can be derived from marine protected areas.

Recommendation 7

New legislation should explicitly recognise the special relationship of tāngata whenua with the marine environment, and provide opportunities to enable the exercise of customary rights within the marine protected area network.

The MPA Policy recognises that Māori customary fishing tools may be included as part of a marine protected area network and this should be given statutory recognition.

Marine protected area establishment process

Recommendation 8

New legislation should retain the ability for the Director General of Conservation to make proposals for the establishment of individual marine protected areas and to receive proposals from other parties. The provision should be broadened to permit any person to make a proposal.

The current legislation enables some specified parties to propose marine reserves. This has resulted in the establishment of reserves for which there is a strong sense of community ownership and responsibility, and the mechanism is highly valued. Equally it is appropriate that the government should be empowered to establish individual marine protected areas.

Recommendation 9

New legislation should provide for a process to develop a comprehensive and representative marine protected area network within the territorial sea and exclusive economic zone. There should be a timeframe for its completion (at the latest by 2020) and requirements for the responsible Minister to report to Parliament periodically on its progress.

The MPA Policy provides for a strategic approach to the development of a marine protected area network. However, it has no statutory force and has been implemented in only two bioregions. New legislation should require the objectives of the MPA Policy to be achieved and set out a means by which it should be done.

Recommendation 10

New legislation should set out a process for the involvement of stakeholders in the development of the marine protected area network, including provision for either comprehensive consultative processes or the adoption of a collaborative decision-making process, where appropriate.

Any initiative for developing the marine protected area network must include strong stakeholder involvement. In order to move past the adversarial approach to marine management which currently exists in New Zealand, more use should be made of collaborative decision-making, which seeks to produce proposals

which are acceptable to all stakeholders. Where collaborative approaches are not appropriate, full engagement of stakeholders and the public should be facilitated through consultation.

Recommendation 11

Detailed consideration should be given to the process for stakeholder engagement, and the outcomes expected of any collaborative process. Consideration should also be given to the means by which the problems with collaborative processes, identified in this paper, can be overcome.

Experiences show that collaborative processes can have significant benefits but there are a range of challenges which can be a barrier to success. As a result, collaborative processes should not be embarked upon without detailed planning, sufficient resourcing and careful consideration of the potential pitfalls.

Recommendation 12

The final decision on the establishment of new marine protected areas should be made by the administering authority (the Minister of Conservation) on the basis of recommendations from an independent board of inquiry. Where proposals are the product of a collaborative process, the board of inquiry's terms of reference should be limited to those matters on which the stakeholder group cannot achieve consensus.

An independent board of inquiry process similar to that used in the preparation of national policy statements under the Resource Management Act would ensure that both conservation and resource use interests can be considered in an objective way. If collaborative decision-making is to be used, limiting the board of inquiry's terms of reference would give greater meaning to the collaborative process, and incentivise participants to achieve agreement.

Management, monitoring and enforcement

Recommendation 13

New legislation should provide a strong framework for ongoing management of established marine protected areas. This should include the gathering of baseline

data upon establishment and on an ongoing basis, provisions for adaptive management and effective enforcement measures.

It is important that resources are invested in the gathering of data that can assess whether management measures are effective and consequently, whether they should be modified. Equally, the establishment of marine protected areas is largely useless if the rules governing them are not enforced.

Recommendation 14

Where appropriate, provision should be made for stakeholders involved in the identification of marine protected areas, and tāngata whenua, to be involved in the ongoing management of areas.

This would enable continued use of the social capital generated through stakeholder involvement in marine protected area design, and encourage local ownership of marine protected areas.

Longer-term oceans reform

Recommendation 15

Government should undertake reform of marine management, to provide for the management of marine protection and marine activities that is integrated across sectors.

Although new marine protected areas legislation will be a step towards better management of the marine environment, more substantial progress can be made through the development of a new approach to oceans management, which considers the whole marine area in a holistic way.

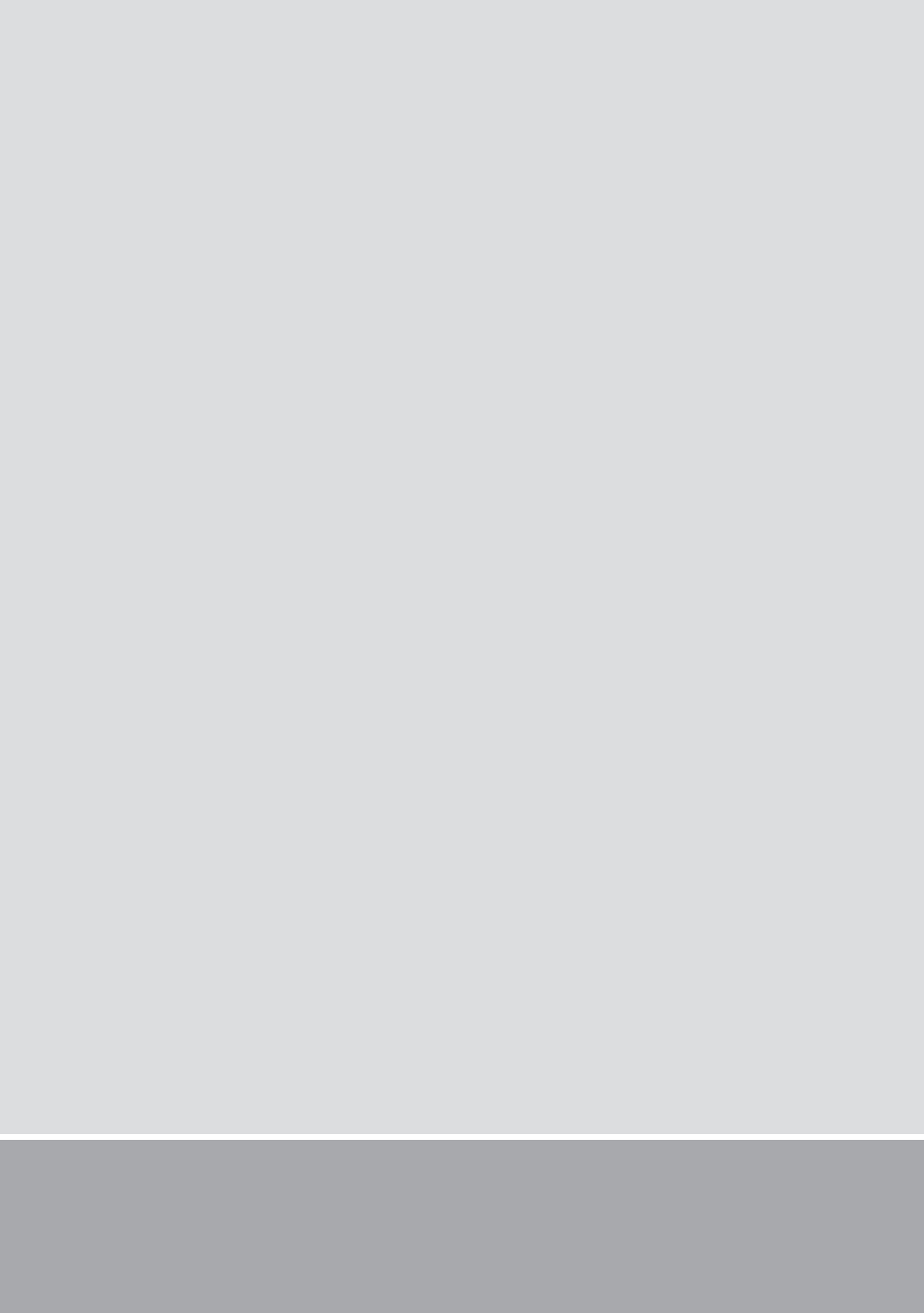
Recommendation 16

Oceans reform should include the provision of a legislative framework for marine spatial planning and better integration of marine management with catchment management, to allow terrestrial threats to the marine environment to be more effectively addressed.

Marine spatial planning will provide for better management of the range of marine activities, and in turn, better management of their impacts on marine protected areas. However, management of marine activities needs to be supplemented with improved management of coastal activities. Sedimentation, nutrient runoff and pollution are significant threats to the marine environment that cannot be addressed without this integration.

PART THREE

CASE STUDIES



chapter twelve



English marine conservation zones

Introduction

The United Kingdom's Marine and Coastal Access Act, passed in 2009, aims to completely overhaul management of the United Kingdom's marine area. One of the key measures contained in the Act is provision for a new type of marine protected area which, together with other existing types, is intended to create an ecologically coherent network consistent with the country's obligations at the European Union and international levels. England, Wales, Scotland and Northern Ireland have adopted different methods of identifying the new marine protected areas.¹

The process adopted in England, where the new marine protected areas are to be called 'marine conservation zones', is described and evaluated in this chapter. The analysis is based on publicly available information and documents, together with interviews conducted with regional project managers, agency officials and other relevant experts.

In England, regional stakeholder groups were formed with responsibility for recommending appropriate marine conservation zones. The stakeholder groups were required to make decisions in line with ecological guidance provided by the government, and with regard to the socioeconomic interests of stakeholders. In September 2011, the regional stakeholder groups submitted their final recommendations to the Department of Environment, Food and Rural Affairs. They were also submitted to the government's science advisory panel for scrutiny. In November 2011 the panel published its report on the proposals.

The government's nature conservation advisory bodies also considered the proposals and provided their final advice in July 2012. Government intends to undertake formal public consultation on the recommendations by the end of 2012. The first marine conservation zone designations are planned to take place in 2013, with additional designations occurring in stages. Although those involved in the process have met with some challenges, and continue to do so, the stakeholder-led approach to site identification has proved innovative and ambitious. It provides useful lessons to those interested in best practice methods for development of marine protected area networks.

Context

Economic benefits

The United Kingdom's marine area² extends over 867,400 square kilometres, which is roughly three times the nation's land area. The area is particularly diverse, comprising a large variety of habitats, and supporting up to 8,000 different species. It provides considerable revenue: income generated by activity in the marine area was estimated at £46 billion in 2006³ (the most recent year for which such an estimate has been made), which is a significant proportion of the total £1.29 trillion national gross domestic product.⁴

The main revenue-generating industries are oil and gas (£37 billion), maritime transport (£4.7 billion), leisure and recreation (£1.3 billion), coastal tourism (£5.3 billion), aggregate extraction (£550 million), naval defence (£470 million) and fisheries and aquaculture (£400 million). In addition, the European Union Renewable Energy Directive,⁶ which requires the United Kingdom to generate 15 per cent of its energy by renewable means by 2020, means that large swathes of the country's marine area are being set aside for renewable energy schemes. In particular, provision is being made for offshore wind farms, which require significant amounts of space. The varied and growing pressures on the marine area mean that implementing more effective protection for marine biodiversity is a pressing need if the health of the country's marine ecosystems is to be protected.

History of marine protection

Informal or voluntary marine reserves existed in United Kingdom coastal waters for many years, until a mechanism to create marine protected areas with statutory protection was put in place in 1981. The Wildlife and Countryside Act 1981 provided for the establishment of 'marine nature reserves' to conserve marine flora and fauna and geological or physiographical features of special interest, while providing opportunities for study of the systems involved. Marine nature reserves could be established anywhere within territorial waters, but there was no provision for them to be established within the exclusive economic zone. Only three marine nature reserves were ever formally designated (Lundy Island in England, Skomer Island in Wales, and Strangford Lough in Northern Ireland) due at least in part to the complexities of the designation process.

Since the 1990s, requirements at European level have provided further impetus for the establishment of marine protected areas in United Kingdom waters. The

1992 Habitats Directive⁷ requires all European member states to create 'special areas of conservation' to protect certain specified habitats and species, including marine. Together with the Birds Directive, adopted in 1979,⁸ which requires the creation of 'special protection areas' to protect specific birds, these constitute a mechanism for the creation of a European network of protected areas called Natura 2000.

The Conservation (Natural Habitats &c) Regulations 1994 first transposed the Habitats Directive into national law in England, Scotland and Wales.⁹ This regulation applied to the territorial sea only, and the directive was only applied to offshore areas in 2007 (by the Offshore Marine Conservation (Natural Habitats &c) Regulations 2007). So, to date, more effort has been directed towards designating sites in the inshore area than offshore.

The United Kingdom currently has 96 'special areas of conservation' for marine habitats or species, covering 4.8 per cent of the country's marine area, of which 84 are located in the territorial sea. Ten sites are wholly in offshore waters and two sites straddle inshore and offshore waters. The country also has 107 special protection areas for birds which have a marine component, but only three of these are comprised entirely of marine space. The identification of special protection areas and special areas of conservation in United Kingdom waters is undertaken by the government's statutory nature conservation advisory bodies, which are the Joint Nature Conservation Committee (comprised of representatives from a range of statutory conservation agencies as well as independent experts) and the country conservation agencies (in England, this is Natural England).¹⁰

In parallel to the development of the Natura 2000 network, the United Kingdom also signed up to international agreements to create an ecologically coherent network of marine protected areas. Together with the other signatories to the Convention for the Protection of the North East Atlantic (OSPAR), the United Kingdom agreed to establish an ecologically coherent network of marine protected areas in the north-east Atlantic by 2010.¹¹ The country also committed to the creation of an ecologically coherent network of marine protected areas by 2012 under the Convention on Biological Diversity. Although the United Kingdom made some progress towards meeting its international commitments, this was slow until the entry into force of the European Union Marine Strategy Directive. This requires a broad range of actions on the part of member states, to improve the health of their marine environments, including the creation of ecologically coherent marine protected area networks.

The Marine Strategy Framework Directive came into force in July 2008 and was transposed into United Kingdom legislation in July 2010.¹² It requires European member states to undertake the following:

- An assessment of the current state of the sea by 2012
- A detailed assessment of what ‘good environmental status’ means for the sea, and associated targets and indicators by 2012
- Establishment of a monitoring programme to measure progress toward good environmental status by July 2014
- Establishment of a programme of measures for achieving good environmental status by 2016

The Directive sets out 11 high level qualitative descriptors for determining ‘good environmental status’. These include ensuring that populations of fish and shellfish are within safe biological limits, maintaining the biological diversity of marine habitats and species, and limiting contaminants to the marine environment to levels which do not cause pollution.

The Directive expressly states that the establishment of a marine protected area network, which may incorporate areas that have already been designated, will be an important contribution to the achievement of ‘good environmental status’. This means that new marine protected areas should be developed as part of the programme of measures implemented under the Directive. Member states are required to have in place a coherent and representative network of marine protected areas by 2016.

Legislative Framework

Reform of marine management in the United Kingdom

The existing marine protection mechanisms (the Wildlife and Countryside Act 1981, the Habitats Directive and the Birds Directive) were not enough to enable the government to fulfil the full extent of the country’s international commitments to create an ecologically coherent network of marine protected areas. Nor were they sufficient to meet the obligation to achieve ‘good environmental status’. This was, in particular, because of the restricted range of habitats and species to which the European Union directives apply, and the fact that the marine nature

reserve mechanism had been little used and was out-of-date. To address these shortcomings, the United Kingdom government and the devolved administrations embarked on a programme of reform of the management of the marine area.

Building on a 2002 commitment to achieve 'clean, healthy, safe, productive and biologically diverse oceans and seas', in 2009 the United Kingdom government, the Welsh Assembly government, the Northern Ireland executive, and the Scottish government published 'high level marine objectives' for the United Kingdom marine area. This document sets out the outcomes that the administrations are seeking to achieve in the marine area and how they will achieve them.¹³ The high level objectives underpin the country's marine policy. The objectives include the following provisions:

- Biodiversity is protected, conserved and where appropriate, recovered, and loss has been halted
- Healthy marine and coastal habitats occur across their natural range and are able to support strong, biodiverse biological communities and the functioning of healthy, resilient and adaptable marine ecosystems
- Oceans support viable populations of representative, rare, vulnerable and valued species

The Marine and Coastal Access Act 2009

The Marine and Coastal Access Act 2009 creates new mechanisms for realising these objectives. The Act mandated the drafting of a 'marine policy statement' (published in March 2011) which sets out all the policies that will contribute to the achievement of sustainable development in the United Kingdom marine area. All decisions made in respect of the marine environment must be consistent with it. In relation to marine protected areas, the marine policy statement provides that *"the UK administrations are substantially committed to completing an ecologically coherent network of MPAs by 2012 as part of a broad based approach to nature conservation. It will be made up of both national ... as well as European designations such as Special Areas of Conservation (as designated) and Special Conservation Areas (as classified under the Birds Directive and sites of international importance (Ramsar Sites))."*¹⁴

Provision for a new type of national marine protected area was included in the Marine and Coastal Access Act 2009. The provisions for the creation of marine protected areas under this legislation differ for each of the United Kingdom's

constituent countries. As stated above, this paper considers in detail the mechanism introduced in England.

In England, the legislation provides for the creation of marine conservation zones, replacing the little used marine nature reserve mechanism (the one marine nature reserve in England has become the country's first marine conservation zone). The Act provides that the Secretary of State (i.e. the Environment Minister) may designate marine conservation zones in any part of the territorial sea or exclusive economic zone for the purpose of conserving marine flora or fauna, marine habitats or types of marine habitat, or features of geological or geomorphological interest.¹⁵

The objective is that the marine conservation zones, together with the conservation sites designated under other processes, will form a network which satisfies the following conditions:

- That the network contributes to the conservation or improvement of the marine environment in the United Kingdom marine area
- That the features which are protected by the sites comprised in the network represent the range of features present in the United Kingdom marine area
- That the designation of sites comprised in the network reflects the fact that the conservation of a feature may require the designation of more than one site¹⁶

A key difference between the European marine sites (special areas of conservation and special protection areas) and the marine conservation zones, is that whereas the European sites are identified on scientific grounds only, in considering whether it is desirable to designate an area as a marine conservation zone the decision-making body may “*have regard to any economic or social consequences of doing so.*”¹⁷ This has enabled the government to involve stakeholders in the site identification process. Whereas European sites are selected by scientists and government officials, solely on the basis of scientific merit, the government adopted a strongly stakeholder-led process for the identification of proposed marine conservation zones.

Marine conservation zone identification process

Structure

Four regional projects were created in which local stakeholders worked together to identify proposed marine conservation zone locations, in accordance with government guidance. The design of the process was strongly influenced by the experiences of a pilot project which had been running in the south-west region since 2005. The pilot project had started with a very broad brief. It was simply instructed to develop a regionally-focused means of identifying potential marine protected area sites which involved stakeholders. The project team studied international best practice and other marine protected area identification projects around the world, in particular the California Marine Life Protection Act initiative, to develop a methodology appropriate to England which would enable stakeholders to participate in a marine protected area identification exercise in a meaningful way.

On the basis of the pilot project's experience, the Joint Nature Conservation Committee and Natural England funded the creation of the four regional projects, covering the south-west seas (Finding Sanctuary), the Irish Sea (Irish Sea conservation zones), the North Sea (Net Gain) and the south-east seas (Balanced Seas). The Joint Nature Conservation Committee and Natural England developed 'project delivery guidance'¹⁸ which outlined the required basic structure of the regional projects and the process they should follow.

Each regional project was required to have a 'project board' with ultimate responsibility for the effective delivery of the project, within budget and in accordance with the guidance. The board, which had no involvement in the marine conservation zone decision-making process, principally comprised the project manager, the project host and representatives of the conservation advisory bodies. The board was responsible for financial management of the project.

A regional project team was responsible for the day-to-day running of the projects. A team of full-time staff was contracted. These staff members came from a variety of backgrounds, in project management, marine protected area planning, marine ecology, GIS, stakeholder liaison, economics, communications and social science, in order to encompass the full range of skills that were required for such a broad-based and novel project. Each project team was required to contract an independent facilitator who was tasked with designing and running a stakeholder engagement process, including undertaking a stakeholder analysis, achieving

consensus on membership of the stakeholder groups, developing the decision-making process, resolving disputes and facilitating constructive discussion.¹⁹

The regional stakeholder groups, made up of the individuals identified by the stakeholder analysis, were responsible for the substantive work in identifying proposed marine conservation zone sites. It was intended that the groups should comprise representatives from all significant sectors as well as from the conservation advisory bodies and the Marine Management Organisation (an independent statutory body established to manage the marine area) who provided technical expertise and information. The inclusion of a wide range of stakeholders meant that the stakeholder groups could potentially be too big to allow for effective decision-making. For that reason they were empowered to break themselves down into smaller working groups.

Each regional project was permitted to design the structure and decision-making processes of the stakeholder groups, and each project adopted a slightly different model. For example, the stakeholder analysis for the Finding Sanctuary identified 41 stakeholders, which formed the 'steering group'. 'Inshore' and 'offshore' working groups were nominated from the steering group, to undertake the detailed planning work on the project, and they met for several days every month. The steering group itself met every three to four months to review the progress of the working groups.

Having found that a preoccupation with process issues (whether the right people were represented in the process, how often they should meet, etc.) hampered decision-making in the early stages, the steering group also created a 'process group', which had responsibility for membership, meeting frequency, location and other similar issues. This meant that the other working groups could leave the time consuming process issues to another group with a mandate to deal with them, and focus on the substantive work. The project team also worked with 'local groups', some of which were established from groups already in existence, to gather local knowledge, suggestions and feedback that could be fed into the decision-making process at the regional group level.

In contrast, the 'Balanced Seas' project formed a regional stakeholder group comprised of around 30 individuals representing regional interests, who were responsible for overall consideration of the issues and options and development of the final recommendations. The project also established three geographically based 'local groups', as advisory bodies to the regional stakeholder group, which met regularly. These were intended to enable the wider stakeholder community to contribute regularly to the project. They were made up of representatives of

organisations with a regional, sub-regional or local perspective, who had been invited to participate by the project team.

Each of the regional projects was able to designate stakeholders as 'named consultative stakeholders', which included those who were not able to attend all the meetings, but who wanted to have some involvement in the process. Large organisations such as international fishing bodies took up this opportunity. These stakeholders were able to play a less resource intensive role in the process by being asked for their views at key stages.

Although the majority of the substantive decision-making was undertaken by the regional projects, they were also able to call on advice from economists based in the Department of Environment, Food and Rural Affairs and from scientists on a national 'science advisory panel'. The panel was an independent body, comprising eight well-respected national and international scientists appointed by the Environment Minister, and was specifically formed to provide scientific knowledge, advice and judgment to the process.

Process

Central government commenced the process in January 2009, with the preparation of the basic principles and governance structures. The regional stakeholder groups were formed in February and March 2010. The project delivery guidance document set out the basic process for each of the regional projects. A relatively short time frame was allowed for completion of the projects.

Each regional project was required to begin data collection immediately and to compile the available data into a draft regional profile. Data came from various sources, including government, stakeholders and scientists contracted by the Department of Environment, Food and Rural Affairs to provide additional information that would inform the process. The stakeholder groups then began identifying proposed marine conservation zones.

Three iterations of the proposed network were submitted for comment by the Joint Nature Conservation Committee, Natural England and the science advisory panel over the following months, and these were slowly refined into draft, then final recommendations. The final recommendations were originally to be submitted to the Department of Environment, Food and Rural Affairs by mid-April to May 2011. In reality, there were some delays, and they were ultimately submitted in September 2011.

The final recommendations submitted by each project comprised a map showing the recommended marine conservation zones, a proposed name for each site, descriptions of the habitats, species and geological and geomorphological features of each site, boundary coordinates, conservation objectives for each site (see below), an impact assessment for each regional package of marine conservation zone recommendations and assessment of cumulative impacts, and a consensus log for each site, recording any outstanding objections to particular sites from stakeholders and the reasons for them.

The science advisory panel then assessed the recommendations against the requirements of the ecological network guidance and their own informed views about how such a project should be undertaken. The intention was that the Joint Nature Conservation Committee and Natural England would then build on the recommendations to deliver final advice and an impact assessment to government by January 2012. However, the science advisory panel found that although the proposed network, if designated in its entirety, would meet the majority of the requirements for an ecologically coherent network, supporting evidence for the proposals was lacking. This meant that the recommendations in their current state were not ready to be finalised.²⁰

As a result, the Department commissioned additional work to gather more data and assess that used by the regional projects. The Joint Nature Conservation Committee and Natural England delivered their final advice to government in July 2012. The advice covered the science behind the recommendations, the quality of the ecological data, and their views on the overall regional marine conservation zone project process. The Environment Minister will consider the advice, including any information pertaining to continuing opposition from stakeholders, and adopt a proposed network of marine conservation zones for public consultation.

The Marine and Coastal Access Act provides that the Environment Minister is required to consult the public prior to designating a marine conservation zone (unless there is an urgent need to protect an area, in which case the relevant authority can designate a marine conservation zone without consultation, which will remain in force for up to two years). It is envisaged that the first marine conservation zone designations will take place in 2013. The Environment Minister has recently indicated that a phased approach to designation will be adopted, where early management measures are put in place to protect sites which have already been designated, while work to build the evidence base for future designations continues.²¹

Network Design

Ecological network guidance

In undertaking the marine conservation zone identification process, regional stakeholder groups were required to work consistently with detailed 'ecological network guidance' produced by the Joint Nature Conservation Committee and Natural England, which set out what the proposed networks need to achieve. The guidance recognises that 'ecological coherence' is an emerging concept, but uses work undertaken by the OSPAR Convention and international best practice documents to inform the design principles and considerations that should be taken into account when trying to achieve it.²²

The guidance sets out seven network design principles (based on policy of the Department of Environment, Food and Rural Affairs and the OSPAR Convention), to which the stakeholder groups are to give effect:

- *Representativity* – the marine protected area network should represent the range of marine habitats and species through protecting all major habitat types and associated biological communities present in our marine area
- *Replication* – all major habitats should be replicated and distributed throughout the network. The amount of replication will depend on the extent and distribution of features within seas
- *Viability* – the marine protected area network should incorporate self-sustaining, geographically dispersed component sites of sufficient size to ensure species and habitats persistence through natural cycles of variation
- *Adequacy* – the marine protected area network should be of adequate size to deliver its ecological objectives and ensure the ecological viability and integrity of populations, species and communities (the proportion of each feature included within the marine protected area network should be sufficient to enable its long term protection and/or recovery)
- *Connectivity* – the marine protected area network should seek to maximise and enhance the linkages among individual marine protected areas using the best current science. For certain species this will mean that sites should be distributed in a manner to ensure protection at different stages in their life cycles

- *Protection* – the marine protected area network is likely to include a range of protection levels, ranging from highly protected sites or parts of sites where no extractive, depositional or other damaging activities are allowed, to areas with only minimal restrictions on activities that are needed to protect the features
- *Best available evidence* – network design should be based on the best information currently available. Lack of full scientific certainty should not be a reason for postponing proportionate decisions on site selection.²³

These principles aim to ensure the resilience of the marine protected area network, so that the ecosystem(s) can resist and recover from disturbances and damage. The guidance requires that where possible all seven network design principles should be met, in order to ensure the ecological coherence of the network. Each principle is to be given equal priority, except for connectivity. Connectivity is a secondary concern because it can only be considered in cases where there is sufficient data available to make a proper assessment of the linkages between sites.

The ecological network guidance provides the regional projects with reasonably detailed instructions about how to achieve the requirements of the network design principles. The guidance instructs that the principles of representativity, replication and adequacy should be addressed by the designers of the network first.

The guidance identifies that in English waters there are 23 broadscale habitat types, 22 habitats of conservation importance, 29 low or limited mobility species and three highly mobile species of conservation importance, all of which should be protected by marine conservation zones. Where any of these features are present in the jurisdiction of a regional stakeholder group, examples must be protected in two or more marine conservation zones, with an adequate amount of each feature protected to enable its long term protection and/or recovery.

The remaining network design principles, and some further ecological and practical considerations which are also set out in the guidance, are then to be used to assist the stakeholder groups to choose between different options that are comparable in meeting the principles of representativity, replication and adequacy.

Although the regional stakeholder groups were required to achieve all the objectives set out in the ecological network guidance, they could also ‘have regard to’ economic and social considerations. This meant that, in practice, the task of the regional stakeholder groups was to find locations for the required marine conservation zones which had the least impact on stakeholder interests. The process was one of compromise, which was intended to ensure that all participant

stakeholders were affected by the marine conservation zone designations as minimally as possible, within the context of the requirements of the ecological network guidance.

Nevertheless, there were occasions where it was not possible to achieve a particular conservation requirement with consensus from all stakeholders. Where the best possible option had been identified, which still gave rise to objections from individual stakeholders, these were submitted with the final proposal. It will be the Environment Minister's task to consider whether achievement of the requirements of the ecological network guidance has been effectively balanced against the requirement to 'have regard to' economic and social interests and make recommendations for designation of sites accordingly.

Management measures

The legislation does not specify that particular types of activity will be prohibited or regulated within marine conservation zones, as would be the case in a zoned system. The management measures that are implemented for each marine conservation zone will be specifically designed to achieve the particular conservation objectives identified for the site, or protection of particular features of the marine protected area, rather than to implement blanket protection of the entire area.

For each proposed marine conservation zone, the regional stakeholder groups were required to identify the features (i.e. species or habitats) requiring protection and the 'conservation objectives' of the site. A conservation objective is 'a statement describing the desired ecological/geological state (quality) of a feature for which a marine conservation zone is designated.'²⁴

The regional stakeholder groups were required to identify whether the activities present at the site put a feature at risk of not achieving a favourable condition (a 'vulnerability assessment'), and thus whether the conservation objective should be to 'maintain' or to 'recover' the feature at the site. This would indicate whether activities were able to continue as they were or should be managed to reduce anthropogenic pressures.

In the absence of real data about what impact activities were having on the features, the regional projects were required to make some assumptions about the effects that particular activities have on the marine environment, and thus whether they should be managed.

The Joint Nature Conservation Committee, Natural England and central government will have responsibility for deciding what management measures are required at a particular site. The aim will be to implement measures which achieve the required level of protection whilst having the smallest possible impact on economic and social interests.

Reference areas

The stakeholder projects were required to identify some marine conservation zones, or parts of marine conservation zones, as 'reference areas'. The ecological network guidance required that "*[e]ach broadscale habitat and FOCI [Feature of Conservation Interest] should have at least one viable reference area within each of the four regional MCZ project areas where all extraction, deposition, or human derived disturbance is removed or prevented.*"²⁵

Areas in 'reference condition' provide an opportunity to demonstrate the unimpacted state of marine features, in the context of the prevailing environmental conditions. Reference areas are therefore intended to help the government and the science advisory bodies to refine concepts such as 'good environmental status' under the Marine Strategy Directive.

Implementation, monitoring and enforcement

Section 140 of the Marine and Coastal Access Act sets out a general offence of damaging a marine conservation zone which can be enforced as soon as the site is designated. If sites require further management measures, responsibility for implementing them will lie with the inshore fisheries and conservation authorities and the Marine Management Organisation, which will have the power to regulate activities through the creation and enforcement of bylaws for the purpose of furthering the conservation objectives.

At the time of writing, the United Kingdom government had not decided on the exact manner in which new management measures will be implemented, monitored or enforced.²⁶ It is expected that implementation will be undertaken using a risk-based approach, so that the most damaging activities are managed first.

The Marine Management Organisation will be empowered to create bylaws to manage unregulated activities in marine conservation zones out to 12 nautical miles. A person contravening the bylaws will be liable for a fine. Bylaws may restrict or prohibit activities in, or entry into, the marine conservation zone. The

Marine Management Organisation expects that it will take around 12 months to implement a new bylaw. But emergency bylaws can be created, and these can be in place within six weeks, for a period of up to 12 months. It is expected that bylaws for the protection of marine conservation zones will be reviewed periodically, but the exact process has not yet been established. The Environment Minister will have the power to revoke bylaws.

It will be necessary for reference areas to be dealt with differently, as the Marine Management Organisation is not empowered to make general bylaws to create blanket protection for a marine conservation zone. Each activity will therefore have to be investigated separately and on a site-by-site basis. Each bylaw will have to be specific to each activity to be restricted, so a number of bylaws may be required.

The Marine Management Organisation will work with the inshore fisheries and conservation authorities, the Environment Agency, police and others to share evidence, undertake joint patrols and carry out other activities directed towards enforcement of the bylaws. Enforcement will be undertaken by “*applying a risk based approach to ensure it can make best use of its resources and target inspection activity at those areas identified as being subject to the highest risk of non-compliance with the regulations.*”²⁷

The Marine Management Organisation employs three officers specifically to implement management measures, and has contracts for aerial and maritime surveillance, currently comprising one surveillance aircraft and three Royal Navy fishery protection squadron vessels. These will be available to support the work of the coastal office and marine enforcement officers based at the organisation's headquarters. The Organisation also runs a vessel monitoring system for fishing vessels over 15 metres in length. These vessels are currently required to ‘ping’ every two hours, but the government is investigating the potential for increasing the ping rate to improve the accuracy of the data.

The Marine Management Organisation and the inshore fisheries and conservation authorities will also be able to manage fishing activities within marine conservation zones through their existing functions for fishing licensing. The inshore fisheries and conservation authorities will be primarily responsible for the regulation and management of the effects of fishing activities in marine conservation zones out to six nautical miles, whilst the Marine Management Organisation has responsibility for most fishing licensing between six and 12 nautical miles.

Commitments under the European Union Common Fisheries Policy mean that fisheries activities outside the territorial sea (12 to 200 nautical miles), and outside six nautical miles where non-United Kingdom fishers have historic fishing rights, will be regulated at the European Union level through the Common Fisheries Policy.

The effects on marine conservation zones of non-fishing activities which require a marine license will be managed by the Marine Management Organisation through the licensing process, for which it is already responsible. It is expected that this mechanism will provide the necessary protection from non-fishing activities for marine conservation zones outside 12 nautical miles, as the majority of non-fishing activities that occur there are licensed. The Marine Management Organisation does not expect that any unregulated activities occurring outside 12 nautical miles will have an impact on a marine conservation zone, but suggests that were any to arise, these would have to be managed by way of voluntary agreements.

The Department of Environment, Food and Rural Affairs will be required to review and report to Parliament on its progress in achieving an ecologically coherent network every six years, starting in 2012. The report must set out any further measures that are required to enable the network or an individual marine conservation zone to meet its objectives. Orders creating a marine conservation zone can only be revoked or amended by a further order by the Secretary of State. The government has committed to keeping its decisions under review and take any measures necessary to make changes to the boundaries, conservation objectives or management measures in relation to a marine conservation zone where the evidence indicates that this is required.

This adaptive process is intended to enable the government to respond to new data on the marine environment and provides for changes required to comply with new law and policy. Scientific monitoring will be undertaken by the Joint Nature Conservation Committee and Natural England. The precise nature of the monitoring programme has yet to be decided.

Relationship with marine spatial planning

In addition to the marine conservation zone process, the Marine and Coastal Access Act also provided for a marine spatial planning process to be undertaken over the entire United Kingdom marine area. Whereas the marine conservation zone project is based on collaborative decision making, the spatial planning project is

being undertaken by the Marine Management Organisation. It was intended that the marine conservation zone recommendations would be completed prior to the start of the spatial planning process, so that they could be incorporated into the marine spatial plans. A key reason for separating the two processes was to ensure that marine protection measures were not diluted by a broader planning process which seeks to balance the interests of all other activities in the marine area.

Analysis

Was it successful?

Many staff and stakeholders of the regional projects considered that the time allowed to complete the process of gathering data and making recommendations was far too short. The science advisory panel published its report on the regional stakeholder groups' recommendations in November 2011, and as a result of this, the government concluded that there is significantly more work to be undertaken before marine conservation zones are designated.

The regional project design was inspired by processes undertaken overseas, particularly in California, but was also clearly shaped to fit English requirements. The regional projects all succeeded in creating a functioning stakeholder group which delivered a final proposal, within the required time frame. The groups kept broadly to schedule, with the final proposals being delayed by only a few months. This was a significant achievement given the fact that nothing similar had ever been undertaken in the country, and there was significant opposition to the project from some stakeholders.

Data

The science advisory panel reviewed and evaluated the data used by the regional projects to support their proposals on a site-by-site basis. The panel found that the data used was generally inadequate, or inadequately referenced. Specifically, the panel found that overall the proposals exhibited *“considerable weakness in the range and use of literature sources presented in the site descriptions.”*²⁸

The panel emphasised, that it expected a rigorous academic approach to evidence to be adopted, to enable informed decisions on designations and associated conservation objectives and management measures to be made. This was not entirely consistent with the methodology adopted by the stakeholder

groups, which of course were driven by stakeholders rather than scientists. A range of local knowledge, unpublished survey data, photos and videos were all used by the groups to identify recommendations.

Evaluation of the proposed network

The science advisory panel noted that gaps remain in adequate coverage of features and information on site characteristics, which may mean that it is necessary to re-engage with stakeholders. It observes that *“the identification of locations for protection has relied greatly on socio-economic considerations with biodiversity often of secondary consideration or taken into account late in the process.”*²⁹ The panel acknowledges that this is an inevitable product of the stakeholder driven process. It is unsurprising that the panel has found that in many cases the regional stakeholder groups have seen the minimum requirements as a target, rather than the absolute minimum level of biodiversity protection required. At present it is unclear how the government will choose to address this issue.

Uncertainty regarding process

Anecdotal evidence indicates that throughout the process, the regional stakeholder groups have been concerned about a lack of clarity surrounding what would happen to their proposals once submitted. Now that the science advisory panel has reported on significant problems with the proposals, it is even less clear what changes will be made to the proposed networks prior to designation.

The Joint Nature Conservation Committee and Natural England will now be responsible for further work to ensure that the proposals are accompanied by adequate supporting evidence and that they meet the requirements of the ecological network guidance. At this stage, it is unclear to what extent changes will be proposed, and whether this work can be completed within the allocated timeframe.

Once the Joint Nature Conservation Committee and Natural England have provided the Environment Minister with their formal advice, the legislation requires that there will be a formal public consultation, a standard governmental procedure, following which the Environment Minister will make decisions about whether to designate the proposed marine conservation zones.

There is concern that some stakeholders may have cooperated in the regional projects in the knowledge that they could oppose the proposals later on, perhaps believing that it was better to participate in the early stages, in order to ensure that

they fully understand the decisions that have been made. This was in order to be in a better position to contest them when the real decision-making takes place at central government level.³⁰

The Environment Minister must ensure that the final designated network of marine conservation zones is representative and ecologically coherent, but in doing so is able to have regard to economic and social issues. It is therefore possible that evidence provided during the public consultation will lead the Environment Minister to decide that it is appropriate that some recommendations are altered to reflect economic and social needs that have not adequately been taken into account in the proposals submitted by the marine conservation zone projects.

This is particularly the case, given that a large number of proposed marine conservation zones remain highly contentious, and the science advisory panel's criticism of the evidence base for the proposals may be said to undermine arguments for their designation. It is very unclear how the Minister will approach objections from stakeholders, and how alterations to individual marine conservation zones will affect the coherence of the network.

Another concern is that public consultation will jeopardise the work that has already been done with stakeholders. Although far from perfect, a delicate consensus was achieved amongst stakeholders during the projects, and there is a fear that more consultation will polarise them again, creating opposition to the process. Some have suggested that the stakeholder-led process was a form of consultation in itself, so the government should not have to undertake another one. The novelty of the stakeholder-led approach has meant that the process arguably does not fit neatly into normal legislative process. It is unclear at this stage whether this will cause problems.

What is certain, however, is that the regional stakeholder projects have given stakeholders a strong sense of ownership over the proposals. If the Joint Nature Conservation Committee, Natural England or the Environment Minister decide to make significant changes to the proposed networks, as appears may be likely following the science advisory panel's report, stakeholders may feel betrayed or that the project was a waste of time. This may have a real impact on the effectiveness of implementation.

Uncertainty over management measures

Stakeholders still do not know what management measures will be implemented in each of the marine conservation zones, because this was not part of the regional

projects' mandate. Although the regional stakeholder groups were required to propose conservation objectives for each marine conservation zone, with suggestions about how they might be realised, the actual decisions about what activities need to be restricted or prohibited will be made by central government.

In practice, however, anecdotal evidence indicates that it was near impossible to conduct the stakeholder-led process without making some assumptions about what protecting particular features would mean for ocean users. The key thing that individual stakeholders wanted to know during the process was “*if we put a marine conservation zone in this location, how will it affect the activity that I undertake there.*”³¹ The regional projects dealt with this in different ways: the south-west regional project came up with a set of assumptions about the management measures that would be necessary to protect particular features, which were formally adopted for the purposes of the project – for example, if a benthic habitat is to be protected, bottom trawling will have to be prohibited in that area. The south-east regional project purposely avoided setting out any assumptions, but it was clear that stakeholders had formed their own views, about the management measures that would likely be required.³²

Of course, the risk is that the final management measures implemented will not be the same as those expected by stakeholders. In this event, it is possible that stakeholders will become disillusioned with the process, and implementation will be made more difficult.

Conservation objectives

The science advisory panel was critical of the approach set out in the ‘conservation objective guidance’ whereby the regional stakeholder groups were required to decide whether the objective for a site should be to ‘maintain’ or to ‘recover’. The panel noted that it is dangerous to assume that the removal of specific anthropogenic pressures from a site will achieve recovery to any (unspecified) state, or that if there is no increase in existing anthropogenic pressures, that the site will be maintained in a particular state.

Furthermore, the panel highlighted the fact that by definition, an objective of ‘recovery’ means that the environmental managers must have an idea about the state to which a site will be recovered to, but the conservation objective guidance gives no advice as to how this should be identified.³³

Anecdotal evidence suggests that stakeholders found the approach confusing and counter intuitive – rather than deciding whether activities should be permitted

to take place at a particular site, they were required to identify in the abstract its vulnerability. The groups did not, and could not, have adequate information or expertise to decide whether a site should be ‘recovered’ to a particular standard of health, or what that standard of health should be.

Opposition to reference areas

Reference areas were not specifically required in the legislation, but were identified as necessary by the Joint Nature Conservation Committee and Natural England in the ecological network guidance. The guidance expressed the need for these areas where “*extraction, deposition, or human derived disturbance is removed or prevented*” mainly in terms of their value to scientists in monitoring the impact of human activities on the marine area as a whole, as implied by their name.

The regional project teams found that the identification of reference areas was one of the most challenging aspects of the process. Some stakeholders supported the idea of marine conservation zones where the *most destructive* activities are managed, but were less willing to support the establishment of areas where *all* extractive activities are prohibited. Given that many stakeholders were concerned, mainly with protecting their right to undertake their particular activity, inevitably reference areas met with the majority of opposition.

Furthermore, the need for reference areas was expressed mainly in terms of the value of the areas to scientific research and the value of recovering the sites to a ‘pristine’ state. This concept was problematic, as the notion of closing areas to return them to some undefined earlier baseline is scientifically questionable. Those opposed to reference areas found this particularly difficult to accept: “*There is absolutely no need for draconian areas of ‘no-man’s land’ for the purpose of measuring ecological improvement in MPAs which would be readily evident from any general time-series programme of monitoring. Such proposals are ill thought through and show a careless disregard for people’s livelihoods in order simply to give a free reign to conservation scientists’ experiments.*”³⁴

The fact that reference areas or no-take zones were not specifically provided for in the legislation was perhaps unhelpful, as it meant that those opposed to their development could argue that the regional projects had no mandate to create them. Although the four regional projects succeeded in identifying reference areas for the majority of required features, their final proposals were qualified by significant numbers of objections from stakeholders.

The science advisory panel was highly critical of the reference area proposals submitted by all four projects. It found that proposed sites were too few and too small, and had failed to meet the requirements for viability of sites. The panel recommends that the Department of Environment, Food and Rural Affairs, the Joint Nature Conservation Committee and Natural England should “*return to the design of the Reference Area network, perhaps initially by testing the outcome, physical feasibility and socioeconomic consequences of expanding the recommended RAs.*”³⁵ It is thus unclear how the government will choose to proceed, given the clear conflict between the levels of protection required in the ecological network guidance, and the amount of opposition from stakeholders.

Key reasons for success

Despite the difficulties, which may be resolved as time progresses, there have been many positive aspects of the process so far, in particular in relation to the level of cooperation exhibited by stakeholders in the process. Those involved in the projects reported that, after initial reluctance, stakeholders from different sectors found that they could work constructively together.

Central government delegated significant power to the regional projects to decide how to structure themselves, and how to go about the decision-making process. As a result, the projects all worked in slightly different ways, which was a reflection of the particular needs of stakeholders in that region.

Some of the projects found that the physical act of all sitting around a map together, and physically moving proposed boundary lines, was the most effective means of getting everyone involved and communicating with each other. Having tried maps on walls and electronic methods, the south-west project found that A2 maps in acetate with marker pens were the most effective means of working – people were drawn in around the maps, which seemed to encourage cooperation and enthusiasm.³⁶

It was apparent that given good instructions, the stakeholder groups were able to understand complex concepts about how to achieve an ecologically coherent network. The project teams have reported that, initially there were problems with communication between the science advisory bodies and the stakeholder groups, because the scientific language used in the instructions was not easy for non-scientists to understand. However, given the right sort of materials including visual explanations, the stakeholders were able to grasp the complex requirements and meet them in their proposals.

The fact that stakeholder groups were given specific tasks, (following the detailed ecological network guidance rather than simply ‘identify marine conservation zones in this region’) meant that they could all focus on a goal together. The level of detail contained in the ecological network guidance document (together with the ‘conservation objective guidance’ and the ‘project delivery guidance’) ensured that the stakeholder groups understood what was required to achieve an ecologically coherent network. This was despite the fact that many of the stakeholders had vested interests in limiting the impact of the proposed network on their particular activity. The detailed guidance material was very helpful in making progress.

The project teams commented that, by the end of the process, individual stakeholders had become extremely knowledgeable and had developed a level of cross-sectoral communication that is rarely seen in relation to marine activities. One project manager commented that the regional groups have essentially now become representative regional bodies, with a deep understanding of the reasons for the proposed marine conservation zones, and that the groups should therefore continue to be harnessed in the implementation and management phase.³⁷ It appears that the government has no plan to do this, however.

Conclusion

The Marine and Coastal Access Act reflects an ambitious attempt to meet the United Kingdom’s commitments to establish a representative network of marine protected areas. Obligations at the European level are a key driver. Significant resource has been invested in collaborative decision-making, in processes closely modelled on the Californian initiative. Despite some challenges, the collaborative initiatives were able to recommend new marine protected areas which broadly met the scientific guidance. The extent to which the process is successful will depend on sustained government commitment to implementing the recommendations.

Endnotes

- 1 The Marine and Coastal Access Act provides for the designation of marine conservation zones in English and Welsh territorial waters and United Kingdom offshore waters. The identification of marine conservation zones in Welsh territorial waters is to be undertaken centrally by the Welsh Assembly government in collaboration with the Countryside Council for Wales. In Scotland, the provisions of the Marine and Coastal Access Act 2009 were transposed into Scottish law by the Marine (Scotland) Act 2010. That Act makes provision for the designation of 'Scottish MPAs', which will be identified through a collaborative science-based process with Marine Scotland, Scottish Natural Heritage, the Joint Nature Conservation Committee and stakeholders. At the time of writing, Northern Ireland is still to pass equivalent legislation
- 2 The territorial sea and the offshore marine area as defined in the Offshore Marine Conservation (National Habitats &c) Regulations 2007
- 3 Pugh D, 2008
- 4 Office for National Statistics, www.statistics.gov.uk (accessed 31 July 2012)
- 5 www.defra.gov.uk/environment/marine (accessed 31 July 2012)
- 6 Council Directive 2009/28/EC on the Promotion of the Use of Energy From Renewable Sources and Amending and Subsequently Repealing Directives 2001/77/EC and 2003/30/EC
- 7 Council Directive 92/43/EEC of 21 May 1992 on the Conservation of Natural Habitats and Wild Flora and Fauna
- 8 Council Directive 79/409/EEC of 2 April 1979 on the Conservation of Wild Birds, Codified in Council Directive 2009/147/EC of 30 November 2009 on the Conservation of Wild Birds
- 9 The Conservation (Natural Habitats &c) Regulations (Northern Ireland) 1995 transposed the Habitats Directive into Northern Irish Law.
- 10 Because of the complexities of devolution in the United Kingdom, offshore waters and inshore waters are the responsibility of different bodies. Broadly speaking, the country conservation agencies have responsibility for inshore waters and the Joint Nature Conservation Committee has responsibility for offshore areas.
- 11 Recommendation 2003/3 on a network of marine protected areas
- 12 United Kingdom Marine Strategy Regulations 2010
- 13 Her Majesty's Government, 2009
- 14 Her Majesty's Government, 2011, para 3.1.1
- 15 Section 117(1), Marine and Coastal Access Act 2009
- 16 Section 123 (3), Marine and Coastal Access Act 2009
- 17 Section 117(7), Marine and Coastal Access Act 2009
- 18 Davis C and C Herbon, 2010
- 19 Although the Balanced Seas and Finding Sanctuary projects reported that the work of the facilitator was invaluable to ensure the project moved along smoothly (Sue Wells and Tom Hooper, *pers comms*) the Net Gain project dismissed its facilitator early on in the process and did not seek a replacement, finding that the project team could undertake stakeholder engagement themselves.
- 20 Science Advisory Panel, 2011a
- 21 Written ministerial statement on conservation zones, 2011, Department of Environment, Food and Rural Affairs, 15 November
- 22 Aish A *et al*, 2010, 17
- 23 Aish A *et al*, 2012, 18
- 24 Joint Nature Conservation Committee and Natural England, 2010, 6
- 25 Guideline 16, Aish A *et al*, 2012, 18
- 26 Joint Nature Conservation Committee and Natural England, 2011
- 27 Joint Nature Conservation Committee and Natural England, 2011
- 28 Science Advisory Panel, 2011b, 7
- 29 Science Advisory Panel, 2011b, 4

- 30 Any management measures which restrict fisheries in the exclusive economic zone and areas of the territorial sea where other European Union member states have historic fishing rights will also have to be debated and accepted or rejected at European Union level under the Common Fisheries Policy
- 31 Tom Hooper, Project Manager, Finding Sanctuary, *pers comms* ; Sue Wells, Project Manager, Balanced Seas, *pers comms*
- 32 Sue Wells, Project Manager, Balanced Seas, *pers comm*
- 33 Science Advisory Panel, 2011b, 27
- 34 Dale Rodnell, Assistant Chief Executive, National Federation of Fishermen's Organisations (member of the Marine Protected Area Fishing Coalition), quoted in 'NFFO urges rethink on MCZ reference areas', 2011, 9 March, www.fishnewseu.com (accessed 31 July 2012)
- 35 Science Advisory Panel, 2011a, 36
- 36 Conversely, the Balanced Seas project preferred working with electronic resources
- 37 Sue Wells, Project Manager Balanced Seas, *pers comm*

chapter thirteen



California Marine Life Protection Act

Introduction

The Marine Life Protection Act 1999 mandates a statewide system of marine protected areas within the 14,374 square kilometres of state waters which extend along California's 1,770 kilometre-long coastline.

To help implement the new legislation, the California Natural Resources Agency, California Department of Fish and Game, and Resources Legacy Fund Foundation formed a public-private partnership known as the Marine Life Protection Act Initiative. The Initiative brought stakeholders, scientists and policy-makers together in a public process, which culminated in the adoption of a network of marine protected areas by the California Fish and Game Commission, and the passing of implementing regulations.

The Initiative adopted a region-by-region approach along the entire Californian coast, enabling learning and adaptation from each successive planning process. At the time of writing, marine protected area networks had been adopted for three out of five regions, resulting in a significant increase in the total coverage of marine protected areas, including those which prohibit all extractive use.

The Initiative is an example of a successful stakeholder-driven planning process which resulted in a network of marine protected areas spanning a large geographic area, designed in accordance with strong science, and managed as part of a statewide system. This chapter describes and analyses the process by which the Marine Life Protection Act has been implemented to date in California.

Context

Economic context

The California coastline features densely populated urban areas such, as Los Angeles in southern California, as well as small rural communities, such as Bodega Bay in northern California. These coastal economies all depend on the ocean to varying degrees. Over three-quarters of California's population live in coastal areas.

Ocean-related industries such as commercial fishing, aquaculture, offshore oil and gas, port activities, and coastal tourism and recreation contribute significantly to California's economy. In 2000, these sectors contributed around US\$42.9 billion

to the economy through direct and indirect impacts. They also created almost 700,000 jobs.¹

Tourism and recreation is by far the largest economic sector in California's coastal economy, generating over US\$22 billion in 2000. This was closely followed in economic importance by transportation (\$18 billion). Fishing was a much smaller sector, generating just under \$300 million during that year.²

History of marine protection

The use of spatial marine protection measures has a long history in California, dating back to the early 1900s. Over time, different designations were established on a case-by-case basis, through legislative and administrative actions, and by public referendum. By the 1990s, there were 18 classifications and sub-classifications that could be applied to marine protected areas in California's waters. These included areas of special biological significance, coastal sanctuaries, ecological reserves, marine refuges, marine reserves, Marine Resource Protection Act ecological reserves, natural reserves and parks.

In 1997, there were 104 marine protected areas along the Californian coast, of which only nine were fully protected from extractive use.³ This myriad of marine protected areas had been created without any coherent overall plan to ensure effectiveness or that the most representative or unique marine areas were included.⁴

In the years preceding the enactment of the Marine Life Protection Act, the State of California initiated a critical review of its marine management practices. The California Natural Resources Agency's landmark report *California's Ocean Resources: An Agenda for the Future*⁵ found that California's array of marine managed area classifications was complex and often confusing. Specifically, the report identified uncertainty about the requirements of, restrictions within, and boundaries of the array of sites and classifications.

The agency subsequently convened an 'interagency marine managed areas workgroup'. This produced a report recommending that the array of ocean and coastal managed area classifications, and the designation and management of individual sites, needed to be redesigned as an organised system.⁶ The Marine Managed Areas Improvement Act 2000 (discussed in the following section) drew on the report and clarified and simplified the variety of existing designations for ocean and coastal managed areas.

There were also concerns, prior to the implementation of the Marine Life Protection Act, about the effectiveness of California's marine protected areas. A 2002 study, focused on central California, found that the exceedingly small size of existing marine reserves prevented them from achieving many of the goals and benefits attributed to marine reserves in the scientific literature. The study also found that existing reserves in central California protected a variety of shallow water habitats and species, but did not provide reserve benefits for animals living in deeper water.⁷

These issues were later echoed in reports prepared by the Marine Life Protection Act Initiative. Marine protected areas along the north-central coast were approximately one square mile or less in area, and did not extend to deeper offshore waters, with the exceptions of areas at Manchester and Arena Rock, Salt Point, and the Farallon Islands.⁸ Similarly, with the exception of the Channel Islands national marine sanctuary, all but five of the existing south coast marine protected areas were located in shallow waters, and were under one square mile in area.⁹

In 1999, the State government recognised the need to redesign California's existing marine protected area system. A number of reasons for this were identified in the Marine Life Protection Act¹⁰ which was subsequently promulgated, including:

- California's marine protected areas were established on a piecemeal basis rather than according to a coherent plan and sound scientific guidelines
- The lack of clearly defined purposes, effective management measures and enforcement of existing marine protected areas
- The asset value that marine biodiversity provides to the state and nation including contributing to public health and well-being, ecological health and ocean-dependent industry
- The threats posed by coastal development, water pollution, and other human activities to the health of marine habitat and the biodiversity found in California's ocean waters
- The role of marine protected areas in sustaining marine habitats and fisheries
- The role of marine protected areas in expanding knowledge, by providing baseline information, and improving our understanding of ecosystems where minimal disturbance occurs

- The role of marine protected areas in protecting habitat and ecosystems, conserving biological diversity, providing a sanctuary for fish and other sea life, enhancing recreational and educational opportunities, providing a reference point against which scientists can measure changes elsewhere in the marine environment, and potentially in helping rebuild depleted fisheries
- The small portion of the marine area currently in protection (only 36 of the 570,000 square kilometres of combined state and federal ocean water off California, or six thousandths of one percent, were set aside as genuine no-take areas)

Legislative Framework

Marine Life Protection Act

The Marine Life Protection Act was passed by the California State legislature in 1999. In accordance with Californian state jurisdiction, the Act only applies to the waters between the mean high tide line and three nautical miles offshore.

The Act contains the legislative finding that *“it is necessary to modify the existing collection of marine protected areas to ensure that they are designed and managed according to clear, conservation-based goals and guidelines that take full advantage of the multiple benefits that can be derived from the establishment of marine life reserves”*.¹¹

The Act establishes a process for the redesign of the marine protected area system in California, which culminates in the California Fish and Game Commission adopting and implementing a ‘marine life protection program’ and ‘plan’ with associated regulations.

The marine life protection program is to be prepared in accordance with a set of goals which are specified in the Act. These include the conservation of biological diversity and the health of marine ecosystems; recovery of wildlife populations; improvements to recreational and educational opportunities consistent with biodiversity conservation; protection of representative and unique habitats for their intrinsic value; ensuring that marine protected areas have defined objectives; effective management and enforcement; a design based on sound science; and

ensuring that marine protected areas are managed, to the extent possible, as a network.¹²

The Act allows the program to include areas with various levels of protection. These include ‘marine life reserves’, which are areas where all extractive activities are prohibited, as well as areas where specified forms of commercial and/or recreational fishing is allowed. However, the Act does require the program to include an ‘improved marine life reserve’ component.¹³

The ‘master plan’ is designed to guide the adoption and implementation of the program. It is prepared by the California Department of Fish and Game. It contains recommendations on the siting of new marine protected areas, modifications of existing marine protected areas, and management, enforcement, monitoring, and funding of California’s marine protected areas.¹⁴

The master plan is required to use the best readily available scientific information and to organise the location-specific contents, where feasible, by biogeographic region.¹⁵ Biogeographical regions are defined in the Act and refer to three specific regions collectively covering the entire Californian coast. However, there is provision for an alternative set of boundaries to be established during the process.

The Department of Fish and Game must obtain the advice of specific parties in preparing the master plan. These include fisheries representatives, marine conservationists, marine scientists and other interested persons. The Department is also required to confer with government bodies including the California Fish and Game Commission, the Pacific Fishery Management Council, the National Marine Fisheries Service, the United States Navy, the United States Geological Survey’s national biological survey, staff from national marine sanctuaries off California, Sea Grant researchers, marine advisers and national parks personnel.

The Department may also engage other experts to contribute to the master plan (including scientists and GIS experts), and commercial and recreational fishermen, divers and other individuals knowledgeable about the state’s underwater ecosystems, the history of fishing effort or marine protected area management, or other relevant subjects.

A ‘master plan team’ is to be convened by the Department to advise and assist in the preparation of the plan. The team members must have expertise in marine life protection and be knowledgeable about the use of protected areas as a marine ecosystem management tool. The members must also be familiar with underwater ecosystems found in California waters, with the biology and habitat

requirements of major species groups in the state's marine waters, and with water quality and related issues.

The master plan team must be composed of:

- Staff from the Department of Fish and Game, the Department of Parks and Recreation and the State Water Resources Control Board
- Five to seven members who are scientists, one of whom may have expertise in the economics and culture of California coastal communities
- One member, appointed from a list prepared by Sea Grant marine advisers, who has direct expertise in ocean habitat and sea life in California marine waters

The Department of Fish and Game and master plan team are required take into account relevant information from local communities and to solicit comments and advice from interested parties on a range of issues including: practical information on the marine environment and the relevant history of fishing and other resources use; areas where fishing is currently prohibited; water pollution in the state's coastal waters; socioeconomic and environmental impacts of various alternatives; design of monitoring and evaluation activities; and methods to encourage public participation in the stewardship of the state's marine protected areas.

The legislation requires workshops with interested parties to be held in each biogeographic region, and to the extent practicable, near major working harbors. The workshops are to review the alternatives for marine protected area networks and to provide advice on a preferred siting alternative.

The legislation provides that the workshop and consultation process will culminate in the Department of Fish and Game and master plan team developing a preferred siting alternative which forms part of the plan. The Act establishes the following guidelines, which the preferred siting alternative must be designed in accordance with:¹⁶

- (1) Each marine protected area shall have identified goals and objectives. Individual marine protected areas may serve varied primary purposes while collectively achieving the overall goals and guidelines of this chapter
- (2) Marine life reserves in each bioregion shall encompass a representative variety of marine habitat types and communities, across a range of depths and environmental conditions

- (3) Similar types of marine habitats and communities shall be replicated, to the extent possible, in more than one marine life reserve in each biogeographical region
- (4) Marine life reserves shall be designed, to the extent practicable, to ensure that activities that upset the natural ecological functions of the area are avoided
- (5) The marine protected area network and individual marine protected areas shall be of adequate size, number, type of protection, and location to ensure that each marine protected area meets its objectives and that the network as a whole meets the goals and guidelines of this chapter

The preferred siting alternative must also be consistent with the broader goals of the marine life protection program. Once the draft master plan has been completed, the legislation makes provision for a public review process. The Department of Fish and Game then submits a draft master plan and program to the Fish and Game Commission. The Commission is required to hold at least two public hearings on the documents prior to their adoption.¹⁷

Following adoption of the master plan and program the Commission must submit the documents to the Joint Committee on Fisheries and Aquaculture for review. The Commission is required to consider all recommendations of the Joint Committee and may amend the program to incorporate its recommendations. It is then required to implement regulations based on the master plan and implement the program to the extent funds are available.¹⁸

Marine Managed Areas Improvement Act

The Marine Life Protection Act provides for the establishment of marine protected areas with various levels of protection. However, it does not define the various different types of marine protected area. In 2000, the State of California adopted the Marine Managed Areas Improvement Act.¹⁹ This law sought to clarify and simplify the variety of existing designations for marine managed areas, including marine protected areas.

Prior to the legislation there were 18 classifications and sub-classifications of marine areas. The new legislation provided that there would be only six classifications:

- State marine reserve – where all take and consumptive use (commercial and recreational, living or geologic) is prohibited. Scientific research and non-consumptive uses are allowed
- State marine park – where commercial take is prohibited but selected recreational harvest may be allowed. Scientific research and non-consumptive uses are allowed
- State marine conservation area – where selected recreational and commercial harvest may be allowed. Scientific research and non-consumptive uses are allowed
- State marine cultural preservation area – where it is unlawful to damage, take or possess any cultural marine resource. Complete integrity of the cultural resources is sought and no structure or improvements that conflict with that integrity is permitted. No other use is restricted
- State marine recreational management area – where it is unlawful to perform any activity that, as determined by the designating entity or managing agency, would compromise the recreational values for which the area may be designated. Recreational opportunities may be protected, enhanced or restricted, while preserving basic resource values of the area. No other use is restricted
- State water quality protection area – where point source waste and thermal discharges are prohibited or limited by special conditions. Non-point source pollution is controlled to the extent practicable. No other use is restricted

The legislation provided that a marine protected area “*is a named, discrete geographic marine or estuarine area seaward of the mean high tide line or the mouth of a coastal river, including any area of intertidal or subtidal terrain, together with its overlying water and associated flora and fauna that has been designated by law or administrative action to protect or conserve marine life and habitat. MPAs are primarily intended to protect or conserve marine life and habitat, and are therefore a subset of marine managed areas (MMAs).*” The legislation further provided that marine protected areas included the first three of the classifications - State marine reserves, State marine parks and State marine conservation areas.

Early attempts at implementing the Marine Life Protection Act

Between 1999 and 2004 there were two efforts at implementing the Marine Life Protection Act. In the first attempt, the Department of Fish and Game and the master plan team developed a set of initial proposals for a statewide network of marine protected areas. They were developed without significant stakeholder input. The draft proposals were intended by the master plan team as a concept only, in order to generate input from fishermen and other stakeholders with local knowledge. However, stakeholders, particularly from the fishing community, perceived this as a case of scientists crafting marine protected areas without stakeholder input. The strong opposition to the proposals caused the first attempt to come to a halt.

The second attempt, launched in 2002, was more inclusive of stakeholders, but suffered from a lack of staff and funding. This effort was designed with seven regional working groups and relied on assistance from a highly regarded private sector mediation organisation. It also focused on the entire California coast. Scientists from the same master plan team were assigned to support each stakeholder group. The Department secured over US\$1 million in funding from a variety of sources to support the effort. After each stakeholder group had held three initial meetings, however, this attempt gradually lost momentum and effectively came to a halt by December 2003. The immediate causes were a lack of funding to pay for facilitation and loss of departmental staff positions.²⁰

After these unsuccessful attempts, state legislators and the Department of Fish and Game realised that such a complex and controversial process required significant resources and time to implement and evaluate successfully. In 2004, the California Resources Agency and the Department partnered with the Resources Legacy Fund Foundation and others in a new initiative to achieve the Marine Life Protection Act goals. This public-private partnership is known as the California Marine Life Protection Act Initiative.

The California Marine Life Protection Act Initiative

Overview

The California Marine Life Protection Act Initiative was launched in August 2004 pursuant to a memorandum of understanding between the California Resources Agency, the Department of Fish and Game, and the Resources Legacy Foundation.

The memorandum of understanding noted that, based on prior and ongoing efforts to prepare a draft master plan, the Department had decided that it would be most effective to prepare it in phases. Specifically, the Department intended, as part of the first phase, to prepare a ‘master plan framework’ for the purpose of providing a framework for developing succeeding phases of the master plan. This framework was to include a timeline to design and implement marine protected areas in phases by region, beginning with the central coast region.

The memorandum of understanding also recorded that the California Resources Agency and Resources Legacy Foundation would assist the Department in preparing the master plan framework and in developing alternative networks of marine protected areas in the central coast region. Parties also agreed to a detailed organisational structure set out in the document. This included the establishment of a ‘Blue Ribbon Task Force’, a ‘master plan science advisory team’, regional stakeholder groups and a steering committee.

The Initiative’s work in the central coast region culminated in the Fish and Game Commission approving a set of central coast marine protected areas. Regulations to implement the marine protected areas came into effect in September 2007.

In another memorandum of understanding, effective January 2007, the same parties agreed to fund and implement the second phase of the Marine Life Protection Act process. One of the objectives that parties sought to achieve under the second memorandum of understanding was to “*refine the MLPA Initiative process to benefit from and be responsive to the lessons learned in the first phase of the MLPA process for the Central Coast*”. Parties also agreed to continue to incorporate the Blue Ribbon Task Force, the science advisory team and the regional stakeholder group in the second phase of the Marine Life Protection Act Initiative.

Soon after the 2007 memorandum of understanding was signed, the Secretary for Resources announced that the next region for Marine Life Protection Act implementation was to be the north central coast region. In 2008 the memorandum

was extended to set out the parties' agreement to progress implementation for the south coast, north coast and San Francisco Bay regions.

At the time of writing, marine protected area networks have been adopted for three of the five study regions (north central coast, central coast and south coast) and the marine protected area network proposed for the north coast was pending adoption by the Commission. Options for the final region, San Francisco Bay, are currently under development (see Figure 13.3 for summary statistics for each study region).

Organisational structure, roles and responsibilities

The following organisations were involved in the Marine Life Protection Act Initiative (see Figure 13.1):

- California Fish and Game Commission – which is the ultimate decision-making authority for implementation of the Marine Life Protection Act. The Commission makes all final decisions on the master plan and regional marine protected area proposals
- California Natural Resources Agency – which provides state policy leadership and direction, including coordination with state agencies. The secretary of the agency selects the chair and other members of the Blue Ribbon Task Force, and convenes and charges the members with meeting their objectives. The agency provides policy direction for coordinating funding and staffing, and seeks current and future funding for agency and department personnel committed to the Initiative and for completing future phases of the Marine Life Protection Act
- California Department of Fish and Game – which serves as the lead agency for the design and implementation of the Marine Life Protection Act master plan and statewide network of marine protected areas. The Department appoints the members of the master plan science advisory team. Through the Initiative's steering committee, the Department assists with the development of the draft master plan framework and proposals for marine protected areas. The Department also provides biological, enforcement and other relevant information, participates in meetings as appropriate, and reviews working documents
- Blue Ribbon Task Force – which was to be composed of members 'known for their integrity, intellect and experience in public policy and concern

for common good’ and who are selected by the secretary of the Natural Resources Agency. The key role of the Blue Ribbon Task Force is to guide the development of alternative marine protected area proposals presented to it by the regional stakeholders group, and to recommend the Commission a range of alternative proposals, and a preferred marine protected area alternative proposal for each region. In this way, the task force provides a ‘buffer’ between the stakeholders and the political decision-makers. Secretary Chrisman described the task force appointed for the central coast region as follows:

“This group represents a wide range of perspectives and is highly regarded for having good judgment. Their track record of results and breadth of experience in statewide and national policymaking is going to play a huge role in the success of this effort. This group has been assembled to look objectively at the history, the science related to marine protected areas, and the process to ensure it remains open, will be accessible and is considerate of all viewpoints.”

- Resources Legacy Fund Foundation – which is an independent not-for-profit organisation. The Foundation focused on obtaining, coordinating and administering philanthropic investments to supplement public funding for the Marine Life Protection Act Initiative. Staff of the Foundation had ongoing responsibility for managing consultant contracts, including initial contracting and reimbursements
- Master plan science advisory team – which is convened for each regional process. The team is composed of the range of members required by the Marine Life Protection Act as described above. It is charged with reviewing and commenting on scientific papers relevant to the implementation of the Act, developing scientific guidelines for marine protected area design, reviewing alternative marine protected area proposals, reviewing draft master plan documents, addressing scientific issues presented by those documents, and addressing scientific questions raised by the Blue Ribbon Task Force or stakeholders. A science advisory sub-team serves each study region by working directly with stakeholders and the Department to help develop scientifically sound alternatives
- Regional stakeholder group – which provides local knowledge for informing the Marine Life Protection Act planning process, evaluates existing marine

protected areas, develops alternative marine protected area proposals, and conducts outreach to constituent groups. The chair of the Blue Ribbon Task Force and the director of the Department of Fish and Game solicit nominations and select individuals to serve on the group

Figure 13.1: Overview of the Marine Life Protection Act organisational structure



Overview of process

On a general level, the marine protected area design process established by the Marine Life Protection Initiative is composed of four tasks: regional marine protected area planning, assembling alternative marine protected area proposals, evaluating alternative marine protected area proposals and finally consideration and action on marine protected area proposals, by the Commission.²¹ The range of activities to be undertaken for each of these steps is set out in Figure 13.2.

Figure 13.2: Overview of the Marine Life Protection Act Initiative process

Task 1: Regional marine protected area planning	Department of Fish and Game staff assemble regional information (biological, oceanographic, socioeconomic, governance and tribal uses) which forms the regional profile. They then identify inadequacies in existing marine protected areas in meeting the goals and objectives of the Marine Life Protection Act. The Departmental director and chair of the Blue Ribbon Task Force convene a regional stakeholder group. The Director also convenes the science advisory team and forms a sub-team for the region The regional stakeholder group, science advisory sub-team and departmental staff identify issues requiring additional advice. The science advisory sub-team and staff then prepare draft advice on these issues. The regional stakeholder group and science advisory sub-team then review information in the regional profile and consider comments from stakeholders. The Blue Ribbon Task Force, Fish and Game Commission and science advisory team review additional advice for designing marine protected areas in the study region. The task force incorporates this additional advice into planning and guidance documents. The science advisory sub-team develops a list of key or critical species and documents their regional distribution. The Department prepares a statement of feasibility criteria and provides it to the Blue Ribbon Task Force, regional stakeholder group, and science advisory team. The regional stakeholder group and the science advisory sub-team determine: • The distribution of representative and unique habitats in the region • Wildlife populations, habitats and various human uses that may negatively impact the populations and habitats in the region • Activities that may affect populations and habitats • Which key or critical species (identified above) are most likely to benefit from marine protected areas in the region They identify key locations in the region where marine protected areas may help achieve the Marine Life Protection Act goals and contribute to an overall network.
Task 2: Assemble draft regional alternative marine protected area proposals	The regional stakeholder group and science advisory sub-team prepare a range of alternative proposals for marine protected areas within the region. Each proposal is intended to achieve the goals and objectives of the Marine Life Protection Act. The regional stakeholder group reviews each revised or potential new marine protected area and identifies initial objectives for each marine protected area to help meet the goals and objectives of the Marine Life Protection Act. The alternative proposals are presented to the Blue Ribbon Task Force and science advisory group for review and evaluation.

Task 3: Evaluate alternative marine protected area proposals	The science advisory team and sub-team prepare preliminary analyses of the habitats within marine protected areas, marine protected area sizes, and marine protected area spacing for each alternative proposal. These analyses provide a relative comparison of how well each proposal meets specific goals of the Marine Life Protection Act. They also, with departmental staff, prepare a socioeconomic analysis of potential impacts of each alternative proposal, including the maximum potential impact of each proposal to existing fishing. The regional stakeholder group reviews the analyses and revises proposals, as necessary, to more fully meet the goals, objectives and design considerations. The regional stakeholder group and science advisory sub-team identify potential monitoring and evaluation indicators to evaluate progress toward achieving goals and objectives. The Blue Ribbon Task Force forwards alternative proposals for marine protected areas, initial evaluations, and the general management plan, together with its own evaluation and a preferred alternative, to the Fish and Game Commission for its consideration and actions. The Department of Fish and Game provides specific comments on the Blue Ribbon Task Force's preferred alternative. It also provides the Commission with a feasibility analysis (for example, ability to enforce, monitor, manage and fund the proposal). The Department, with assistance from the science advisory team, designs a general management plan for marine protected areas in the region, including specific plans for monitoring, enforcement, costs and financing, and periodic review of effectiveness. This plan may be forwarded to the Commission along with the specific area proposals or separately during the decision-making process.
Task 4: Commission consideration and action	The Commission reviews the alternative marine protected area proposals, hears public testimony at multiple hearings, and determines whether to request that the Department begin the formal regulatory process. The science advisory team evaluates any alternative marine protected area proposals or changes considered by the Blue Ribbon Task Force and the Commission up until final adoption of a proposal by the Commission. The Commission hears public testimony at multiple hearings on the alternative marine protected area proposals and on any legal analysis. It then adopts its preferred alternative marine protected area proposal.

Spatial decision support system

The Marine Life Protection Act Initiative invested in the development of a spatial decision support system to deliver key spatial data and analytical results to stakeholders, scientists, and decision-makers in a usable format. The system included a centralised and internet accessible geospatial database that contained

over 400 spatial datasets for California's marine environment. These were compiled from state and federal agencies, scientists and non-governmental organisations. The system also included web-based map services and desktop GIS tools and analytical products.

In the south coast and north coast regions the Initiative used a decision support system known as 'MarineMap'. This replaced an earlier and less effective decision support tool named 'Doris' that was used in the north central coast.

To assist the design of the Californian marine protected area network, the Marine Life Protection Act Initiative contracted the MarineMap consortium (a partnership between the Marine Science Institute at the University of California Santa Barbara, Ecotrust, and the Nature Conservancy) to develop MarineMap. MarineMap was used to display the following key data layers during the planning process for the south and north coasts:

- Marine protected area designs – draft marine protected area proposal boundaries
- Biological – species distributions
- Existing management areas – such as essential fish habitat and fishery management boundaries
- Cultural – cities, coastal access points and shipwrecks
- Physical – bathymetry, hydrography, canyons and pinnacles
- Habitat – such as predicted substrate, eelgrass and kelp

Stakeholders were able to use MarineMap to draw multiple scenarios for prospective marine protected areas. They were able to generate analytical reports containing data on size, habitat representation and economic impact on fisheries, to determine the extent to which the proposals meet the Initiative's science advisory team guidelines.

MarineMap was used by stakeholders with a wide variety of backgrounds, most who had little or no experience with traditional GIS software. For this reason, a great deal of attention was paid to the usability and responsiveness of the application through, for example, developing a purpose-built user interface and refining the application as the planning process evolved and feedback was gathered from stakeholders.²²

Funding

The legislature recognised, when enacting the Marine Life Protection Act, that funding would be critical to its implementation. For example, when signing the Bill, Governor Davis stated that he was encouraging the proponents and the Department of Fish and Game “*to seek assistance from private resources to help implement the provisions of the bill.*”²³ The Marine Life Protection Act itself only requires that the Commission implement the program “*to the extent funds are available*”.²⁴ It also states that “*recommendations for funding sources to ensure all marine protected area management activities are carried out and the Marine Life Protection Program is implemented*” must be included in the master plan.²⁵

The funding history of the Initiative began with a 2004 public-private partnership between the Resources Agency, the Department of Fish and Game, and the Resources Legacy Fund Foundation. The role of the Foundation was to obtain, coordinate and administer philanthropic investments to supplement public funding for the Initiative.

In 2006 the Initiative estimated the total cost of implementing the Marine Life Protection Act for the fiscal year 2006 to 2007 would range from US\$6.1 to US\$20.9 million, with an average of US\$12.9 million. Costs were expected to increase over subsequent years as new study regions were designated and became operational. Costs were expected to stabilise when all regions were designated.²⁶

In addition to state government funding, a number of private entities provided significant contributions towards the implementation of the Marine Life Protection Act. These included the Gordon and Betty Moore Foundation (which provided approximately US\$13 million from 2004 to 2008), the David and Lucile Packard Foundation, and the Marisla Foundation.

Monitoring and review

The Marine Life Protection Act requires the ‘marine life protection program’ to include provisions for monitoring, research and evaluation to facilitate adaptive management of marine protected areas and to ensure that the system meets the goals of the Act.²⁷ The master plan is required to contain recommendations for monitoring, research, and evaluation in selected areas of the preferred alternative, to assist in adaptive management of the marine protected area network.²⁸

The California Ocean Science Trust leads the monitoring of marine protected areas implemented under the Marine Life Protection Act through its ‘monitoring enterprise programme’. The Trust is a quasi-state institution, established pursuant

to the California Ocean Resources Stewardship Act 2000. Its purposes include promoting more effective coordination of Californian ocean resource science useful to management agencies. It describes itself as a policy-neutral non-profit organisation. It does not advocate for or against the implementation of marine protected areas, but rather ensures that policymakers, resource managers and the public have impartial, timely and useful information to support marine protected area management decisions.

The Trust develops a regional monitoring plan, sequentially, as planning is completed for each region. The master plan states that, while the regional monitoring programmes are developed separately, integrating them into a coherent state-wide program is essential to ensure that the resulting data can be used to inform state-wide policies.²⁹

Marine protected area monitoring is implemented in two steps: first through undertaking an initial baseline program, and secondly through long-term monitoring. A baseline program is implemented for newly designated regional marine protected area networks and continues for two or three years following implementation. Long-term monitoring builds on the broad foundation established through the baseline monitoring, to assess the marine protected area network's performance in meeting the goals of the legislation, and to inform future adaptive management decisions.

Completed monitoring plans are submitted to the Fish and Game Commission for adoption. For example, the north central coast monitoring plan was adopted by the Commission in April 2010, and approved for inclusion in the Marine Life Protection Act master plan, thereby formally establishing it as part of the policy guiding implementation.

In the monitoring plan for the north central coast, nine ecosystem features were selected to provide the basis for assessing the condition of the region's ecosystems, including their human inhabitants. For each ecosystem feature, key attributes were identified to be collectively used to assess the condition of the ecosystem feature.

In addition, simpler sampling protocols will be used to monitor a set of 'vital signs'. For example, one of the ecosystem features identified in the north central coast was estuarine and wetland ecosystems. Vital signs for this ecosystem feature include the areal extent of eelgrass and harbor seal abundance. A key attribute that is to be monitored is the trophic structure, which requires monitoring of a range of indicator species, including the abundance of various shrimp and clam species.

Incorporating traditional use

The Marine Life Protection Act did not provide any special recognition for tribal groups located along the coast. Nor did the Marine Managed Areas Improvement Act provide any classification of marine managed areas which specifically recognised or provided for tribal use. In the north coast region there were 27 federally recognised tribes and numerous tribal communities. For many of these, the ability to exploit marine resources, was an integral part of their culture and traditions.³⁰

Little information was available to the north coast region planning process on tribal resource use. Some tribes wrote qualitative accounts of their activities which were included in the regional profile. Seven tribal groups were represented in the regional stakeholder group and through this forum, were able to communicate their interests to some extent. Other tribal groups directly met with Marine Life Protection Act Initiative staff. A group of students from the University of California Santa Barbara undertook a research project which documented species which tribal groups wished to continue targeting and the type of fishing gear used.³¹

The only legal mechanism available to provide for tribal use within marine protected areas was to include this activity within the broader category of recreational use. This meant that if a marine protected area was to allow tribal use, it would also have to allow recreational use, which significantly increased the potential impact of extractive uses on marine resources.

Despite this difficulty, a review of the effectiveness of integrating tribal resource use into the marine protected area proposals, concluded that tribal groups had a strong influence on the design of the final north coast marine protected area proposals submitted to the Fish and Game Commission and that their concerns and issues dominated the north coast process. However, the lack of a specific legal mechanism to address tribal concerns, resulted in marine protected area proposals that accommodated all recreational users to the greatest extent possible, and thereby potentially undermining conservation goals.³²

Analysis

Was it successful?

This analysis will only address the success of the Marine Life Protection Act Initiative at a general level, as each regional process (while similarly structured

in terms of the public-private partnership model, funding structure and decision making process) was different due to varying political, social and natural environments.

At the time of writing, marine protected area networks had been adopted for three of the five study regions (north central coast, central coast and south coast). The marine protected area network proposed for the north coast is pending adoption by the Commission and the values provided in the following section are based on the north coast preferred alternative. This may be subject to change depending on the final north coast marine protected areas that are adopted. Options for the final region, San Francisco Bay, have been developed for consideration at a future date.

At the time of writing, the statewide coastal network includes 123 marine protected areas. They cover approximately 2,196 square kilometres (16 per cent) of state waters (excluding San Francisco Bay). There are 48 state marine reserves which represent approximately nine per cent of state waters (excluding San Francisco Bay).

Figure 13.3 displays the percentage of the total area of each region occupied by marine protected areas before and after implementation of the Marine Life Protection Act. It also provides the percentage of the total area of each region occupied by state marine reserves. Figure 13.4 displays statistics of areal coverage of each marine protected area type for each study region after implementation of the Marine Life Protection Act. Post implementation of the Act, all regions experienced a substantial increase in the areal coverage of both marine protected areas and state marine reserves which are full protected. Therefore, at a substantive level, the Initiative has undoubtedly achieved the requirement of the legislation to implement an ‘improved marine life reserve component’.

One commentator has suggested that the results of the Marine Life Protection Act Initiative failed to live up to the promise of the legislation. The emphasis, during the process, on obtaining broad-based agreement among a high number of stakeholders meant that “*powerful fishing interests carried the day*” and as a result there was an emphasis on “*fishery-based science*” rather than on “*multi-species ecosystem-based planning*”.³³

The Initiative should also be assessed on a procedural level. The memoranda of understanding emphasise the importance of transparency and openness to the public in decision-making. The legislation also requires the Department of Fish and Game and master plan team, to take into account relevant information from local communities, and to solicit comments and advice for the master plan

from interested parties on a number of issues. Therefore the extent to which the Initiative process was one that the public and stakeholders felt was transparent and satisfactory is a critical measure of success.

Figure 13.3: Summary of the results of implementation of the California Marine Life Protection Act

	Imple- mented	Total Area of Region (km ²)	% marine protected areas of total area		% State marine reserves of total area	
			Pre imple- mentation	Post imple- mentation	Pre imple- mentation	Post imple- mentation
Central Coast	2007	2,963	3.76	17.86	0.65	7.35
North Central Coast	2010	1,976	3.52	19.96	0.04	11.03
South Coast	2011	6,089	7.71	15.08	6.7	10.27
North Coast (values pending final adoption by Commission)	Expected 2012	2,660	0.3	13.33	0.2	4.98

A number of measures contributed to a transparent public process. The Blue Ribbon Task Force and science advisory team convened in publicly notified, open meetings. Meetings were available as a live web cast and past meetings were available in video and audio formats. The Department of Fish and Game also hosts a website which displays information for each study region, including key documents, process updates and press releases.

The involvement of the community and interested parties is undoubtedly a strength of the Initiative process. It was the stakeholders themselves, through the regional stakeholder groups, that were tasked with developing regional goals and objectives, evaluating existing state marine protected areas, and developing multiple marine protected area network proposals for the region to support decision-making by the Blue Ribbon Task Force on a preferred alternative.

The extent to which stakeholders were satisfied with the process varied by region. In the central coast, stakeholders were surprised and frustrated by the way the Blue Ribbon Task Force pursued an independent staff-derived proposal, and directed changes to stakeholder proposals. Members of the central coast regional stakeholder group expected the Task Force to select its preferred alternative from

amongst the three options prepared by the group, rather than either putting forward its own option or tinkering with the group's proposals.

There were also regional differences in the level of convergence reached by stakeholders in relation to the alternative proposals put forward. In the central coast there was limited success in reaching a common solution. Members of the regional stakeholder group staked out more extreme package positions than either the Blue Ribbon Task Force or the Department of Fish and Game ended up recommending.

An evaluation of the stakeholder process in the central coast indicates that the Blue Ribbon Task Force had limited success in *“cajoling the Package 1 and 2 supporters to move toward the middle by telling Package 1 supporters to better meet the SAT [Science Advisory Team] guidelines and Package 2 and 3 supporters to reduce socio-economic impacts”*.³⁴

The process for the north central coast appears to have experienced a higher level of convergence than that for the central coast. The lessons learned report states proposals *“reflected a joint effort by consumptive, conservation, and other non-consumptive representatives to find a balance”* and that there was a relatively narrower band of disagreement within the regional stakeholder group about an acceptable solution.³⁵ This is evidenced by the letter sent by a group of 20 north central coast regional stakeholder group members representing different interests to the Fish and Game Commission, asking that it adopt the Blue Ribbon Task Force's preferred alternative without modification.

The role of scientific expertise in the implementation of the Marine Life Protection Act is explicit in the legislation. It appears that each regional process to date has had a strong science base for the adoption of marine protected areas. Extensive scientific input is evident in the regional profile documents prepared for each region at the start of the planning process. Regional profiles are intended to support the marine protected area planning process by providing background information. The documents are highly detailed and comprehensive. They include information on marine habitats and communities, important regional species, areas of biodiversity significance, effects of land use on coastal waters, population projections, commercial fisheries, aquaculture, recreational fisheries, cultural uses and non-consumptive uses, amongst other things.

Scientific analysis of alternative marine protected area proposals is also a critical part of the planning process for each region. For example, in the north coast the science advisory team evaluated proposals for habitat representation, habitat replication, marine protected area size, marine protected area spacing,

potential commercial and recreational fishery impacts, bioeconomic aspects (such as biomass and fishery yields), and other considerations such as effects on birds, mammals, water and sediment quality.

Adequate public and private funding

The Marine Life Protection Act provides no dedicated source of funding, and given the California State government's budget deficit, private funds were essential to the Initiative's success. Prior to the Initiative being developed, the Department of Fish and Game had tried twice to implement the Marine Life Protection Act using existing funding sources. Neither effort succeeded. The public-private partnership between the State of California and the Resources Legacy Fund Foundation enabled significant private funds to be obtained.

However this funding model is not without critics. A lawsuit was brought challenging the private funding aspect of the Initiative, in which it was claimed that the process opens the door for the Legislature to underfund programs, which in turn will motivate special interests to bid against one another in order to assume legislative and regulatory drafting power.³⁶

Region-by-region approach

The region-by-region approach enabled the central coast process to inform the north central coast process, and for that to inform the other regional efforts in succession. This approach presented opportunities for adaptation and learning. 'Lessons learned' reports have been completed for the first two regional efforts.

A number of changes were made in the north central coast region based on the central coast experience. For example three iterations of marine protected area proposals and alternatives (with science advisory team evaluation, departmental input and Blue Ribbon Task Force guidance after each round) occurred during the north central coast process whereas only two occurred during the central coast process. Each round was intended to improve the quality of proposals, reduce the number of proposals, and promote convergence among the remaining proposals.

There were also changes in the process for appointment of regional stakeholder group members. The process for appointments was more deliberate in the later north coast initiative than for the central coast. Appointments were based on specific criteria published in the call for nominations, and public workshops were held for meeting and observing potential regional stakeholder group members, as well as interviews of nominees by Initiative staff.

A legal mandate and political will

The passing of the Marine Life Protection Act in 1999 was critical in providing the legal mandate for planning a state-wide marine protected area network.

Role of MarineMap

MarineMap enabled the stakeholder group in the north and south coast regions to view data layers, create individual marine protected area concepts, receive basic feedback on how well these met guidelines for marine protected area design, and submit marine protected area arrays to staff as proposals. MarineMap was also useful in increasing public participation in marine protected area planning as it is a tool which can be used by those with little or no experience with traditional GIS software. In this way it was seen as levelling the playing field between stakeholders.

Role of the Blue Ribbon Task Force

There is no mention of a body such as the Blue Ribbon Task Force in the Marine Life Protection Act, but this particular body played an innovative and valuable role in the Initiative process. The Task Force's decision-making role eliminated the need for stakeholders to resolve all of their conflicts within a single marine protected area proposal, and in doing so, took some of the pressure off of the stakeholders and scientists involved.³⁷ The Task Force has been described as a 'buffer' for the Fish and Game Commission.³⁸ The need for the Task Force to be open and transparent in its decision-making, and to be trusted by the regional stakeholder group members to make its decision without bias or predetermination, was raised as a key factor for a successful stakeholder process.³⁹

Role of the science advisory team

The outcome of the planning process – though not dictated by science alone – was strongly guided and influenced by the science guidelines, best available data and input from scientists. The science advisory team's role as marine protected area reviewers was distinct from the role of the regional stakeholder group as developers of marine protected area proposals. The separation of these processes was important in the Initiative as it helped to ensure that scientists were perceived by the public as undertaking their role without making any value judgments. Identifying data gaps and uncertainty, and communicating this effectively to non-scientists, was also identified as an important role of the science advisory team.⁴⁰

Clear science guidelines for marine protected area design

The specific set of science guidelines for marine protected area design, recommended by the science advisory team, provided stakeholders with a set of unambiguous yardsticks by which respective marine protected area proposals were to be evaluated. The science guidelines focused on habitat representation, habitat replication, and size and spacing of marine protected areas to promote connectivity. They had a strong influence on regional marine protected area network design.

Effective public communication methods

The Initiative undertook a number of measures to make information accessible to the public. Meetings were available as a live web cast and past meetings were available in video and audio formats. The Department of Fish and Game also hosts a website which displays information for each study region, including key documents process updates and press releases.

Conclusion

The California Marine Life Protection Act committed California to the establishment of a network of marine protected areas. Several initiatives to implement the Act were attempted and success was achieved once greater levels of resourcing were committed. The Initiative has acted as a blueprint for other jurisdictions since and has succeeded in greatly increasing the level of protection for California's marine environment.

Figure 13.4: Summary statistics of marine protected area implementation for each region of the California coast

	Imple- mented	Total Area (km ²)	SMR		SMCA		SMP		SMRMA		Special closures	
			Count	Area (km ²)/ % of total area	Count	Area (km ²)/% of total area	Count	Area (km ²)/% of total area	Count	Area (km ²)/% of total area	Count	Area (km ²)/% of total area
Central coast	2007	2,963	13	217.9 7.35%	15	303.5 10.24%	0	0 0%	1	7.8 0.26%	0	0 0%
North central coast	2010	1,976	10	217.8 11.03%	12	175.0 8.86%	0	0 0%	3	1.5 0.08%	7	2.9 0.15%
South coast	2011	6,089	19	625.5 10.27%	30	292.6 4.80%	0	0 0%	0	0 0%	2	4.6 0.08%
North coast (provisional)	Expected 2012	2,660	6	132.5 4.98%	13	220.0 7.97%	0	0 0%	1	2.0 0.08%	7	0.5 0.02%

The table uses the following abbreviations:

State Marine Reserve (SMR): Prohibits all take and consumptive use (commercial and recreational, living or geologic). Scientific research and non-consumptive uses are allowed.

State Marine Park (SMP): Prohibits commercial take but may allow select recreational harvest to continue. Scientific research and non-consumptive uses are allowed.

State Marine Conservation Area (SMCA): May allow select recreational and commercial harvest to continue. Scientific research and non-consumptive uses are allowed.

State Marine Recreational Management Areas (SMRMA): Provides subtidal protection equivalent to a marine protected area, while legal waterfowl hunting, scientific research and non-consumptive uses are allowed.

Special closures: Are geographically specific areas and prohibit human entry. Special closures are generally smaller in size than marine protected areas and are designed to protect breeding seabird and marine mammal populations from human disturbance.

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chapter fourteen



Australian Commonwealth
marine reserves

Introduction

Australia has the third largest marine area of any country in the world. It contains extremely high levels of biodiversity, including the world's largest areas and highest species diversity of tropical and temperate seagrasses, the world's largest coral reef system, diverse mangrove species and high levels of endemism.

The main pressures facing Australia's marine area are fishing, introduced marine pests and diseases, tourism, recreation, pollution and sedimentation.

Australia has taken a particularly progressive approach towards the development of marine protected areas, and although the initiatives have not always been a complete success, it now has a relatively well-developed and robust network. In addition, the Commonwealth government currently has plans to develop the world's largest marine protected area in the Coral Sea, off the north-east coast of Australia.

Generally, the State and Territory governments have responsibility for the establishment and management of marine protected areas out to three nautical miles, whilst the Commonwealth government manages the area extending from the State or Territory limit to the edge of the Australian exclusive economic zone.

The key aspects of the Commonwealth government's approach to the creation and management of marine protected areas are set out in this chapter. This review has been based on a desk top study of relevant literature available from the internet including websites, reports and journal articles.

Context

Economic context

Australia's marine estate provides an estimated A\$44 billion per year in revenue. Recent analysis by the Commonwealth government identifies the following activities as the most important sources of marine revenue in Australia: offshore oil and gas exploration and production (A\$24.2 billion per year), marine tourism (A\$11.1 billion per year), boat building, repair and maintenance and infrastructure (A\$6.4 billion per year), and commercial fishing and aquaculture (A\$2.3 billion per year).¹

The Australian Centre for Policy Development has estimated that marine ecosystem services provide an additional A\$25 billion per year to the Australian economy. Included in this figure is the A\$1.85 billion spent by recreational fishers on fishing-related activities and equipment. Approximately 19.5 per cent of Australians

over the age of five fish at least once a year. Also included are a range of non-market benefits, such as the contribution of the ocean to Australian culture, sport and recreation, regulation of carbon dioxide, and control of pests and diseases.²

History of marine protection

There are currently 27 Commonwealth reserves in Australia (including the Great Barrier Reef Marine Park which is managed by the Great Barrier Reef Marine Park Authority). These include the south-east Commonwealth marine reserve network, which itself comprises 14 marine protected areas, covering 226,458 square kilometres. Some of the Commonwealth reserves (such as the Great Australian Bight Marine Park and the Ningaloo Marine Park) cover both State and Commonwealth waters, and so are managed through cooperative arrangements. The south-east Commonwealth reserves network was created subsequent to the completion of the south-east regional marine plan. Within the network, 42 per cent of the total area is designated 'sanctuary zone', and is managed for scientific research and passive uses. Thirty-six per cent of the network is designated 'special purpose zone' and is closed to commercial fishing. Twenty-one per cent of the network is multiple use, where low impact fishing methods and activities are permitted. The remaining one per cent comprises a 'benthic sanctuary zone' and a 'recreational use zone' where some fishing is permitted in accordance with regulations.

Legislative Framework

Environment Protection and Biodiversity Conservation Act

Under the Environment Protection and Biodiversity Conservation Act 1999, Commonwealth marine areas are 'matters of national environmental significance'. This means that proposed activities in the marine area which will, or are likely to, have a significant impact on the environment must be referred to the Environment Minister and must undergo an environmental assessment and approval process.

Australia's definition of a marine protected area is: *"An area of ... sea especially dedicated to the protection and maintenance of biological diversity and of natural and associated cultural resources, and managed through legal or other effective means"*.³ Outside State and Territory waters, marine protected areas take the form of 'Commonwealth reserves.' The Governor General is authorised to create Commonwealth reserves in the marine area by the Act. The legislation also

empowers the Governor General to create ‘conservation zones’ for the purpose of protecting biodiversity in an area whilst it is being assessed for inclusion in a Commonwealth reserve.

Under the Act, the Governor General can declare a Commonwealth reserve in an area of sea that is:

- In a ‘Commonwealth marine area’ – which essentially comprises the area from the edge of State and Territory coastal waters (normally three nautical miles from the coast) out to 200 nautical miles from the coast
- An area outside Australia, which Australia has international obligations to protect under an agreement with one or more other countries

A Commonwealth reserve includes the seabed under the declared sea area. The declaration can also include in the reserve the subsoil of the seabed (to a depth stated in the declaration).

Declaration process

The process leading to the declaration of a Commonwealth reserve under the Environment Protection and Biodiversity Conservation Act is shown in Figure 14.1:⁴

Figure 14.1: Process to declare an Australian Commonwealth reserve

1. The Director of National Parks publishes a notice inviting the public to comment on the proposal to declare a Commonwealth reserve over the area, allowing a minimum period of 60 days for comments. This notice includes a statement of the proposed name of the reserve, the proposed boundaries of the reserve and any zones within the reserve, the purpose for which the reserve is to be declared, the management category that the reserve (and any zones) will be assigned to, and the purposes for which it is intended to manage and use the reserve.
2. Any native title holders, registered native title claimants and native title representative bodies for the area are notified of the proposed declaration, and given an opportunity to comment, in accordance with the requirements of the Native Title Act 1993.
3. The Director of National Parks provides the Minister for Sustainability, Environment, Water, Population and Communities (the Minister) with a report on the Commonwealth reserve proposal. The report must include any comments received and the Director’s views on the comments.
4. If necessary a regulation impact statement examining any impacts that declaration of the proposed Commonwealth reserve would have on business is prepared.
5. The Minister considers the report from the Director of National Parks.

6. The Minister decides not to proceed and thus the declaration process ends here or the Minister is satisfied a reserve should be established and the Governor General is advised accordingly.
7. The Governor General makes a Proclamation declaring the area to be a Commonwealth reserve. The proclamation names the reserve; states the purposes for which it is declared; states the depth of any seabed included in the reserve; and assigns the reserve to a management category.
8. The Proclamation is registered on the Federal register of legislative instruments.

Management

The management measures implemented in Commonwealth reserves, and consequently the level of protection from human activity, vary depending on the particular feature identified for protection. As a result, some areas allow multiple uses and some are no-take zones. An individual Commonwealth reserve may contain a number of different areas permitting different levels of activity.

The Environment Protection and Biodiversity Conservation Act (and the Environment Protection and Biodiversity Conservation Regulations 2000) require that each reserve, and zone within it, should be allocated to a particular management category. The Commonwealth government has essentially adopted the IUCN's set of seven management categories (although the numbering system is different). The categories are not arranged in a hierarchy of levels of protection, but describe the purpose of protection for a particular area, and the types of activity that will be permitted in that area. Figure 14.2 sets out the categories and their functions.⁵

The Director of National Parks is the statutory authority directly responsible for managing marine protected areas as specified by the Environment Protection and Biodiversity Conservation Act. The Director carries out his/her responsibilities assisted by Parks Australia, which is a division of the Department of the Environment. The Marine and Biodiversity Division of the Department is responsible for the management of Commonwealth marine protected areas on behalf of the Director of National Parks.

Figure 14.2 Categories for Australian Commonwealth reserves compared to IUCN categories

Australian Category No.	Type of Area	IUCN	Explanation
1	Strict nature reserve	Ia	Managed primarily for scientific research or environmental monitoring
2	Wilderness area	Ib	Protected and managed to preserve its unmodified condition
3	National park	II	Protected and managed to preserve its natural condition
4	Natural monument	III	Protected and managed to preserve its natural or cultural features
5	Habitat/species management area	IV	Managed primarily, including (if necessary) through active intervention, to ensure the maintenance of habitats or to meet the requirements of specific species
6	Protected landscape/seascape	V	Managed to safeguard the integrity of the traditional interactions between people and nature
7	Managed resource protected area	VI	Managed to ensure long-term protection and maintenance of biological diversity with a sustainable flow of natural products and services to meet community needs

Under the Environment Protection and Biodiversity Conservation Act, all Commonwealth reserves must have a management plan. These are prepared by the Director of National Parks, with public input, and approved by the Minister for the Environment. Management plans can last up to ten years. The plans provide for the protection and conservation of the Commonwealth reserve. They must set out how it is to be managed, what activities will be allowed within the reserve, and how they must be carried on. Management must be consistent with the relevant management principles.⁶

The management plan of each Commonwealth reserve generally specifies if a permit is required in order to carry out a particular activity. The Environment Protection and Biodiversity Conservation Act (and division 12 of the Environment Protection and Biodiversity Conservation Regulations) prohibits some activities being carried out in a Commonwealth reserve unless they are expressly provided for by a management plan for the area. These include actions that affect native

species such as scientific research, commercial fishing, mining operations and other commercial activities.

Permits are issued by the Director of National Parks, apart from in respect of mining, where applications must be addressed to the Governor General. The Director of National Parks also has the discretion to vary, revoke or cancel a permit.

The Environment Protection and Biodiversity Conservation Act forms an overarching framework for the management of Australia's environment and so other parts of the legislation can also be relevant to activities in a Commonwealth marine protected area. Under the Act, actions that have a significant impact on a 'matter of national environmental significance' will be subject to the assessment and approval provisions set out in the Act.

'Matters of national environmental significance' include the environment in a Commonwealth marine area and some specified threatened or migratory marine species. Applications to undertake activities which have a significant effect on the marine area, or the listed species, may therefore be subject to an additional decision-making process.

The Act provides for a number of assessment approaches, and the Commonwealth Environment Minister decides on the most appropriate one, depending on the significance of the project and how much information is already available. Each option involves considering information assembled by the proponent and comments made by the public.⁷

Commonwealth marine protected areas program

Although the creation and management of marine protected areas within Australian waters is the responsibility of various government bodies, the Commonwealth government has put in place measures to coordinate them into a coherent marine protected area network. The 'Commonwealth marine protected areas program', initiated in 1991, is intended to develop a national representative system of marine protected areas across Commonwealth, State and Territory jurisdictions.

The primary goal of the programme is *"to establish and manage a comprehensive, adequate and representative system of marine protected areas to contribute to the long-term ecological viability of marine and estuarine systems, to maintain ecological processes and systems, and to protect Australia's biological diversity at all levels."*⁸

The following secondary goals are designed to be compatible with the primary goal:

- To promote the development of marine protected areas within the framework of integrated ecosystem management
- To provide a formal management framework for a broad spectrum of human activities, including recreation, tourism, shipping and the use or extraction of resources, the impacts of which are compatible with the primary goal
- To provide scientific reference sites
- To provide for the special needs of rare, threatened or depleted species and threatened ecological communities
- To provide for the conservation of special groups of organisms, e.g. species with complex habitat requirements or mobile or migratory species, or species vulnerable to disturbance which may depend on reservation for their conservation
- To protect areas of high conservation value including those containing high species diversity, natural refuges for flora and fauna and centres of endemism
- To provide for the recreational, aesthetic and cultural needs of indigenous and non-indigenous people.

The 'Marine Protected Areas Working Group' is a national committee tasked with coordinating the development of policy and planning related to the establishment of the national representative system of marine protected areas. The working group is a subsidiary body of the Natural Resource Management Ministerial Council, which was set up to promote integrated management of natural resources in ways which would not otherwise be possible because of the obstacles caused by the constitutional division of powers in Australia.⁹ Members of the working group are from Commonwealth, State and Territory agencies responsible for marine conservation and fisheries management.

Since 1992, the working group and its predecessors have been developing a planning and policy framework to support the development of the national

representative system of marine protected areas. In addition, the group provides technical assistance to individual jurisdictions. Although the working group provides assistance with policy coordination, it remains the responsibility of each jurisdiction to identify and establish marine protected areas within waters under its control. The working group has developed a national framework for the marine protected area system.

Individual marine protected areas which are included in the national system must have the following characteristics:

- Has been established especially for the conservation of biodiversity (consistent with the primary goal of the national representative system of marine protected areas)
- Is able to be classified into one or more of the six IUCN protected area management categories reflecting the values and objectives of the marine protected area
- Must have secure status which can only be revoked by a Parliamentary process
- Contributes to the representativeness, comprehensiveness or adequacy of the national system

The working group is guided by a number of principles for identifying and developing the network. Firstly, the planning framework uses ecosystems as the basis for determining representativeness, setting out a spatial framework for classifying Australia's marine environment into bioregions that make sense ecologically, and are at a scale that is useful for regional planning.¹⁰ The working group aims to ensure that the full range of ecosystems is recognised at the appropriate scale within the network, with adequate levels of protection to ensure the ecological viability of the ecosystem. The network is required to include some highly protected areas (i.e. those in IUCN Categories I and II). The working group is also guided by the notion that the processes of identification and selection of marine protected areas should include effective and high quality public consultation with appropriate community and interest groups, addressing current and future social, economic and cultural issues.

These principles are given effect in the 'guidelines for establishing the national representative system of marine protected areas',¹¹ which are intended to assist government agencies to develop their marine protected areas in a way which can

contribute to the development of the national system. The strategic plan of action for the national marine protected area system¹² was drafted in 2001, and sets out how the aims of the national marine protected area system will be achieved.

The basis of the plan of action was that the Commonwealth government would first identify the marine bioregions that could inform regional planning. Then each jurisdiction would develop priorities for its particular region, using analysis of gaps, values, threatening processes and other factors such as socio-economic criteria.¹³ The jurisdictions would use these identified priorities to help them identify candidate areas within regions, consulting with other jurisdictions where appropriate. Consistency of approach is the responsibility of the working group.

The government has invested significant resources into classifying Australia's marine environment into bioregions. It uses a spatial framework called the 'integrated marine and coastal regionalisation of Australia'. The classification process sets out ecosystem boundaries and processes, and assists marine managers in understanding the specific management needs of particular ecosystems. The bioregions thus provide the basis for design of the network of marine protected areas.

Marine bioregional planning

Since 2005, when regional marine planning was brought under the Environment Protection and Biodiversity Conservation Act, the national representative system of marine protected areas programme has been incorporated into the marine bioregional planning process. Plans for the marine area are now created as bioregional marine plans under section 176 of the Act. Under the new approach, the Minister for Sustainability, Environment, Water, Population and Communities is responsible for making marine bioregional plans for large parts of the Commonwealth's marine area, with the National Oceans Office overseeing the creation of the plans.

Section 176 of the Act provides that a bioregional plan may include provisions about:

- The components of biodiversity, their distribution and conservation status
- Important economic and social values
- Heritage values of places

- Objectives relating to biodiversity and other values
- Priorities, strategies and actions to achieve the objectives
- Mechanisms for community involvement in implementing the plan
- Measures for monitoring and reviewing the plan

In broad terms, the bioregional plans set out the Australian government's objectives for conservation and environmental management measures such as marine protected areas, listed marine species protection and sustainable fisheries. The purpose of the marine plans is to guide the Minister, sectoral managers and industry about the key issues and priorities in the region, rather than to act as a regulatory tool.¹⁴

The plans do not have the status of regulations, but the Commonwealth Environment Minister must have regard to a bioregional plan when granting environmental approvals under the Environment Protection and Biodiversity Conservation Act. These approvals are required for 'matters of national environmental significance' which include all activities which are likely to have a significant impact on the environment of Commonwealth marine areas.

In 2007 the Department of the Environment and Water Resources released goals and principles for the establishment of the 'national representative system of marine protected areas in Commonwealth waters'.¹⁵ These set out four goals and principles which related to the location, selection, size, shape and zoning of marine protected areas. They are reproduced in Appendix 2.

Marine bioregional planning is being implemented in six large bioregions – the south-east, south-west, north-west, north, temperate east and coral sea marine regions. There are two major parts to the marine bioregional planning process: first, the identification of regional conservation priorities; and secondly the identification of marine reserves to fulfil the Australian government's commitment to establishing the national representative system of marine protected areas.

Regional conservation priorities are identified from an analysis of the threats to the conservation values of each marine region. The marine bioregional plans identify measures to address regional conservation priorities and are designed to facilitate better decision-making under national environment law.

The planning process involves the creation of a bioregional profile, and the development of a draft plan, which goes out for public consultation before its finalisation. The bioregional profiles describe the ecological and biophysical features and the conservation values of each region, together with the human

activities that occur there. In some regions, they also identified ‘areas for further assessment’. The purpose of these is to narrow down the areas that warrant closer examination in the reserve identification process, because they are representative of the conservation values of the region.

In May 2011 draft plans were released for five regions, (the plan for the south-east region being completed in 2004 – see Figure 14.3). Formal 90-day public consultations on the draft plans were undertaken in each region between May 2011 and February 2012. Across Australia, 245 public and stakeholder meetings were held, and 566,377 submissions were received. After considering the public submissions, together with detailed socio-economic assessments, the government released final proposals in June 2012.

A further 60-day public consultation process was then undertaken (see Figure 14.3). After the final marine bioregional plans have been approved, the proposed network of new marine protected areas will be proclaimed through a separate statutory process. The proclamation of the proposed national marine reserves network will increase the total number of Commonwealth marine reserves to 60, representing examples of all of Australia’s different marine ecosystems and habitats.¹⁶ The largest of the proposals is a marine reserve covering 989,842 square kilometres of the Coral Sea. All mining and petroleum development is to be prohibited in this area, whilst a sanctuary zone in which commercial fishing is prohibited will cover approximately half of the reserve area.

Following proclamation, the Director of National Parks will be required to draft management plans for the new reserves, setting out how the reserves are to be managed and what activities will be permitted within the reserves. These plans will be subject to two rounds of public consultation, after which they are approved and become operational for a ten year period.¹⁷

South-east Commonwealth marine reserves network

The south-east Commonwealth marine reserves network, established as a result of the south-east marine planning exercise, was the first temperate deep sea network of marine reserves in the world. It aims to cover representative examples of all key marine features found in the south-east Australian marine region. The network uses a slightly modified version of the IUCN management categories to indicate which activities are permitted and which are prohibited. Figure 14.4 sets out the zoning rules.

Figure 14.3: Process to establish Australian Commonwealth marine protected areas through bioregional planning

Step	Action
1.	Release of Commonwealth marine reserve network proposal
2.	90-day consultation on initial proposal
3.	Final proposal for Commonwealth marine reserve network released publicly
4.	Intention to declare the Commonwealth marine reserve network published by the Director of National Parks
5.	60-day statutory consultation period on final marine reserve network proposal
6.	New Commonwealth marine reserve network declared
7.	Consultation on and development of interim management arrangements
8.	Invitation to comment on proposal to prepare a draft management plan
9.	Minimum 30-day statutory consultation period
10.	Draft management plan released
11.	Minimum 30-day statutory consultation period on draft management plan
12.	Management plan approved and becomes operational

Forty-two per cent of the total marine reserve network is ‘sanctuary zone’ managed for scientific research and passive uses. Thirty-six per cent of the network is ‘special purpose zone’ and is closed to commercial fishing. Twenty-one per cent of the network is multiple use, where low impact fishing methods and other activities are permitted. The remaining one per cent comprises a benthic sanctuary zone and a recreational use zone, where some fishing is permitted in accordance with regulation.

Provision for indigenous rights

Section 359A of the Environment Protection and Biodiversity Conservation Act exempts indigenous people from compliance with the restrictions in Commonwealth [marine] reserves in relation to traditional uses of the area which involve hunting or food gathering for subsistence.

The scoping and consultation process for the development of the marine bioregional plan for the north region incorporated the preparation of three ‘sea country plans’. These were prepared by groups of traditional owners, and set out the “*key aspirations of Indigenous peoples within the Northern Marine Planning*

Area.”¹⁸ The plans provided a mechanism for traditional owners to provide constructive and considered input into the planning process.

Figure 14.4: Zoning rules applying to the south-east Commonwealth marine reserves network

Zone	Rules
Sanctuary zone	No commercial or recreational fishing or mining* activities are permitted. The area can only be used for scientific research and passive recreational activities. Vessel transit is allowed.
Benthic sanctuary zone	Some commercial and recreational fishing is permitted. However no fishing or other extractive use in the area from 500 metres below sea level to 100 metres beneath the seabed is allowed.
Recreational use zone	This zone allows for recreational and charter fishing if carried out in accordance with the relevant State fishing arrangements. It prohibits commercial fishing and mining activities.
Special purpose zone	Recreational fishing and mining activities are permitted. Charter fishing may be permitted providing the operator has applied for an approval from the Director of National Parks. Each application will be assessed on a case-by-case basis. No commercial fishing is allowed in this zone.
Multiple use zone	Recreational fishing and mining activities are allowed. Some forms of commercial fishing, excluding demersal trawl, Danish seine, gill netting (below 183m) and scallop dredge, are allowed providing the operator has registered under the approval from the Director of National Parks and abides by the conditions of the approval.

*Mining includes oil and gas exploration and extraction and geosequestration of carbon. Where seismic surveying and transit are allowed, they are permitted under general approval from the Director of National Parks. Other mining activities are permitted on a case-by-case assessment.

Conflict resolution mechanisms and compensation for affected parties

Section 359 of the Environment Protection and Biodiversity Conservation Act provides that any restrictions on activity put in place in a Commonwealth reserve will not affect ‘usage rights’ held by a person in relation to the land or seabed immediately before it was included in the reserve.

Usage rights are defined broadly for these purposes and include estates (i.e. a freehold or leasehold interest), privileges, authorities, licences or permits. As a result, the holder of a usage right is allowed to continue to exercise the right despite the restrictions. However, if the right expires (for example in the case of a fixed term permit) it will only be renewed with the Minister's consent. Because the wording of the provision makes it clear that the prior usage right applies to rights over *land or seabed*, it is likely to apply to mining interests and sea dumping carried out under permit, but probably does not apply to fishing rights, as these do not relate to rights over land or seabed in the obvious sense.¹⁹

The government has made some efforts to address the needs of fishers affected by the creation of Commonwealth reserves. The government's draft 'Marine protected area and displaced fishing: A policy statement' of January 2004 sets out "*the agreed Government position in relation to marine protected areas and displaced fishing effort*". First, it makes clear that there is no constitutional or legal requirement for the Australian government to provide compensation to commercial fishers impacted by new or re-zoned marine protected areas. But nevertheless, it indicates that the government is willing to consider issues of fairness and equity in relation to the impacts on individual fishers and those reliant on the fishing industry.

The policy statement adopts the beneficiary pays approach, in that it emphasises that the costs of marine protected areas are 'likely to be borne in the short term by a small group of private individuals and communities in marine-related industry sectors', while "*the public good benefits flow to all Australians from marine protected areas*". Therefore, where marine protected areas create additional requirements for the reduction of fishing effort beyond that required to achieve fisheries management objectives, the government will consider funding adjustment assistance to support the additional adjustment to fishing effort and mitigate impact on local communities.

Decisions on adjustment assistance are to be made on a case-by-case basis, following an assessment of the socio-economic impacts of the marine protected area, which will involve an industry and community consultation process. The policy states that if assistance is provided, the industry and community should also be involved in the design of the package. The policy acknowledges that where marine protected areas displace fishing activity, resulting in increased activity in the rest of the fishery, management changes required in those areas may be attributable to the creation of the marine protected area and should be taken account of in the assistance package. It also warns that, it is important to separate

the effects of fishing measures introduced to achieve fisheries management outcomes, from those related to the creation of marine protected areas. Importantly, the document makes it clear that the intention of the scheme is not to provide compensation for fishers and others whose industry is affected by the creation of a marine protected area, but to help them to restructure their industry.

The most prominent examples of applications of the policy are the rezoning of the Great Barrier Reef Marine Park in July 2004 and the creation of the Cod Grounds Commonwealth marine reserve in 2007. The process to develop the south-east marine reserve network in 2006, initially included a structural adjustment package, but ultimately this was not applied. This was because a wider adjustment package to address changes in fisheries management was being developed at the time, and this could include an element related to the creation of the marine protected area network.

Great Barrier Reef structural adjustment package

When the Great Barrier Reef Marine Park zoning plan came into effect in 2004, the government announced a 'structural adjustment package' guided by the policy on displaced fishing, which was intended to assist fishers, fishery-related businesses, employees and communities adversely affected by the zoning plan. In addition, it was designed to manage, in the most cost effective manner, any displaced fishing effort that had unsustainable ecological or economic impacts

The package provided A\$213.7 million to 1,782 fishers, seafood processors and upstream providers to the fishing industry.²⁰ The package comprised seven components, of which the main two were assistance to operators wanting to exit the industry and business restructuring assistance to assist operators remaining in the industry to adjust to the new zoning arrangements. Other components included support to operators to obtain professional business advice, as well as support to fishing industry employees and to fishing communities.

Thirteen commercial fisheries were identified as being affected by the rezoning. Of these, six fisheries were considered fully fished, so that areas outside the marine protected area were deemed incapable of accommodating displaced effort without having unsustainable ecological and/or economic impacts. For these fisheries, effort reduction targets were estimated and expressed as units of effort, using the best available data.

A voluntary competitive tender was held where applicants were asked to tender a price for the purchase of their fishing entitlements (not their vessels, gear

or equipment). Tenders were evaluated according to the best value for money defined as the amount of units of effort being removed for the price sought. Preferential treatment was not given to those most affected by the rezoning, as it was assumed that fishers who chose to remain in the fishery could fish elsewhere. In total, 122 fishing licences were bought out at a cost of A\$33 million. In addition, a total of A\$195 million for the other financial assistance has been paid out, of which the biggest component has been for business restructuring.²¹

The structural adjustment package was reviewed in 2008 following complaints from the fishing industry that it had not compensated them adequately. The review found that, although the policy statement document was “*an excellent foundation on which to build a more comprehensive marine reserves compensation or structural adjustment policy*”,²² the haste with which the package was implemented meant that the clear statements of intent and directives were only partially implemented.

Before the zoning plan was put into place there was little quantitative assessment of the impacts that it would have on industry sectors, and there was no detailed socio-economic analysis of the impacts of the package on industry after it was implemented. It was therefore difficult to quantify the impact that the zoning plan had had, apart from by listening to industry perceptions. The package was intended to assist the industries to restructure themselves after the rezoning, rather than to compensate them, but relatively little effort was made to evaluate what the industries should look like.

The review found that many of the complainants were unable to “*conceptually or financially separate the impacts of the Zoning Plan from the significant negative impacts from a number of external factors*”, and saw the package as compensation rather than as an attempt to support the structural adjustment of affected industries. In addition, businesses were reluctant to cooperate so individual applications had to be assessed on their merits, leaving the same number of operators competing for less business. Combined with a broad scope and vague goals, this meant administering the package was somewhat disorganised, with scope creep common.

The review suggested that in future “*structural adjustment associated with the development of marine protected areas should aim to effectively adjust levels of fishing effort and/or catch to account for reduced access to resources, and to minimize the impacts of displaced effort. Where available fishery management measures do not prevent activation of latent effort, structural adjustment should not be attempted until such time that the latent potential is removed. In this event, autonomous adjustment is recommended.*”²³

Cod Grounds structural adjustment package

The Cod Grounds Commonwealth marine reserve was declared on 10 May 2007, mainly to protect the critically endangered grey nurse shark. This marine protected area is located approximately four nautical miles from the New South Wales coast, covering a total area of approximately three square kilometres. It has been assigned to management category 1a (no take) sanctuary zone, meaning fishing is prohibited.

After the proposal for the creation of the marine protected area had been devised, the Bureau of Rural Sciences was engaged to undertake a socio-economic impact assessment of the reserve. The assessment was intended to inform the government's decision on the reserve and its management of displaced interests, consistent with the displaced fishing policy. The Bureau's report was completed in mid-2004. It identified the fishers who would lose income as a result of the creation of the Commonwealth reserve. It also found that other fisheries in the area were fully exploited, meaning that there was little capacity for displaced fishing effort to move to other fisheries. The report noted that most fishers did not want to leave the industry and that lack of skills and job opportunities in the region meant that they would find it difficult to obtain alternative employment.

On the basis of this report, the government prepared a structural adjustment package in accordance with the structural adjustment policy. It consisted of the voluntary acquisition of six fishing licences from the owner-operators that regularly fished the Cod Grounds area. Assistance was also provided to fishing cooperatives adversely affected by the creation of the reserve. No upper limit was placed on the assistance to be provided. The government originally estimated that the licence buy-out would cost A\$1.25 million and that the assistance to the fishing cooperatives would cost up to A\$600,000. However, that figure has been significantly exceeded.

Analysis

Was it successful?

The process to create Commonwealth marine protected areas through bioregional planning has yet to be completed. One plan for the south-east was completed in 2004. Five other plans have been completed in draft, public submissions have closed, and the government has released final proposals. These are now being

considered through the statutory process. So at the time of writing the final outcome was not clear. However, the Commonwealth government has been successful in completing draft plans for all Australia's bioregions which is a major achievement given the very large size of Australia's oceans. The entire process has taken 12 years to date (with the south-east plan process commencing in 2000).

It is less certain the extent to which the proposed marine protected areas meet the goal of achieving a *fully* representative network. The first bioregional plan, which was developed for the south-east region in 2004, fully protected 10 per cent of the region. The five draft bioregional plans released over seven years later show a wide divergence in the proportion of each region which is proposed to be fully protected. This ranges from 3 per cent in the north to 51 per cent in the Coral Sea (See Figure 14.5).

In relation to the south-west region proposals, Professor Hugh Possingham from the University of Queensland suggests that they are deficient. The area has a very high level of diversity and endemism and he argues it is more biologically important than the Great Barrier Reef. Although the proposal to protect 22 per cent of the area 'looks good on paper', in his opinion it *"misses the mark on the issue of representing all the habitats properly"*. Only 3.5 per cent of the shelf area, which is heavily used by fishing, has been fully protected. Although the process drew on a large amount of data, including information on 2,000 different organisms, Possingham suggests that the government was so keen to avoid impacts on industry, that it missed out on achieving the scientific goals. In Possingham's view, the government appears to have been *"rolled by industries such as fishing and mining"*.²⁴

Strong guidance for marine protected area design

The government developed detailed guidance on the location and design of the marine protected area networks which helped to focus the process. However, despite this guidance, the imperative to minimise socioeconomic costs appears to have compromised the representativeness of the network (with only three and four per cent of the total marine area protected in the north and temperate east respectively and criticisms about the poor representativeness of the south-west proposals).

Flexible approach

The more recent bioregional plans have taken a flexible approach to marine protection. A relatively large proportion of the regions' total area have been identified as marine protected areas (between 19 and 100 per cent), but with a smaller percentage of the network being fully protected. The marine protected area network has therefore been designed as a flexible marine management tool which accommodates a range of uses within the overall purpose of marine protection.

Top-down approach

The approach taken to identifying the proposed networks of marine protected areas in Australia has been largely top-down. The draft proposals were developed by a government agency, the National Oceans Office, and then made available for public comment. Although stakeholders were consulted during the plan preparation process, it was very much the government agency in the driving seat. This contrasts with the approach taken in England and California where stakeholder groups very much led the task of initial identification of a network of marine protected areas, although in most cases they were unable to reach consensus.

Bioregional approach

The focus of the planning process on large regions enabled the government to comprehensively plan for a very large area in a relatively short timeframe. Using bioregions as the organising principle for identifying the different planning regions also enabled the characterisation of the regions to follow scientific principles.

Single jurisdiction

The planning process did not include State waters, and thereby avoided the complication of needing to reach agreement with multiple State governments. However, it also means that activities within State waters and catchments might impact on the Commonwealth marine protected area network.

Conclusion

Australia has taken a particularly progressive approach to the development of marine protected areas. The 27 existing Commonwealth marine reserves are soon to be supplemented with a further 33 reserves, representing all of Australia's marine ecosystems and habitats. This has been undertaken within a bioregional planning framework for the entire marine area. The adoption of the IUCN marine protected area classifications provides for a comprehensive system in which a range of activities can be provided for within marine protected areas. This comprehensive framework has made it possible for the Australian government to meet its objectives.

Figure 14.5: Summary of marine protected areas proposed under Australian Commonwealth bioregional planning

Region	Consultation closed	No. submissions	Total area (km ²)	No. of marine reserves proposed	Area proposed for marine protected area network (km ²)		Area proposed for full protection (km ²)	
* South-east			1,600,000	14	226,458	14%	154,046	10%
South-west	Aug 2011	39,266	1,292,015	8	538,226	42%	286,290	22%
North-west	Nov 2011	19,702	1,070,000	10	377,297	35%	123,528	12%
North	Aug 2011	12,861	625,000	8	121,723	19%	18,475	3%
Temperate east	Feb 2012		1,466,792	9	371,114	25%	63,240	4%
Coral Sea	Feb 2012		989,842	1	989,842	100%	507,487	51%

*Plan completed in 2004

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chapter fifteen



New South Wales marine parks

Introduction

The New South Wales State government manages the marine area generally out to three nautical miles. The New South Wales system of marine protected areas encompasses six multiple use marine parks, 12 aquatic reserves and 62 national parks and reserves with marine components. Marine parks are large areas established under the Marine Parks Act 1997 with the prime aim of biodiversity protection. Aquatic reserves are much smaller areas, established under the Fisheries Management Act 1994, and aim 'to conserve the biodiversity of fish and marine vegetation'.

Since the passing of the Marine Parks Act 1997, six new marine parks have been established incorporating almost 3,451 square kilometres of ocean, or almost 35 per cent of New South Wales's marine area (see Figure 15.1). The parks range in size from 220 to 972 square kilometres. Biodiversity protection is the primary purpose for these large areas, which include many no-take sanctuary areas as well as areas where recreational and commercial fishing and other activities are permitted.¹

Since at least 2007, controversy over the marine parks has grown and it became a significant issue in the March 2011 New South Wales State election. This prompted an independent scientific audit of the New South Wales marine park system to be undertaken. Results of the audit were released in early 2012.

This case study has been based on a web-based search of relevant material, including that on the website of the New South Wales Marine Parks Authority, and also draws on the 2012 report of the independent scientific audit.

Context

In 1991 the Australian Commonwealth government established the Commonwealth marine protected areas program which was intended to develop a national representative system of marine protected areas across Commonwealth, State and Territory jurisdictions as described in chapter 14. This prompted the New South Wales government to take action, and it subsequently released a regional-scale policy setting out goals for marine biodiversity conservation. This ultimately led to the passage of the Marine Parks Act 1997. Prior to this new legislation, there were nine marine reserves in New South Wales waters covering only 7.1 square kilometres.²

Figure 15.1: New South Wales marine protected areas

Source: New South Wales Department of Primary Industries



Legislative framework

Marine Parks Act 1997

Marine parks in New South Wales are established under the Marine Parks Act 1997. The objectives of the Act are:³

- a) To conserve marine biological diversity and marine habitats by declaring and providing for the management of a comprehensive system of marine parks
- b) To maintain ecological processes in marine parks
- c) Where consistent with the preceding objects:
 - i. to provide for ecologically sustainable use of fish (including commercial and recreational fishing) and marine vegetation in marine parks, and
 - ii. to provide opportunities for public appreciation, understanding and enjoyment of marine parks.

The Act provides that the Governor may declare an area, which may include any part of the marine area, land adjacent to and islands within the marine area, to be a marine park.⁴ If land above mean high water springs is to be included in the park, the Governor must seek the consent of the owner or the Minister administering the Crown Lands Act 1989.

Declarations establishing marine parks may only be revoked by an Act of Parliament, but they can be varied by proclamation. If an area is to be removed, the relevant Ministers must certify that it is no longer required to achieve the objectives of the Act.

Marine Parks Authority

The Act provides for the establishment of a 'Marine Parks Authority', a statutory body representing the Crown, tasked with

- Investigating, assessing and recommending where marine parks should be established or where variations to marine parks should be established
- Making recommendations on zoning measures within marine parks
- Preparing an operational plan in respect of each marine park
- Managing activities in marine parks so as to protect the marine environment

- Providing for and regulating the ‘ecologically sustainable use’ (including commercial and recreational fishing) of marine parks
- Public education
- Carrying out and encouraging research on how marine parks affect the marine environment and nearby communities.

The Authority reports jointly to the Minister for Primary Industries and the Minister for the Environment. It is chaired by the Director General of the Department of Premier and Cabinet. The other two members are the Director General of the Department of Industry and Investment, and the Chief Executive of the Office of Environment and Heritage in the Department of Premier and Cabinet.

Marine Parks Advisory Council

The Act provides for the establishment of a ‘Marine Parks Advisory Council’, consisting of the Director General of the Department of Environment and Climate Change, the Director General of the Department of Industry and Investment, one member representing the Commonwealth government, two members representing the interests of marine conservation, one member with expertise in marine science, and one member each to represent the interests of Aboriginal people, the tourism industry, commercial fishers, recreational fishers and scuba divers.

The Advisory Council, on the request of the relevant Ministers or the Marine Parks Authority, advises on a range of matters from a state-wide perspective including on proposals for marine parks, the conservation of marine biological diversity within parks, and their use.

Management and zoning

Some activities are expressly prohibited in marine parks – for example, under section 18 of the Marine Parks Act it is unlawful to prospect or mine for minerals in a marine park except where expressly authorised by an Act of Parliament. Sand extraction is allowed, provided that it is undertaken for the purpose of furthering conservation objectives, or to prevent the risk of serious injury to a person or to the environment, and is undertaken in accordance with a consent. Discharge of ballast water from outside the marine park is prohibited. All commercial activity is

prohibited in the park except fishing in accordance with the regulations and other activities under a consent.

The Act provides that the regulations will provide for the establishment of a zoning plan which will govern the use and management of the marine park. The Marine Parks Authority is required to draft a zoning plan for the marine park within 12 months of its establishment. The Authority submits the draft to the relevant Ministers, together with any comments from the advisory committee. The Ministers then have three months to either undertake public consultation or return the draft back to the Authority for amendment. The primary focus of public consultation in relation to marine parks is on the zoning plans.⁵

The Marine Parks (Zoning Plans) Regulation 2009 (which replaces the Marine Parks Zoning Regulations 1999) sets out the measures governing the zoning system. Under the regulations, there are a number of zones which may be included in a marine park as shown in Figure 15.2. Figure 15.3 shows how the zones have been applied in practice within the Batemans marine park.

Zoning plans must be reviewed five years after commencement and every ten years after that. Amendments to the Marine Parks Act 1997 in mid-2008, introduced clearer procedures for reviewing and amending marine park zoning plans. Subsequent amendments to zoning plans, to increase protection of biodiversity, proved controversial and were ultimately abandoned.

The 2012 audit of the marine park system noted that the identification of fully protected sanctuary areas, as part of the zoning plans, was in some cases based on valid information, but in other cases such information was notably lacking.⁶ It recommended that the zoning and rezoning of marine parks should more explicitly and transparently consider the assessment of risks and that more focus should be placed on social-optimal zoning of marine parks. The scientific auditors argued that social and economic benefits and impacts should be an integral part of the zoning and management process.

Figure 15.2: Zones which can be applied to New South Wales marine parks

Sanctuary zone

The objective is to provide the highest level of protection for biological diversity, habitat, ecological processes, natural features and cultural features in the zone. It also seeks to provide opportunities for recreational, educational and other activities which do not harm the features, and to provide opportunities for scientific research. Harming plants or animals, or damaging or interfering with habitat are prohibited in a sanctuary zone (except with the consent of the Ministers, for the purposes of research, environmental protection, public health or public safety). The cleaning of fish and fishing gear is also prohibited. Aquaculture is prohibited within sanctuary zones, as is dredging.

Habitat protection zone

The objectives are to provide a high level of protection for biological diversity, habitat, ecological processes, natural and cultural features, and where consistent with these objectives, to provide opportunities for recreational and commercial activities (including fishing), scientific research and educational activities so long as they are ecologically sustainable and do not have a significant impact on any fish populations or other animals, plants or habitats.

Harming any plants or animals (apart from fish), or damaging or interfering with habitat within a habitat protection zone is prohibited except under a consent. This may be granted for research, environmental protection, public health, traditional use or public safety, or for the purposes of an ecologically sustainable use that does not have a significant impact on fish populations within the zone or other plants, animals or habitats. Within habitat protection zones, certain types of fishing activity are permitted: hook and line, hand, scoop net or landing net, spear fishing, and other methods permitted by the zoning plan. Aquaculture is permitted within the zone, under consent.

General use zone

The objectives are to provide protection for biological diversity, habitat, ecological processes, natural features and cultural features, and where consistent with these objectives, to provide opportunities for recreational and commercial activities (including fishing) scientific research, educational activities and other activities so long as they are ecologically sustainable. In general use zones, consent must be obtained to harm animals other than fish, plants and habitat. Consent will only be given for research, environmental protection, public health, traditional use or public safety purposes, or for the purpose of an ecologically sustainable use.

Fishing is permitted in the general use zones, but must be consistent with the particular measures set out in the zoning plan for the marine park. Aquaculture is permissible with consent.

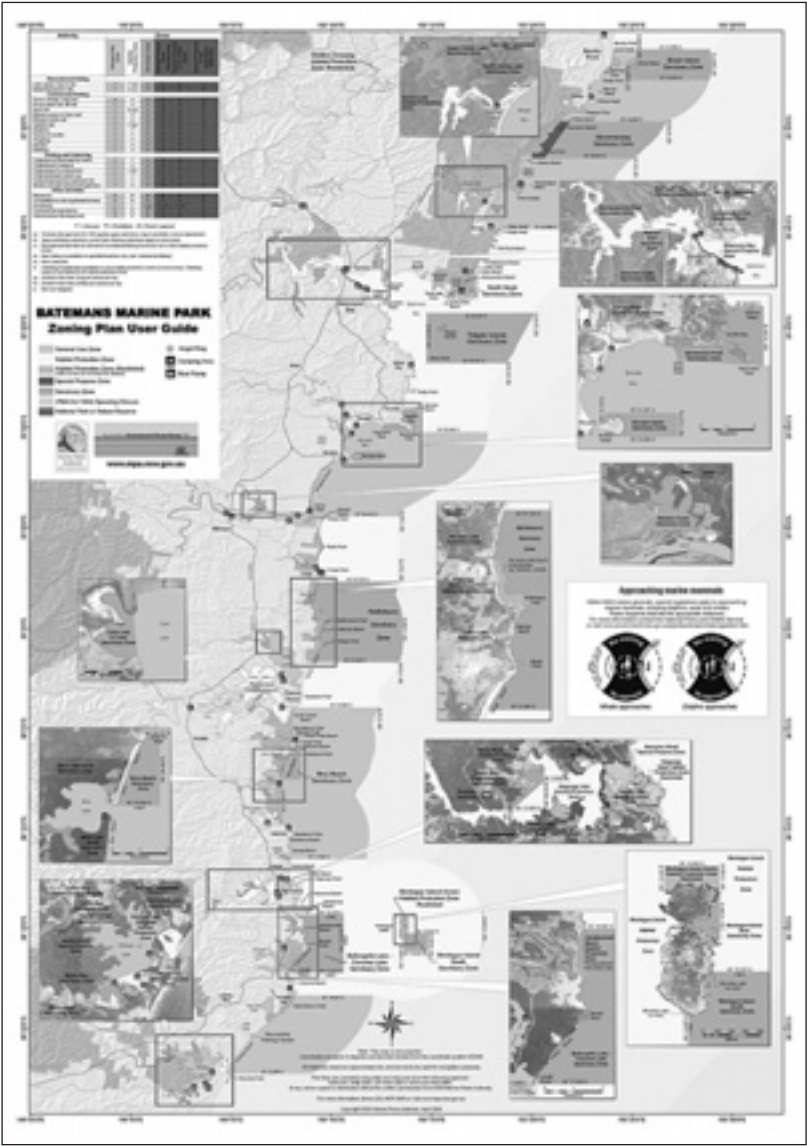
Special purpose zone

The objectives are to provide for the management of biological diversity, habitat, ecological processes and natural cultural features in the zone, where phenomena, sites or items in the zone warrant special management, and to cater for special facilities and features in the zone such as slipways, breakwaters, berthing facilities and shipwrecks and such objects as may be set out in the zoning plan for a marine park.

In special purpose zones, consent must be obtained to harm animals other than fish, plants and habitat. Consent will only be given for research, environmental protection, public health, traditional use or public safety purposes, or for the purpose of an ecologically sustainable use.

Figure 15.3: Zoning map for Batemans marine park

Source: New South Wales Department of Primary Industries



Consenting

Applications for consents to undertake activities within marine parks are governed by the Marine Parks Regulation 2009. The Ministers must have regard to the assessment criteria set out in the regulation when deciding whether or not to grant a consent. These are:

- a) The objects of the Act
- b) The objects of the zone in which the activity is proposed to be carried out
- c) The activities that are permissible in the zone in which the activity is proposed to be carried out (as specified in the relevant zoning plan)
- d) Any operational plan for the marine park adopted by the Authority
- e) Any threatened species or other protected flora or fauna under the Fisheries Management Act 1994, the National Parks and Wildlife Act 1974 or the Threatened Species Conservation Act 1995 that may be affected by the proposed activity
- f) The form of transport to be used to gain access to the zone in, on or from which the activity is proposed to be carried out, having regard to the adequacy of facilities for parking, mooring and landing vehicles, vessels and aircraft, and for loading and unloading them
- g) The type of equipment to be used in connection with the proposed activity
- h) The arrangements that have been made for the making good of any damage to the marine park that arises from the proposed activity
- i) Such other requirements as the relevant Ministers consider appropriate to the proposed activity

The Ministers are entitled to prioritise certain types of applicants and to set a limit on the number of permits that can be issued in respect of an activity in a marine park or in a zone in a marine park.

Operational plan

Once the zoning plan has been adopted, the Marine Parks Authority is required to establish an operational plan which sets out the strategies, actions and activities that the Authority intends to undertake to operate the park and give effect to the zoning plan.

Monitoring and research

Once the marine parks were established, the Marine Parks Authority developed a 'Strategic Framework for the Evaluation and Monitoring of Marine Parks in New South Wales'.⁷ This identified research priorities amongst other things. The research priorities were further fleshed out in the 'Marine Parks Strategic Research Framework 2010-2025'. The purpose of this document was to provide guidance to the research community on what research and monitoring would best assist the management of the marine parks.

An independent review of marine park marine science, which was undertaken in 2009, recommended that the focus of research should shift away from habitat characterisation and assessment, and towards monitoring the effectiveness of the marine park system including the extent to which it is protecting marine biodiversity in New South Wales. The reviewers also recommended that greater emphasis be put on researching the socio-economic and heritage aspects of the marine park system.⁸

The 2012 scientific auditors concluded that research needed to better incorporate social and economic data into decision-making. In addition, more resilience and multi-stressor research was required to better understand the response of marine ecosystems to combinations of threats.

In order to improve the quality and relevance of marine science in New South Wales, the scientific auditors recommended that a 'scientific committee' be established. This was to be independent of government agencies, and would oversee strategic research in the entire marine estate of New South Wales, (i.e. not just the marine park system). The ambit of the science included would include marine sciences, economics and social science.

The scientific committee would be tasked with reviewing the five yearly and annual work plans for science in the entire New South Wales marine estate and would make specific recommendations relating to the adoption and/or modification of plans.

Provision for indigenous rights

There is some provision for indigenous engagement with the marine park programme. The Marine Park Authority in some cases has established special purpose zones for customary use. The Authority has entered into agreements with Indigenous groups to provide for cultural resource use, and special permits have been provided from time to time for community events.⁹

In some marine parks, the Authority has also appointed indigenous liaison officers and established marine skills training programs so that members of the local Aboriginal community can develop the skills to undertake marine-related activities.

Despite these initiatives, the 2012 scientific audit of the marine park system concluded that *“there is a lack of commitment and capacity to ensure that Indigenous knowledge and expertise of land and sea management is incorporated into marine protected area decision making at all levels.”*¹⁰ To help rectify this problem, the scientific audit team recommended that each marine park employ an Aboriginal liaison officer and that an Aboriginal Cadet Program be supported in each marine park.

Implementation

The first marine park under the new legislation was created in 1998. This was the Solitary Islands marine park which covered 722 square kilometres. The zoning plan for the marine park was completed in 2002 and fully protected 12 per cent of the marine park in a sanctuary zone. The extent of this zone was 86.5 square kilometres (see Figure 15.4).

The second marine park at Lord Howe Island was created a year later, in 1999, and the zoning plan finalised in 2004. The park covered 480 square kilometres and 27 per cent of this (125 square kilometres) was protected with a sanctuary zone.

In 2001, the New South Wales government prepared a document setting out how it intended to implement the Marine Parks Act and create a representative system of marine protected areas. The document articulated the government's goal which was *“to establish a comprehensive, adequate and representative system of marine protected areas that includes a full range of marine biodiversity at ecosystem, habitat, and species levels.”*¹¹

The document outlined that the first step was to undertake bioregional assessments to identify the conservation values of the bioregion and possible candidate marine protected sites. Marine bioregions for New South Wales had previously been defined from regional analyses of geomorphology, oceanography and species distributions.¹² In total, five bioregional assessments were carried out for the Tweed-Moreton,¹³ Manning Shelf, Hawkesbury Shelf, Batemans Shelf and Twofold Shelf bioregions.¹⁴

The assessments mapped and analysed areas of estuarine and coastal habitats and ecosystems as well as available data for fishes, seabirds, waders, mammals,

and estuarine and coastal vegetation. The assessments also included reviews of literature, analyses of the area, abundance and diversity of habitats and species, and scenario modelling of potential marine protected area designs.

Spatial indices of coastal condition and vulnerability were also derived from a range of coastal and catchment use databases. The data was incorporated into multiple criteria models and these, together with supporting information, were used to compare alternative options for marine parks.

The second step was to undertake site assessments. These would focus on site-specific conservation values and effects on coastal communities and other stakeholders. Site assessments included more detailed mapping of habitats using sidescan and geoswath sonar, underwater video, discussions with fishers to collect local knowledge, local surveys of fishes and invertebrates, social and economic surveys of stakeholder values, and random statistical surveys of the regional population. This information was mapped, analysed in decision support tools including Cplan and Marxan, and used in participatory GIS workshops with stakeholder representatives to develop options for marine park zoning and calculate buy-out of commercial fishing licences.

Two marine parks were created in 2002 (Jervis Bay and Cape Byron), and another two marine parks created in 2005 and 2006 (Port Stephens-Great Lakes and Batemans). By 2006, almost 3,500 square kilometres of New South Wales's marine area was within marine park boundaries, or almost 35 per cent of the State's entire marine area. Of this, 19 per cent (or 116 square kilometres) was fully protected in sanctuary zones.

The marine parks have been very popular and have attracted large numbers of visitors. Both the Jervis Bay and Cape Byron Marine Parks attracted an estimated two million people during the two-year period covering 2004 and 2005.¹⁵

Community consultation has focused on the zoning phase, rather than on the creation of the park in the first instance. Once a new marine park has been established, the Marine Parks Authority advertises for expressions of interest from local community groups to join a park advisory committee. This committee is involved in the development of the zoning plan, and once the plan is completed, continues to be involved in providing advice on the day-to-day management of the marine park.

Figure 15.4: Summary of New South Wales marine parks

Marine park	Date created	Size km ²	Fully protected km ²	Fisheries buy-back budget	Zoning plan
Solitary Islands	1998	722	86.5 (12%)		Finalised 2002; new zoning plan approved 1 March 2011 which increased full protection to 19%, then abolished 26 May 2011
Lord Howe Island	1999	480	125 (27%)		Finalised 2004; zoning plan reviewed in 2011 and unchanged
Jervis Bay	2002	214.5	430 (20%)		Finalised 2002; new plan which made minor changes approved 1 March 2011; then abolished 26 May 2011
Cape Byron	2002	220	61 (27.5%)	A\$4.3m	Finalised 2005
Port Stephens-Great Lakes	2005	987	173.7 (17.6%)	A\$10m	Finalised 2007
Batemans	2006	850	116.5 (19%)	A\$8.5m	Finalised 2007
TOTAL		3,473.5	836.6 (24%)		

In the case of the Lord Howe Island Marine Park, which was established in 1999, the Marine Parks Authority spent four years consulting with the public over the development of the zoning plan. Public comment was sought on two successive drafts. Almost half of the Lord Howe community lodged written submissions on the drafts, indicating the high level of engagement in the process. Of these, the majority of local residents either supported the proposed zoning or wanted higher levels of protection.¹⁶ When the plan was reviewed in 2010, the Parks Authority only received 169 submissions and the plan was retained unchanged. Other zoning plans have been more controversial. When the draft zoning plan for the Cape Byron Marine Park was released the Marine Parks Authority received more than six thousand submissions.

Scientific audit

Zoning plan reviews for Jervis Bay and Solitary Islands marine parks commenced in 2008. Proposals were developed to increase the sanctuary zone in the Solitary Islands from 12 to 19 per cent and to ban prawn trawling in Jervis Bay. These provisions were hotly contested but were adopted by the Labor Government on 1 March 2011.

Just two months later, in May 2011, the new coalition government abolished the changes to the zoning, arguing that decision-making about marine park zoning had been ‘politicised’ by the former government. A media release stated that “[t]he New South Wales Government is committed to common-sense marine parks policy that is based on science, not politics,” and “we will disallow these most recent changes, overturning Labor’s politically motivated plans ...”¹⁷

The government supported a bill proposed by the Shooters and Fishers Party which culminated in the Marine Parks Amendment (Moratorium) Act 2011. This imposed a moratorium on the declaration of additional marine parks, or the alteration or creation of sanctuary zones within existing marine parks, for a period of five years. It also required Ministers to arrange for an independent scientific audit of the New South Wales marine parks.

The scientific audit was a commitment of the Liberal and National Parties prior to the election, and aimed to ensure that the government “*strike the right balance between sustainable fishing and conservation of marine biodiversity in our Marine Parks policy.*”¹⁸ Associate Professor Robert Beeton from the University of Queensland was appointed Chair of the audit team. It was tasked with reviewing the effectiveness of marine parks in protecting different habitat types and providing recommendations for further action and, if appropriate, alternative management approaches.

The audit report was released in February 2012. Overall, the audit team supported the system of marine parks in New South Wales and also recommended that mechanisms be found to address gaps in coverage of two bioregions for which no marine parks had been implemented despite recommendations from previous biodiversity assessments.¹⁹

However, the audit team recommended a fundamental shakeup of the institutional structure applying to the New South Wales marine area. It formed the view that the entire area, including the marine parks, should be managed by a single body under a new consolidated piece of legislation. The audit team’s rationale for this recommendation was that “*the best design for a system that can*

achieve the sustainable management of the New South Wales coast is one that is managed as a single and continuous system” and that this is “a better way of co-ordinating many competing interests and fostering a stewardship ethic whilst managing for a plurality of legitimate concerns.”²⁰

The specific design features of this proposed new system include:²¹

- The Marine Parks Authority, the fisheries management agency (New South Wales Fisheries) and other bodies to be replaced by one integrated agency called the Coastal and Marine Management Authority
- The new Authority would manage the entire New South Wales marine area (including the marine parks) as well have control over land-based development that had the potential to affect the marine area
- The new Authority would be given ‘real powers’ to coordinate activities with other management authorities
- Management of the marine area would be split between five sections of the marine area which coincide with catchment management authorities

A mechanism that the audit team suggested could be used to give the new Authority control of land-based impacts would be to require any land-based proposal that is within a prescribed distance upstream from a marine park to be automatically referred to the Authority for assessment of potential impacts. The new Authority could also have a concurrence role in the subsequent local government planning decision. This would presumably be in addition to mechanisms within the current Marine Parks Act which require consenting within the locality of a marine park to take into consideration the objects of the Marine Parks Act, and where there is likely to be an effect on animals, plants or habitat, to consult with the Marine Parks Authority and take into consideration its advice.

Analysis

Was it successful?

The New South Wales marine park system was established over a very large area, incorporating over a third of the state’s marine area, with a very short period of time – within nine years of the legislation being enacted. Although the marine

parks were designed to accommodate multi-uses, a quarter of the entire area was protected in no-take sanctuary zones.

However, the extent of the marine park system provoked strong opposition which culminated in a moratorium on any further expansion. It also prompted a scientific audit, which, although coming out in support of the marine park system, proposed to do away with the dedicated Marine Parks Authority, and combine marine park management with fisheries and other marine management functions. If adopted by the government, this could serve to dilute the focus of the dedicated Marine Parks Authority on marine conservation.

Clear legislative guidance

The objects of the Marine Parks Act provided a clearly prioritised goal for marine parks in New South Wales and a process for their creation and management. Subsequent regulations set out four potential zones which could be applied to areas within marine parks, their objectives, and the rules that would be attached to them. The Act was nested within broader Commonwealth government policy which provided significant guidance on the design of the national marine protected area network of which the New South Wales marine parks would form a component.

‘Top down’ approach

Similar to the Australian Commonwealth government’s approach to bioregional plans, the New South Wales initiative to create a representative network of marine parks was strongly led by the State government agencies. Although there was extensive public consultation, the Marine Parks Authority led the development of the marine park proposals and zoning plans. This approach avoided the need to seek consensus amongst all the stakeholder groups, or the adoption of the lowest common denominator. It also resulted in the successful establishment of a large network within a short period of time.

Focus on large multi-use marine parks

The focus on establishing large multi-use marine parks, rather than on a network of smaller and more numerous marine reserves, helped to diffuse opposition to marine protection in the initial stages. Stakeholder engagement and opposition to marine protection was focused on the detail of the zoning plan, rather than on the initial establishment of the marine parks themselves.

Adaptive management

Marine park locations and zoning plans were based on the best available scientific evidence to represent marine habitats and species while providing for sustainable use. This work includes some of the State's first regional mapping of many marine ecological, social and economic values and application of tools to assess this information in alternative planning strategies and provide public education.²²

The establishment of large marine park areas, within which zoning plans were developed and regularly reviewed, enables flexibility and the application of adaptive management. If the zoning controls were found not to be achieving the biodiversity outcomes sought, they could be amended as part of the review. This happened with the Solitary Islands Marine Park, where the sanctuary zoning was increased from 12 to 19 per cent (although this was later overturned after a new government gained power).

Conclusion

New South Wales has established a number of marine parks in its waters and has a comprehensive framework for their establishment and management. The concept of zoning within them is well established and understood. In recent years, the politicisation of the issue has made it less likely that new marine parks will be established, but independent scientific advice suggests that the current framework is at least a good basis for further development.

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chapter sixteen



Fiordland Marine Guardians

Introduction

The ‘Guardians of Fiordland’ began as a stakeholder-led initiative in 1995. Local citizens and representatives from various sectors of the community became concerned about the impacts of tourism in the region and the sustainability of local fisheries.¹ United by their interest in preserving Fiordland’s exceptional qualities as a unique marine environment and valuable resource, the group developed a collective vision, “... *that the quality of Fiordland’s marine environment and fisheries, including the wider fishery experience, be maintained or improved for future generations to use and enjoy.*”²

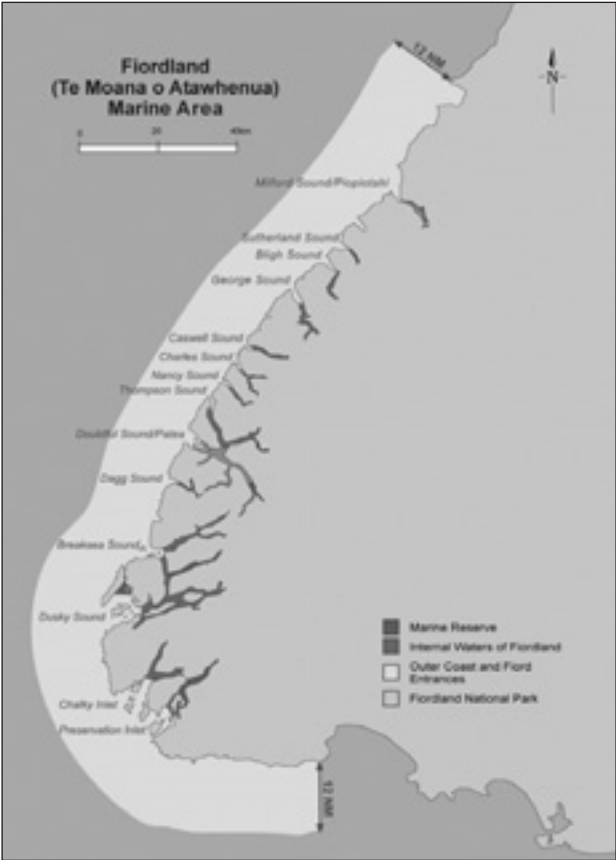
This vision guided the drafting of the ‘Fiordland Marine Conservation Strategy’ which, backed by the Ministry for the Environment, gained support not only from the community but from government agencies alike.³ To implement the Strategy, special legislation was drafted in the form of the Fiordland (Te Moana o Atawhenua) Marine Management Act 2005, formalising what some regard as one of New Zealand’s most successful example, of a cooperative and integrated management scheme.⁴

A total of 103 square kilometres of inner fiord marine habitat is currently protected in marine reserves. An independent Ministerial review of the initiative was completed in 2010, lifting a moratorium on the creation of further marine reserves. There is now an opportunity to assess whether the protection measures in place in Fiordland are adequate and appropriate for the high ecological value of the region.

The research for this case study has been conducted using information and documents publically available on the Fiordland Marine Guardians and the Ministry for the Environment websites. Further insights were gained from research conducted by Jason Mize, the results of which were published in a report titled ‘Stakeholder engagement strategies for designating New Zealand marine reserves’.⁵ In addition, material was drawn from an independent review conducted by Allen and Clarke Policy and Regulatory Specialists on the effectiveness of the management of the Fiordland Marine Area,⁶ which was required five years after the commencement of the Act.

Figure 16.1: Fiordland marine reserves

Source: Department of Conservation



Context

Social and economic environment

Māori exploration of Fiordland began around 800 years ago. It is thought that two significant battles were fought between Ngāti Mamoe and Ngāi Tahu around 1780 in Preservation Inlet, which was called Rakituma, or ‘the threatening sky’ by

Māori. Ngāti Mamoe were escaping to the remote south-western corner of Fiordland whilst being pursued by Ngāi Tahu.⁷

The first European interest in Fiordland stemmed from Captain Cook's voyages on the *Endeavour* through Dusky and Doubtful Sounds in 1770 and 1773. Sealers returned in 1791, and began killing with such intensity that fur seals were driven to the brink of extinction by the early 19th Century. In 1829, a whaling station at Preservation Inlet was established, but after exhausting the resource, the station was deserted in 1838.⁸

From 1952, a host of explorers made various expeditions, surveyed, took up settlement, established sheep runs and searched for gold. By the early 1900s, however, activity slowed and the area returned to a more undisturbed state. In 1945, crayfishing began and rapidly grew in the 1950s and 60s, becoming a major export industry which persists today although at much reduced levels.⁹

In addition to fishing, other commercial activities in the region include live deer recovery, commercial deer shooting, farming along the fringes of the National Park and tourism.¹⁰ The most significant of these is tourism, beginning with the opening of the Milford Track in 1890. The urban hub of Fiordland is the town of Te Anau, which most visitors either stay in or pass through on their way to a range of tourist activities including tramping, boat cruises, sea kayaking, diving, fishing, scenic flights or exploring glow worm caves. The permanent population of Fiordland is around 3,000, although this increases to around 10,000 in the summer months.¹¹

Natural environment

Fiordland, nestled in the south-western pocket of New Zealand, is a spectacular terrestrial and marine environment which is recognised as globally unique. Inscribed with the name 'Te Wāhipounamu/South West New Zealand' on the World Heritage List, the area is characterised by its steep mountains, abundant waterfalls and dense forest reflected across the water surface within deep narrow fiords carved by ancient glaciers.¹²

The area experiences exceptionally high levels of rainfall, with up to seven metres received annually, compared to the 0.6 to 1.6 metre average in other areas of New Zealand.¹³ This fresh water runoff from the mountains carries tannins, humic acid and dissolved organic substances which stain the water its characteristic brownish tea colour.¹⁴

Underwater sills formed by glacial terminal moraines shield the inner waters from the open ocean, maintaining the unusual colouring of the surface waters due

to a lack of turbulence and mixing in the fiords.¹⁵ The dark coloured freshwater floats on top of the denser seawater, reduces the light and encourages typically deep water species such as sea pens, black coral and sponges to grow nearer the surface and making them unusually accessible to divers.¹⁶

Coral communities include *brachiopods* (lamp shells), *antipatharians* (black corals) and *gorgonians* (horny corals including sea fans and red coral). Rock lobster, blue cod, paua and Jock Stewarts (sea perch) are also abundant in the area.¹⁷ Kelp generally cannot grow on the inshore rocks within the fiords, due to the poor light conditions and low salinity in surface waters, and are excluded to the more turbulent areas beyond the fiord entrances and outer coast.¹⁸ This interesting dynamic creates distinctly different environments between the inner fiords and the outer coastline, with implications for the marine species occupying these areas.

Fiordland is home to a number of special marine species including bottlenose dolphins, New Zealand fur seals (kekeno), Fiordland crested penguins (tawaki) and blue penguins (korora). Whales also swim close to the edge of the continental shelf here.¹⁹ There are around 200 bottlenose dolphins in the area which frequent the narrow fiords.²⁰

There are a wide range of marine-related interests in the area. Diving is popular both for recreational and research purposes. Tourism plays a major part in the local economy, with charter and tour operators benefiting from the high ecological values of the region.²¹ Both recreational and commercial fishers have a strong interest in the ongoing use of Fiordland's fishing resources. Twenty per cent of New Zealand's commercial rock lobster harvest comes from Fiordland, as does 12 per cent of the paua harvest.²² The sheltered fiords make recreational fishing more accessible for small boat owners.

Māori customary use of Fiordland has traditionally been seasonal. The remote location led to temporary residence for mahinga kai (food gathering) which was typical of South Island Māori. Also of significance is the collection of pounamu, important for cultural practices and trade.²³ Ngāi Tahu represents interests in both the commercial and non-commercial sectors.²⁴

Existing marine protection

The Fiordland National Park was established in 1952, although it encompasses the land area only. Due to its special character, Fiordland was successful in becoming a World Heritage Site in 1990, but again the designation stops at mean high water springs.²⁵

Prior to the establishment of the Fiordland Marine Guardians, two marine reserves were established in 1993 by the Fiordland Fisherman's Association which was part of the New Zealand Federation of Commercial Fishermen. These were the Piopiotahi (Milford Sound) Marine Reserve and the Te Awaatu Channel (The Gut) Marine Reserve. This was the first initiative to prevent further decline in these fiords and throughout the greater region.²⁶

There are mixed perceptions on what the underlying drivers for these applications were. Publically, local fishers say that the application arose solely from the observed decline in fish stocks and genuine concerns about deteriorating conditions. But fishers also admit that the action was in part strategic, designed to pre-empt environmentalists, through fishers retaining control over the placement of protected areas and avoiding areas of fishery significance.²⁷

Southland Regional Coastal Plan

The Southland Regional Coastal Plan became operative on 10 September 2008.²⁸ It describes the values and management issues for Southland's coast, sets out fundamental principles, and then deals with specific matters in its various sections. The Plan contains general rules that permit, limit or prohibit various activities. For example, these may place restrictions on disturbances such as noise levels, protect against threats such as bio-invasion, meet Treaty of Waitangi obligations, or identify areas of outstanding natural character.

There are some rules in the plan that pertain to the Fiordland coastal marine area and which address the specific needs of that special area. These rules include anchoring prohibitions in places containing particularly fragile species, marine farming prohibitions, and rules controlling the conduct of cruise ships and surface water activities.

Stakeholder body

Establishment

The Fiordland Marine Guardians initiative did not begin as a marine reserve designation process, nor was there any original intent to create a network of marine reserves. The increasing visitor interest in Fiordland had led to a corresponding increase in risk to the marine ecosystem due to damage from vessels anchoring, reckless diving practises, over-extraction of fish and bio-invasion of species

brought in on the hulls of visiting boats. A charter boat could deliver 15 divers to an area who could take 600 to 800 crayfish within legal recreational limits. These activities, in combination, were having a major effect on the inner fiord and concerned locals began looking for a solution.²⁹

Resident fishers observed that a 1994 working group approach in Southland, had successfully developed a fisheries plan for the Paterson Inlet (Stewart Island), assisted by the Ministry of Fisheries and the Department of Conservation. This initiative included various voluntary measures but eventually led to the establishment of a marine reserve in 2004.³⁰ Representatives from the fishing community approached the Ministry to assist with a similar initiative in Fiordland.

In 1995, the Guardians of Fiordland's Fisheries was established with representatives from commercial fishing, recreational fishing, charter boat operators and Ngāi Tahu.³¹ The group was able to secure Laurel Tierney as a facilitator. She had also facilitated the Stewart Island initiative in her capacity as Ministry of Fisheries Southern Regional Manager.³²

In 1999, the Guardians of Fiordland Fisheries became an incorporated society. In May 2005, the organisation changed its name to Guardians of Fiordland's Fisheries and Marine Environment Incorporated, reflecting its broader scope.

The group finalised the Fiordland Marine Conservation Strategy in 2003, which was later implemented through special legislation – the Fiordland (Te Moana o Atawhenua) Marine Management Act 2005. This Act not only implemented the Strategy, but it formally established the Fiordland Marine Guardians under section 12.

Purpose

The Guardians adopted a shared vision at their first meeting that provided them with a common purpose. The vision endured and was retained, word for word, in their Conservation Strategy:³³

“That the current quality of Fiordland's marine environment and fishery, including the wider fishery experience, be maintained for future generations to use and enjoy”

Once the Guardians were established as an incorporated society they agreed to the following objectives to be contained in their rules:

- To ensure sustainable utilisation of the finite fisheries resources, having regard to the special nature of the fiord marine environment

- To support the current fisheries management framework
- To ensure the rights of tāngata whenua, recreational, charter operators, commercial and other user groups are identified and recognised and that these groups are involved in fisheries management decisions
- To ensure an equitable allocation of access to the fishery resource for harvesting groups
- To prevent uncontrolled expansion of effort/harvest by any one group
- To identify information requirements/research needs
- To adopt a cautious and responsible approach to developing any new fisheries

On 29 November 2001 the objectives contained in the rules were altered slightly to read as follows (changes shown in italics):

- To ensure sustainable utilisation of the finite fisheries resources, having regard to the special nature of the fiord marine environment
- *Ensure the ongoing integrity of areas, habitats and communities of special significance*
- Support *overarching* fisheries management framework
- Ensure the rights of tāngata whenua, recreational, charter operations, commercial and other user groups identified and recognised are involved in fisheries management decisions *including access to the fisheries resources*
- *Encourage voluntary compliance and reinforce the view that non-compliance is unacceptable behaviour*
- Prevent uncontrolled expansion of effort/harvest by *all* groups
- *Take a proactive role in identifying and advocating research needs to obtain the necessary information for advancing the Guardians' objectives*
- Adopt a cautious and responsible approach to proposals *for new developments* including fisheries

The progression of name changes and rewording of objectives shows how the group matured and gained broader perspectives and a deeper understanding of the issues about which they were concerned.

The above objectives were superseded by the Fiordland (Te Moana o Atawhenua) Marine Management Act 2005. The purpose of the Act, among other matters, was to establish the Fiordland Marine Guardians to provide advice on fisheries management, biosecurity, sustainable management and marine preservation and protection. Section 13 of the Act specifies the Guardians' functions, which include providing advice and recommendations on relevant matters such as the effectiveness of management measures in the Fiordland Marine Area, facilitating and promoting integrated management in the Fiordland Marine Area, assisting management agencies in various roles and performing other functions as assigned.

Membership

Members were originally derived from those groups that were already active in Fiordland's fisheries such as commercial fishing organisations, the Southland Recreational Fisheries Association and charter operators. Selection was based on knowledge and experience of Fiordland's fisheries and the marine environment, a commitment to looking after the resource, a willingness to work with other interests and the time to invest in the group's operations.³⁴

The Guardians developed a draft 'conservation strategy' which they presented to the Minister of Fisheries and the Minister for the Environment in order to gain support and to investigate options for implementation. The strategy was initially rejected and improvements were demanded in two particular areas.³⁵

First, it required greater inclusion of conservation values and recognition of the entire environment. The strategy had not considered putting areas aside for reserves and had completely overlooked marine mammals.³⁶ Secondly, the process had excluded certain representatives, namely science and environmental interests. The Guardians justified this by arguing that inclusion of 'extreme' views, such as the environmentalists, would have prevented the cooperation and cohesion that the Guardians achieved and would have slowed down the whole process.³⁷ They believed it was better to have a good process rather than complete representation.

To resolve this issue, in June 2000 the Department of Conservation selected marine scientist Steve Wing to represent science issues, and then Dr Alan Mark was also invited onto the Guardians in 2002.³⁸ Although he was an alpine botanist, Mark was a member of Forest and Bird, had been involved in the 'Save Manapouri' campaign in the 1970s and was a Southlander. This gave him credentials to be

an environmental representative and acceptable to the Guardians.³⁹ Further representational changes included the appointment of Ian Buick, a helicopter operator, and Irene Barnes who represented community interests.⁴⁰

Unlike the statutory processes for the west coast of the South Island and the Sub-Antarctic Islands, where the community forums were appointed for a fixed term, the commitment by the Guardians is ongoing. Therefore, representatives change in order to share the workload across the community. All representatives live in the Fiordland area, the exception being the science representative. When an appropriately qualified science representative is not available locally, he or she will be selected from somewhere in the wider southern region.

Figure 16.2: Fiordland Marine Guardians members

Name	Role	Resident or Non-resident	Interest
Stewart Bull	Member Guardian (since inception)	Resident	Māori customary uses
Jerry Excell	Member Guardian	Resident	Commercial fishing (CRA8 rock lobster)
Ken Grange	Member Guardian	Non-resident (Nelson)	Marine science (NIWA)
Alan Key	Member Guardian (since inception)	Resident	Recreational fishing
Malcolm Lawson	Chairperson	Resident	Commercial Fishing (CRA8 rock lobster)
Alan Mark	Member Guardian	Non-Resident	Conservation and science (botany)
Anne McDermott	Member Guardian	Resident	Recreational use (diving, boating and fishing)
Mark Peychers	Member Guardian (since inception)	Resident	Commercial fishing and tourism (charter cruises)

Process

Overview

The Fiordland Marine Guardians process was characterised by the ‘ground rules’ adopted at the suggestion of their facilitator. These included a willingness to share information and knowledge, listen to others, discuss issues rather than positions, be committed to sustainability of the resource, cooperate and devote sufficient time to the process.⁴¹ The facilitator required the group to adopt a shared vision that would guide the group and keep members working towards the same goal.

Role of stakeholder body

The Fiordland Marine Guardians meet approximately every two months, usually in the Southland Regional Council offices. Their meetings are open to the public, and representatives from each of the supporting agencies attend, although voting and decision-making rights are vested only in the Guardians themselves.⁴² The meetings provide an opportunity for the key agencies involved in management of the marine space to meet, get updates on progress in other areas, discuss key issues and build relationships.

A significant rule or strategy was the concept of ‘gifts and gains’. This approach required members to offer a concession or offer up something that would benefit the group before they asked for a concession from another member or from the group. This approach helped neutralise potential conflict and maintained better group cohesion through the negotiation process.⁴³

Early concessions led to commercial fishers agreeing to stay out of the inner fiords completely and recreational fishers considerably reducing catch limits. In some ways, the concessions were not great, as there was little commercial fishing conducted in the inner fiords anyway and marine reserves were still not on the agenda.⁴⁴ The process minimised impact on existing users, which was also a key feature of the MPA Policy.

Role of statutory agencies

The Guardians were initially supported by the Ministry of Fisheries, but this support was short-lived as the Ministry focused on the quota-management system to manage fisheries and considered a community-led initiative to be outside of its mandate.

A significant source of funding came from the Ministry for the Environment through the Sustainable Management Fund. The decision to pursue support through the Ministry for the Environment rather than the Department of Conservation was partially because funding was more readily available through the Ministry, but the Guardians also thought that it provided a better organisational fit. The Department of Conservation was already involved in the management of the Fiordland National Park and individuals from that agency had expressed support for either extending the National Park boundaries or including the marine area as part of the World Heritage Site.⁴⁵ The Guardians did not want to be pressured by any party that had a preconceived 'agenda'. Even still, the affiliation with the Ministry for the Environment forced the Guardians to broaden their focus to managing the marine environment rather than just sustaining fisheries.⁴⁶

The administrative costs for the Guardians are borne by the Ministry for the Environment, whilst the other supporting agencies undertake their roles as part of their regional operating budget. The Guardians have an informal protocol to outline the relationship they have with the supporting agencies. Generally, the agencies take the lead in their area of expertise, whilst the Guardians will raise specific concerns about relevant issues, gather community opinion, and make recommendations where appropriate.⁴⁷

Information and science

The Guardians initially gathered their own information to respond to the issues identified, which the members then used to produce a number of community guides and codes of practice. The members published 'Beneath the Reflections: Caring for Fiordland's Fisheries', a code for responsible fishing practice.⁵³ The group later invited a number of scientists to collaborate in order to publish a survey of Fiordland's resources titled 'Beneath the Reflections: A Characterisation of Fiordland's Fisheries' (1999). The Guardians also published a listing of research publications which was called 'Beneath the Reflections: Fiordland's Fisheries and the Marine Environment, a Bibliography' (2001).⁵⁴

Figure 16.3: Summary of agency support

Agency	Involvement
Ministry for the Environment	Provides administrative support to the Guardians and the administering agency under the Fiordland (Te Moana o Atawhenua) Marine Management Act. It also leads the communication plan. ⁴⁸
Ministry for Primary Industries (formerly Ministry of Fisheries)	Leads compliance for fisheries regulations and is responsible for ensuring the sustainable use of fisheries resources. The Ministry also collects some scientific data in the Fiordland Marine Area. ⁴⁹
Biosecurity New Zealand (located within the Ministry for Primary Industries)	Primarily responsible for implementing measures that will reduce the risks of invasive organisms, one of the most serious risks to Fiordland fishing. ⁵⁰ A biosecurity plan co-ordinates measures taken by different agencies and individuals.
Department of Conservation	Responsible for the Fiordland Marine Area monitoring plan which determines methods by which the health of the marine environment can be assessed. It will also indicate how successful the management strategy is. The Department of Conservation is responsible for managing marine reserves under the Marine Reserves Act 1971. ⁵¹
Environment Southland	Responsible under the Resource Management Act for the sustainable management of the Fiordland marine areas' natural and physical resources. Environment Southland also assists the Ministry for the Environment in education and information distribution in accordance with the communication plan. ⁵²

As part of the information gathering process, it was necessary for the fishers in the group to share anecdotal information about their fishing practises. Although there was a considerable amount of information available already, and there was ongoing research being conducted on Fiordland in general, the information was insufficient considering the group's strong focus on fisheries sustainability.⁵⁵ To overcome this information gap, members volunteered details on their own fishing activities, and gathered further anecdotal evidence through their constituents and communities. This kind of knowledge was exceptionally valuable and enabled the Guardians to develop an intimate knowledge of their marine space. Members were able to identify areas of high biodiversity and those with fragile species termed 'china shops'. Proposed protection measures for these areas included anchoring prohibitions and diving regulations.⁵⁶ Some suggested that this local knowledge meant that they were able identify areas that needed protecting better than Wellington-based central government officials.⁵⁷

One of the key tasks to meet the requirements of the strategy was to define the habitat lines. Some boundaries between the inner fiords and the outer habitats were not clear cut so a project was initiated to develop a GIS model to indicate where the habitat lines would be. Five features were used to distinguish between the inner and outer habitat. These were: (1) surface salinity, (2) wave exposure, (3) bathymetry, (4) slope of the rock wall, and (5) aspect. Ground truthing was conducted in 2002 through sampling. The final lines were confirmed through the discussion of further information and management practicalities.⁵⁸

Public consultation

The 'Draft Integrated Management Strategy for Fiordland's Fisheries and Marine Environment' was made available for public comment in September 2002 through a series of public meetings. The final strategy was released in September 2003 as the 'Fiordland Marine Conservation Strategy: Te Kaupapa Atawhai o Te Moana Atawhenua'.⁵⁹

Outcomes

Stakeholder recommendations

In the strategy the Guardians identified around 45 issues that affected fisheries and the marine environment. These issues were grouped into four general components that were: (1) provisions to address fisheries management, (2) commitments to protect values of special significance, (3) identification of potential risks and measures to respond to them, and (4) expression of kaitiakitanga, recognising Māori ties to the area.⁶⁰

The strategy saw commercial fishers withdraw completely from habitat lines in the inner fiords, recreational fishers reduce bag limits, and a two-year closure of the blue cod fishery in Doubtful and Milford Sounds. Areas of special significance were protected with a range of restrictions including anchoring prohibitions, fishing closures and a code of practice for divers. There were also areas designated as no-take, although marine reserves were still not identified as the appropriate tool to achieve this.⁶¹

Fishing restrictions

The strategy includes a number of fishing restrictions. In Milford and Doubtful Sounds there is no commercial fishing inside the habitat lines. There is also a Fisheries Act section 186B temporary closure of the blue cod fishery for a two-year period that has been rolled over for another two years. This closure continues to be monitored and will be maintained if required. The groper daily recreational take is limited to two, with no accumulation, as is the rock lobster recreational take. In the remaining inner fiords, there is no commercial fishing inside habitat lines, and the daily recreational limit for blue cod, groper and rock lobster is three.

In the fiord entrances and outer coast, commercial fishing is controlled by the quota management system. The recreational limit for blue cod is set at 20 with no accumulation, groper is set to five with no accumulation and rock lobster is limited to six with an accumulation limit of 15 for three days. There are bag limits for scallop, paua, groper and sea perch for the fiords and outer coast. The total finfish take cannot exceed 30 and there are restrictions on various fishing methods including no scallop dredges, no set nets, no cod and rock lobster pots and limits on Dahn lines. Cod pots cannot be stored in china shops, but they can be stored in marine reserves outside of the season.

Government response

The Guardians initially looked to provisions in the Fisheries Act to implement the strategy, although the range of measures available were not well integrated to control impacts beyond those of fishing. Traditional customary fishery management measures were initially seen as having potential, but were later deemed inadequate for long-term protection. Eventually the Guardians met with the Minister for the Environment and the Minister of Fisheries to seek support.

The strategy was launched at a formal ceremony in Te Anau on the 6th of September 2003. In this public forum the Ministers committed to implement the strategy by September 2005 through special legislation.⁶² The Ministers delivered on this commitment which saw the creation of the Fiordland (Te Moana o Atawhenua) Marine Management Act 2005.

The Fiordland (Te Moana o Atawhenua) Marine Management Act 2005

The Act, containing three parts, establishes the Fiordland Marine Area with a unique management framework. This had the effect of formalising the Guardian's

strategy into law. The legislation included the creation of eight new marine reserves to protect the representative areas identified in the strategy. Through this Act, the Fiordland Marine Guardians are assigned an enduring role as government-appointed advisors.

Functions of the Guardians

Section 13 of the Act details the functions of the Guardians. These statutory obligations include providing advice and making recommendations to central and local government management agencies⁶³ and respective Ministers on: (1) management procedures in the marine area; (2) activities having an effect on the marine area; and (3) threats to the marine area. The Guardians are also tasked with facilitating and promoting integrated management, obtaining, sharing and monitoring information, assisting management agencies, conducting reviews when invited and performing other functions when requested. Any recommendations or advice must be ‘taken into account’ when exercising powers under the Act. Section 27 also requires an agreement or protocol to be entered into between the Guardians and the management agencies. This provides a basis for the Guardians to raise any procedural concerns with the Minister for the Environment.

Relationship to other legislation

The marine reserves within the Fiordland Marine Area are established in accordance with section 4(1) of the Marine Reserves Act 1971 and are therefore subject to the same high level of protection and preservation as any other marine reserve established under that Act.

The Guardians are to be appointed in the same way as any committee established under section 56 or section 6N(2)(b) of the Conservation Act 1987. Commercial fishing in the Fiordland marine area is governed by the Fisheries (Southland and Sub-Antarctic Areas Commercial Fishing) Regulations 1986.

Moratorium on further protection

The Fiordland (Te Moana o Atawhenua) Marine Management Act 2005 imposed a moratorium on creating any further marine reserves under the Marine Reserves Act 1971, until such time as seven years had passed since the commencement of the Act or a Ministerial review was conducted (scheduled to occur five years after

the commencement of the Act). The Ministerial review was conducted in the latter half of 2010. With the review complete, the moratorium is deemed to be expired.

Marine reserves

The eight marine reserves established by the Act are:

- Te Hapua (Sutherland Sound) – 4.49 square kilometres
- Hawea (Clio Rocks) – 4.11 square kilometres
- Kahukura (Gold Arm) – 4.64 square kilometres
- Kutu Parera (Gaer Arm) – 4.33 square kilometres
- Taipari Roa (Elizabeth Island) – 6.13 square kilometres
- Moana Uta (Wet Jacket Arm) – 20.07 square kilometres
- Taumoana (Five Fingers Peninsula) – 14.66 square kilometres
- Te Tapuwae o Hua (Long Sound) – 37.62 square kilometres

The two previously existing marine reserves are:

- Piopiotahi (Milford Sound) – 6.90 square kilometres
- Te Awaatu Channel (The Gut) – 0.93 square kilometres

The total marine area within Fiordland protected as marine reserves therefore totals 103 square kilometres, which is 1.01 per cent of the total area of the Fiordland bioregion. All the marine reserves are within the fiords. In comparison 12,000 square kilometres is protected in the National Park area.⁶⁴

Future

The Guardians continue to contribute to the management of the Fiordland marine area in various ways. An array of guides and publications such as user guides, annual reports, newsletters and media releases are available on their website as well as on the Ministry for the Environment's website.

In 2007, and then again in 2010, the Guardians conducted a user survey to investigate what activities people are conducting in Fiordland and to understand their perceptions and values associated with the area. The Guardians, in conjunction with support agencies, have a number of monitoring projects underway. The Fiordland Marine Area monitoring programme involves research commissioned in several areas over both biological and social issues, as shown in Figure 16.4.

Figure 16.4: Fiordland Marine Area monitoring programmes

Research	Lead Agency	Issue
Blue Cod closure	Ministry of Fisheries	Biological
Baseline biodiversity survey and biosecurity checks	Department of Conservation	Biological
Milford port survey	Biodiversity New Zealand	Biological
Biosecurity risk assessment	Biodiversity New Zealand	Social and biological
Catch/diary survey	Ministry of Fisheries	Social and biological
Fisheries management area user survey	All Agencies	Social

In 2008 the Guardians published 'Beneath the Reflections: A User's Guide to the Fiordland (Te Moana o Atawhenua) Marine Area'. This is a practical user guide for visitors to Fiordland containing information about the area's history, the environment, the establishment of the Fiordland Marine Area and all the marine regulations, including a fiord-by-fiord guide complete with maps and descriptions.

In response to the lifting of the moratorium, in 2011 the Guardians conducted a review of the rules governing the Fiordland marine area. The review considered the eight objectives of the strategy,⁶⁵ what was originally intended and what had happened since. The Guardians then sought public feedback and formal submissions to gain community views on what needs to be done. Submissions closed on 29 April 2011. The Guardians received no strong indications that change was required, and have not indicated an intention to make any radical changes to the measures they already have in place.⁶⁶ Therefore no applications for further marine reserves are anticipated in the near future.

The ongoing role of the Guardians in managing the Fiordland marine area is seen as important. As set out above, the Guardians provide a mechanism through which to integrate the management agencies and act as an intermediary with the community and general public. Their community involvement has a positive effect on compliance, reducing the enforcement burden, which can be a logistical challenge for agencies due to the remote nature of the region.⁶⁷ This increases the efficiency and effectiveness of the supporting agencies and enhances the cohesion of the central and local government agencies.

The Guardians are likely to continue to be supported as long as there are clear benefits being observed. The indication from agency officials is that they are

still seen to be contributing to the overall effectiveness of marine management in Fiordland and will continue to serve a purpose.⁶⁸ One of the key issues they face, however, is continual representation. There is pressure to keep rotating membership and to have new people appointed, but the number of people with appropriate skills, knowledge, experience, long-standing in the community and the time to contribute is limited. It is therefore difficult to keep bringing in new people.⁶⁹ There is also a risk that members will become ‘burnt out’ if they are required to remain on the Guardians for an extended period of time, for lack of an appropriate replacement.

Strengths of the approach

Process and consensus-building approach

There is potential for conflict wherever competing or overlapping interests exist. Within the Guardians, some of the strongest contention arose between the commercial and recreational fishers. They each placed a high value on continued use of the resource, but realised that the only way forward was to concede some of their usage. One of the notable approaches to consensus-building adopted by the group was the process by which stakeholders were prepared to relinquish benefits in the interests of ensuring the quality and sustainable management of the Fiordland marine environment and its fisheries. This strategy is referred to as ‘gifts and gains’ and enabled agreement to be reached on proposals and conservation strategies to progress.

The gifts and gains approach was only applied because Laurel Tierney, the Guardians’ independent facilitator, convinced the Guardians to adopt this technique. Her role was pivotal to the group achieving consensus. Using this approach, the commercial fishers withdrew beyond the habitat lines to the open coast and recreational fishers reduced their bag limits and abolished accumulated catch. The Guardian representatives then worked exceptionally hard to sell their strategy to the community.⁷⁰

Broad focus

The Fiordland Marine Guardians are different from the other marine protection initiatives canvassed, in that the creation of marine protected areas is not their primary focus. Their role is broader and ongoing, with continuous involvement

in implementation of the strategy and management of the Fiordland marine area. The Guardians approach keeps the community involvement strong and stimulates a sense of stewardship.

Broad stakeholder involvement during the early stages has been important to the ongoing success of this model. If not from the beginning, the Guardians model now involves wide representation from across the district including commercial and recreational fishers, divers, charter boat and tourism operators, environmentalists, scientists, community representatives and tangata whenua (Ngāi Tahu). This community-led approach, with strong local involvement, ensures stakeholder 'buy in' and agreement on major issues. Enforcement requires resources and the more 'self-enforcement' that occurs the more efficient the system will be.

The management approach considered a variety of conservation tools, including a large number of non-statutory measures to achieve the vision. Initiatives included voluntary measures that in combination contribute to the overall effectiveness of the scheme. These include the establishment of 'china shops' to protect small, discrete areas with species of special significance; habitat lines that distinguish between inner and outer fiords to prevent commercial fishers from fishing inner fiords; no-anchoring areas to protect particularly fragile communities and habitats which could be damaged by anchors, chains or ropes; and the 'Doubtful Sound dolphin protection zones' which offer non-statutory protection for bottlenose dolphins that reside in Doubtful Sound. Their insistence on developing new approaches overcomes the lack of statutory tools considered suitable for marine protection.

Weaknesses of the approach

Lack of conservation values

The process was principally led by commercial fishers, who may have had a number of motivations to initiate marine protection in their region. Firstly, recreational fishing and charter boats were not governed by the quota management system. Significant increases in boat numbers and people visiting the area, meant that overall numbers of fish being taken was increasing, whilst remaining within legal bag limits.⁷¹ This rise in activity was putting pressure on the fishery and highlighted tension between competing interests.

John Steffens, previous Chairperson of the Guardians, acknowledges that the ultimate motivation for forming the Guardians was to do something before more restrictive conservation measures were imposed on them.⁷² After the IUCN awarded Fiordland National Park world heritage status, there were suggestions that a marine park should be created that encompassed the coastal environment out to 200 nautical miles.⁷³ The implications for commercial fishers were immense. Royal Forest and Bird Protection Society was one of the environmental organisations which strongly advocated for total protection and hence from the outset was regarded by commercial fishers as radical. It was thought that ‘positional’ stances would compromise the consensus and halt the entire process.⁷⁴ Forest and Bird were initially excluded from the Guardians and from community dialogue about marine protection.

The perception that an environmental interest was too extreme to be accommodated within the Guardians had a direct effect on representation. The Guardians portrayed themselves to be an inclusive stakeholder-led initiative that represented all interests, but they were reluctant to include anyone with marine science or environmental interests until they were eventually directed to do so by the government.⁷⁵

It was also clear that even when Dr Alan Mark was added to the Guardians, as a newcomer and only one voice, little would change in the conservation strategy and any ideas of extending the national park status of Fiordland beyond mean high water springs were out of the question.⁷⁶ Of concern was the attitude taken towards submissions that mentioned either Forest and Bird or marine reserves. These were all side-lined as ‘extreme’ viewpoints and largely dismissed, despite often being based on extensive scientific research.⁷⁷

Dominance of use rights

There were concerns raised by the commercial fishers that people were ‘forced out’ of the industry. The rock lobster fishery has suffered severe decline in the past 15 years forcing the quota to be cut by 65 per cent and reducing the commercial fleet from 240 boats to 60.⁷⁸ Anecdotal evidence from older fishers indicates that the cray fishery in the 1950s was vastly different to what it is now. Fishing in an unregulated virgin fishery, boats would fill to capacity. However, within three years they estimate the fishery was reduced by half and within ten years fishers were recovering 70 per cent less than their original 1950s catch.⁷⁹ The announcement of the quota management scheme in the 1980s sent fishers into a harvest frenzy,

so that by the time quota management was introduced, stocks were so low fishers could not even catch their quota. Commercial fishers acknowledge the degradation that has occurred, and aim to reverse it, although regard trying to return the fishery to its original biomass as unrealistic.⁸⁰

Fishers go so far as to argue that taking the productive areas of the outer coast from them would be like taking land away from farmers.⁸¹ There is a failure to recognise that marine protection measures are also required for regeneration and have positive effects that may be beneficial to fishers and other users.

Effectiveness of tools

The effectiveness of the protection measures described in the strategy deserves closer scrutiny. Despite the establishment of ten marine reserves, there are none positioned in any of the fiord entrances or on the outer coast. The areas are also small. Therefore, protection of the full range of habitats and scientific baseline information has not been achieved.⁸² The commercial fishers argue that the quota management system sufficiently protects the fishery in these unprotected areas and nothing more is required.

Conclusions of the independent evaluation

A Ministerial review on the effectiveness of the management of the Fiordland marine area was completed in September 2010, as required in section 25(1) of the Act. The review evaluated how effective the advisory body (the Guardians), marine management measures and the protocol between the Guardians and management agencies have been in: (1) contributing to integrated management, and (2) in achieving the intent of the Act. Overall the review found very few shortcomings and many successes to report.

To improve, the reviewers suggested a more strategic approach was required to set priorities and make the Guardians less reactive. They also suggested further attention to succession planning for Guardians membership, in particular, considering scientific input.

The review found local awareness, understanding and support for the Guardians to be strong. They are highly regarded by the community and have a positive relationship with the management agencies and their respective Ministers. Subsequently, a high level of compliance from users and co-operation

with management regulations is experienced in the Fiordland marine area. The review also found that the high level of collaboration and cooperation between management agencies was essential to the successful operation of the Guardians. The usual 'turf battles' between agencies concerned about losing control or influence did not exist in this situation.

Two factors were identified by the review as being critical to the Guardians' success. The first was the 'gifts and gains' approach set out above which allowed for balance and negotiation. The second was a concept referred to as 'fish for a feed', which represents a shift in approach to recreational fishing. Instead of taking as much as possible on any occasion, fishers only take what they and their family require for a meal.

The review concluded that to really determine how successful the Fiordland Marine Guardians have been in the preservation, protection, and sustainable management of the marine environment and biological diversity of the marine area, monitoring results would need to be evaluated to determine if there had been improvement. A monitoring plan is in place, although the review found that insufficient research and monitoring information is available to draw any conclusions. More time and research is required to fully determine the effectiveness of the marine management measures.

Conclusion

Fiordland's Marine Guardians are the product of a local community's desire to better manage their marine resource. Although its focus was initially upon fisheries management measures, this expanded to include marine protection when the group received statutory recognition. The Guardians' work resulted in the development of a number of marine reserves and an ongoing management strategy for the Fiordland marine area. The ongoing commitment to long-term management helps to ensure that rules are enforced and management measures are responsive to changes in the marine environment.

Figure 16.5: Chronology of key events for marine protection in Fiordland

Date	Event
1993	Te Awaatu Channel (The Gut, Doubtful Sound) and Piopiotahi (Milford Sound) marine reserves formally established. Initially proposed by the New Zealand Federation of Commercial Fishermen.
1995	Guardians of Fiordland's Fisheries and Marine Environment (the Guardians of Fiordland) established.
2002	Draft Fiordland Marine Conservation Strategy published in October.
2003	Guardians of Fiordland present finalised strategy to the Minister of Fisheries and the Minister for the Environment who make a commitment to implement the strategy within in two years (by September 2005). The Ministers appoint an 'investigative group' (comprising representatives of central and regional government and the Guardians of Fiordland) to report on how best to implement the strategy.
2004	Government agrees to develop special legislation to give effect to recommendations of the investigative group's report.
2005	Commencement of Fiordland (Te Moana o Atawhenua) Marine Management Act 2005 (date of assent 20 April)
2010	A Ministerial review on the effectiveness of the management of the Fiordland marine area completed in September 2010

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chapter seventeen



West Coast Marine
Protection Forum

Introduction

The West Coast Marine Protection Forum was established in 2005, as a non-statutory body, by the West Coast Conservator of the Department of Conservation.¹ The purpose of the Forum was to consider how best to achieve integrated marine protection on the west coast of the South Island and to develop recommendations that satisfied the objectives in the New Zealand Biodiversity Strategy² and the principles of the MPA Policy.³

The geographical area to be considered by the Forum covered the West Coast Regional Council coastal boundary which extends 600 kilometres from Kahurangi Point in the north to Awarua Point in the south. The seaward boundary extends to the 12 nautical mile limit of the territorial sea and the onshore boundary is to mean high water springs.⁴ The area is known as the West Coast coastal marine area as defined by section 2(1) of the Resource Management Act 1991. This is also the boundary defined by the West Coast Regional Coastal Plan, in section 1.3 of the plan.

The research for this case study was based on information and documents publicly available on the West Coast Marine Protection Forum website.⁵ Further insights were gained through interviews conducted with Forum members, agency officials and other relevant experts. A full list of interviewees can be found in Figure 17.5 located at the conclusion of this chapter.

Context

Social and economic environment

The West Coast region has a land area of approximately 23,000 square kilometres and a population base of around 35,000 people.⁶ The area is divided into three territorial authorities:

- Grey District, containing the smallest land area but largest population
- Westland District, containing the largest land area and smallest population
- Buller District

The commercial centre of the region is Greymouth, with the other major towns being Westport and Hokitika. Human settlement began hundreds of years ago when Māori migrated south for mahinga kai (food gathering) and pounamu (New Zealand greenstone).⁷ Later, Europeans explored the coast leading to the West

Coast gold rushes of the 1860s bringing an influx of people. Settlements tended to concentrate around the river mouths, as they provided an important mode of transport. Coastal shipping remains important on the West Coast with commercial ports based in Westport and Greymouth.⁸

The West Coast has an unusually high proportion of land administered by the Crown. Seventy-eight per cent of the region's land area is allocated to Crown conservation estates and provides the largest area of unmodified indigenous forest in the country. The rest of New Zealand has only 26 per cent of land administered in this way.⁹ Only 21 per cent of the land is fully or partly ratable, creating significant challenges for local authorities in raising revenue for public services. The West Coast is the highest rated region in the country per head of population.¹⁰

Rich in natural resources, a substantial proportion of the West Coast's economic base comes from primary and extraction industries such as farming, mining, fishing and forestry. Agriculture, forestry and fishing are the region's largest employers and can be adversely affected by a strong kiwi dollar as most of the produce is exported.¹¹ Other industries include the sale of pounamu goods, the propagation of sphagnum moss and coal mining.

There is a quickly growing tourism sector that is becoming increasingly important, accounting for over nine per cent of employment and six per cent of gross domestic product. This is higher than the national average which are 5.2 and 4.1 per cent respectively.¹² The West Coast is rich in scenic attractions such as the Pancake Rocks, Franz and Fox Glaciers, rivers, lakes, heritage trails and national parks, all with high recreational value.¹³

The West Coast Marine Protection Forum included representation from Te Rūnanga o Ngāti Waewae and Te Rūnanga o Makaawhio which are both Ngāi Tahu Tai Poutini rūnanga (governing council of a Māori subtribe).¹⁴ Historically, the coastal area provided a traditional food resource as well as a means of transport. Today, the coastal marine area contains places of strong cultural value to Māori including wāhi tapū (sacred spiritual) sites, taonga (treasures), areas of food gathering such as reefs, islands and fishing grounds, tauranga waka (waka launching/landing places) and pā sites.¹⁵

Continued access to areas of cultural value is important to Māori to provide for their cultural, economic and social wellbeing. Tāngata whenua desire the freedom to exercise their role as kaitiaki (guardians), a responsibility that involves taking active measures to protect or conserve the resource in a traditional manner.¹⁶ Several taonga species and particular sites are recognised formally in the Ngāi Tahu Claims Settlement Act 1998. These species include marine mammals and

birds such as the southern right whale (tohora), blue penguin (korora) and titi; plants such as flax (harakeke) and bull kelp (rimurapa); and fish and invertebrates such as pipi.¹⁷ Customary fishing is regulated through the Kaimoana Customary Fishing Regulations 1998.

The West Coast is part of a much larger 'Challenger' Fishery Management Area, which is also known as the Fisheries Management Area seven (FMA 7).¹⁸ This encompasses the top of the South Island, the Marlborough Sounds, Golden Bay, Tasman Bay and the West Coast. It encompasses an ocean area out to the 200 nautical mile exclusive economic zone.¹⁹

FMA 7 contains three fisheries: (1) challenger finfish, (2) challenger scallops and oysters, and (3) challenger shellfish. In 2010, the total allowable commercial catch for finfish was around 29,045 tonnes, but the actual total reported commercial catch was much less than this, at only 19,656 tonnes.²⁰

Port facilities, proximity to fishing grounds and abundance of sealife result in an expanding fishing fleet during various fishing seasons as vessels from other ports base themselves on the West Coast during that time.²¹ Inshore fisheries include blue cod, snapper, gurnard, John dory and tarakihi, whilst deepwater takes include hoki, tuna, squid and orange roughy.

For recreational fishers, this area is regulated by the Fisheries (Challenger Area Amateur Fishing) Regulations 1986 (SR 1986/224). These regulations set the maximum daily number of fish which can be taken (known as the bag limit) minimum fish length, minimum size mesh for nets, closed seasons and net mesh size.²² In the Challenger area, it is estimated that 20 per cent of the resident population are recreational fishers.

Natural environment

The West Coast has a unique marine ecology, with physical differences that are distinguishable from other parts of New Zealand.²³ The 600 kilometre stretch of coastline from Kahurangi Point to Awarua Point is highly exposed, subject to prevailing westerlies, heavy mountain (orographic) rainfall and high energy wave action. The beaches are dominated by sedimentation originating from glaciers.²⁴ The numerous rivers and streams carry a high sediment load and supply the beaches with sand and gravel.²⁵

The biological characteristics of the region are directly influenced by the interaction of a variety of geophysical factors which include:²⁶

- Ocean and coastal currents and freshwater inflows

- Tides, wave action, weather patterns and sea temperature
- Seabed and coastal topography
- Geology and geomorphology including sedimentation and river discharges

In the northern part of the coastline the seabed extends gradually out to the Challenger Plateau. However, south of Hokitika a series of submarine canyons divide the topography, the most significant of these being the Hokitika and Cook Canyons.²⁷

The West Coast geology is layered with basement rock dating back 300 – 540 million years to the Gondwana continent. More recent sedimentary formations partially cover this basement layer. This softer rock is more susceptible to weathering, creating unique features such as the ‘pancake rocks’ seen near Punakaiki.²⁸ Offshore, the seabed consists of Cretaceous-Tertiary sedimentary rocks, overlaid by sediment discharged from rivers.

The West Coast is home to an abundance of large marine mammals such as whales, dolphins and seals including the endangered southern right whale, the endangered Hector’s dolphin (three quarters of the total population), the endangered sub-population of bottlenose dolphin, as well as the southern elephant seal and orca both listed by the Department of Conservation as critical.²⁹ Recreational and commercial set netting is seasonally closed in many areas to reduce incidental by-catch of Hector’s dolphins.³⁰

Birdlife is also plentiful, ranging from albatross and petrels to the threatened Fiordland crested penguin. The beaches, estuaries and rocky shores provide habitat for shorebirds such as terns, gulls and oystercatchers, whilst estuarine tidal flats and salt marshes provide roosting and feeding grounds for wading birds and waterfowl such as godwits, stilts, ducks and heron.³¹

Fish species in the nearshore shelf environment number over 90 whilst the coastal reef and estuarine areas boast 86 species, including nine threatened species. The species of invertebrates present are estimated to be in the hundreds.

Seaweed provides food and shelter for invertebrates and fish. The largest of these is bull kelp, but there are many smaller varieties. Vascular plants exist in the coastal margins and intertidal estuarine environments. Free-swimming plankton gathers in certain sea conditions and forms an essential component of the marine foodweb.³²

The MPA Policy identifies depth, substrate and exposure (energy) as being of most importance to determining different marine habitats. The ecosystem types that result from this classification system can be numerous and complex, therefore, the West Coast Marine Protection Forum utilised a simplified classification of the

physical environment and identified four 'environmental domains': estuarine, intertidal (open coast), shallow subtidal (open coast), and deep nearshore.

Existing marine protection

The West Coast Marine Protection Forum asked the management agencies to assess the level of protection afforded to existing protected areas against the MPA Policy protection standard. This 'gap analysis' was completed prior to the drafting the 'consultation document'. Four sites were identified as meeting the protection standard; Saltwater Lagoon, Five Mile Lagoon, Cook River mouth and Three Mile Lagoon

Apart from Cook River Lagoon, these sites are all 'coastal protection areas' identified in the Regional Coastal Plan. There are no specific rules associated with these areas in the plan, only objectives to provide management guidance.

The beds of Three Mile Lagoon, Five Mile Lagoon and Cook River are all within the Westland Tai Poutini National Park and are therefore protected by the National Parks Act 1980. Section 4(2) of the Act ensures that national parks are administered and maintained to preserve them in their natural state (so far as possible), and their value as soil, water and forest conservation areas is maintained. The bed of the Saltwater Lagoon is a scenic reserve and is also afforded sufficient protection to meet the policy standard.

Stakeholder body

Establishment

The West Coast Marine Protection Forum was established in 2005 by the West Coast Conservator of the Department of Conservation, pursuant to section 53(2) (b) of the Conservation Act 1987 which states (under powers of the Director General), *"the Director-General may ... conduct or commission research or study"*.

The West Coast was chosen as a test case for the incoming MPA Policy for a number of reasons. The area is less densely populated compared to many other regions and there were no existing marine reserves or any other form of marine protection. In addition, staff from the West Coast Tai Poutini Conservancy office, were enthusiastic about getting a process underway and were optimistic that it could work.³³ They approached local mayors to get the political support to proceed.³⁴

The Forum first convened in April 2005, prior to the MPA Policy being drafted and released in December 2005. The Forum had been issued with a terms of reference to assist with procedural matters. New terms of reference were then issued in February 2008 with the release of the 'Marine Protected Areas Classification, Protection Standard and Implementation Guidelines' by the Ministry of Fisheries and the Department of Conservation, well after the Forum had commenced its process. The Forum decided to retain its original terms of reference, however, for continuity and consistency.³⁵

Purpose

The purpose of the West Coast Marine Protection Forum is derived from the higher objective of the New Zealand Biodiversity Strategy; to build a comprehensive network of marine protected areas. The principal objective of the Forum was to consider how best to achieve integrated marine protection on the West Coast, consistent with the objectives of the New Zealand Biodiversity Strategy and the principles of the MPA Policy. Therefore, the Forum was to make recommendations for a range of protection measures to be used to safeguard marine and coastal habitats and ecosystems that are representative, outstanding or rare.³⁶ The terms of reference also stipulate that a marine reserve is to be used to protect at least one representative example of each habitat or ecosystem type.

The Forum adopted the following mission statement which was included in its terms of reference: "*That the character and quality of the West Coast – Tai Poutini marine environment be protected, maintained or improved for present and future generations to use and enjoy.*"³⁷

The Conservator made it clear to the Forum that it was to work with the management agencies (Department of Conservation and Ministry of Fisheries) and with the community of the West Coast to achieve its objective.³⁸ To achieve this, the Forum was to consult the wider community, including stakeholders, user groups and tāngata whenua, and to facilitate opportunities for all those who may express an opinion on marine protection, regional priorities and the various options to be considered.³⁹

Membership

The Forum comprised a selection of representatives that came from the stakeholder community, but did not necessarily represent all users or interest groups in the area. The past and current Forum members are listed in Figure 17.1.

Some members were not able to remain for the whole process and were replaced by representatives from their interest groups. Overall, the Forum had a good level of continuity, with 12 of the 14 members remaining throughout the process, which took five years and three months to complete.

The West Coast Tai Poutini Conservancy office in Hokitika orchestrated the selection process for the Forum. Staff approached individuals in the community who represented key interests, who had influence and connections in the community, and preferably who represented more than one interest.⁴⁰ The composition of the Forum would have been slightly different if the 2008 Terms of Reference had been issued from the outset, as these require the inclusion of a both a science and a mining industry representative.⁴¹

The Forum was advised by Murray Reedy (Department of Conservation), Campbell Robertson (Department of Conservation), Don Neale (Department of Conservation) and Nicola Pindur (Ministry of Fisheries). Administrative support was provided by the West Coast Tai Poutini Conservancy of the Department of Conservation and by Andrea Jackson (Andrea Jackson and Associates, environmental planning consultancy).

Figure 17.1: Members of the West Coast Marine Protection Forum

Name	Role	Location	Interest
Bruce Hamilton	Chairperson	Westport	Community
Ian McKenzie	Member	Westport	Commercial fishing industry
Stuart Thompson	Member	Greymouth	Commercial fishing industry
Eugenie Sage	Member	Banks Peninsula	Royal Forest and Bird Protection Society
Geoff Rowling	Member	Motueka	Recreational Fishing
Emily Arthur	Member	Christchurch	West Coast Tai Poutini Conservation Board
Carrol Browne	Member	Fox Glacier	Community and tourism
Kerry Eggeling	Member	Okuru	Commercial fishing industry
Dennis Shanahan	Member	Dobson	West Coast Regional Council
Bruce Watson	Member	Hokitika	Environment and community
Brad Stenhouse	Member	Greymouth	Recreational fishing and diving
Helen Rasmussen	Member*	Greymouth	Te Rūnanga o Makaawhio, Ōkuru

Te Whe Weepu	Member*	Hokitika	Te Rūnanga o Ngāti Waewae, Arahura Pa
Carol Scott	Member	Nelson	Commercial fishing industry
Ben Hutana	Member**	Ross	Te Rūnanga o Ngāti Waewae, Ross
Richard Wallace	Member**	Hokitika	Te Rūnanga o Makaawhio, Hokitika
Rick Barber	Member*	Greymouth	Te Rūnanga o Ngāti Waewae, Greymouth/Mawhera
Peter Coburn	Member***	Westport	Community and education

* Original members of the Forum, but stood down midway through the process

** Current members of the Forum who were appointed mid-proceedings

***Original Chairperson of the Forum (2005-2006), but stood down midway through the process

Process

Overview

Based on a consultative process, the Forum was expected to evaluate a range of options that achieve the main objective. These options would include mechanisms for statutory protection available under Fisheries Act regulations, the Marine Reserves Act or other legislation if appropriate.⁴² The process was undertaken in three phases:

- Phase one: information gathering and facilitation of stakeholder engagement
- Phase two: evaluation of the advice and viewpoints provided in order to develop options
- Phase three: reporting options back to the Department of Conservation, the Ministry for Primary Industries and other agencies as required⁴³

Role of stakeholder body

The Forum first met in April 2005 and then a further 29 times before producing its 'recommendation report' in July 2010.⁴⁴ Meetings would typically last one to two days and would often include a tour or field trip.⁴⁵ Members of the Forum would

receive a meeting fee and reimbursement of travel costs from the Department of Conservation in accordance with the State Services Commission fees framework for members appointed to bodies in which the Crown has an interest.⁴⁶

This payment recognised that one or two meetings every month was a significant commitment by the Forum members who had to take time out from their primary occupations to attend.⁴⁷ There was also a certain amount of informal consultation that needed to be conducted between meetings. Participants agreed that they could not have committed any more time than this.

The Forum was able to use an independent facilitator to assist in progressing discussion, decision-making and achieving consensus. Sarah Wilson of Adventure Coast Consultancy Limited worked with the Forum from the beginning through to the production of the final recommendation report. Wilson has a masters degree in natural resource management and is a *Zenergy* trained facilitator. She has extensive experience facilitating conservation and environmental processes and in working with both the Department of Conservation and the Ministry for the Environment.⁴⁸

The facilitator's role was to work with members of the Forum to educate them on how to make trade-offs and negotiate. The facilitator was used principally at the beginning of the process, and later when agreements and decisions needed to be made, prior to public consultation. Wilson described the West Coast Marine Protection Forum as one of the more difficult projects she had been involved in, as the nature of the issues were complex, and required a number of trade-offs to be made.⁴⁹

Having a facilitator allowed the Chairperson to step back from his role as Chair and to be involved in Forum discussion. The facilitator was able to urge Forum members to refocus, to tackle difficult issues and to make decisions when they were reluctant to do so. She also provided technical assistance in drafting the consultation and recommendations reports.⁵⁰

The facilitator convinced a majority of the Forum to present only one option for each proposed site in the consultation document. It was anticipated that this approach would attract more meaningful submissions. This was because the alternative of presenting several options, would likely polarise submitters into selecting either greater or lesser protection, depending on how they were philosophically aligned. Unfortunately, the same agreement could not be achieved for the recommendation report. Discussions broke down between commercial fishing and environmental representatives just prior to producing recommendations for the Ministers.⁵¹

Role of statutory agencies

The Forum was allocated assistance from a number of organisations and individuals to inform or support the process and aid decision-making. Both the Department of Conservation and the Ministry of Fisheries assigned multiple staff members (six to seven each) in a support role. These government agencies were jointly responsible for developing and implementing the MPA Policy. Department and Ministry representatives provided science and policy advice where requested and were regarded as *ex officio* members of the Forum.

The Department of Conservation acted as the lead agency, resourcing a majority of the Forum's operational and administrative needs, whilst the Ministry of Fisheries acted as a partner agency meeting its own costs. Having the agencies work together was considered an advantage and Forum members agreed that the officials from both these agencies worked hard to provide everything the Forum needed to operate.

Although individual Ministry of Fisheries officials were highly praised for their contribution, a combination of internal restructuring, high staff turnover and restrictions imposed by confidentiality agreements, left many Forum members frustrated.⁵² The Ministry could not always provide information about fishing activity to the level of detail the Forum desired.

The West Coast Regional Council had a role to play through the preparation of the Regional Coastal Plan (which applies from mean high water springs out to the 12 nautical mile extent of the territorial sea) in accordance with Schedule 1 and Schedule 1A of the Resource Management Act 1991. The Act states that a Regional Coastal Plan must exist at all times.⁵³ Generally, activities that produce adverse effects in the coastal marine area require a resource consent, unless expressly permitted by a rule in the plan. Activities that have only minor effects such as swimming, kayaking, fishing and surfing do not need a resource consent unless there is a specific rule in the plan saying that it does.

Information and science

The first task undertaken was to gather all the information and knowledge available on the West Coast coastal and marine area, to produce a compendium of facts, prior to the commencement of deliberations. Don Neale, Marine Technical Support Officer for the Department of Conservation, acted as lead author and assembled the majority of information.⁵⁴ Additional staff from the Department of Conservation and the Ministry of Fisheries acted as contributing authors,

researchers and advisors. The Forum also formed a steering committee to assist with the development of the report.⁵⁵ The draft document, although not peer reviewed, was put out for public consultation. Further information was included following submissions and the final document adjusted.

Although this was time consuming and laborious, taking around 18 months to complete, the process gained buy-in from Forum members and enabled members to trust the information provided. It also provided a medium for discussion and the exchange of ideas before any difficult decision-making needed to be done.⁵⁶ The 'environment report', which was completed in April 2007, became an essential guide and was referred to frequently throughout the process.⁵⁷

There was a period of time where the process halted, due to the MPA Policy not having been issued, as the supporting agencies were still negotiating the wording of the policy and protection standard. The Forum was established in April 2005 and the MPA Policy was released in December 2005. This held up the overall process by several months. There was pressure on the Forum to make progress. However, members decided that it would be prudent to halt proceedings until clear direction had been provided and the policy confirmed.⁵⁸

The principal science advisor to the Forum was Don Neale from the West Coast Conservancy. Although not a marine scientist, he had a strong science background in coastal geography and around 20 years experience with the Department of Conservation on the West Coast.

The Department of Conservation provided all the GIS product required by the Forum.⁵⁹ The GIS team, managed by Gary Eason, faced numerous challenges to gather sufficient data to produce even basic maps of the marine area. Maps overlaying exposure, depth and substrate were required to classify and identify habitats according to the MPA Policy.⁶⁰

Very little digitised information was available on the marine area. The GIS team often encountered data that was in an unsuitable format. The Department had to scan and digitise old maps and work with maps of differing scales and co-ordinate systems. Some of the substrate data was extrapolated from beach surveys conducted from the shore by the Department of Conservation.⁶¹

Although NIWA had additional information, it was unable to provide raw data due to commercial issues, as the NIWA database of seabed information is owned by TUMONZ (The Ultimate Map of New Zealand). Prospectors were happy to give copies of maps they had, as they provided their mapping information to the Ministry of Business, Innovation and Employment anyway. The GIS team found that, in general, inter-agency support was good and if Land Information New

Zealand or the Ministry of Fisheries had information available, they would pass it on.⁶²

Public consultation

To begin with, the Forum held five public meetings in the main centres on the West Coast to introduce the key issues and objectives to the community. These were well attended. The environment report also went through a public consultation process before being finalised.⁶³

In preparation for the main consultation phase, eight public meetings were held in July 2009, informing the public on both substantive and procedural issues. Other methods employed to engage the public during the consultation phase included mailouts to Treaty of Waitangi partners and key stakeholders, public notices, placement of key documents in public places such as libraries, Department of Conservation and Ministry of Fisheries offices, and information centres, and by placement of all documentation on the Forum's and agencies' websites.

The Forum received a total of 1,130 submissions. Hearings were held in late September in Nelson, Westport and Hokitika so that submitters had the opportunity to provide further information or elaborate on their written submissions.⁶⁴ Individuals sent 800 submissions using a standard form from the Greenpeace website. The remainder were from recreational fishing and hunting interests (39 per cent), individuals and community groups (36 per cent), environmental interests (11 per cent), the fishing industry (nine per cent), Māori (three per cent), local government (one per cent) and the minerals industry (one per cent).

Two-thirds of the community and individual submitters supported the proposals, although some suggested that the protection was inadequate and needed to be larger. A third opposed the proposals on the grounds that they denied use rights. Recreational fishers and hunters opposed certain proposals, namely Taranga Bay and Ōkārito. Ngāi Tahu supported most of the proposals, although it had concerns where they affected mātaihai reserves. Other Māori organisations raised concerns about a failure to actively protect customary non-commercial fishing and opposed the use of marine reserves as a preferred tool. Environmental non-governmental organisations, individuals and Conservation Boards supported the proposal at Kahurangi, although they thought that the other areas were too small and did not reflect international best practice. Members of the commercial fishing industry opposed many aspects of the proposals. They

claimed that the process lacked transparency, the process was not inclusive, the Forum failed to present alternatives, the educational showcases were outside the Forum's scope, there was no economic impact analysis, the process was not coordinated, and there would be adverse effects on certain fisheries.⁶⁵

Outcomes

Stakeholder recommendations

The Forum made three levels of recommendations to the Ministers in its recommendation report. It first made recommendations on four primary locations for marine protected areas at Kahurangi, Punakaiki, Ōkārito and Gorge, shown in Figure 17.2. Alternative protection options were presented for each of these locations. Secondly, it presented five additional educational showcase sites, which were intended to increase public understanding and appreciation of the marine area. Finally, the Forum made some general recommendations unrelated to locations or protection measures, but regarding administration and procedures.

Kahurangi

The Forum chose Kahurangi as a primary site for protection, because of its diversity of fish and invertebrate species, and importance for marine mammals. It also adjoins Kahurangi National Park and the Heaphy River catchment maximising an area that is already protected on land.

Two options were presented for this location. Option A was supported by the major environmental groups, non-governmental organisations and many individuals. It proposed the creation of a large marine reserve and was the same as the option proposed in the consultation document. Option 2 was proposed by a commercial fishing stakeholder and was supported by most submitters from the fishing, petroleum and minerals industries. This would see the creation of a much smaller marine reserve and the use of a Type 2 marine protected area using Fisheries Act regulations.

Figure 17.2: Locations of proposed West Coast marine protected areas and educational showcase sites

Source: West Coast Marine Protection Forum



Punakaiki

Punakaiki was chosen as a primary location due to its marine habitats and ecosystems but also for its international significance and tourism value due to its proximity to the Pancake Rocks, Dolomite Point blowholes and Paparoa National Park.

Three options were presented at this location. Option A was the same as the option in the consultation document and was supported by many on the Forum and by submitters as a reasonable compromise. It consisted of both a marine reserve and an additional area protected by Fisheries Act regulations. Option B

was proposed by an environmental representative on the Forum. Many submitters valued the importance of having at least one protected area that extended from the mean high water springs to the seaward boundary of the territorial sea. This included a large marine reserve and two areas controlled by Fisheries Act regulations. Option C differed from the consultation document also and was proposed by a commercial fishing representative. This option allowed the paua industry access to the rocky reefs at the northern end of the site. It included a smaller marine reserve and two areas to be controlled by Fisheries Act regulations.

Ōkārīto

Ōkārīto was chosen for its range of marine habitats and support for a range of representative marine life. It is also adjacent to the Ōkārīto Lagoon and to part of Tai Poutini National Park and Te Wāhipounamu South West New Zealand World Heritage Area. In the approximate vicinity are a major Department of Conservation visitor centre and the Franz Josef glacier. Parts of the area are fairly remote and large areas remain unmodified. The biodiversity value is high and the lagoon is a spawning and nursery area for fish.

Two options were presented for Ōkārīto. The first, Option A, extended 26 kilometres from Waitahi Bluff to the southern end of Waiho Beach from mean high water springs to two nautical miles offshore. It did not contain any protection for the lagoon due to user opposition, leaving that habitat type unrepresented in the Forum's proposals. However, it was agreed by the Forum that Three Mile Lagoon, Five Mile Lagoon and Saltwater Lagoon already met the MPA Policy protection standard. There was a mātaītai reserve application over the lagoon area submitted by Te Rūnanga o Makaawhio to the Minister of Fisheries. Option B was proposed by a commercial fishing stakeholder representative, which offered less protection, approximately seven kilometres along the coast from Five Mile Lagoon to the southern end of Waiho Beach.

Gorge

This location was selected as an area for marine protection for its examples of southern latitude West Coast marine habitats and notable wildlife including marine mammals and birds. This particular area is adjacent to protected land that is part of the Te Wāhipounamu South West New Zealand World Heritage Area. It is considered remote and is difficult to access from the land.

Two options were presented for Gorge. The first option was similar to that consulted on and included a marine reserve from Hackett River to Long Ridge Point as well as a Type 2 marine protected area from Awarua Point to around Barn Bay, although not all this area is connected to the coast. The second option was similar except that the area of marine reserve would be much larger, extending out to the 200 metre isobath.

This location would be the only area within mainland New Zealand where a marine reserve or some form of protection would extend to the edge of the continental shelf and the deepwater canyons. This would provide a unique opportunity for research across the spectrum of depths from intertidal to deepwater. There is also continuity with mountain to sea protection connecting to marine protection.

Education sites

In addition to the primary sites the Forum proposed small marine reserves at five sites: Siberia Bay (Cape Foulwind), Mahinapua Creek/Tuwharewhare, Ship Creek, Hapuka Estuary and Jackson Head. They were selected for their proximity to populated areas and accessibility to the general public. The rationale for education sites was to raise awareness of the benefits of marine protected areas and to increase community support for them.

Government response

On receiving the recommendation report, the Ministers requested that their officials draft an advice paper before making a decision.⁶⁶ This paper was completed on 2 June 2011 and the Ministers' decision was released through a media statement on the 26 August 2011.⁶⁷

The officials' advice was that Kahurangi B, Punakaiki A, Ōkārito B and Gorge A options provided the best balance between biodiversity protection and minimising adverse impacts on existing users of the West Coast marine environment. These options were all the smallest options in each of the primary sites presented by the Forum. The Ministers were advised that the educational sites were additional to the requirements of the MPA Policy but that they could consider them on their merits.

The Ministers announced that the new reserves would be Kahurangi (84.66km², Option B), Punakaiki (35.58km², Option A), Ōkārito (46.41km², Option A) and Gorge (8.47km², Option A). They decided in favour of one educational site located

at Ships Creek near Haast (0.16km²). This site was reported in the recommendation report to have had the least objections to it.

A total of 175.28 square kilometers of Type 1 marine protected area protection was endorsed by the Minister. Once established, Kahurangi and Ōkārito will be the two largest marine reserves on New Zealand's mainland coastline. Type 2 marine protected areas, covering another 95.57 square kilometers, will also adjoin the Punakaiki and Gorge marine reserves using Fisheries Act tools to protect them from bottom trawling, dredging and Danish seining.⁶⁸

The overall effect was to increase full (Type 1) marine protection of the West Coast bioregion from zero to 1.3 per cent. The impact on New Zealand's target to achieve 10 per cent protection of the territorial sea was to increase the area of Type 1 marine protected area protection from 6.9 to seven per cent.⁶⁹

Future

The proposed marine protected areas will now be implemented through their respective statutory processes. The Department of Conservation project manager in Hokitika, Campbell Robertson, is responsible for overseeing completion of this task, which is estimated to take another 12 months. The marine reserve proposals were formally notified under the Marine Reserves Act on 22 June 2012. The proposed fisheries regulations were notified at the same time. Submissions on all the proposals closed on 22 August 2012.

Strengths of the approach

Stakeholder forum

It was advantageous to have all stakeholders 'around the table'. The commercial fishers, especially the owner-operators, proved invaluable in providing the information about commercial trawl activity, including size and locations, that the Ministry of Fisheries was unable to provide. Representational imbalance was further reduced by the Forum's drive for consensus rather than majority voting.⁷⁰

The fishing representatives on the Forum did not represent homogenous interests. Three of the commercial fishing members represented small owner-operators from local areas, who either own quota or purchase annual catch entitlements, whilst the fourth member represented Challenger Finfish, the commercial stakeholder group representing all finfish quota owners in FMA 7

and 8. Commercial rock lobster, paua and eel fishers were not represented on the Forum. Recreational fishers were concerned about use rights to areas that competed with commercial fishing. Likewise, customary fishing interests had separate concerns. Iwi felt that in supporting the Forum process, the Crown failed to protect their interests because it did not include a component dedicated to establishing customary fishing areas.⁷¹ In response, Ngāi Tahu continued to further its own project to protect customary fishing which involved the submission of mā taitai applications to the Minister of Fisheries.⁷²

When offshore commercial fishing representatives requested a place on the Forum, this placed significant pressure on the process.⁷³ Until this point, the Forum had only considered impacts on inshore fishing due to the seaward limit being the 12 nautical mile territorial sea. However, when considering the areas in and around the submarine canyons, some parts were within the 12 nautical boundary, but deeper than 200 metres. Commercial fishing interests in this situation are associated with offshore fisheries. Offshore fishers wanted to be represented on the Forum if their 'patch' was to be discussed and they requested places for three representatives. The Forum was concerned that expanding membership would increase the complexity of the issues and make negotiations even more difficult. The Forum therefore made a decision to limit the scope of their deliberations to the 200 metre depth contour, excluding deep water areas from their consideration, thereby eliminating the need to include offshore fisheries.⁷⁴ The deep water areas may be considered in a separate process, most likely using a technical advisory group.

Independent facilitator

The appointment of an independent facilitator was beneficial to keep the Forum focused on its tasks and to assist with decision-making. There was unanimous support amongst Forum members and agency officials for the use of an independent chairperson or facilitator. Although the Forum exceeded its time frames, and did not reach consensus, it may not have progressed to delivering recommended options had there not been an independent facilitator. The role of the facilitator was to apply the policy and keep the Forum on track towards achieving its outcomes. It was important that the facilitator be independent and have relevant knowledge and skills in the area.

The Chairperson, as a member of the community, was not independent. The facilitator was able to take over the running of the meeting at times, which allowed the Chairperson to participate in the discussion and express an opinion.

The facilitator also strongly encouraged members to reach agreement on four primary sites with protection measures that all members of the Forum could at least 'live' with. There was mixed opinion over whether the five educational sites should proceed, but due to their small size, these were less of an issue. As a result, the Forum was able to go into the public consultation phase with some degree of agreement.

Commitment of members and agencies

The commitment of the individual participants to remain involved in the process is noteworthy. The members of the Forum were involved for five years, meeting once every month or two during this time. Many of them travelled a significant distance to meetings, from as far as Christchurch and Nelson. Twelve out of an original 14 remained from start to finish. The local knowledge that the Department of Conservation used from within its West Coast Conservancy office when selecting people enabled it to find those with a high level of commitment and perseverance.

Forum members reported that they were well supported by the officials assigned from the Department of Conservation and the Ministry of Fisheries providing policy and science advice. The process was predominantly funded by the Department of Conservation which also provided technical support such as GIS mapping. The policy framework was the area in which the Forum needed the most technical guidance. Don Neale presented and advised on matters pertaining to science and was an appropriately experienced and knowledgeable individual to do so. His professionalism and dedication was praised by many. Murray Reedy and Campbell Robertson of the Department of Conservation also played pivotal roles in getting the process initiated on the West Coast and making it work.

Weaknesses of the approach

Imbalanced representation of stakeholders

The West Coast Marine Protection Forum was disadvantaged from the outset by the fact it did not receive initial direction on Forum representation requirements. The MPA Policy, MPA Guidelines and official terms of reference were released at varying stages after the Forum had already been established and was deliberating

over issues. West Coast conservancy staff used their own discretion to select a wide range of people to represent the key interests and stakeholders in that area.

The Forum had greater numbers representing fishing interests than any other interest group. There were four commercial fishing representatives and two recreational fishers, but only one member representing an environmental group and one other member representing conservation and community interests. An environmental representative commented that at times she felt like a 'lone voice' on the Forum.⁷⁵

Marine science was not represented on the Forum, contrary to the terms of reference drafted by the Government, which outlined the need for science as an interest to be represented. It might be argued that scientists should not be represented on the Forum, due to their need to be impartial in giving science advice. However, a science representative could play a role in advocating for adequate areas to be set aside for scientific research. The purpose of the Marine Reserves Act is to protect areas in their natural state for scientific study. At least one member of the Forum, therefore, should have been appointed to defend these interests and to be able to give a 'users' perspective on which areas were preferred for this purpose, just as fishing, tourism or recreational users were entitled to do so.

Furthermore, Te Ohu Kai Moana (the Māori Fisheries Trust which represents Māori commercial and non-commercial fishing interests) expressed concern that it was not allowed to participate in the Forum.

The Forum lacked representatives from the mining industry, although this omission was not contentious. When the Forum was considering areas that had potential mining interests, it engaged with industry representatives on those issues. The mining industry was satisfied that, even if protection measures were applied, companies would still be able to apply for mining access in those areas although they would have to satisfy greater environmental standards.

Lack of detailed marine science

A lack of sufficient information and data was frequently mentioned as a hindrance to the Forum's progress. Comprehensive work from Don Neale (Department of Conservation) to produce the environment report overcame the initial need for background information on the marine environment, although this took many months. Had this information been available from the outset, the process might have been fast-tracked another 18 months. Forum members made use of this

time to establish trust and come to terms with their roles, whilst a working group contributed to the development of the report.

The lack of information reflects the relatively low priority and resourcing allocated to the marine environment. A former Marine Unit Manager at the Department of Conservation estimates that marine environmental research and protection is 20 to 30 years behind its terrestrial equivalent.⁷⁶ The value of the marine environment, especially on the West Coast, is not immediately obvious to all. It is easier to justify in a place like Fiordland where the underwater environment is so unique and special, but the West Coast to many, is 'just another coast'.⁷⁷

An essential element of the Forum's task was to identify habitats for protection. Insufficient information is available for the marine environment to identify habitats, so the classification system used substrate as a proxy. However, even substrate was difficult to map. The GIS team at the Department of Conservation had to manually digitise old maps, use incomplete or inadequate data, or extrapolate land based surveys of substrate into the marine environment.⁷⁸ At times, it was difficult to define boundaries and the local fishers were able to assist with their knowledge and experience.

Lack of science independence

The primary science advisor to the Forum was provided by the Department of Conservation. The department's Marine Technical Support Officer, Don Neale, had extensive knowledge and experience on the West Coast and was highly regarded. He provided dedicated support to the Forum, attending every meeting. However, by virtue of the fact that he was employed by the Department, he was not a completely independent advisor and his advice was challenged at times.⁷⁹ This issue was partially overcome by conducting a public consultation process for the environment report to validate his advice, however, the situation might have further been improved by enlisting an independent marine scientist.

Constraints on fishing information

The Forum experienced frustration with the lack of specific information provided by the Ministry of Fisheries. The Forum had an expectation that the Ministry would be able to provide all the fishing information it required, including fishing locations, users and catch volumes. However, the Ministry was constrained in its support by its limited budget, restructuring and confidentiality agreements.⁸⁰ The West Coast Marine Protection Forum was a Department of Conservation-led

initiative that invited the Ministry of Fisheries to participate. The process was kicked off at a local level before the MPA Policy had been finalised, therefore the Ministry of Fisheries office at Nelson was required to support the process out of its routine funding in the absence of central government direction.⁸¹

In general, information released from the Ministry of Fisheries is aggregated to avoid identifying specifics about individual fishers.⁸² The Ministry has three criteria for withholding data: where the data is personal or private, where it is at a fine scale level, or where information about the catch is from less than three vessels. The Ministry maintains that its requirement to withhold data is due to commercial sensitivity and the need to avoid prejudicing future supply. If the Ministry breaches the relationship it has with the commercial sector, fishers may cease to supply accurate data.⁸³

The Ministry changed its reporting requirements on 1 October 2007, to include trawl start positions.⁸⁴ The data doesn't indicate trawl direction or finish location, but this is still an improvement on the data collected prior to 2007. Fishers are obliged to complete catch returns, which include such information as vessel, statistical area, target species, catch, bycatch, processing done on board the vessel, date and duration of the voyage. However, whilst some forms are being completed in electronic form, many are still paper-based, making any sort of statistical analysis a laborious task. It appears that information gathering processes are improving, although, not to a level that would provide the kind of information on catch volumes and locations that would better assist a marine protected area forum.

Policy ambiguity

The Forum was uncertain over the correct interpretation of particular phrases and objectives of the MPA Policy, which led to disagreement. When it requested clarity through the policy advisors, it did not receive definitive answers as definitions were still being discussed within the agencies.⁸⁵

Balancing the planning principles of the MPA Policy created division within the Forum. Planning principle five ensures that adverse impacts on existing users of the marine environment should be minimised when establishing the marine protected areas. However, planning principle two also states that *“the management tool(s) used at a site must be sufficient to meet the protection standard”* and the overall objective of the MPA Policy is to protect marine biodiversity. The Forum found it difficult to determine an appropriate balance between environmental protection and use rights. The policy needed a stronger

overall objective, a hierarchy within the planning principles, or a mechanism to provide an environmental bottom line.

The definition of a marine protected area, and the corresponding protection standard contained in planning principle two, was also challenging to understand and apply. A marine protected area needs to provide ‘adequate protection’, for the ‘maintenance and/or recovery of biological diversity’. Adequate protection is achieved by putting in place a management regime that meets the protection standard. Planning principle two includes a detailed description of what is required to achieve maintenance and recovery, but there still remains a significant amount of analysis to identify what to protect, how much of it to protect and with which tools. For example, it is easy to assume that a marine reserve will automatically meet the protection standard. However, each proposed marine protected area needs to be assessed on a case-by-case basis, because a marine reserve in the wrong place or one that is too small, would not necessarily do so. The absence of available data on the biodiversity of the region hindered the Forum’s ability to meet these requirements.

Insufficient time allocated

The MPA Guidelines, released in 2008, contains a timeline for completion. It allows six months preparatory work by the Department of Conservation and the Ministry of Fisheries, six months to establish the Forum and consult with the community, six months to develop recommendations and a final six months to conduct a formal public consultation process and deliver a recommendation report to the Ministers. This equates to 18 months for the Forum to complete its part and 24 months for the Ministers to receive recommendations. The West Coast Forum took significantly longer than this – five years – to be concluded. Although delays in the issuing of guidance meant that, it was perhaps inevitable that it would take longer, the time required was still much more than anticipated.⁸⁶

This reflects a lack of experience in running collaborative processes. The complexity of the issues and the length of time required to work through each phase were not fully appreciated. Forum members commented that they could not have met more often than they did, as they were all contributing to the Forum in their own time and still had jobs and personal commitments. If the time frame is to be reduced, more preparatory work needs to be undertaken by the agencies to facilitate the process.

Lack of integration with cultural processes

Mātaïtai applications that were submitted by Ngāi Tahu just prior to the recommendations being drafted, put further pressure on the Forum. From the perspective of the commercial fishers, they were already giving up access to fishing areas and the mātaïtai applications were asking for more concessions. This was not a deliberate attempt by Ngāi Tahu to upset the process, but unfortunate timing. The applications had been developed over an extended period of time, well before the Forum began its process, and were finally ready for submission during the Forum's deliberations. The lack of integration of the statutory tools has been noted by Department of Conservation officials and by the Forum as a hindrance to the implementation of the MPA Policy.⁸⁷

No scope for compensation

The option of compensation was raised in the interests of commercial fishers. It was considered that the establishment of the new marine protected areas would unfairly disadvantage commercial fishers due to a loss in fishing access. The Forum wrote to the Ministers stating that an impasse had been reached and requested that compensation be considered. The Ministers responded indicating that compensation was not 'on the table' and urging the Forum to seek the 'least cost' option.⁸⁸ In its final report, the Forum reiterated that the availability of compensation options for fishers would assist consensus-building in future negotiations on the West Coast or elsewhere.⁸⁹

No consensus

The most difficult challenge the Forum faced was accommodating the diversity of viewpoints within the community and the conflicting interests of representative members. The Forum found that balancing impacts on users with environmental protection, and interpreting the requirements of the MPA Policy with respect to this issue, was complicated. Conflict between commercial fishing and environmental interests dominated the Forum's debate. Although the Forum was able to agree on the four primary locations for marine protection, members could not achieve consensus on the size of the area to protect or which combination of tools to use at each location. There was also disagreement over whether the Forum was protecting the habitats with the greatest biodiversity or highest ecological value due to a lack of scientific information.⁹⁰

As a result of disagreement, the Forum presented a series of options in its recommendation report. Originally, the Forum provided one set of recommendations that went to public consultation. However, following submissions, a commercial fishing representative insisted that smaller options were included in the recommendation report, to reduce the impacts on fishers. To counter this proposal, an environmental representative requested larger options be included, which is notable particularly at Kahurangi and Punakaiki. Neither representative was willing to compromise, so as a result the recommendation report included between two and three options for each primary location.

Duplicated effort

There was a significant level of duplicated effort and risk contained within the process. Following recommendations by the Forum, the full statutory process including public consultation for each of the proposed tools must be undertaken.

Because the process to identify the marine protected areas has been community-led, and stakeholder support has already been established, it is hoped that little or no opposition should be encountered. The process is still open to objection though, and if parties choose to oppose it, they can. If there are delays experienced, stakeholders could claim they were not consulted on the issues of the day. In addition, as the Ministers have already announced that marine reserves are to be established on the West Coast, it could be argued that they have prejudiced their own proceedings.

The process has already taken six years. If there are further delays, or if the process is halted altogether due to objections, the resources invested will have all been wasted. There exists a gap in the legislation and policy for implementation to overcome this major operational risk.

Conclusion

The West Coast Marine Protection Forum was the first regional initiative undertaken under the MPA Policy. It took significantly longer than the policy anticipates, as it was initiated before the government had finalised its advice on the process. The Forum's ability to operate was hampered by this, and an absence of available data about the marine area, but the process was completed and options were submitted. The Forum found it hard to achieve consensus, and the

final recommendations could only be submitted in the form of different options. As such, the Forum achieved some successes, but also highlighted issues which could be improved in future initiatives.

Figure 17.3: Timeline of key events in West Coast marine protected area planning process

Date	Event
2000	New Zealand Biodiversity Strategy published
2004	Draft MPA Policy released by the Department of Conservation and Ministry of Fisheries, 8 November
2005	West Coast Marine Protection Forum formed to implement the MPA Policy on the West Coast, April
2006	Release of final MPA Policy, January
2007	Initial report for the West Coast Marine Protection Forum published, April
2008	Release of MPA Guidelines, February 2008
2008	Addenda report published following public comment
2008	Completion of the West Coast marine and coastal classification and marine protected area inventory
2008	Forum identification of candidate marine protected areas
2009	Letter from the West Coast Marine Protection Forum to the Minister of Fisheries providing a progress update (January). Letter indicates an impasse has been reached between the commercial fishers and conservationists regarding the Kahurangi site. It also requests compensation as an option to consider
2009	Hon Phil Heatley, Minister of Fisheries, written response dated 5 March 2009, urges the Forum to seek compromise and to make recommendations that meet the requirements of the MPA Policy. It states that compensation is not an option and therefore a 'least cost' solution is required. It iterates that once recommendations are made the government will disband the Forum
2009	Hon Phil Heatley, Minister of Fisheries letter to Forum, dated 5 May, notes 11 mātaihai applications put forward by Ngāi Tahu. It is noted that mātaihai reserve applications and marine protection areas undergo different statutory processes. It encourages the Forum to discuss options with local iwi and investigate if one fisheries tool can achieve multiple goals. The Minister is clear that he must follow statutory obligations with regard to mātaihai applications
2009	Hon Tim Groser letter to the Forum dated 5 May, to congratulate it on progress and acknowledge the proposal for nine areas of marine protection prior to entering the public consultation process. It notes the concurrent mātaihai applications underway and supports the approach to minimise overlap and conflict by engaging with Ngāi Tahu

2009	Forum consultation on package of proposed marine protected areas. It releases public consultation document 'Proposed areas for the South Island's West Coast Te Tai o Poutini' on 29 July
2009	Submissions close on 21 August. Hearings held and submission analysis completed by October
2009-2010	Forum reconvenes to consider all new information and decide on locations and tools for marine protected areas, including options to present to Ministers, October 2009 to January 2010
2010	Recommendations report completed and presented to Ministers, July
2011	Ministers announce their decision that five options put forward by the Forum should be progressed, August
2012	Department of Conservation and Ministry for Primary Industries (formerly Ministry of Fisheries) undertakes public notification and consultation of proposals to establish proposed marine reserves and fisheries regulations (submissions closed 22 August)

Figure 17.4: Evaluation of options in the West Coast Marine Protection Forum's recommendation report

Site/ Option	Type of Protection	Area	Planning Principle 1	Planning Principle 2	Planning Principle 3	Planning Principle 5	Planning Principle 9
Kahurangi A	Marine reserve (type 1)	665km ²	Fully protects representative areas of 9 habitats in a marine reserve	The tools proposed meet the protection standard	Some opposition from Ngāi Tahu and local rūnanga	Some effect on commercial fishing. Allowed areas for recreational fishing	Boundaries are simple straight lines
Kahurangi B	Marine reserve (type 1); Fisheries regs (type 2)	85km ² 0.19km ²	Provides protection to 8 habitats in a marine reserve and 1 under fisheries regs	The tools proposed meet the protection standard	Iwi indicated they could support this option	Supported by commercial fishing users	Compliance unlikely to be a concern due to the minimal impact on existing activities
Punakaiki A	Marine reserve (type 1); Fisheries regs (type 2)	35.58 km ² 3.26 km ²	Provides protection to 7 habitats	The tools proposed meet the protection standard	Iwi supported this option	Most impacts on users were minimised or avoided	The boundaries are in straight lines and shoreline boundaries can be easily marked. Location is close to a field centre and visitor centre encouraging compliance

Site/ Option	Type of Protection	Area	Planning Principle 1	Planning Principle 2	Planning Principle 3	Planning Principle 5	Planning Principle 9
Punakaiki B	Marine reserve (type 1); Fisheries regs (type 2)	240.53km ² 3.26 km ²	Provides protection to 9 habitats	The tools proposed meet the protection standard	Iwi opposed this option	Most impacts were minimised although there would be some impact on local inshore fishing	The boundaries are in straight lines and shoreline boundaries can be easily marked. Location is close to a field centre and visitor centre encouraging compliance
Punakaiki C	Marine reserve (type 1); Fisheries regs (type 2)	34.65km ² 4.19 km ²	Provides protection to 7 habitats	The tools proposed meet the protection standard	Iwi opposed this option	This option has minimal impact on existing uses	The boundaries are in straight lines and shoreline boundaries can be easily marked. Location is close to a field centre and visitor centre encouraging compliance

Site/ Option	Type of Protection	Area	Planning Principle 1	Planning Principle 2	Planning Principle 3	Planning Principle 5	Planning Principle 9
Ōkārito A	Marine reserve (type 1); Fisheries regs (type 2)	46.41km ² 59.47km ²	Provides protection to 7 habitats	The tools proposed met the protection standard	Ngāi Tahu supported this option when the lagoon was removed	Affects the activities of commercial fishers. Mining can still occur at the discretion of the Minister of Conservation and protection over the lagoon was removed to reduce impact on other users	The proposed boundaries are straight and the site is near a settlement and tourism destination, Franz Josef
	Marine reserve (type 1)	28.63km ²	Provides protection to 4 habitats	The tools proposed met the protection standard	Ngāi Tahu supported this option	This option avoids the impact on fishers, duck shooters, mining and other users of the lagoon	The proposed boundaries are in straight lines and the site is fairly accessible from land
Gorge A	Marine reserve (type 1); Fisheries regs (type 2)	8.47km ² 113.98km ²	Provides protection to 9 habitats	The tools proposed and the area meet the protection standard	Ngāi Tahu were able to support this option provided that the commercial and non-commercial take of pounamu was provided for	This option has little to no effect on existing commercial and recreational fishing. It deliberately avoids the reef at Longridge Point which is important for crayfishing	The boundaries are in straight lines and commercial fishers should be equipped with sufficient navigation equipment to avoid this area

Site/ Option	Type of Protection	Area	Planning Principle 1	Planning Principle 2	Planning Principle 3	Planning Principle 5	Planning Principle 9
Gorge B	Marine reserve (type 1); Fisheries regs (type 2)	28.37km ² 94.07km ²	Provides protection to 9 habitats	The tools proposed and size of the area meet the protection standard	Ngāi Tahu were able to support this option provided that the commercial and non-commercial take of pounamu was provided for	Due to the remote location there is little to no effect on recreational fishing. The larger size of the marine reserve includes areas that are important for crayfishing and longlining. This option would affect commercial trawl, long line and set net fisheries	The boundaries are in straight lines and commercial fishers should be equipped with sufficient navigation equipment to avoid this area

Figure 17.5: List of interviewees for West Coast Marine Protection Forum case study

Name	Role on Forum	Current occupation
Randall Bess	N/A	Policy Advisor, Ministry for Primary Industries
Gary Eason	GIS support	GIS Manager, Department of Conservation
Bruce Hamilton	Chairperson	CEO, Buller Development Company
Murray Hosking	Marine reserves facilitator	Marine Conservation Unit, Department of Conservation (now retired)
Kim George	N/A	Information Centre Manager, Ministry for Primary Industries
Don Neale	Science Advisor	Marine Technical Support Officer, Department of Conservation
Nikki Pindur	Policy Advisor (fisheries)	Policy Advisor, Department of Conservation
Campbell Robertson	Project Manager	Project Manager, Department of Conservation
Eugenie Sage	Member	List Member of Parliament, Green Party of Aotearoa New Zealand
Sarah Wilson	Independent facilitator	Adventure Coach Consultant

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chapter eighteen



Sub-Antarctic Marine
Protection Planning Forum

Introduction

The Minister of Conservation, Kate Wilkinson, and Minister of Fisheries, Phil Heatley, announced in January 2011 that three large marine reserves covering a total of 4,351.63 square kilometres would be established around the Sub-Antarctic Islands of Antipodes Island, Bounty Islands and Campbell Island. They would encompass 100 per cent, 58 per cent and 39 per cent of their respective territorial seas.¹ These marine reserves, along with additional Fisheries Act regulations, arose from recommendations made by the Sub-Antarctic Marine Protection Planning Forum, which was responsible for considering marine protection options for the Islands.

There are five island groups in the Sub-Antarctic region, including the Auckland Islands, the Snares Islands, Campbell Island, Antipodes Island and the Bounty Islands (see Figure 18.1). The Auckland Islands are already fully protected by a nature reserve, a marine mammal sanctuary and a marine reserve covering all the territorial sea, so were not considered for further protection by the Forum.² The Snares were also not included, but were to be considered at a later date within another biogeographic region. All five island groups and their territorial seas are significant due to high biological diversity. This is reflected in their status as a World Heritage site.³

The Sub-Antarctic Marine Protection Planning Forum was established in February 2008 by the Ministers of Fisheries and Conservation.⁴ Its purpose was to provide recommendations to government for marine protected areas in the Sub-Antarctic biogeographic region. The Forum consisted of appointed members representing relevant stakeholder interests, including commercial fishing, environment/conservation, tangata whenua and science. Tourism and the mining sector were not represented because they were not considered to be affected by the process.

The members of the Forum were unable to achieve consensus and instead presented two options for each island to the Ministers of Fisheries and Conservation in their recommendation report. The Forum also stated that the islands' status as a World Heritage site influenced the options that were developed, and that similar recommendations were unlikely to apply in other biogeographic regions under the MPA Policy.⁵ A regulatory impact statement, compiled by Ministry of Fisheries and Department of Conservation officials, gave further advice on implementation options before the Ministers made their final decision on marine protection. The marine reserves are to be implemented using special legislation in the form of the Sub-Antarctic Island Marine Reserves Bill 2011, which was introduced into

Figure 18.1: Map of New Zealand including Subantarctic Islands biogeographic region

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Context

Social and economic environment

Although there has not been a continuous human presence in the Sub-Antarctic Islands, small settlements of Māori, Moriori and European origin have existed in various parts of the Islands at different times; the two former from the 13th Century and the latter from the early 19th Century.⁶ Ngāi Tahu is recognised as tāngata whenua for the region and the special relationship the tribe has with particular species in the area is acknowledged in the Ngāi Tahu Claims Settlement Act 1998.⁷ Oral accounts of various hunting and settlement activities passed on through generations are supported by archaeological remains.⁸

Europeans have been associated with the area for around 200 years through various pursuits including hunting, exploration, wartime coast watch, farming and research. Sealers, and later whalers, brought people of mixed races and introduced terrestrial mammals such as rabbits, goats, pigs, rats, sheep and cattle. As a result of hunting, seals and sea lions in the region were driven almost to extinction and human activities modified the land and marine ecosystems to some extent.⁹ The overall effects were limited as the settlements were concentrated in the Auckland Islands and Campbell Island, with Antipodes Island virtually unmodified by humans.¹⁰

The commercial fishing management area for the region is FMA (Fisheries Management Area) 6 with individual transferrable quota for stocks allocated by the Minister for Primary Industries. The region represents 23 per cent of New Zealand's total exclusive economic zone, so there is significant economic value in the Sub-Antarctic commercial catch.¹¹ Between 71,000 and 97,000 tonnes are caught annually which corresponds to a value of NZ\$114-163 million (2007 export prices). Squid represents around half the catch, with southern blue whiting, hoki and ling comprising the majority of the remainder.¹² The most commonly used fishing method is trawling, followed by bottom long lining.¹³

A tourism industry exists for the Auckland and Campbell Islands, although it is strictly controlled by the Department of Conservation. The number of visitors to each of the islands is limited (per day and per season) and there are rules governing where people can go.¹⁴ The remoteness of the islands makes monitoring of illegal landings and enforcement difficult, but it also limits visits from small private vessels. There is no recreational fishing interest in the area.

Natural environment

There are a total of five island groups in the New Zealand Sub-Antarctic region, which lie to the south-east of New Zealand, in the Southern Ocean. Each island group is separated by almost 1,000 kilometres, so they all have a fully enclosing territorial sea, as can be seen in Figure 18.2.

The islands rest on the southern plateau which extends 1,100 kilometres from the southeast tip of the South Island and includes the Campbell Plateau, Bounty Plateau, Pukaki Rise and Auckland Shelf.¹⁵ This undersea structure creates a large zone of water shallower than the rest of the Southern Ocean, with depths of approximately 600 to 1,000 metres across an area of 433,620 square kilometres. The islands have differing geological origins, with the Snares and Bounty Islands formed from basement granite and metamorphic rock of origins dating back 180 million years, whereas the Auckland Islands, Campbell Island and Antipodes Island are formed on a base of volcanic rock.¹⁶ Campbell Island and the Auckland Islands possess deep inlets and harbours carved out by glaciers.¹⁷

The islands lie between the subtropical front, which defines the northern limit of the Southern Ocean, and the Antarctic circumpolar current. Sub-Antarctic waters surround the area, which are cooler and less saline than northern parts of the Southern Ocean.¹⁸ The region is exposed and aptly described as “*cold, wet and windy*”¹⁹, subject to high winds (prevailing westerlies) and the coolest average annual temperatures in New Zealand, ranging between six and eleven degrees Celsius. Average annual rainfall is around 1,360 millimetres, although this varies throughout the islands. Snow fall will also occur at any time of the year, but is heavier in the winter season.²⁰

The large distances between the islands have led to distinct ecosystems and high levels of endemism on individual island groups. The biological diversity of the area is high, although less is known about the marine environment than about the terrestrial environment. On many of the islands the vegetation is rich and diverse with a variety of vascular plants and forests. Notable species include the southern-most tree fern and unique megaherbs. The Bounty Islands has virtually no soil or vegetation and only one species of vascular plant. Under the sea, several varieties of kelp and seaweed are recognised as notable and endemic.²¹

The abundance of seabirds and marine mammals attract great interest and the islands are also home to a number of rare and endangered species. Campbell Island supports several vulnerable and endangered albatross and a rare snipe, with bird populations improving since introduced rats were eliminated.²² Under threat

is also the world's rarest cormorant (the Bounty Island shag), the world's rarest penguin (the yellow-eyed penguin) and the world's rarest duck (the Campbell Island teal).²³

The New Zealand sea lion (formerly known as the Hooker's sea lion) found on the Auckland Islands and Campbell Islands is listed by the Department of Conservation as 'nationally critical', partly due to highly constrained breeding grounds but also because of the population decline over the last decade at the Auckland Islands.²⁴ Southern Right Whales are recovering after facing near extinction due to whaling and are drawn to the shallower waters of the region for breeding and calving.²⁵

Existing marine protection

The terrestrial ecological communities of the islands enjoy a high level of protection as nature reserves under section 20 of the Reserves Act 1977.

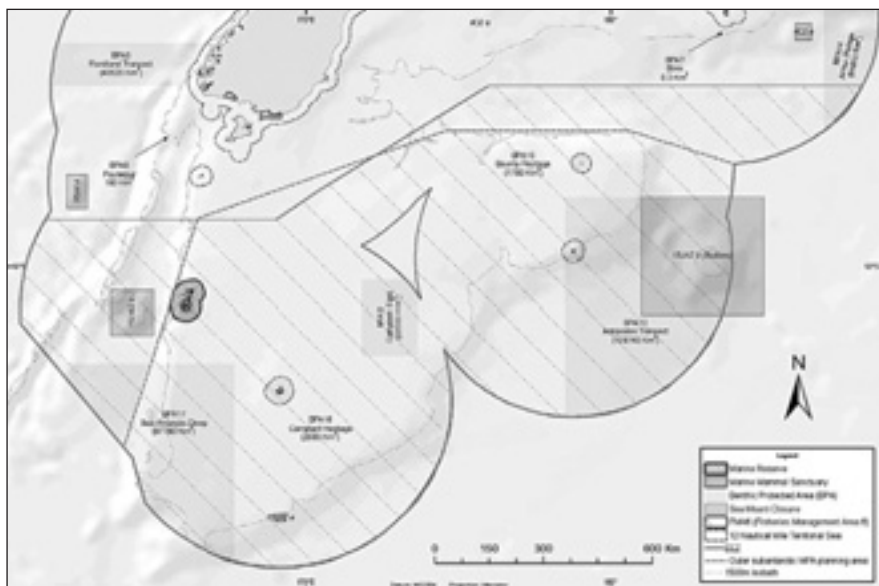
A marine mammal sanctuary and marine reserve are established around the Auckland Islands. One of the key drivers for the establishment of the sanctuary was the level of incidental mortality of New Zealand sea lions resulting from the squid trawl fishery. Protection measures now prohibit fishing within 12 nautical miles of land and by-catch is monitored by the Department of Conservation in conjunction with the Ministry for Primary Industries.²⁶ The Department of Conservation administers a five-year species management plan, with the timeline of the current plan extending until 2014.

Benthic protection areas are established around Campbell Island, Antipodes Island and the Bounty Islands which prohibit bottom trawling and dredging.²⁷ Within all the island's territorial seas there are further fisheries regulations prohibiting commercial fishing by any vessels longer than 46 metres in length or foreign charter vessels.²⁸

The Resource Management Act 1991 assigns the functions, duties and powers of a regional council and territorial authority to the Minister of Conservation. The 'Proposed Regional Coastal Plan: Kermadec and Sub-Antarctic Islands' was publically notified on 15 January 2011, well after the Forum completed its recommendations to the Ministers. Prior to this, the 'Sub-Antarctic Islands Conservation Management Strategy (1998-2008)' was in place and considered the protection of the marine environment, although no statutory action had been taken through this means.²⁹

Figure 18.2: Existing protection tools in the Sub-Antarctic Islands

Source: Sub-Antarctic Islands Marine Protection Planning Forum



In 1998 the Sub-Antarctic Islands of New Zealand were formally inscribed by the United Nations Educational Scientific and Cultural Organisation as a World Heritage site. According to the ‘Convention Concerning the Protection of the World Cultural and Natural Heritage’, natural heritage may be “*physical and biological formations*” that have “*outstanding universal value*” (from either a scientific or aesthetic point of view) or areas that “*constitute the habitat of threatened species of animals and plants of outstanding universal value from the point of view of science or conservation*”.³⁰

The World Heritage site applies to the Snares Islands, the Bounty Islands, Antipodes Island, the Auckland Islands and Campbell Island as well as their respective territorial seas. The application was justified on the following basis:

- The New Zealand Sub-Antarctic Islands display a pattern of immigration of species, diversifications and emergent endemism, offering particularly good opportunities for research into the dynamics of island ecology.
- The New Zealand Sub-Antarctic Islands are remarkable for their high level of biodiversity, population densities, and for endemism in birds, plants and

invertebrates. The bird and plant life, especially the endemic albatrosses, cormorants, landbirds and ‘megaherbs’ are unique to the islands.³¹

The Convention imparts a duty on the respective state to ‘identify’, ‘protect’ and ‘conserve’ heritage areas for future generations. They are obliged to ‘do all to the utmost of their resources’ to achieve this. They must be seen to be taking ‘active and effective measures’. A country is expected to achieve this through the creation of policy, the establishment of services for protection and conducting research.³²

Stakeholder body

Establishment

The Sub-Antarctic Marine Protection Planning Forum was established as a non-statutory body in February 2008. The chairperson and members of the Forum were appointed by the Ministers of Conservation and Fisheries and provided with terms of reference by which they were to conduct themselves.³³

Purpose

The Forum's purpose was to provide options and recommendations to government for marine protection in the Sub-Antarctic biogeographic region.³⁴ The specific matters the Forum was to consider and present in its report to the Ministers were:³⁵

- Classification and inventory of information
- Consultation with existing users and interests in the area
- Sites and potential tools for area-based protection of biodiversity
- Any consensus on proposed areas to be set aside as protected areas
- Consultation conducted on protection options and recommendations to the Ministers

Membership

The Forum consisted of 14 appointed members representing commercial fishing, environment/conservation, Māori and scientific interests. Figure 18.3 details the membership of the Forum and the respective interests represented. Members of the Forum were expected to have strong links with the region, be able to negotiate,

work well with others, engage with their respective interest sectors and put forward these views during decision-making.³⁶

The Chairperson, Paul Beverley of Buddle Findlay, was appointed by the Ministers. He is an expert in conservation law as well as Māori interests under environmental legislation and has previously advised the Crown on environmental and conservation legal issues. He leads the environment and resource management team and the Māori Law team at Buddle Findlay, Wellington.³⁷

Figure 18.3: Sub-Antarctic Marine Protection Planning Forum members

Name	Role	Organisation	Interest
Paul Beverley	Chairperson	Buddle Findlay	
Richard Wells	Member	Deepwater Group	Non-profit, commercial fishing, fisheries sustainability
Andrew Bond	Member	Scampi Quota	Commercial fishing
Andy Smith	Member	Crab Company	Commercial fishing
Kate Bartram	Member	Seafood Industry Council	Commercial fishing
Tania McPherson	Member	Te Ohu Kai Moana	Māori fisheries
Nigel Scott	Member	Ngāi Tahu	Tāngata whenua
Gail Thompson	Member	Southland Kaitiaki Roopu (Awarua Runanga)	Tāngata whenua
Rebecca Bird	Member	World Wildlife Fund	Non-profit environmental
Kevin Hackwell	Member	Royal Forest and Bird Protection Society	Non-profit environmental
Viv Shaw	Member	Southland Conservation Board	Environmental management
Don Robertson	Member	NIWA (now retired)	Science
John Booth	Member	Marine science	Non-profit, science
Barry Weeber	Member	Environment and Conservation Organisations	Non-profit, environmental

Process

Overview

The Forum was established as a collaborative stakeholder body to consider how marine protection could be achieved in the Sub-Antarctic biogeographical region. It was expected to meet the requirements of the MPA Policy and Guidelines and comply with the terms of reference provided by the supporting agencies. The Forum was directed to carry out the assessment in two phases:

Phase 1: Consider protection options for the territorial seas of the Campbell, Antipodes and Bounty Island groups

Phase 2: Consider the remainder of the Sub-Antarctic region to the extent of the 200 nautical mile exclusive economic zone

The Forum was originally expected to have Phase 1 completed by August 2008 and Phase 2 was to commence after its recommendations had been delivered to the Ministers. A meeting schedule was outlined in the terms of reference for the completion of Phase 1 as shown in Figure 18.4.

Figure 18.4: Meeting schedule for the completion of Phase 1

Date	Meeting Number	Meeting Objective
February 2008	First meeting of Forum	Information and process outline
March 2008	Second meeting of Forum	Marine protected area discussion
March/April 2008	Third meeting of Forum	Marine protected area site selection and development of recommendations to Ministers
April 2008	N/A	Ministerial review of Forum progress
May 2008	Fourth meeting of Forum	Consultation on recommendations and analysis
July 2008	Fifth meeting of Forum	Finalisation of recommendations report for referral to Ministers by August
August 2008		Ministerial review of Forum's progress
August 2008		Meetings after this point would be as required to conclude phase 2.

The Forum did convene in February 2008 and met seven times for full-day meetings during that year. When the time allocated to Phase 1 had passed, the members

had completed only a fraction of the tasks required by the MPA Policy and their terms of reference.³⁸ They had only just come to terms with the information and policy provided and had conducted a number of discussions around stakeholder interests.

The Forum's process began with mapping existing management tools.^{39,40} At this point, the Forum members decided to elect a smaller working group including four representatives of the Forum and the independent Chairperson to prepare the consultation document. The participants of this working group were Kevin Hackwell, Barry Weeber, Tania McPherson, Kate Bartram and Paul Beverley as Chairperson.⁴¹ The consultation document was released on 5 June 2009 with a period of two months for submissions.⁴²

The terms of reference indicated that Phase 2 would begin immediately after the recommendations for Phase 1 were delivered. As marine reserves are confined to the territorial sea, it suggested that the Forum would make recommendations that could be implemented once appropriate legislation, providing necessary tools for the exclusive economic zone, was drafted. Alternatively, an expert panel would consider protection within this area from 2013.⁴³

Role of stakeholder body

The Forum was responsible for the conduct of its own meetings. The Chairperson's main responsibility was to convene and facilitate meetings, but also to ensure that the requirements of the MPA Policy and the terms of reference were met.⁴⁴

The Chairperson had some experience with the Department of Conservation as a senior lawyer and had a good understanding of the statutory tools and processes under the relevant environmental and conservation legislation. He also had extensive experience in general legal proceedings and negotiations.⁴⁵ Forum members reported he was a capable facilitator and crucial to the process. He led the group through the process and drafted key documents and correspondence with the Ministers.

The Forum met for day-long meetings and worked through the tasks in a systematic fashion. Members discussed matters as a group and then formed a smaller working group to prepare the consultation document. Once the consultation period was over, the Forum resumed large group meetings on 14 September 2009 and developed the final recommendations.⁴⁶

Role of statutory agencies

The Department of Conservation and the Ministry of Fisheries were jointly responsible for developing and implementing the MPA Policy. Representatives from these agencies were regarded as *ex officio* members of the Forum, but were also there to provide information and support where required.⁴⁷ Officials provided national policy guidance to the Forum, along with planning and technical advice, and their support was consistent with the MPA Policy.

The Department of Conservation also provided funding to enable Forum members to be reimbursed for reasonable and actual expenses incurred from their attendance at meetings. Members were not paid a meeting fee as most were attending in a professional capacity as the representative from their stakeholder bodies.⁴⁸ The Forum received secretarial support to assist the organisation and conduct of meetings. This support also aided communication and distribution of information.⁴⁹

The Sub-Antarctic region had been identified in advance by the Department of Conservation as a priority area for marine protection. As a result, the Department had undertaken some preparatory work prior to the Forum being established. This included developing a resource report called a 'background resource document', which contained information about the region in general and specific information about each of the islands. It included ecological, climatic, oceanic and biological characteristics, threats and issues. It also outlined the current management regime.⁵⁰

Information and science

The Department of Conservation and the Ministry of Fisheries were jointly responsible for providing scientific, policy and technical advice to the Forum. The Forum members interviewed reported that they received a high standard of support from the agencies, with the Department of Conservation providing a majority of the general support required and the Ministry of Fisheries providing specialist information and advice when requested.

High quality GIS product was an essential component of the Forum's process. The majority of the mapping was provided by the Department of Conservation, although some mapping showing fishing effort was provided by the Ministry of Fisheries.

One of the first tasks was an analysis of the protection tools available and an assessment of whether they would satisfy the MPA Policy requirements. This

analysis was requested at the first Forum meeting and was presented by the Department of Conservation at the second meeting.

The Department of Conservation indicated that marine reserves and most Fisheries Act tools would meet the protection standard for marine protected areas. Benthic Protection Areas, however, would likely not meet the standard due to the limited protection they provide and the permitting of fishing methods that would, under usual circumstances, be excluded from a marine protected area.⁵¹

Several documents were provided by supporting agencies to guide and inform the process. These were in addition to the MPA Policy and Guidelines that were completed before the Sub-Antarctic Islands Protection Forum began their process:

- *Terms of Reference (2008)* provided at the commencement of the process to guide the Forum in their process and conduct.
- *A Background Resource Document (2006)* developed by the Department of Conservation to provide the necessary background information for the Forum to better understand the social, economic and ecological context of the Sub-Antarctic region. It included general regional information as well as more detailed descriptions of the three islands being considered for protection.
- *Bibliography: Sub-Antarctic Islands Marine Biogeographic Region (2006)* produced to accompany the *Background Resource Document*. It contains a comprehensive list of research and publications relevant to the Sub-Antarctic region.
- *Analysis of Protection Tools for Sub-Antarctic Marine Protection Planning Forum (2008)* provided at the second Forum meeting on 26 March 2008 by the Department of Conservation. This was an inventory of existing protection measures already in place in the Sub-Antarctic region and included an assessment of whether they would meet the requirements of the MPA Policy.
- *Analysis of Protection Tools against the Protection Standard (2008)* provided at the fourth meeting of the Forum to advise on whether nature reserves and benthic protection areas met the protection standard as set out in the 2008 Guidelines.

Mapping was an essential tool for the Forum members to understand the spatial aspects of their task. Maps were provided by the supporting agencies on request from the Forum. Some key examples of maps that were produced are:

- *Bathymetry* – displaying the depth contours of the waters surrounding the islands
- *Composite maps* – combining exposure, sediment and bathymetry (depth) to be used as a proxy to classify habitats. The locations of corresponding habitats were also displayed graphically.
- *Fishing effort* – the Ministry of Fisheries provided maps of fishing effort. To avoid commercial sensitivity, the maps showed pixelated coloured areas of fishing intensity rather than specific trawl lines or gear sets.

The Forum produced two key documents during their tenure. The first was the consultation document (2009). This was publically released on 5 June, 2009, allowing a period of two months for submissions. The document contained background information about the Sub-Antarctic Islands and about the formation and purpose of the Forum. It presented two options for protection for each of the islands that were basically the same as those options later recommended to the Ministers. The second document was the recommendation report (2010). This was drafted by the Chairperson of the Forum, and addressed to the Ministers of Conservation and of Fisheries. It contained a summary of the Forum's process, submissions received during public consultation, and a brief on the recommended options for protection.

Public consultation

Consultation with key stakeholders began soon after the Department of Conservation initiated the Sub-Antarctic Marine Protection Project in 2003.⁵² This was a scoping exercise to contemplate marine protection in the Sub-Antarctic region.⁵³ A workshop was held by the supporting agencies in 2004 to begin stakeholder engagement. This provided an opportunity to adjust the proposed membership of the Forum and ensure that all stakeholders were fairly represented. There was an expectation that members of the Forum would actively engage with their stakeholder groups, but also with those who may have an interest in, or could be affected by, marine protected areas.⁵⁴ There was no residential community to consult within the Sub-Antarctic islands. Therefore a formal consultation process to reach the general public was important.

The terms of reference directed that consultation must be conducted on the recommendations being made to the Ministers, allowing a minimum of 40

working days for submissions to be made.⁵⁵ The Forum also visited a Southland marae prior to commencing the formal consultation process.⁵⁶

The Forum released the public consultation document on 5 June 2009 and received 65 submissions. A summary of submissions prepared by the Forum indicated that a majority were in favour of full protection (52 in favour of full protection, eight in favour of partial, one in support of no protection and the remainder unclear).⁵⁷ The only submissions received in favour of partial protection were from commercial fishers and other extractive users. Two submissions were received from iwi, with one in support of full protection and one in support of partial.

As the marine reserves are to be implemented using special legislation, there will be a further opportunity for submissions through the select committee process.

Outcomes

Forum recommendations

The members of the Forum were unable to reach consensus and resolved to present six options, two for each island.⁵⁸ In each island scenario, the first option provided a combination of marine protected area tools for partial protection of the territorial sea and the second option provided full protection of the territorial sea with a marine reserve. The recommendation report advised the Ministers that they could select any combination of the six options.

For the Antipodes Island the Forum presented two options as shown in Figure 18.5:

- Option A1: Protect part of the territorial sea. A single marine reserve would extend to the edge of the territorial sea in the south-west quadrant. On the short side, it would extend a minimum of 5.8 kilometres from the island group. The area not in the marine reserve was to be protected by Fisheries Act prohibitions on Danish seining and existing benthic protection area prohibitions on bottom trawling and dredging. The area selected protects all intertidal and subtidal habitats and contains representative examples of all known deep water habitats in the territorial sea.

- Option A2: Protect the whole territorial sea with a marine reserve. This would provide protection for all marine habitats in the territorial sea.

For the Bounty Islands the Forum presented two options as shown in Figure 18.6:

- Option B1: Protect part of the territorial sea. A single marine reserve would extend to the edge of the territorial sea in the south. On the short side, the boundary would be extended a minimum of 6.8 kilometres from the island group. The remainder of the area was to be protected by Fisheries Act prohibitions on Danish seining and existing benthic protection area prohibitions on bottom trawling and dredging. The area selected protects all intertidal and shallow subtidal habitats and contains representative examples of all known deep subtidal and deep water habitats in the territorial sea. The fisheries prohibitions provide protection to benthic habitats.
- Option B2: Protect the whole territorial sea with a marine reserve. This would provide protection for all marine habitats in the territorial sea.

For Campbell Island the Forum presented the two options shown in Figure 18.7:

- Option C1: Protect part of the territorial sea. A single marine reserve would extend to the edge of the territorial sea to the west. The remainder of the area was to be protected by Fisheries Act prohibitions on Danish seining and benthic protection area prohibitions on bottom trawling and dredging. The orientation was selected to include an area that had the highest percentage of habitats in water deeper than 200 metres. The area protects all intertidal and shallow subtidal habitats as well as representative examples of all known deepwater habitats, including those known not to be present in the Auckland Island marine reserve. The Fisheries Act prohibitions in the remainder of the area within the territorial sea was to protect the remaining benthic habitats.
- Option C2: Protect the whole territorial sea with a marine reserve.

All options met the requirements of network design principle 1 as they protect examples of the full range of known or expected habitats and ecosystems, or the entire range of habitats and ecosystems out to the territorial sea. The full marine reserve options have a greater impact on current users, although this was mainly relevant for Bounty Islands, as the Campbell and Antipodes Islands had no established fisheries in the territorial sea. It was considered there would be only positive effects for Treaty obligations due to the remoteness of the islands and the high level of protection it would provide for the species identified as taonga.

Source: Sub-Antarctic Marine Protection Planning Forum



Source: Sub-Antarctic Marine Protection Planning Forum



Source: Sub-Antarctic Marine Protection Planning Forum



Government response

The government made its announcement in January 2011 that three large marine reserves would be established around the Sub-Antarctic islands of Antipodes Island, the Bounty Islands and Campbell Island.⁵⁹ The announcement corresponded to options A2, B1 and C1 of the recommendation report, with some minor differences.

The Antipodes Island is to be fully protected with a marine reserve covering the full territorial sea (option A2). The Bounty Islands included an expanded version of option B1 with 58 per cent of the territorial sea protected by a marine reserve (Type 1 marine protected area) and the remainder protected by Fisheries Act prohibitions on Danish seining and benthic protection area prohibitions on bottom trawling and dredging (Type 2 marine protected area). The proposal for Campbell Island allows for the development of a potential crab fishery. An independent review is to be conducted five years after the establishment of the marine reserve. The review is to assess the value and impact on the marine environment of the marine reserve to determine whether to increase the size of the reserve, potentially expanding it to cover the entire territorial sea.⁶⁰

Ministry officials influenced implementation through a 'regulatory impact statement' prepared by the Department of Conservation and Ministry of Fisheries. In this document, officials provided two options each for the implementation of the proposed marine reserves and for proposed Fisheries Act restrictions. The two options involved either implementing the protection measures through respective statutory processes or through special legislation.

The regulatory impact statement recommended that the marine reserves be implemented through special legislation whilst the Fisheries Act restrictions be implemented through the normal statutory process. The officials argued that the Marine Reserves Act 1971 is only designed to process single marine reserve applications and the lack of statutory time frames within the Act leads to protracted processes (the average time for implementing a marine reserve is 12 years). Special legislation would implement the reserves as a package, be more efficient, and less costly. The report also recommended that the Fisheries Act regulations should follow the usual statutory process as it would be consistent with existing restrictions in place and provide greater flexibility and responsiveness for future adjustment.

Future

The Sub-Antarctic Marine Reserves Bill 2011 was introduced to Parliament on 7 July 2011, but at the time of writing it had not yet been passed into law. The purpose of the Bill is to create three new marine reserves in the territorial sea of Antipodes Island, Bounty Islands and Campbell Island.

Clause 8 directs that an independent review of the Campbell Island marine reserve is to be conducted at least five years after the Bill's commencement. This review is to consider the effects of fisheries on the environment and the proposed expansion of the marine reserve. It provides the Minister of Conservation with authority to recommend the extension of the reserve through an Order in Council, with the agreement of the Minister for Primary Industries (formerly the Minister of Fisheries). The extension must occur no later than seven years after the Bill's commencement. The Bill also assigns Commanding Officers of the New Zealand Defence Force to be rangers under the Marine Reserves Act 1971.

Strengths of the approach

Stakeholder representation

The representation on the Forum contained an even balance between commercial fishing and environmental interests and included a majority of the key stakeholders, although none from the tourism or mining industries. Interests that were represented included commercial fishing, Māori, environment and science. There were several environmental non-government organisations and scientists on the Forum; significantly more than on the West Coast Marine Protection Forum.

Agency support and resourcing

The agencies provided a good level of technical support and resourcing. The Forum members were provided with guidance, advice, information and GIS mapping products whenever requested. The Department of Conservation, especially, was regarded as integral to holding the process together and provided the majority of support.⁶¹ Paul Beverley, the Chairperson for the Forum, commented that in order for a process involving stakeholder fora to be successful, it must be adequately resourced.⁶²

Independent chairperson

The appointment of an independent chairperson was crucial to the process and eliminated the need to have an independent facilitator. Members of the Forum reported that the chairperson was highly skilled and important to the process.⁶³

Legal approach

The MPA Policy provided a useful mechanism to keep the Forum focused on working towards recommendations. When disagreement appeared to halt progress, the Chairperson was able to refer back to the policy and the Forum's terms of reference to work through issues. He could use the policy to map out decision points and prepare a logical process for the Forum to follow towards its desired outcome. If there had not been a process laid out, it would have been much more difficult to proceed.

Weaknesses of the approach

The Sub-Antarctic Islands were identified for protection whilst the policy was still under development. Due to its status as a world heritage site, and high level of biological diversity value, it was an area that was considered to need priority attention. In terms of fishing interest, there was much less competition for space than on mainland New Zealand as there were no inshore operators, recreational fishers, or customary fishers and the offshore fisheries were well defined. In the words of one Forum member, "*it should have been a no-brainer*".⁶⁴ However, the Forum members were unable to achieve consensus, took much longer than expected to provide recommendations, and the size of the marine reserves announced by the Ministers at the Bounty and Campbell Islands seem small for an area recognised as internationally significant.

Insufficient time allocated

The terms of reference allocated insufficient time, just six months involving seven meetings, for the Forum to develop recommendations for the Ministers. In practise, the stakeholder process took three years. The Forum was constantly under pressure to produce results, but the supporting agencies underestimated the complexity of the policy and issues involved.⁶⁵ The Forum members reported

they needed at least a whole day to become familiar with the policy alone. Time was also required to establish trust and develop useful dialogue between members.⁶⁶

Stakeholder barriers to consensus building

Forum members describe a definitive split between environmental advocates and commercial fishing representatives.⁶⁷ The Te Ohu Kai Moana and Seafood Industry Council representatives preferred the use of a range of tools, rather than reliance on marine reserves. These representatives were concerned about setting a precedent for future marine reserve applications. Their argument was that if the commercial fishing industry was seen to give in too easily, and the Sub-Antarctic region was afforded complete protection, they would have less bargaining power in future negotiations in the other biogeographic regions of New Zealand.⁶⁸

The Forum was not able to achieve consensus for the consultation document or the recommendation report. The options in the consultation document revealed two main positions, with submitters either in favour of a range of protection tools, or full protection using marine reserves. Presenting only one option for each island may have drawn a more varied response.

Forum members commented that the matters raised in submissions were as expected, therefore little was changed in the recommendations made to the Ministers.⁶⁹ Consensus was never achieved at any stage of the process and after the submission process, environmental advocates argued more strongly for full protection. The approach became less collaborative and more a battle of 'use' versus 'environmental protection'. If members had been better positioned to negotiate, more progress might have been made.

Not all sectors represented

Not represented on the Forum were the tourism and mining sectors. Both these sectors made submissions on the consultation document, with tourism being in favour of larger marine reserves and the mining industry in favour of smaller reserves, or wanting reassurance that there will still be potential for mining, if large marine reserves were established. Neither of these stakeholders caused disruption in the Forum's process. However, Forum members commented that their presence may have diffused the environment protection – commercial fishing impasse.⁷⁰

Legislation

The regulatory impact statement produced by the Department of Conservation and the Ministry of Fisheries highlighted a number of deficiencies in the Marine Reserves Act 1971, opting in favour of special legislation to implement the marine reserves, instead of the usual statutory process. The argument in favour of special legislation covered key points as follows:

- A lack of statutory time frames can lead to a long processing period under the Marine Reserves Act 1971
- Fisheries Act regulations can be progressed relatively quickly compared to marine reserves applications, leading to a fragmented approach
- Given the timeframes imposed on the Forum to deliver recommendations, it would be appropriate to implement the package in a timely fashion also
- The consultation process is duplicated in the Marine Reserves Act 1971, and requires concurrence from the Minister for Primary Industries and the Minister of Transport, which can involve further consultation
- Special legislation is considered more efficient and less costly than applications under the Marine Reserves Act 1971 and carries less risk

However, the processing time for the special legislation is dependent on government legislative priority. The fishing industry representatives considered that Ministers acted in bad faith by introducing special legislation – industry participation in the Forum was on the understanding that any proposed marine reserves would be progressed through the Marine Reserves Act.⁷¹

Conclusion

The Sub-Antarctic was identified as an appropriate region in which to continue the implementation of the MPA Policy because of its remoteness and consequent perceived lack of complex stakeholder interests. Its World Heritage status provided further impetus for the implementation of effective management measures. However, stakeholder positions on the Forum were highly polarised and the members were unable to achieve consensus. Nevertheless, the initiative resulted in proposals for new marine reserves, although at the time of writing these have not yet been finalised.

Figure 18.8: Timeline of key events in the Sub-Antarctic marine protection planning process

Date	Event
1910	Adams Island in the Auckland Islands declared a 'reserve for the protection of fauna and flora'
1934	Same designation extended to all of the Auckland Islands group
1954	Same designation applied to Campbell Islands
1961	Same designation applied to Antipodes, Bounty and Snares Islands
1978	All five island groups reclassified as nature reserves under the Reserves Act 1977 and subsequently accorded national nature reserve status by the United Nations Environment Programme
1993	Marine mammal sanctuary established in Auckland Islands under the Marine Mammals Protection Act 1978, to protect the breeding population of New Zealand sea lion
1998	UNESCO listed the Sub-Antarctic islands and their territorial seas as World Heritage sites
2004	Auckland Islands / Motu Maha marine reserve declared out to edge of 12nm territorial sea
2006	Fishing industry recommends the establishment of a series of benthic protection areas
2007	Minister of Fisheries approved the proposals through the Fisheries Act regulations (Fisheries (Benthic Protection Areas) Regulations 2007) which prohibit contact with seabed by dredge or trawling methods
2008	Ministers of Fisheries and Conservation appoint 14-member Sub-Antarctic Marine Protection Planning Forum
2009	Consultation document published and released, June
2009	Public submissions close, 31 July
2009	Forum reconvenes on 14 September to consider submissions and to develop final recommendations
2010	Forum delivers final report and recommendations to the Ministers on 12 January
2011	The Ministers announce the formation of three new marine reserves around Antipodes Island, Bounty Islands and Campbell Island
2011	Sub-Antarctic Islands Marine Reserves Bill 2011 introduced to Parliament on 7 July, but at time of writing not yet passed

Figure 18.9: List of interviewees for the Sub-Antarctic Marine Protection Planning Forum case study

Name	Role on the Forum	Current Occupation/Organisation
Paul Beverley	Chairperson	Lawyer, Buddle Findlay
Kevin Hackwell	Member	Advocacy Manager, Royal Forest and Bird Protection Society
Richard Wells	Member	Fisheries Specialist, Deepwater Group
Nigel Scott	Member	Ngāi Tahu

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33. Sub-Antarctic Islands Marine Protection Planning Forum, 2009, 5
34. Sub-Antarctic Islands Marine Protection Planning Forum, 2009, 29
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37. <http://www.buddlefindlay.com/our-people/paul-beverley> (accessed 1 August 2012)
38. Beverley P, 2010, 1
39. Sub-Antarctic Islands Marine Protection Planning Forum, 2009, 14
40. The terrestrial area of the five islands and their territorial seas are also recognised as a World Heritage Site
41. Kevin Hackwell, August 2011, *pers comm*
42. Sub-Antarctic Islands Marine Protection Planning Forum, 2009, 6
43. Sub-Antarctic Islands Marine Protection Planning Forum, 2009, 30
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45. Buddle Findlay, n.d., *Paul Beverley*, retrieved from: <http://www.buddlefindlay.com/our-people/paul-beverley>
46. Beverley P, 2010, 3-4
47. Sub-Antarctic Islands Marine Protection Planning Forum, 2009, 29
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61. Kevin Hackwell, September 2011, *pers comm*
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65. Richard Wells and Kevin Hackwell, September 2011, *pers comm*
66. Richard Wells and Kevin Hackwell, September 2011, *pers comm*
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69. Richard Wells, September 2011, *pers comm*
70. Richard Wells, September 2011, *pers comm*
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chapter nineteen



Ngātiwai and marine reserves

Ngātiwai presently has four marine reserves within its rohe (territory). New Zealand's first marine reserve, the Cape Rodney-Okakari Point reserve at Leigh, was created during 1975 in the rohe of Ngātiwai's hapū, Ngāti Manuhiri. The site of the reserve, surrounding Motu Hawere (Goat Island), is significant to Ngātiwai as the place of landfall of the historic waka, Moekakara. Although it has been nearly 40 years since the creation of the marine reserve, there has been no direct involvement of the tāngata whenua in its management, and no benefits specific to the hapū or iwi. Recently a second marine reserve in the immediate vicinity has been created at Tawharanui.

Poor Knights Islands marine reserve

When the Poor Knights Islands marine reserve was proposed, Ngātiwai, in correspondence with the Department of Conservation, did not oppose it in principle but insisted that their customary rights in the reserve must not be extinguished. The establishment of the reserve took place in 1981, but there was no explicit recognition of the tribe's customary right in the gazettal.

Ngātiwai have been very protective of their customary rights. The Ngātiwai Trust Board, on clear instruction from the Ngātiwai iwi, refused to sign the Sealords deal for settlement of fisheries assets because Ngātiwai's customary rights would be compromised. When the Sealords deal was enshrined in statute, Ngātiwai with two other iwi, challenged through litigation the impact on their customary rights. The case was lost, but it has enabled Ngātiwai to assert that it has never ceded customary fishing rights.

Subsequent to the gazettal of the Poor Knights marine reserve in 1981, recreational fishing was permitted in more than 95 per cent of the reserve area, and this also enabled iwi to fish in the area under the recreational fisheries regulations. In 1997, the Minister of Conservation announced that the entire marine reserve would be closed to fishing.¹ Before the new regulations could come into effect, Ngātiwai challenged the Minister's decision in the High Court. The Court held that the Minister's decision had been unlawful *"to the extent [and in no other respect] that it related to customary Māori fishing interests"* and directed the Minister to reconsider. This was on the basis that the iwi *"had a legitimate expectation that its customary rights should have been considered"*.²

The High Court decision highlighted the fact that the Marine Reserves Act does not explicitly address Māori customary fishing rights. If provision was to be made

for their continuance within marine reserves, this could only be achieved through the Minister of Conservation exercising his or her general discretion under section 3(3) if the Act to authorise fishing within the reserve through a *Gazette* notice. The Minister can only authorise such fishing after having regard to the purpose of marine reserves which is 'preserving' the areas 'for the scientific study of marine life'.

Soon after the High Court decision, the Northland Conservancy asked Ngātiwai to identify the customary rights in question, so that the Department could appropriately provide for them. However, determining what customary rights are in any instance is not a simple matter. Although considerable commentary on customary rights has been written, for example in relation to fisheries rights and foreshore and seabed rights, there has been no systematic determination of the actual nature and extent of those rights.

Customary rights are determined in practice through tikanga, are variable and changeable, and seldom have written records. It would have been no simple exercise for Ngātiwai to codify those rights in relation to the Poor Knights marine reserve. A later attempt was made to resolve the issue through a longer process, but this also failed to produce a result, and the customary rights associated with the marine reserve remain undetermined today.

Aotea (Great Barrier Island) marine reserve proposal

In 1989, the Department of Conservation initiated informal discussions about marine protection around Aotea (Great Barrier Island), again within Ngātiwai's rohe. In 1994, a draft application was prepared which covered a small area of sea adjacent to the north-east coastline of the island.³ Tāngata whenua agreed to this in principle. The area was not one where they often fished, but was open to commercial fisheries. It was felt that closure of that area would not affect customary fishing rights unduly, and had potential benefits. This application was subsequently not pursued due to competing Departmental priorities.

In 2000, the Department revived the idea of a marine reserve at Aotea and recommenced consultation and investigations. In August 2004, a proposal was notified for a much larger marine reserve, which extended along around half of the east coast of the island, and out to the 12 nautical mile extent of the territorial sea. This would have had significant impacts on the customary fishing rights of

Ngāti Rehua, the hapū on Aotea. Despite opposition from Ngātiwai, the Department decided to progress the application.

Ngātiwai expressed concern to the Ministry of Fisheries, and held a series of hui throughout the rohe. A report from those hui was sent to the Minister of Fisheries. As well as the general issues of customary fishing rights, that report emphasised the importance of access to traditional fishing grounds for an island community. As a result, the Minister declined to give his concurrence, and the marine reserve application failed.

The Ngātiwai responses to marine protection illustrate the issues that arise for iwi in response to marine protection proposals. Customary rights can have high importance, but their nature is little understood by most non-Māori, and are also not easy for Māori to articulate. Only tāngata whenua have interests in all formally defined fishing sectors – customary, commercial and recreational; all of which can be impacted by marine protection measures. Further, iwi recognise the tension between commercial drivers and kaitiaki perspectives. Most other parties addressing marine protection projects come from sector groups such as commercial fishing or environmental non-governmental organisations. Iwi often need to resolve and reconcile internal tensions across these issues before addressing them publicly.

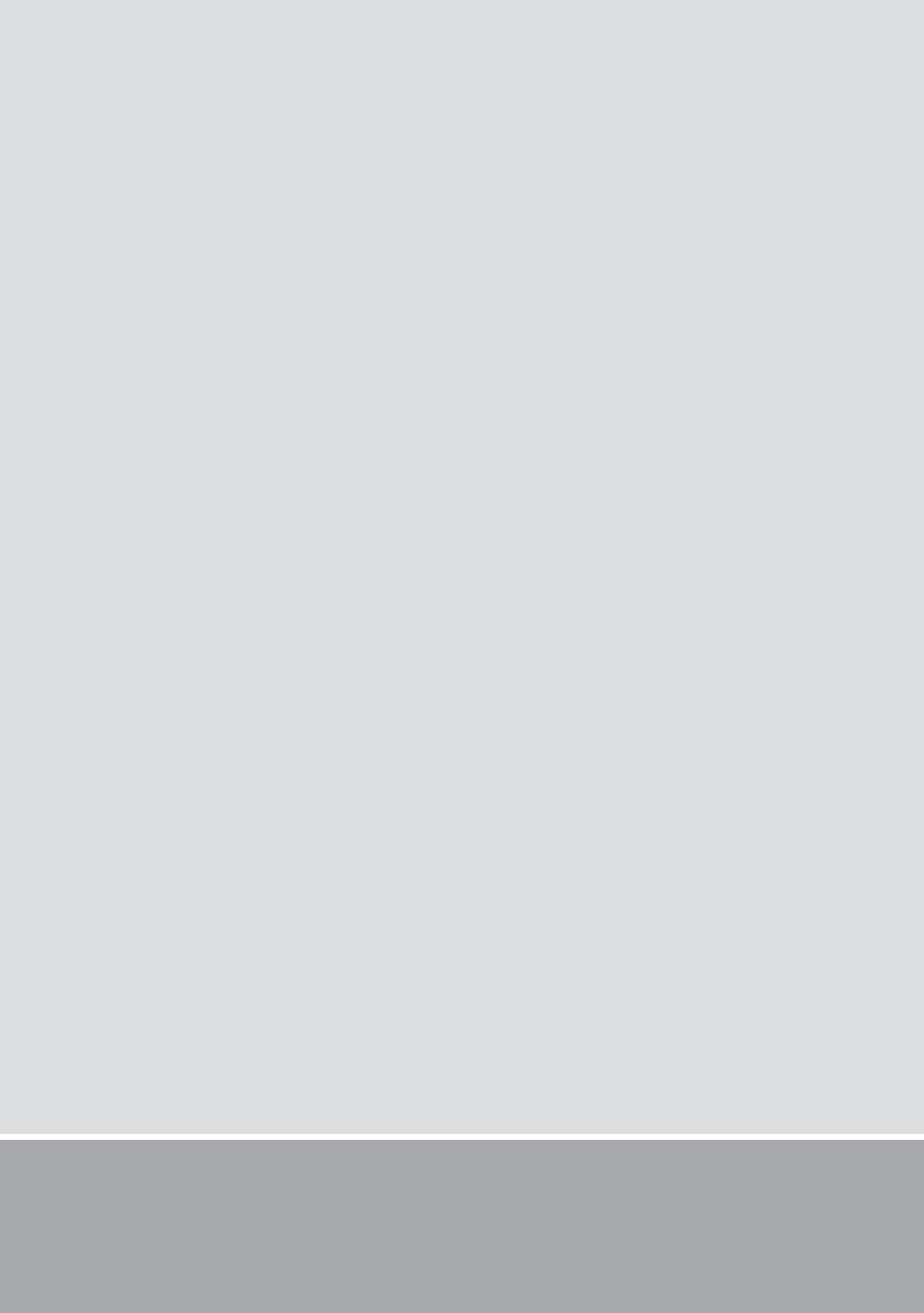
Further marine protection proposals

Currently Ngātiwai are responding to two further marine protection proposals in their rohe – conversion of a marine park at Mimiwhangata into a marine reserve; and a marine park with restricted take in the area between the Bay of Islands and Whangarei Harbour. In each case creative solutions to conflict resolution are possible and need to be sought.

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APPENDICES





New Zealand marine protected areas
principles and protection standard

Network design principles¹

1. The MPA network will protect examples of the full range of natural marine habitats and ecosystems.
2. MPAs should be designated based on a consistent approach to classification of habitats and ecosystems.
3. The MPA network should be viable.
4. National priorities for additions to the MPA network will be developed, and reviewed on an annual basis.
5. An evaluation programme will be undertaken.
5. A monitoring programme will be undertaken.

Planning principles

1. Every MPA should be designed on the basis that it is representative of one or more habitats or ecosystems, and in a manner consistent with the national network priorities and the MPA implementing principles.
2. The management tool(s) used at a site must be sufficient to meet the protection standard.
3. The special relationship between the Crown and Māori will be provided for, including kaitiakitanga, customary use and mātauranga Māori.
4. MPA establishment will be undertaken in a transparent, participatory, and timely manner.
5. Adverse impacts on existing users of the marine environment should be minimised in establishing MPAs.
6. The management tools used to establish MPAs should be consistent and secure in the long term, subject to any necessary changes to allow them to better achieve the MPA Policy objective, taking into account natural dynamics.
7. Best available information will be taken into account in decision-making.

8. Decision-making on management actions will be guided by a precautionary approach.
9. The MPA management regime must be enforceable.
10. MPA research will be effectively planned and co-ordinated.

Marine protected areas protection standard²

2.4 Maintenance and recovery of physical features and biogenic structures

This aspect of the MPA protection standard looks to ensure the seabed in an MPA is protected from physical damage. To ensure this, activities that may cause significant damage to the seabed and associated biodiversity should be prohibited from an MPA.

Whether a particular activity causes damage to the seabed will depend on the nature of the seabed. Those seabed habitats that are particularly fragile will be damaged more easily than those that are exposed to natural disturbance.

When considering the effect of fishing activity, the main fishing methods used in New Zealand waters were ranked according to the relative damage they cause to the seabed. As a result, bottom trawling, dredging and Danish seining were considered not to allow maintenance and recovery of physical features and biogenic structures. As such, the presumption is that these methods would not be permitted within an MPA.

Benthic netting and potting were considered to cause only moderate damage. These methods would be allowed in an MPA unless the seabed was comprised of particularly fragile biogenic habitats. Other methods may be deemed acceptable but would need also to be considered as part of the second half of the MPA protection standard.

2.5 Ecological systems, natural species composition and trophic linkages

This second aspect of the MPA protection standard looks to ensure any activity within an MPA does not unduly disturb ecological systems, natural species composition and trophic linkages.

When considering the effect of fishing activity, it is difficult to set a level of extraction that would ensure the MPA protection standard is met. Setting an acceptable quantity of extraction would require large amounts of information about the species present in an area and how they contribute to the associated ecological system. There are also considerable problems with compliance when setting catch limits at small spatial scales. Because of these difficulties, fishing methods have been used as a proxy for extraction from potential MPAs.

2.6 Coastal MPAs

It is considered that purse seining, midwater trawling, midwater gillnetting and benthic netting either extract large quantities of fish over short time periods and/or are relatively unselective in nature. Many of the species harvested by the methods in coastal areas could have close affinities to the benthic environment. As such, these methods will probably not be permitted within an MPA.

Other methods such as benthic longlining, potting, pelagic longlining and hook and line fishing do not generally extract such quantities of fish over short time frames and are more selective. These methods may be allowed within an MPA subject to the case by case analysis describe below.

Case by case analysis

A case by case approach is necessary for two reasons. First, using fishing methods as a proxy may not accurately reflect the actual extraction from a site as much depends on the frequency and intensity with which that method is used.

Second, there are statutory requirements in the Fisheries Act that mean such an analysis is necessary prior to any method prohibition.

The factors that would be considered in a case by case analysis are further outlined in the implementation section.

2.7 Deepwater MPAs

In November 2007, government established 17 Benthic Protection Areas, primarily in New Zealand's Exclusive Economic Zone (EEZ). These areas protect about 30% of the seabed in the EEZ. Because of the contribution these protected areas make to benthic protection, government has chosen not to implement the MPA Policy in the EEZ until 2013. Implementing the MPA Policy will concentrate on Territorial Sea until then.

Prior to implementing the MPA Policy in the EEZ, government will revisit both the classification system and protection standard to incorporate improved knowledge and research conducted between now and 2013.

Design guidelines used to identify and select potential protected areas³

Site identification and protected area design guidelines

Protect whole habitats and ecosystems – It is desirable that sites be selected on the basis that whole habitats or ecosystems can be protected, particularly where a habitat or ecosystem represents a relatively small mapped unit. For example it would be desirable to incorporate a whole reef in a protected area rather than establishing a boundary that cuts across a reef.

Size of protected areas – Protected areas may be of various shapes and sizes but should be of sufficient size to provide for the maintenance of populations of plants and animals. For the same amount of area to be protected it is desirable to protect fewer, larger areas rather than numerous smaller areas. This helps maintain healthy self-sustaining populations resilient to ‘edge effects’ resulting from use of the surrounding/adjacent areas. This also allows for more efficient and cost effective compliance and law enforcement.

Maximise connectivity – The design of the protected areas network should seek to maximise and enhance the linkages among individual protected areas, groups of protected areas within a given biogeographic region, and across biogeographic regions.

Represent latitudinal and longitudinal variation – Many processes create latitudinal and longitudinal (cross-shelf) differences in habitats and ecosystems. This diversity is reflected partly in the distribution of the biogeographic regions, but care should be taken to identify potential protected area sites that include differences in habitats and ecosystems that cover both latitudinal and longitudinal or cross-shelf ranges. It may be convenient to extend protected areas from the intertidal zone to deep waters offshore.

Consider sea and adjacent land uses in planning protected areas – Placement of protected areas should take into account the adjacent terrestrial environment (including islands) and associated human activities. Past and present uses may have influenced the integrity of biological communities, and designers should consider these effects, where known, when proposing the location of protected areas. For example, existing no-take protected areas and areas adjacent to terrestrial national parks are likely to have greater biological integrity than areas that have been used heavily for resource exploitation.

Keep boundaries simple and aim for low boundary to area ratio – To achieve this, protected area design should aim for simple shapes and reduced fragmentation of areas. This can be achieved by using straight boundary lines and minimising the perimeter-to-area ratio. Protected areas should also be designed so they can be realistically enforced. Users and surveillance staff find straight lines much easier to find and follow than lines following depth contours or distance from land or reefs. Squares are easier for users and compliance staff to find and work with than odd shapes. Boundaries should follow major latitude and longitude lines where possible. This makes it easier for users to match with charts. For coastal zones, clear sight lines on-shore or using other fixed objects are good alternatives to areas defined by coordinates.

Site selection guidelines

Primary considerations

Protect the full range of marine habitats and ecosystems – The MPA Policy calls for the protection of “the full range of marine habitats and ecosystems” as well as those which are rare, distinctive or internationally or nationally important. Within each biogeographic region, the approach to the classification of habitats and ecosystems should be used as a pragmatic guide to the representation needed to achieve this goal.

Cultural use – Consider information on traditional use, values, current economic value and Treaty settlement obligations.

Adverse impacts on users – Where there are choices of several sites that would add a similar ecosystem or habitat to the protected area network if protected, the site(s) chosen should minimise adverse impacts on existing users and Treaty

settlement obligations. Where there is a choice to be made among minimum impact sites, selection may also be guided by:

- Accessibility for management and enforcement requirements; and
- Benefits such as educational, diving and tourism opportunities.

Social and economic interests – When choosing among potential sites, information related to social and economic interests should be considered to minimise adverse impacts on existing users. Such information may include: current and potential use for the purpose of extraction or exploration, or contribution to economic or intrinsic value by virtue of its protection.

Secondary considerations

Number of protected areas – The number of potential habitat and ecosystem types, defined by the classification and mapped within a biogeographic region, does not equate to the number of protected areas required to protect the full range of natural marine habitats and ecosystems. Multiple habitats should be protected within each protected area.

Have fewer larger (versus numerous smaller) protected areas – It is beneficial to have fewer larger protected areas representative of more than one habitat or ecosystem than a large number of small protected areas.

Susceptibility to degradation – Incorporate information on the location of, for example, coastal structures, dredging or dumping sites that potentially may impact on the integrity of the site.

Compatibility with adjacent land-use – It is desirable to design protected area boundaries to align with other protected areas. This includes national parks on land and other protected waters, such as fish habitat. This allows opportunities for collaborative compliance efforts between agencies.

Replication – Consideration should be given to whether the site provides replication of habitats and ecosystems in a biogeographic region.

Tool selection guidelines

The size of the MPA – Larger MPAs will be more likely to compensate for any higher level of biological extraction when compared to smaller MPAs. As such, higher

quantities of biological extraction would be acceptable in larger MPAs compared to those of smaller size.

The likely level of biological extraction from an MPA (from all sources) – If the biological extraction from a potential MPA is having an adverse effect on the aquatic environment or creating a sustainability concern, then that level of extraction is not consistent with the protection objective of the MPA Policy. Method prohibitions would be put in place to increase the biomass to levels acceptable under the Fisheries Act.

The frequency of extraction – A method such as recreational line fishing may not extract large quantities of species on any one occasion. However, where such a method is used frequently and/or by a large number of people, this may lead to a similar result as would large scale methods such as trawling.

The type of species being extracted and its ecological importance – Because more mobile species cannot be constrained within the boundaries of MPAs, MPAs are better at protecting species that are sedentary or have limited mobility. For this reason, case by case analyses will consider those mobile species that have some seasonal affinity with the areas but will focus on sedentary species or those with limited mobility.

Endnotes

- 1 Extracted from Department of Conservation and Ministry of Fisheries, 2005, 16-20
- 2 Extracted from Ministry of Fisheries and Department of Conservation 2008, 10-12
- 3 Extracted from Ministry of Fisheries and Department of Conservation 2008, 19-22



Goals and principles for the
establishment of the National
Representative System of Marine
Protected Areas in Australian
Commonwealth waters

Goals

Goal 1: Each provincial bioregion occurring in the marine region should be represented at least once in the marine reserve network. Priority will be given to provincial bioregions not already represented in the National Representative System.

Goal 2: The marine reserve network should cover all depth ranges occurring in the region or other gradients in light penetration in waters over the continental shelf.

Goal 3: The marine reserve network should seek to include examples of benthic/demersal biological features (for example, habitats, communities, sub-regional ecosystems, particularly those with high biodiversity value, species richness and endemism) known to occur in the marine region at a broad sub provincial (greater than hundreds of kilometres) scale.

Goal 4: The marine reserve network should include all types of seafloor features. There are 21 seafloor types across the entire exclusive economic zone. Some provincial bioregions will be characterised by the presence of a certain subset of features, such as continental slope or seamounts

In developing options that meet the four goals, the following location principles will be applied:

1. Marine reserves will be located taking into account the occurrence and location of existing spatial management arrangements (for example, existing protected areas and sectoral measures) that contribute to the goals.
2. The goals should be met with the least number of separate marine reserves (that is, a smaller number of larger marine reserves rather than many small marine reserves) to maximise conservation outcomes.

Where different options that meet the Goals exist, the following selection principles should be considered in selecting areas suitable for inclusion in the National Representative System of marine protected areas:

3. The capacity of a marine reserve to mitigate identified threats to conservation values.
4. The occurrence of spatially defined habitats for and/or aggregations of threatened and/or migratory species.
5. The occurrence of ecologically important pelagic features which have a consistent and definable spatial distribution.
6. The occurrence of known small-scale (tens of kilometres) ecosystems associated with the benthic/demersal environment.
7. Relevant available information about small-scale distribution of sediment types and sizes and other geo-oceanographic variables.
8. Occurrence of listed heritage sites (where inclusion in the marine reserve network would improve administration of protection regimes).
9. Socioeconomic costs should be minimised.

Once the broad location of marine reserves has been determined, the following design principles should be applied to further refine the size and shape of individual marine reserves:

10. Individual areas should, as far as practicable, include continuous depth transects (for example, from shelf to the abyss).
11. Whole seafloor features (such as geomorphic features) should be included.
12. Features should be replicated wherever possible within the system of marine reserves (that is, included more than once).
13. Size and shape should be orientated to account for inclusion of connectivity corridors and biological dispersal patterns within and across marine reserves.

14. Boundary lines should be simple, as much as possible following straight latitudinal/longitudinal lines.
15. Boundary lines should be easily identifiable, where possible coinciding with existing regulatory boundaries.
16. The size and shape of each area should be set to minimise socioeconomic costs.

The following principles will be applied in developing the regional systems of marine reserves:

17. Zoning will be based on the Environment Protection and Biodiversity Conservation Act/IUCN categories of protection
18. The regional marine reserve network will aim to include some highly protected areas (IUCN Categories I and II) in each provincial bioregion.
19. Zoning will be based on the consideration of the threat that specific activities pose to the conservation objectives of each marine reserve.
20. Zoning of marine reserves will seek to ensure that the conservation objectives of the area are protected, taking into account a precautionary approach to threats as well as the relative costs and benefits (economic, social and environmental) of differing zoning arrangements.

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New Zealand is fortunate to have a very large and biodiverse marine area, hosting over 17,000 known marine species, with many thousands yet to be discovered. But this rich variety of marine life is coming under increasing pressure. Three endemic marine mammals are threatened, as are over sixty per cent of ocean-going seabirds and many species of sharks and marine invertebrates.

Historically, New Zealand has achieved significant success in marine protection. But today the country is far behind international best practice. Fiordland is the only mainland coastal biogeographic region which has more than one per cent of its area safeguarded by marine reserves. Several regions have no protection at all. Extensive benthic protection areas have been created in the exclusive economic zone, but these are not representative, and none are fully protected.

This report reviews New Zealand's current marine protection framework, developments in other countries, and international best practice. It identifies weaknesses in New Zealand's current approaches in the context of its international obligation to achieve a representative network of marine protected areas. The report then provides recommendations for the design of new marine protection legislation, which is essential if New Zealand is to effectively manage its oceans.

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